
Abstract Algebra

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Contents

Introduction	3
A Historical Overview	4
Groups	7
Power Rules	10
Group Morphisms	13
Groups and Order	14
Cayley Tables	15
Subgroups	18
Finite Subgroup	21
Cyclic Groups	22
Permutation and Cycles	30
Cycle Notation	32
Transpositions	38
Alternating Group	43
Group Homomorphism	45
Group Isomorphism	46
Cosets	51
Normal Subgroups and Factor Groups	54
Normal subgroup homomorphism & first isomorphism theorem	58
Rank-Nullity Theorem	63
Direct Product	67
Second Isomorphism Theorem	74
Third Isomorphism Theorem	80
Isomorphism Theorems as Commutative Diagrams	80
First Isomorphism Theorem	81
Second Isomorphism Theorem	82
Third Isomorphism Theorem	82
Group Action	83
Orbits and Stabilizers	86
Ring & Field Theory	95
Basic Definitions	95
Elementary Results	103
Ideals	107
Factor Rings	110
Ring Homomorphism	117

Field of Fractions	124
Polynomial Ring	127
Motivation	127
Construction	129
Zeros and Division	138
Polynomial Factorization	142
Content of a Polynomial and Primitive Polynomials	144
Unique Factorization	149
Divisibility in Integral Domains	152
Unique Factorization Domain	160
Extension Fields	167
Simple Extensions	174
Splitting Fields	179
Multiple Zeros of a Polynomial	184
Algebraic Extensions	189
Galois Theory	195
Galois Group	195
Fundamental Theorem of Galois Theory	206
Roots of Unity	211
Radical Extension	213
Insolvability of the Quintic	215
Fundamental Theorem of Algebra	222

Introduction

Presented below is abstract algebra at an undergrad level, covering most of the material usually thought in Abstract Algebra I and the first half of Abstract Algebra II curricula at the Math undergrad level. Namely, we discuss the following topics, starting from first principles:

1. Group theory
2. Ring theory
3. Field theory
4. Galois theory

The discussions all build towards the fundamental theorem of Galois theory and the insolvability of the quintic, and towards the fundamental theorem of Algebra. Other highlights include the first three isomorphism theorems, some useful counting techniques that deal with symmetry, classification of finite groups of prime order, and others.

The only real prerequisite here is linear algebra and elementary real analysis, the latter of which we use to prove the fundamental theorem of algebra, and the former of which we use throughout in many examples, and in proofs that deal with field extensions and Galois extensions. Elementary number theory knowledge also comes in handy, since we use modular arithmetic quite a lot and use some results that deal with primes.

Finally, it should be noted that these notes are based on many different resources, most significant of which are:

1. Prof. James Cook's [Abstract Algebra I](#) taught at Liberty University.
2. Prof. James Cook's [Abstract Algebra II](#), also taught at Liberty U.
3. Dr. Gadi Aleksandrowicz's writing on [abstract algebra](#) in his blog (Hebrew).
4. Abstract Algebra: Theory and Application by Prof. Thomas W. Judson, available freely at [LibreTexts](#).
5. [Contemporary Abstract Algebra](#) by Prof. Joseph Gallian.

A Historical Overview

Consider the following problem: *what number, when multiplied by itself and then divided by 5, gives twice the number?*

When faced with such a word problem, one trained in algebra immediately produces the following equation: $\frac{x^2}{5} = 2x$, which they then solve approximately as follows:

$$\frac{x^2}{5} = 2x \iff$$

$$x^2 = 10x \iff$$

$$x(x - 10) = 0 \iff$$

$$x = 0 \text{ or } x = 10$$

One then declares the solution to be either 0 or 10. This process - of taking a word problem and then phrasing it entirely in terms of symbols and arithmetic operations expressed using those symbols - is an *abstraction* of the problem to the realm of algebra, and has not always been the approach used to tackle such problems. Even though the term *algebra* dates back to the 9th century when it was coined by Al-Khwarizmi, it was not until over 700 years later that algebra in its symbolic form that is thought today become the primary form of algebra. This fully symbolic algebra was presented by Viète in 1591, and later expanded by Descartes in 1637.

Once fully symbolic algebra has been formalized, mathematicians could forgo word problems and instead focus on solving symbolic equations, also known as algebraic equations or *polynomial equations*. Indeed in those centuries before the development of the theories which would later be unified under the umbrella term of abstract algebra, algebra itself would be defined as *the study of polynomials*.

Solving algebraic equations was done either through direct manipulation of the equation (as we did in our first example) or using explicit formulas that produce the solution to algebraic equations of a given form. The formula for the solutions of a second degree polynomial, known as the *quadratic formula*, is perhaps the second most well known mathematical formula, second only to Pythagoras' theorem

$$ax^2 + bx + c = 0 \iff x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Given that $a \neq 0$ (but if $a = 0$ then this is not a quadratic equation, so this isn't a limitation). Deriving this formula is rather straightforward. One way to derive it is presented below:

$$\begin{aligned} ax^2 + bx + c &= 0 \\ x^2 + \frac{b}{a}x + \frac{c}{a} &= 0 \\ x^2 + \frac{b}{a}x &= -\frac{c}{a} \\ x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} &= -\frac{c}{a} + \frac{b^2}{4a^2} \\ \left(x + \frac{b}{2a}\right)^2 &= \frac{b^2 - 4ac}{4a^2} \\ x + \frac{b}{2a} &= \pm \frac{\sqrt{b^2 - 4ac}}{2a} \\ x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \end{aligned}$$

Notice that this derivation was performed via algebraic manipulations of the first equation. Each transition is justified from the properties of $\mathbb{R}$ (suppose this is a polynomial over the reals) and from the fact that certain operations - such as multiplication by a non-zero value and addition - maintain the equality. We intentionally do not dissect each transition with proper rigor since we reserve such discussion for the proper introduction of groups and fields.

Turns out that via algebraic manipulation, one can also derive a formula for the solution of a cubic polynomial - polynomial of degree 3. In fact, there exists one for quartic polynomials as well - these have degree 4. Attempting to devise a similar method for degree 5 (quintic) polynomials remained an open problem in algebra. The problem was eventually solved by the young Evariste Galois in the 19th century, who used a new algebraic structure called *groups* to reason about polynomials. Galois eventually concluded that no such formula exists. This was one of the first steps in the development of *group theory*. Around the same time, first steps towards *ring theory* and *field theory* were also made. These three theories form the pillars of *abstract algebra* or *modern algebra*. Abstract algebra is to algebra as algebra is to word problems - *an abstraction*. Instead of discussing a concrete algebraic equation, we consider algebraic *structures* such as groups, rings and fields, that satisfy certain *axioms*, and from those we produce powerful results which we can later apply to a particular class of algebraic problems.

It is important to note that these structures are defined in such a way that the results derived apply to domains outside of what might be consider classical algebra. For example, the symmetries of a

geometric shape which are geometric in nature turn out to be a *group*, and thus also algebraic in nature.

Coming up with the right abstractions is difficult - too abstract and you cannot draw interesting conclusions, too specific and you lose generality. To find the right abstraction, you first have to consider many different *instantiations* and draw similarities between them, concocting the right abstractions after a long and iterative process. This is an important fact in any field of mathematics: take for example the topic of continuity. Discussions on continuity usually start with a definition, followed by a list of results. The definition for a continuous function in topology is “a function $f : A \rightarrow B$ is *continuous* if for every open set in the image of f , its pre-image is also open”. At first, the definition seems arbitrary and not particularly interesting, and it is only through seeing the results on continuous functions in a topological space that the reader becomes convinced that it is indeed a useful (in the sense that it generalizes key properties of many useful results) and good (in the sense that it makes derivation of those results relatively simple and clean) definition, but to come up with a good definition one must first have a list of results, thus the mathematical process is oftentimes presented in reversed order.

All of this is to say that, while our discussion will follow the usual form of *definition* $\rightarrow$ *results* $\rightarrow$ *new definition* $\rightarrow$ *results* etc, it is important to remember that the definitions are exactly the necessary abstractions for the derivation of the results that follow them, i.e. historically results *precede* definitions. However, once we have good abstractions, we can shift our focus to the study of the abstractions themselves, producing powerful new results.

Groups

Introductory courses in abstract algebra usually start with groups, however I personally prefer to start with monoids.

Definition 0.0.1 (monoid).

A monoid is a structure $(S, \cdot)$ where S is a set and $\cdot$ is a binary operation on the set (usually referred to as multiplication or addition), such that it satisfies the following properties:

1. (Identity) $\exists e \in S$ s.t. $\forall s \in S, e \cdot s = s \cdot e = s$.
2. (Associative) $\forall a, b, c \in S, a \cdot (b \cdot c) = (a \cdot b) \cdot c$

Note that here we use multiplicative notation, but it is also customary to consider additive notation, then we say that $(S, +)$ is a monoid if it has an identity e s.t. $\forall s \in S, e + s = s + e = s$ and write associativity as $a + (b + c) = (a + b) + c$.

Proposition 1 (uniqueness of the identity).

Given a monoid $(S, \cdot)$, its identity element is unique.

Proof. Take $e_1, e_2 \in S$ s.t. both are identities, not necessarily distinct (since $(S, \cdot)$ is a monoid there must exist an identity), then $e_2 = e_1 \cdot e_2 = e_1$ since both are left and right identities, so $e_1 = e_2$, so the identity element is unique. $\square$

Example 0.1.

$(\mathbb{Z}, \cdot)$ where $\cdot$ is the usual multiplication operation on $\mathbb{Z}$ is a monoid.

Proof. We know that 1 is the multiplicative identity on $\mathbb{Z}$, so we have an identity. Next, we know that multiplication on $\mathbb{Z}$ is associative. We conclude that $\mathbb{Z}$ is a monoid under $\cdot_{\mathbb{Z}}$. $\square$

By requiring a third property from the structure, we devise a group:

Definition 0.0.2 (group).

A group is a monoid $(G, \cdot)$ such that:

1. (Inverse) $\forall a \in G \exists a^{-1} \in G$ s.t. $a^{-1} \cdot a = e$.

Proposition 2 (group multiplication is bijective).

Given a group $(G, \cdot)$, then the partial mapping $\cdot_a : x \mapsto a \cdot x$ is a bijection for some $a \in G$.

Proof. Let $h = \cdot_a$. Take any $y \in \text{domain}(h) = G$. Since G is a group, $a^{-1} \in G$ and $a^{-1}y \in G$ since G is closed under multiplication, and $h(a^{-1}y) = a(a^{-1}y)$ by definition, which gives y , so h is surjective.

Take $x_1, x_2 \in G$ s.t. $h(x_1) = h(x_2)$, i.e. $ax_1 = ax_2$. Since $a \in G$ and G is a group then $a^{-1} \in G$, so pre-multiplying both sides by a^{-1} (this preserves the equality by the substitution principle which we discussed at length in [Introduction to Formal Logic](#), we get $a^{-1}(ax_1) = a^{-1}(ax_2)$, and by associativity we eventually get $x_1 = x_2$, so h is *injective*.

We conclude that h is bijective.

□

Proposition 3 (properties of the inverse).

1. The inverse is both a left inverse (as defined above) and a right inverse.
2. The inverse is unique.
3. The inverse is an involution, i.e. $(a^{-1})^{-1} = a$.
4. $(ab)^{-1} = b^{-1}a^{-1}$ (socks-shoes property)

Proof. 1 : Let $a \in G$ and let $a^{-1} \in G$ be inverse of a , so $a^{-1}a = e$. Now consider aa^{-1} multiplied by itself, i.e. $(aa^{-1})(aa^{-1})$, by associativity we have $(aa^{-1})(aa^{-1}) = a(a^{-1}a)a^{-1} = aea^{-1} = aa^{-1}$, so we have $(aa^{-1})(aa^{-1}) = aa^{-1}$, so aa^{-1} acts as the identity element on aa^{-1} , and by **1 uniqueness of the identity** we have $aa^{-1} = e$, so by definition a^{-1} is the right inverse of a . This shows that left inverse implies right inverse. To show right inverse implies left inverse, we follow a similar argument and write $(a^{-1}a)(a^{-1}a) = a^{-1}(aa^{-1})a = a^{-1}ea = a^{-1}a$. As a result we can speak of *an inverse* in both cases and we don't need to specify if we are referring to a right inverse or a left inverse.

2 : Let $a \in G$ and let $x, y \in G$ be inverses of a not necessarily distinct, then from (1) we know that $ax = e = ay$. By **2 group multiplication is bijective** it follows that $x = y$.

3 : Consider $a^{-1}a = e$, then a^{-1} is the left inverse of a which exists by the group axioms. Similarly by definition, a is the right inverse of a^{-1} , and by (1) this implies that $(a^{-1})^{-1} = a$.

4 : Let $ab \in G$. Consider $(ab)(b^{-1}a^{-1})$. From associativity, $(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = aea^{-1} = aa^{-1} = e$, so $b^{-1}a^{-1}$ is an inverse of ab , and from uniqueness of the inverse $(ab)^{-1} = b^{-1}a^{-1}$.

□

Corollary 0.1.

$(G, \cdot)$ is a group if and only if G has a left identity and left inverse for every element under $\cdot$ and $\cdot$ is associative on G .

Proof. $\implies$ is immediate from the definition of a group. $\Leftarrow$ We will show that a left identity implies a right identity given an associative operator with a left inverse. Let e be the left identity on G , then $\forall a \in G, ea = a$. Now consider $aea = a(ea) = aa$ and also $aea = (ae)a$ so we have $(ae)a = aa$. Taking the left inverse a^{-1} and post-multiplying both sides we get $(ae)aa^{-1} = aaa^{-1}$, and by **3 properties**

of the inverse we know that a^{-1} is also a right inverse, so $aa^{-1} = e$, so $(ae)aa^{-1} = ae = aaa^{-1} = a$, so $ae = a$, so e is also a right identity. We see that $(G, \cdot)$ has an identity in e (both left and right), an inverse for every element, and has the associative property by supposition, so it is a group. $\square$

Example 0.2 (groups).

1. From the properties of $(\mathbb{Z}, \cdot)$, we know that $1 \in \mathbb{Z}$ is the identity element, that $\cdot$ is associative, and that $\forall a \in \mathbb{Z}$, its reciprocal $a^{-1} \in \mathbb{Z}$ s.t. $a^{-1}a = 1$, so $(\mathbb{Z}, \cdot)$ satisfies all group axioms, so we conclude it to be a group.

2. Let $GL_n(\mathbb{R})$ be the set of all $n \times n$ matrices with scalars from $\mathbb{R}$ such that $\forall A \in GL_n(\mathbb{R})$, A is non-singular (i.e. $|A| \neq 0$). From linear algebra we know that $|AB| = |A| \cdot |B|$ so it follows that if $|A| \neq 0$ and $|B| \neq 0$ then $|AB| \neq 0$ so $AB \in GL_n(\mathbb{R})$ for every $A, B \in GL_n(\mathbb{R})$ so it is closed under matrix multiplication, i.e. matrix multiplication can be thought of as a binary operator on the elements of $GL_n(\mathbb{R})$. We also know that $|A|$ if and only if A is invertible, i.e. $\exists A^{-1} \in GL_n(\mathbb{R})$ s.t. $A^{-1}A = I$, where I is the identity $n \times n$ matrix and multiplication is the usual definition for multiplication of matrices, so we have a left inverse. We also know that for every matrix A , $IA = AI = A$, so I is the identity element of matrix multiplication. Similarly, we know that $A(BC) = (AB)C$ (if multiplication on those matrices is defined, which is always the case if A, B, C all have the same dimensions, which is the case in $GL_n(\mathbb{R})$), so we have associativity. Thus we conclude that $GL_n(\mathbb{R})$ is a group under the usual matrix multiplication operation.

It is important to note that even though $(a = e) \implies (a = a^{-1})$, the converse is not true, i.e. $(a = a^{-1}) \implies (a = e)$. As an example, consider the subset of **involutory matrices** of $GL_n(\mathbb{R})$, clearly it is not just I_n , yet all such matrices satisfy $A = A^{-1}$.

Theorem 1 (cancellation law).

Let G be a group and $a, b, c \in G$, then $(ab = ac) \implies (b = c)$ and $(ba = ca) \implies (b = a)$.

Proof. Suppose $ab = ac$, then pre-multiply by a^{-1} and from associativity we have $a^{-1}ab = (a^{-1}a)b = eb = b$ and also $a^{-1}ac = (a^{-1}a)c = c$, so $b = c$. To show $(ba = ca) \implies (b = a)$ we post multiply each side with a^{-1} and get $b = a$. $\square$

Definition 0.0.3 (abelian group).

A group is abelian or commutative if it also satisfies the following property:

1. (commutative) $\forall x, y \in G, x \cdot y = y \cdot x$.

Proposition 4.

Let G be a group. If $\forall a, b, c \in G, (ab = ca) \implies (b = c)$, then G is abelian.

Proof. Let $a, b \in G$ and substitute a for a^{-1} , b for ab and c for ba in $(ab = ca) \implies (b = c)$, then we get $(a^{-1}(ab) = (ba)a^{-1}) \implies (ab = ba)$. Consider $a^{-1}(ab) = (ba)a^{-1}$, by associativity we have $(a^{-1}a)b = b(aa^{-1})$, so $b = b$, so the LHS of the implication chain is true, so by MP we have $ab = ba$, so by definition G is abelian. $\square$

Example 1.1 (abelian and non-abelian groups).

1. $(\mathbb{Z}, \cdot)$ is an abelian group since $\cdot$ is commutative.
2. $GL_n(\mathbb{R})$ is not a group under matrix multiplication since generally matrix multiplication does not commute.

Power Rules

Let G be a group and consider some $a \in G$, and assume multiplicative notation. Since G is a group and $a \in G$, then $a \cdot a \in G$, and similarly $(a \cdot a) \cdot a \in G$, etc. Also from associativity, it follows that we can lose the parenthesis and write $(a \cdot a) \cdot a = a \cdot a \cdot a$. We can keep playing this game and multiply $a \cdot a$ with itself, and get $a \cdot a \cdot a \cdot a$, or $a \cdot a \cdot a$ with a and get $a \cdot a \cdot a \cdot a$. This form of writing is rather tedious, so we introduce power notation:

Definition 0.0.4 (power notation).

Let G be a group and let $a \in G$, then a^n for $n \in \mathbb{N}$ is defined recursively as follows:

1. $a^0 = e_G$
2. $a^{n+1} = a^n a$
3. $a^{-n} = (a^{-1})^n$

From this definition, it quickly follows that $a^{-n+1} = a^{-n}a$:

Proposition 5.

Let $a \in G$ a group and let $n \in \mathbb{N}$, then $a^{-n+1} = a^{-n}a$.

Proof. Let G be a group, $a \in G$ and $n \in \mathbb{N}$. By definition $a^{-n} = (a^{-1})^n$. Consider two cases:

1. $0n$, then $a^{-n} = (a^{-1})^n = (a^{-1})^{n-1}a^{-1}$, post-multiplying both sides by a and using the associative property of the group operation we have $a^{-n}a = (a^{-1})^{n-1}a^{-1}a = (a^{-1})^{n-1}(a^{-1}a) = (a^{-1})^{n-1}$, again by definition we have $(a^{-1})^{n-1} = a^{-n+1}$, so $a^{-n}a = a^{-n+1}$.
2. $0 = n$, then $a^{-n+1} = a^{-0+1} = a$, and also $a^{-n} = a^{-0} = a^0 = e$, so $a^{-n+1} = a = ea = a^0a = a^{-n}a$,

$\square$

It can be shown that all regular power rules apply:

Theorem 2 (exponent laws (power rules)).

Let G be a group and $a, b \in G, m, n \in \mathbb{Z}$, then:

1. $a^m a^n = a^{m+n} = a^n a^m$
2. $(a^{-1})^n = (a^n)^{-1}$
3. $(a^m)^n = a^{mn} = (a^n)^m$
4. $(ab)^{-n} = (b^{-1}a^{-1})^n$, and if G is abelian then $(ab)^n = a^n b^n$. This condition may be relaxed to requiring only that a, b commute under the group operation of G .

Proof. 1. Suppose $a^m a^n = a^{m+n}$, since addition in $\mathbb{Z}$ commutes then $a^{m+n} = a^{n+m}$ and by supposition $a^n a^m = a^{n+m}$ so $a^n a^m = a^m a^n$, which takes care of the second part of this proposition. To show $a^m a^n = a^{m+n}$, consider the following:

Suppose $m, n \in \mathbb{N}$, then we prove by induction on n . If $n = 0$ then $a^m a^n = a^m a^0 = a^m e = a^m = a^{m+0}$, similarly if $n = 1$ then $a^m a^n = a^m a^1 = a^m a$ which by definition is $a^{m+1} = a^{m+n}$ so the theorem holds. Suppose it holds for some n and consider $n + 1$, then $a^m a^{n+1} = a^m a^n a^1$, so by induction hypothesis $a^m a^n a^1 = a^{m+n} a^1$, and by definition $a^{m+n} a^1 = a^{m+n+1}$, which completes the proof by induction for the case where $m, n \in \mathbb{N}$.

Suppose $m, n \in \mathbb{Z}$. If $0 \leq m, n$ then this reduces to the case we have just proven, so either both $m, n \geq 0$ or one is negative and the other is non-negative. Consider two cases:

1. $m, n \geq 0$, then $\exists m', n' \in \mathbb{N}$ s.t. $m = -m', n = -n'$, so $a^m = a^{-m'}$ and $a^n = a^{-n'}$, so by **0.0.4 power notation** we have $a^n = (a^{-1})^{n'}$, $a^m = (a^{-1})^{m'}$. Let $b = a^{-1}$, then $a^m a^n = b^{m'} b^{n'}$ with $m', n' \in \mathbb{N}$ which reduces to the previous case, so $a^m a^n = b^{m'+n'} = (a^{-1})^{m'+n'}$ which by definition is $a^{-m'-n'}$, which is just a^{m+n} , which proves the theorem for the case when $m, n \geq 0$.

2. One of (m, n) is negative and the other is not, w.l.o.g suppose $m \geq 0$ and $n \leq 0$, then we can proceed by induction on n as we did for the case when $n, m \in \mathbb{N}$ using **5**.

So in all cases for m, n integers, we find that $a^m a^n = a^{m+n}$, which completes the proof.

2. Let $n \in \mathbb{Z}$. If $0 \leq n$, then we prove by induction on n . For $n = 0$ we have $(a^{-1})^0 = e = e^{-1} = (a^0)^{-1}$. Suppose by induction the proposition holds for some n , and consider $n + 1$, then a^{n+1} is by definition $a^n a$, and by **3 properties of the inverse** we have $(a^{n+1})^{-1} = (a^n a)^{-1} = a^{-1} (a^n)^{-1}$. By the induction hypothesis $(a^{-1})^n = (a^n)^{-1}$, so $a^{-1} (a^n)^{-1} = a^{-1} (a^{-1})^n$, which by definition is $(a^{-1})^{n+1}$, so $(a^{n+1})^{-1} = (a^{-1})^{n+1}$, which completes the proof by induction for $0 \leq n$.

Suppose then that $n \leq 0$, then $(a^{-1})^n = ((a^{-1})^{-1})^{|n|}$ by definition. Write $a^{-1} = b$, then we have $(b^{-1})^{|n|} = (b^{|n|})^{-1}$ (by the previous case), but $b^{|n|} = (a^{-1})^{|n|}$ which by definition is a^n (since $n \leq 0$), so $(a^{-1})^n = (b^{-1})^{|n|} = (b^{|n|})^{-1} = (a^n)^{-1}$.

3. As with the previous proposition we first handle the second part. Suppose $(a^m)^n = a^{mn}$, then since multiplication in $\mathbb{Z}$ commutes $a^{mn} = a^{nm}$, and by supposition $(a^n)^m = a^{nm}$, so $(a^n)^m = (a^m)^n$. We now turn our attention to the first part.

Suppose $n \in \mathbb{N}$, we prove by induction on n . If $n = 0$ then $(a^m)^0 = e = a^0 = a^{mn}$, if $n = 1$ then $(a^m)^1 = (a^m)^1 = a^m = a^{mn}$. Suppose the proposition is true for some $n \in \mathbb{N}$ and consider $n + 1$. By definition $(a^m)^{n+1} = (a^m)^n a^m$, by the induction hypothesis $(a^m)^n = a^{mn}$ so $(a^m)^n a^m = a^{mn} a^m$, and by proposition (1) we have $a^{mn} a^m = a^{mn+m} = a^{m(n+1)}$, which completes the proof.

Now suppose $n \notin \mathbb{N}$, then $n < 0$. If both m, n are negative, then $a^m = a^{-|m|}$ and $a^n = a^{-|n|}$. By definition, $a^m = a^{-|m|} = (a^{-1})^{|m|}$. By proposition (2), $(a^{-1})^{|m|} = (a^{|m|})^{-1}$, so $(a^m)^n = ((a^{|m|})^{-1})^{-|n|}$, which by definition is $((a^{|m|})^{-1})^{-|n|} = (a^{|m|})^{|n|}$, which simplifies to $(a^{|m|})^{|n|}$. Since $0 \leq |m|, |n|$ we can use the result from the inductive proof and get $(a^{|m|})^{|n|} = (a^{|m|})^{|n|} = a^{|m||n|}$, but since $m, n < 0$ then $mn = |m||n|$ so $(a^m)^n = a^{mn}$.

Finally, suppose $n < 0$ and $0 \leq m$, then we write $(a^m)^n = (a^m)^{-|n|} = ((a^m)^{-1})^{|n|} = (*)$. Using proposition (2) again, we get $(*) = (a^{-m})^{|n|} = ((a^{-1})^m)^{|n|}$, and now the result for natural exponents applies and we get $((a^{-1})^m)^{|n|} = (a^{-1})^{m|n|}$, which by definition is $a^{-m|n|} = a^{m \cdot (-|n|)} = a^{mn}$.

4. Let $n \in \mathbb{Z}$. If $0 \leq n$ then by definition $(ab)^{-n} = ((ab)^{-1})^n$, and by **3 properties of the inverse** we know that $(ab)^{-1} = b^{-1}a^{-1}$, so $(ab)^{-n} = (b^{-1}a^{-1})^n$. Otherwise, $n < 0$ so we use the fact that $((ab)^{-1})^{-1} = ab$ and write $a^{-n} = (((ab)^{-1})^{-1})^{-n} = (*)$ and since $n < 0$ then $0 - n$ so by definition $(*) = ((ab)^{-1})^n$, and finally we substitute $(ab)^{-1}$ with $b^{-1}a^{-1}$ and get $(ab)^{-n} = (b^{-1}a^{-1})^n$.

We turn our attention to the second part of the proposition. Suppose G is abelian and $n \in \mathbb{Z}$. If $0 \leq n$, we proceed by induction. For $n = 0$ we have $(ab)^0 = e = ee = a^0 b^0$, similarly for $n = 1$ we have $(ab)^1 = ab = a^1 b^1$. Suppose the proposition is true for some n and consider $n + 1$. By definition, $(ab)^{n+1} = (ab)^n (ab)$ and since G is abelian $ab = ba$ so $(ab)^n (ab) = (ab)^n (ba)$, now from the induction hypothesis and associativity we have $(ab)^n = a^n b^n b a = a^n b^{n+1} a$, and by commutativity $(a^n b^{n+1}) a = a(a^n b^{n+1})$ which by associativity is equivalent to $(a a^n) b^{n+1} = a^{n+1} b^{n+1}$, so the proposition holds for $n \in \mathbb{N}$. Suppose then that $n < 0$, then $ab^n = (b^{-1}a^{-1})^{-n}$ by the first part of the proposition, with $-n \in \mathbb{N}$, so by the inductive argument we have $(b^{-1}a^{-1})^{-n} = (b^{-1})^{-n} (a^{-1})^{-n}$, and by definition we have $(b^{-1})^{-n} = b^n$ (since $n < 0$) and also $(a^{-1})^{-n} = a^n$, so $ab^n = b^n a^n$, and finally by the commutative property we have $ab^n = a^n b^n$, which completes the proof for the second part of the proposition. $\square$

Note that in this section, we assumed the group operation is multiplicative, but the results obviously hold for additive group operation with a change of notation.

Definition 0.0.5 (additive notation).

Let G be a group with an additive operation $+$, then $n \cdot a$ (also juxtaposed as na) for $n \in \mathbb{N}$ is defined recursively as:

1. $0 \cdot a = e$
2. $(n + 1) \cdot a = n \cdot a + a$
3. $(-n) \cdot a = n \cdot (-a)$

The proof for **Theorem 2 exponent laws (power rules)** is identical, with a change in notation (for example, $a^m a^n$ becomes $ma \cdot na$).

Group Morphisms

Group morphisms are mappings between two groups that preserve the group structure.

Definition 0.0.6 (group homomorphism).

A group homomorphism is a function $\varphi : G \rightarrow H$ where $(G, \cdot)$ and $(H, \times)$ are groups, such that $\forall a, b \in G, \varphi(a \cdot b) = \varphi(a) \times \varphi(b)$.

Example 2.1.

Recall that $\mathbb{Q}$ is usually constructed as the quotient set of $\mathbb{Z} \times \mathbb{Z}^+$ under the equivalence relation $\langle a, b \rangle \sim \langle c, d \rangle \iff a \cdot d = b \cdot c$ (where multiplication on the RHS is multiplication as usually defined on $\mathbb{Z}$). Consider a function $E : \mathbb{Z} \rightarrow \mathbb{Q}$ given by $E : z \mapsto \langle z, 1 \rangle$. We have seen in [Introduction to Set Theory](#) that this function satisfies $E(z +_{\mathbb{Z}} w) = E(z) +_{\mathbb{Q}} E(w)$, so E is a group homomorphism from $(\mathbb{Z}, +_{\mathbb{Z}})$ and $(\mathbb{Q}, +_{\mathbb{Q}})$.

Proposition 6 (group homomorphism preserves identity and commutes with inverse).

Let e_G, e_H be the identity elements of G, H respectively, and let $\varphi : G \rightarrow H$ be a group homomorphism, then:

1. $\varphi(e_G) = e_H$
2. $\forall a \in G, \varphi(a^{-1}) = (\varphi(a))^{-1}$

Proof. 1 : Since φ is a group homomorphism then $\varphi(e_G e_G) = \varphi(e_G) \varphi(e_G)$, now since $e_G e_G = e_G$ we have $\varphi(e_G) = \varphi(e_G) \varphi(e_G)$, so by definition $\varphi(e_G)$ is the identity of H , and by uniqueness we have $\varphi(e_G) = e_H$.

2 : We have $\varphi(aa^{-1}) = \varphi(a) \varphi(a^{-1}) = e$, so by **3 properties of the inverse** we have that $\varphi(a^{-1})$ is the inverse of $\varphi(a)$, i.e. $\varphi(a^{-1}) = (\varphi(a))^{-1}$.

□

More generally, a homomorphism (or *morphism* for short) is a structure-preserving mapping, e.g. a ring homomorphism would be a function $\varphi : R \rightarrow T$ where both R, T are rings, such that it preserves the ring structure. We will present concrete examples when we define rings. We have already seen such examples in [Introduction to Set Theory](#).

Groups and Order

Definition 0.0.7 (order of a group).

Let G be a group, then the number of elements in the group, i.e. $|G|$, is called the order of the group. If G is not finite, then we write $|G| = \infty$.

If G is a finite group, then it must be that $\forall a \in G \exists z \in \mathbb{N}$ s.t. $a^z = e$, since otherwise the group would not be finite. For example in the finite group $\mathbb{Z}_3$ whose elements are 0, 1, 2 and the group operation is addition mod 3 (so we use additive notation), we have $e_{\mathbb{Z}_3} = 0$ and $3 \cdot 1 = 3 \equiv 0 \pmod{3}$, and $3 \cdot 2 = 6 \equiv 0 \pmod{3}$.

A formal argument for the general cause is given below:

Proposition 7 (order of elements in finite group is finite).

Let G be a finite group and $e \in G$ be the identity element, then $\forall a \in G \exists z \in \mathbb{N}$ s.t. $a^z = e$.

Proof. Let G be a finite group and let $|G| = n$, then G has exactly n distinct elements. Consider the sequence $(a^0, a^1, \dots, a^n)$. Since each element a^i is given by $a^i = a^{i-1}a$ with $a^0 = e$, it follows that all the elements in the sequence are also members of G , so it must be that not all are unique, since G has exactly n elements, so $\exists m, k \in \mathbb{N}, m, k \in [0, n], m \neq k$ s.t. $a^m = a^k$. Suppose w.l.o.g $k < m$, then post multiply both sides by a^{-k} (by power rules $a^{-k} = (a^{-1})^k$ and since $a \in G$ and G is a group, $a^{-1} \in G$ and also all its powers since it is closed under the group operation) yields $a^m a^{-k} = a^k a^{-k}$, which by exponent laws is equivalent to $a^{m-k} = a^{k-k} = a^0 = e$, and since $k < m$ it follows that $0 < m - k$, so $m - k \in \mathbb{N}$. Denote $z = m - k$, then $a^z = e$. $\square$

This motivates defining the *order of an element*.

Definition 0.0.8 (order of an element).

Let G be a group and $a \in G$, then the order of a , denoted $|a|$, is the least positive integer n s.t. $a^n = e$. If there isn't such n , then we say $|a| = \infty$.

Example 2.2 (order of 2 in $\mathbb{Z}_6^*$).

Recall that $\mathbb{Z}_6^*$ consists of the elements $\{1, 5\}$ (the positive integers up to 6 that are co-prime w.r.t 6). We will find the order of each element:

1. 1 has order 1 since $1^1 = 1$, and 1 is the multiplicative identity. In general for every group G , $|e_G| = 1$.
2. $[5^2] = [25] = [4 \cdot 6 + 1] = [1]$, so 5 has order 2 in $\mathbb{Z}_6^*$.

By **7 order of elements in finite group is finite** and by the well-ordering principle on $\mathbb{N}$ it follows that the order of an element is well-defined for finite groups.

Theorem 3.

Let G be a group and $a \in G$. Then:

1. If $|a| = \infty$ then $\forall i, j \in \mathbb{Z}, a^i = a^j \iff i = j$
2. If $|a| = n$ for some $n \in \mathbb{N}$, then $\forall i, j \in \mathbb{Z}, a^i = a^j \iff n|(i - j)$

Proof. 1. $\Leftarrow$ Suppose $i = j$, then $a^i = a^j$.

$\Rightarrow$ Suppose by contradiction $\exists i, j \in \mathbb{Z}$ s.t. $a^i = a^j$ and $i \neq j$, then $i - j \neq 0$. Since G is a group then $a^{-j} \in G$, and multiplying both sides by a^{-j} we have $a^i a^{-j} = a^j a^{-j}$, which by **Theorem 2 exponent laws (power rules)** simplifies to $a^{i-j} = a^{j-j} = a^0 = e$, and also $(a^{i-j})^{-1} = a^{j-i} = e$, so by **0.0.8 order of an element** $|a| \leq |i - j|$, but we assume $|a| = \infty$, a contradiction, so $i = j$.

2. $\Leftarrow$ Suppose $n|(i - j)$, then $\exists m \in \mathbb{Z}$ s.t. $i - j = mn$, so $a^{i-j} = a^{mn} = (a^n)^m = e^m = e$. Multiplying both sides by a^j we have $a^{i-j} a^j = e a^j$, i.e. $a^i = a^j$.

$\Rightarrow$ Suppose $a^i = a^j$, and multiply both sides by a^{-j} , so we have $a^{i-j} = a^0 = e$. Now by the division algorithm we know that $\exists! r, q \in \mathbb{Z}, q \in [0, n)$ s.t. $i - j = rn + q$. Substituting for $i - j$ we have $e = a^{i-j} = a^{rn+q} = (a^n)^r a^q = e^r a^q = a^q$, so $a^q = e$ and also $0 \leq q < n$, but since $|a| = n$ it cannot be that $0 < q < n$, so it must be that $q = 0$, so $i - j = rn$, so by definition $n|(i - j)$.

□

Corollary 3.1.

If $a^k = e$ for some $k \in \mathbb{Z}$, then $n|k$.

Proof. Recall $a^0 = e$. By **Theorem 3** $a^k = a^0 \implies n|(k - 0)$, so $n|k$.

□

Cayley Tables

In this section, we consider groups of finite order.

Definition 0.0.9 (trivial group).

The trivial group is a group which consists of a set with one element, denoted e , and a binary operation s.t. $e \cdot e = e$, i.e. it is the set that consists only of the identity element. We denote this group $\mathbb{Z}_1$. It has order 1.

Verifying that it is indeed a group is trivial: e is the identity element, e is its own inverse and $e \cdot e = e$ so $e \cdot (e \cdot e) = e \cdot e = (e \cdot e) \cdot e$, so it is associative, so $\mathbb{Z}_1$ is a group. It is also vacuously abelian.

Consider a set of two elements $\{a, b\}$. If we wish to create a group from this set, we need to choose one of the elements as the identity element, so we have a set $\{e, a\}$ where e is the identity under a binary

operation we haven't yet fully defined, however we only know that $e \cdot e = e$, $a \cdot e = e \cdot a = a$ (properties of the identity), so all that remains is to determine $a \cdot a$. Since $a \cdot a$ should be $\in \{e, a\}$, it can be either e or a . Suppose $aa = a$, then the function $x \mapsto ax$ is given by $e \mapsto a$, $a \mapsto a$, i.e. its range is $\{a\}$, which is $\neq \{e, a\}$, so it is not surjective, contradicting that **2 group multiplication is bijective**, so it cannot be the case that $aa = a$ if $(\{e, a\}, \cdot)$ is a group, then it must be that $aa = e$, and indeed we can verify that given the set $\{e, a\}$ and the multiplication operation $\cdot$ given explicitly by $ee = e$, $ae = a$, $ea = a$, $aa = e$, the structure $(\{e, a\}, \cdot)$ is indeed an abelian group and is the only group on two elements (up to an isomorphism).

Before we discuss groups of order three, we should introduce a nice way to visualize how elements of a finite group behave under the group operation: Cayley table.

Definition 0.0.10 (Cayley table).

A Cayley table is a two dimensional table similar to a multiplication table from elementary school, where each row and each column corresponds to an element of a finite group, and every (i, j) cell where i is the row and j is the column corresponds to the result of multiplying the element represented by row i with the element represented by row j .

More concretely, if G is a group of order n , then consider an ordering $G = \{a_1, \dots, a_n\}$ on its elements, then the Cayley table of G consists of a two-dimensional table T where the cell at $T_{i,j} = a_i a_j$.

Example 3.1 (Cayley table of finite group on two elements).

Let $G = \{e, a\}$, i.e. G is the finite group on two elements, then as described above we have $ee = e$, $ae = a$, $ea = a$, $aa = e$, so the Cayley table of G is given by

$\cdot$	e	a
e	e	a
a	a	e

Since G is a group of order 2 and we found that there ought to be exactly one such group (up to a relabeling), and $\mathbb{Z}_2^+$ is a group of order 2, then we find that G and $\mathbb{Z}_2^+$ are essentially the same, i.e. they are isomorphic: $\mathbb{Z}_2^+ \cong G$. We make this connection more formal later.

Cayley tables make it extremely easy to tell whether a group is abelian: the table is symmetric along the diagonal if and only if the group is abelian.

By **2 group multiplication is bijective**, it is apparent that if G is a group and $T(G)$ is its corresponding Cayley table, then each row and each column in $T(G)$ should have each element of G appearing exactly

once. Also, if we maintain that the first row and first column represent the identity element e , then clearly the value of each cell is already predetermined, since $ea = ae = a$ for every $a \in G$.

This motivates a Soduko-like approach to constructing the Cayley table of a group, however caution is warranted in this case: this restriction leaves too many DOFs - it does not capture the associative property of a group, hence it risks thinking of non-group structures as groups since their Cayley table seems to be valid, in the sense that each element appears exactly once along each row and each column. The reader should beware of such cases, and always verify the associative property is satisfied.

As an example of using a Cayley table to understand how a group looks like under the group operation, consider a finite group on 3 elements e, a, b . First, we fill the template for its Cayley table:

$\cdot$	e	a	b
e	e	a	b
a	a		
b	b		

The following restrictions are:

1. Of aa, ab , one should be b and the other e by bijectivity of left multiplication by a
2. Of ba, bb one should be a and the other e .

Suppose $aa = e$, but now we have $ab = b$ and $eb = b$ both in the last column, which violates the restriction that every element appear once, so this case is impossible, so it must be that $aa = b$, and then also $ab = e$.

Likewise, suppose $ba = a$, then $ba = a$ and two rows above it on the same column we have $ea = a$, which is not allowed for the same reason, so it must be that $ba = e$, and then also $bb = a$.

So, filling out the table, we find that there exists a unique group of order 3 up to an isomorphism, and its Cayley table looks like this:

$\cdot$	e	a	b
e	e	a	b
a	a	b	e
b	b	e	a

It is evident that this group is *non-abelian*, since the table is not symmetric along the diagonal. The

reader should also practice caution and verify that multiplication as described in this table is indeed associative on the elements of G .

Consider $|G| = 4$. Playing the same game, we find that two different Cayley tables representing valid groups, however for $|G| = 5$ we again find only one such table. So, there appears that there are two truly different finite groups of order 4, and again a single (up to isomorphism) finite group of order 5.

We spend a great deal of time later in order to create the machinery needed to classify finite groups, but for now the reader should wonder why it is the case that for $|G| = 2, 3, 5$, all of which are prime, we find that there really is only one such group (up to an isomorphism), while for $|G| = 4$, a composite number, this is not the case.

Subgroups

Consider the group $(\mathbb{Z}, +)$. Now consider the subset of $\mathbb{Z}$ that consists of only the even integers $\{2z | z \in \mathbb{Z}\}$ or equivalently via coset notation (which we will formally define soon) $2\mathbb{Z}$. Now consider the structure $(2\mathbb{Z}, +)$. Let us verify that it is indeed a group:

1. First we must verify $2\mathbb{Z}$ is closed under $+$: let $a, b \in 2\mathbb{Z}$, so $a = 2z, b = 2w$ for some $z, w \in \mathbb{Z}$, so $a + b = 2(z + w)$, and since $z + w \in \mathbb{Z}$ we have $2(z + w) \in 2\mathbb{Z}$, so it is closed under addition. This is the same claim as *the sum of evens is even*.
2. Identity: For $z = 0, 2z = 0$ so $0 \in 2\mathbb{Z}$, and $2z + 0 = 2z = 0 + 2z$, so 0 is the identity in $2\mathbb{Z}$.
3. Associativity: Since $+$ is associative on $\mathbb{Z}$ and $2\mathbb{Z} \subset \mathbb{Z}$, $+$ is also associative on $2\mathbb{Z}$.
4. Inverse: If $a \in 2\mathbb{Z}$ then $a = 2z$ for some integer z , and also $2(-z) \in 2\mathbb{Z}$ by definition and $2(-z) = -2z$, now by adding the two we get $2z + (-2z) = 0$, so $\forall a \in 2\mathbb{Z}$ we can find an inverse in $2\mathbb{Z}$.

We then conclude that $(2\mathbb{Z}, +)$ is a group, and call it a *subgroup* of $(\mathbb{Z}, +)$.

Definition 0.0.11 (subgroup).

Let G be a group and $H \subseteq G$. If H is a group under the group operation of G , then H is called a subgroup of G , denoted $H \leq G$.

Note that in our example we found that 0 , the additive identity of $(\mathbb{Z}, +)$, is also the additive identity of $(2\mathbb{Z}, +)$. This is always the case.

Proposition 8 (identity and inverse elements of a subgroup).

Let G be a group and $H \leq G$, then

1. $e_H = e_G = e$ (subgroup shares group's identity)
2. $\forall a \in H, a_G^{-1} = a_H^{-1} = a^{-1}$ (inverse of subgroup element is the inverse of the element in the group)

Proof. 1. Suppose by contradiction $e_H \neq e_G$, and let $a \in H$. Since e_H is the identity on H we have $e_H a = a$ and also $e_G a = a$ when considered in G , so $e_H a = a = e_G a$, so $e_H a = e_G a$, so by **Theorem 1 cancellation law** we have $e_H = e_G$, so we can write $e = e_H = e_G$.

2. Since $a_G^{-1} a = e_G$ and $a_H^{-1} a = e_H$, and by 1 $e_G = e_H = e$ so $a_G^{-1} a = a_H^{-1} a$, so again by **Theorem 1 cancellation law** $a_G^{-1} = a_H^{-1}$.

□

We now present two theorems for testing if a structure is a subgroup.

Theorem 4 (two-step subgroup test).

If $(G, \cdot)$ is a group and $H \subseteq G$, then $(H, \cdot)$ is a group if and only if:

1. H is not empty
2. H is closed under the group operation $\cdot$
3. $\forall a \in H, a^{-1} \in H$

Proof. $\implies$ is trivial. $\Leftarrow$ Suppose H is a non-empty subset of G . Since H is closed under the group operation, we can consider $\cdot$ restricted to H as a binary operator on H , so it remains to show that $(H, \cdot)$ is associative, has an identity and an inverse for every element:

1. Associativity follows directly from the associative property of $(G, \cdot)$ since both share the same operation.
2. Inverse is immediate from (3)
3. For identity, consider some $a \in H$ (H is nonempty so there is always an element a). From (3), $a^{-1} \in H$, so $aa^{-1} \in H$ from (2), but $aa^{-1} = e$, so $e \in H$, hence identity.

So $H \leq G$.

□

Theorem 5 (one-step subgroup test).

If $(G, \cdot)$ is a group and $H \subseteq G$, then $(H, \cdot)$ is a group if and only if:

1. H is not empty
2. $\forall a, b \in H, ab^{-1} \in H$.

More compactly, we say that H is a subgroup of G if and only if it is nonempty and closed under division (if we interpret the group operation in the multiplicative sense. If we interpret it in the additive sense, then H must be closed under subtraction).

Proof. $\implies$ Suppose $(H, \cdot)$ is a group, then $e \in H$ so H is nonempty. Also, $\forall a, b \in H, b^{-1} \in H$ by the inverse property, and so $ab^{-1} \in H$ since H is closed under the group operation.

$\Leftarrow$ Suppose H is a nonempty subset of G .

1. (Identity) Take $a, b \in H$ s.t. $a = b$, then by (2) $ab^{-1} \in H$, but $ab^{-1} = aa^{-1} = e$, so $e \in H$.
2. (Inverse) Take $a \in H$, then by (2) $ea^{-1} \in H$ but $ea^{-1} = a^{-1}$ so $a^{-1} \in H$.
3. (Closure) Take $a, b \in H$, since $b = (b^{-1})^{-1}$ then by (2) with $c = b^{-1}$ we have $ac^{-1} \in H$, but $ac^{-1} = a(b^{-1})^{-1} = ab$, so $ab \in H$, so H is closed under $\cdot$.
4. (Associative) Follows from the properties of $\cdot : G^2 \rightarrow G$.

□

Example 5.1.

$SL(n, \mathbb{R})$ is defined as $\{A \in \mathbb{R}^{n \times n} \mid \det A = 1\}$, i.e. the set of all $n \times n$ matrices with determinant 1. We will show that $SL(n, \mathbb{R}) \leq GL(n, \mathbb{R})$ using **Theorem 5 one-step subgroup test**. First, note that it is a subset of $GL(n, \mathbb{R})$ since $\det A = 1 \implies \det A \neq 0$. Next, we note that $\det I_n = 1$ so $I_n \in SL(n, \mathbb{R})$ so it is nonempty. Now take $A, B \in SL(n, \mathbb{R})$, then from linear algebra we know that B is invertible since it has non-zero determinant and also we know that $\det B^{-1} = (\det B)^{-1} = 1^{-1} = 1$. Now consider AB^{-1} , from linear algebra we know $\det AB^{-1} = \det A \det B^{-1} = 1 \cdot 1 = 1$, so $AB^{-1} \in SL(n, \mathbb{R})$, so by the one-step subgroup test $SL(n, \mathbb{R}) \leq GL(n, \mathbb{R})$.

Proposition 9 (intersection of subgroups is a subgroup).

If G is a group and $\mathcal{H} = \{H_1, \dots, H_n\}$ is a set of subgroups of G , then $\cap_{H_i \in \mathcal{H}} H_i$ is a subgroup of G .

Proof. Denote the intersection A . Since $\forall H_i \in \mathcal{H} (e \in H_i)$, then $e \in A$ and in particular A is nonempty. Let $a, b \in A$, then $\forall H_i \in \mathcal{H} (a \in H_i \wedge b \in H_i)$, and since each H_i is a group it follows that b^{-1} is in all of H_i so $b^{-1} \in A$. Similarly, ab is in all of H_i so $ab \in A$, so A is a nonempty subset of G , is closed under the group operation and has the inverse property, so by **Theorem 4 two-step subgroup test** $A \leq G$. □

We proceed with a couple of definitions which are examples of subgroups:

Definition 0.0.12 (center of a group).

Let G be a group, then $Z(G) = \{x \in G \mid \forall g \in G (xg = gx)\}$ is called the center of G .

Definition 0.0.13 (centralizer of an element).

Let G be a group, then $C(a) = \{x \in G \mid xa = ax\}$ is called the centralizer of a .

Proposition 10.

Let G be a group, then:

1. $Z(G) \leq G$ and is abelian
2. $C(a) \leq G$
3. $Z(G) = \cap_{a \in G} C(a)$

Proof. 1. Let $Z(G)$ be the center of G and let $g \in G$.

- a. (non-empty) Consider the identity e of G . Since $eg = ge$, then by definition $e \in Z(G)$, so $Z(G)$ is nonempty.
- b. (inverse) Let $b \in Z(G)$, then $\forall g \in G (bg = gb)$, and in particular $b(g^{-1}) = (g^{-1})b$ since $g^{-1} \in G$. Taking the inverse of each side, we have $(b(g^{-1}))^{-1} = (g^{-1})b$, which by properties of the inverse is $(g^{-1})^{-1}b^{-1} = b^{-1}(g^{-1})^{-1}$, which is just $gb^{-1} = b^{-1}g$, but since this is true $\forall g \in G$ then by definition $b^{-1} \in Z(G)$.
- c. (closure under group operation) Let $a, b \in Z(G)$ and let $x \in G$, then $ax = xa$ and $bx = xb$, so $(ax)b = (xa)b$, and $(ax)b = a(xb) = a(bx) = (ab)x$, so $(ab)x = (xa)b = x(ab)$, so $ab \in Z(G)$, so $Z(G)$ is closed under the group operation.

So by **Theorem 4 two-step subgroup test**, $Z(G) \leq G$.

Now we show $Z(G)$ is abelian. Let $a, b \in Z(G)$, then since $a, b \in G$ it follows that for $g = b$, $ab = ba$ by definition of the center of a group, so by definition $Z(G)$ is abelian.

2. Let $a \in G$ and let $C(a)$ be the centralizer of a in G . The proof is analogous to (1), only now we fix the group element under consideration.

3. We prove by mutual inclusion: Let $x \in Z(G)$, then $\forall a \in G, xa = ax$, so by definition of $C(a)$ we have $\forall a \in G (x \in C(a))$, so $Z(G) \subseteq \cap_{a \in G} C(a)$. Now let $x \in \cap_{a \in G} C(a)$, so again by definition we have that $xa = ax$ for each $C(a)$, so $\forall a \in G (xa = ax)$, so by definition $x \in Z(G)$, so $\cap_{a \in G} C(a) \subseteq Z(G)$. Thus we conclude by mutual inclusion that $Z(G) = \cap_{a \in G} C(a)$.

□

Finite Subgroup

A finite subgroup is a subgroup that is also finite. We can refine our one-step subgroup test for the case of a finite group as follows:

Theorem 6 (one-step subgroup test for finite group).

Let G be a finite group and $H \subseteq G$, then H is a subgroup of G if and only if:

- 1. H is nonempty
- 2. $\forall a, b \in H, ab \in H$ (i.e. H is closed under the group operation)

Proof. $\implies$ is immediate. $\impliedby$ Suppose H is a nonempty subset of G closed under the group operation of G . Since G is finite then $\forall a \in G$ the order of a is finite, so $\exists n \in \mathbb{N}^+$ s.t. $a^n = e$, so since H is closed under multiplication we have $a^n = e \in H$. Similarly, since $0n$ then $a^{n-1} \in H$, and since $a^{n-1} = a^n a^{-1}$ by **Theorem 2 exponent laws (power rules)** and $a^n = e$ it follows that $a^{n-1} = a^{-1} \in H$,

so $\forall a \in H \exists a^{-1} \in H$ s.t. $aa^{-1} = e = a^{-1}a$. Now let $a, b \in H$, then $b^{-1} \in H$, so $ab^{-1} \in H$ since H is closed under multiplication, so by **Theorem 5 one-step subgroup test**, $H \leq G$. $\square$

Cyclic Groups

Let G be a multiplicative group and let $a \in G$, and consider the set $\langle a \rangle = \{\dots, a^{-2}, a^{-1}, a^0, a^1, a^2, \dots\}$, or more formally the set $\langle a \rangle = \{a^z | z \in \mathbb{Z}\}$. Since G is a group and $a \in G$, then we can use the one-step subgroup test to verify that it is indeed a group.

1. (nonempty) trivially it is not empty since $a \in \langle a \rangle$
2. Take $a^x, a^y \in \langle a \rangle$, so $a^{-y} \in \langle a \rangle$ since $-y \in \mathbb{Z}$, and also $a^{x-y} \in \langle a \rangle$ since $x - y \in \mathbb{Z}$, and by exponent laws $a^{x-y} = a^x a^{-y}$, so by the one-step subgroup test, $\langle a \rangle$ is a subgroup of G .

We call $\langle a \rangle$ the subgroup *generated by* a :

Definition 0.0.14 (generated subgroup).

Given an element a in a group G , the subgroup generated by a , denoted $\langle a \rangle$, is given by $\langle a \rangle = \{a^z | z \in \mathbb{Z}\}$.

We proved earlier that $\langle a \rangle \leq G$. Note that if $\langle a \rangle = G$ then it would still be a group, so we sometimes think of $\langle a \rangle$ simply as the group generated by a . Formally we cannot use a subgroup test to show that such $\langle a \rangle$ is indeed a group without showing it is a subset of the elements of another group with the same operation, but by definition we impose that the elements of the generated group of a comply with exponent laws, so trivially $\langle a \rangle$ satisfies all group axioms. We call a group generated by a single element a *cyclic group*:

Definition 0.0.15 (cyclic group).

If G is a group and it is generated by a single element $a \in G$, i.e. $\langle a \rangle = G$, then G is a cyclic group. We call this a a generator of G .

Proposition 11 (properties of a cyclic group).

1. $\langle a \rangle$ is abelian
2. $|\langle a \rangle| = |a|$ and if $|a|$ is finite and $|a| = n$, then $\langle a \rangle = \{e, a, a^2, \dots, a^{n-1}\}$.

Proof. 1. Let $a^i, a^j \in \langle a \rangle$, then $a^i a^j = a^{i+j} = a^{j+i} = a^j a^i$, so the group operation is commutative on the elements of the group, so it is abelian.

2. If $|a| = n$ for some $n \in \mathbb{N}$ (i.e. $|a|$ is finite), then by writing all exponents z in terms of remainder form w.r.t. n , we have $z = qn + r$ for $q, r \in \mathbb{Z}$ and $r \in [0, n)$, so $a^z = a^{qn+r} = (a^n)^q a^r = a^r$, and since there are only n options for r then by counting there are exactly n distinct values for a , and they are the same as $\{e, a, a^2, \dots, a^{n-1}\}$.

Otherwise if $|a| = \infty$, then suppose by contradiction $\langle a \rangle$ is finite and consider the product of all elements of $\langle a \rangle$ with non-negative exponents, denote it π . Since $\langle a \rangle$ is a group then $\pi \in \langle a \rangle$, and since we suppose $\langle a \rangle$ is finite then $\pi = a^k$ for some finite positive integer k , so it must be that $a^k = a^j$ for some $a^j \in \langle a \rangle$ (since $\pi = a^k \in \langle a \rangle$), but it also must be that $k \neq j$ and in fact that jk by construction, but by **Theorem 3** since $|a| = \infty$ it must be that $j = k$, a contradiction, so $|a| = \infty$ implies that $\langle a \rangle$ is also infinite.

□

Note that not all finite groups are cyclic groups, and not all cyclic groups are finite. We consider a few examples below:

Example 6.1 (infinite cyclic group).

Consider $(\mathbb{Z}, +)$. Clearly $\forall z \in \mathbb{Z}, z = z \cdot 1$, so clearly $\mathbb{Z} = \langle 1 \rangle$ (note here that we use additive notation instead of multiplicative notation), so it is a cyclic group, however $\mathbb{Z}$ is also infinite.

Example 6.2 (generators of a group).

Consider the group of modular multiplication with modulus 9, i.e. $\mathbb{Z}_9^*$. From number theory, we know that the elements of this group are the coprime positive integers w.r.t 9 lesser than 9, i.e. $\mathbb{Z}_9^* = \{1, 2, 4, 5, 7, 8\}$. Consider the generated subgroups of the elements of $\mathbb{Z}_9^*$:

1. $\langle 1 \rangle = \{1\}$, so it is not a generator of the group
2. $\langle 2 \rangle = \{2, 4, 8, 7, 5, 1\} = \mathbb{Z}_9^*$ so 2 is a generator of the group
3. $\langle 4 \rangle = \{4, 7, 1\}$ so it is not a generator
4. $\langle 5 \rangle = \{5, 7, 8, 4, 2, 1\} = \mathbb{Z}_9^*$ so it is a generator of the group
5. $\langle 7 \rangle = \{7, 4, 1\}$ so it is not a generator
6. $\langle 8 \rangle = \{8, 1\}$ so it is not a generator

We conclude that the group is generated by 2, 5 (mod 9), i.e. $\mathbb{Z}_9^* = \langle 2 \rangle = \langle 5 \rangle$, so it is a cyclic group. From this example we see that the same group can have multiple distinct generators.

Example 6.3 (finite non-cyclic group).

Consider $\mathbb{Z}_8^* = \{1, 3, 5, 7\}$.

1. $\langle 1 \rangle = \{1\}$
2. $\langle 3 \rangle = \{3, 1\}$
3. $\langle 5 \rangle = \{5, 1\}$
4. $\langle 7 \rangle = \{7, 1\}$

The group has no generator, so it is not cyclic.

While not every group is cyclic, it is true that every group is the union of its cyclic subgroups. In **6.3 finite**

non-cyclic group, while none of the elements of $\mathbb{Z}_8^*$ is a generator of the group, it is easy to demonstrate that $\cup_{a \in \mathbb{Z}_8^*} \langle a \rangle = \mathbb{Z}_8^*$. We demonstrate this is true in general:

Theorem 7.

Let G be a group, then G is the union of its cyclic subgroups, i.e.

$$G = \cup_{a \in G} \langle a \rangle$$

Proof. Trivially true since $\forall a \in G \exists \langle a \rangle \leq G$ s.t. $a \in \langle a \rangle$, so a is in the union of the cyclic subgroups of G , so the union of all these subgroups consists of all elements of G , so $G \subseteq \cup_{a \in G} \langle a \rangle$. Conversely, if $b \in \cup_{a \in G} \langle a \rangle$, then $b = a^k$ for some $a \in G, k \in \mathbb{Z}$, so $b \in G$ since G is closed under the group operation, so $\cup_{a \in G} \langle a \rangle \subseteq G$, so we conclude the two are the same by mutual inclusion. $\square$

We have found that in $\mathbb{Z}_9^*$, $\langle 2 \rangle = \langle 5 \rangle$. We turn our attention to sufficient and necessary conditions for the equality of two cyclic groups.

Theorem 8.

If G is a group and $a \in G$ s.t. $|a| = n$, and $k \in \mathbb{N}$, then:

1. $\langle a^k \rangle = \langle a^{\gcd(n,k)} \rangle$
2. $|a^k| = \frac{n}{\gcd(n,k)}$

Proof. For brevity, denote $d = \gcd(n, k)$.

1. Trivially both are groups with the same operation since both are generated by an element of G , so it remains to show that they have the same elements. We prove this by mutual inclusion.

$\langle a^k \rangle \subseteq \langle a^d \rangle$: Let $(a^k)^m \in \langle a^k \rangle$. Since $d|k$ we have $k = td$ for some integer t . By exponent laws, $(a^k)^m = a^{km} = a^{(td)m} = a^{(d)tm} \in \langle a^d \rangle$.

$\langle a^d \rangle \subseteq \langle a^k \rangle$: Since $d = \gcd(n, k)$ then by Bezout's identity $\exists x, y \in \mathbb{Z}$ s.t. $xn + yk = d$, so $a^d = a^{xn+yk} = (a^n)^x (a^k)^y$. Since $|a| = n$, then $a^d = (a^k)^y$. Now let $(a^d)^m \in \langle a^d \rangle$, then $(a^d)^m = ((a^k)^y)^m = (a^k)^{ym} \in \langle a^k \rangle$.

Thus by mutual inclusion $\langle a^k \rangle = \langle a^d \rangle$.

2. From (1) and **11 properties of a cyclic group** it follows that $|a^k| = |a^d|$. First we show that $(a^d)^{\frac{n}{d}} = e$, which is trivially true by power rules (note that $\frac{n}{d}$ is an integer since d is a divisor of n , so this operation is well defined using **0.0.4 power notation**) since $(a^d)^{\frac{n}{d}} = a^{\frac{dn}{d}} = a^n$, and since $|a| = n$ then $a^n = e$, so it must be that $|a^d| \leq \frac{n}{d}$.

Suppose by contradiction $\exists m \in \mathbb{N}, 0 < m < \frac{n}{d}$ (if no such m exists, i.e. if $\frac{n}{d} = 1$, then $n = d$ and the proposition is trivially true) s.t. $(a^d)^m = e$, then $a^{dm} = e$ and also $dm < n$, but then $|a| \leq dm < n$, but we assumed $|a| = n$, a contradiction, so $|a^d| = \frac{n}{d}$, so $|a^k| = \frac{n}{d}$.

□

Consider **6.2 generators of a group** again. We have seen that $\langle 4 \rangle = \langle 7 \rangle$ and also that $\langle 2 \rangle = \langle 5 \rangle$. Note that $4^2 \equiv \text{mod } 9 \ 7$ and that $2^5 \equiv \text{mod } 9 \ 5$, so we have $\langle 4 \rangle = \langle 4^2 \rangle$, and also $\langle 2 \rangle = \langle 2^5 \rangle$, and as for the orders of 2, 4 we have $|2| = 6$ and $|4| = 3$. These are not coincidences:

1. For $\langle 4 \rangle = \langle 4^2 \rangle$, observe that $|4| = 3$ and $\gcd(3, 2) = 1$, so $\langle 4^2 \rangle = \langle 4^{\gcd(3,2)} \rangle = \langle 4^1 \rangle = \langle 4 \rangle$ by the theorem.
2. For $\langle 2 \rangle = \langle 2^5 \rangle$, observe that $|2| = 6$ and $\gcd(6, 5) = 1$, so again $\langle 2^5 \rangle = \langle 2^{\gcd(6,5)} \rangle = \langle 2 \rangle$.

We continue our survey with a few corollaries of **Theorem 8** :

Corollary 8.1.

If $|a| = n$, then $\langle a^i \rangle = \langle a^j \rangle \iff \gcd(n, i) = \gcd(n, j)$.

Proof. $\implies$ Suppose $\langle a^i \rangle = \langle a^j \rangle$. By **Theorem 8** $|a^i| = \frac{n}{\gcd(n,i)}$ and $|a^j| = \frac{n}{\gcd(n,j)}$, and by **11 properties of a cyclic group** $|a^i| = |\langle a^i \rangle| = |\langle a^j \rangle| = |a^j|$, so $\gcd(n, i) = \gcd(n, j)$.

$\impliedby$ Suppose $\gcd(n, i) = \gcd(n, j)$, then by **Theorem 8** we know that $\langle a^i \rangle = \langle a^{\gcd(n,i)} \rangle$ and $\langle a^j \rangle = \langle a^{\gcd(n,j)} \rangle$, and since we suppose the two gcds are equal then by transitivity $\langle a^i \rangle = \langle a^j \rangle$.

□

Corollary 8.2.

If $|a| = n$ then $\langle a \rangle = \langle a^k \rangle \iff \gcd(n, k) = 1$.

Proof. Identify that this is simply an instantiation of **8.1** with $i = 1$ and $j = k$.

□

We have seen **8.2** in our example when we say that $\langle 2 \rangle = \langle 2^5 \rangle$ (since $|2| = 6$ in $\mathbb{Z}_9^*$).

Corollary 8.3.

Consider $\mathbb{Z}_n^+$ (i.e. the group of modular addition with modulus n), then $k \in \mathbb{Z}_n$ is a generator of the group if and only if $\gcd(n, k) = 1$.

Proof. Observe that 1 is trivially a generator of the group and $|1| = n$, so the proposition can be restated as $\langle 1 \rangle = \langle k \rangle \iff \gcd(n, k) = 1$, which is exactly **8.2** expressed in additive notation ($k = 1 \cdot k$). □

Consider the case where $|a|$ is prime, then by definition $\forall k \in \mathbb{N}, k|a|, \gcd(k, |a|) = 1$, so by **8.2** every element of $\langle a \rangle$ is a generator of the group:

Corollary 8.4.

If $|a| = p$ where p is prime, then $\forall n \in \mathbb{N}^+, \langle a \rangle = \langle a^n \rangle$.

Example 8.1.

Consider $\mathbb{Z}_7^+$ which is the group of integers under modular addition modulus 7, and has 7 elements: $\{0, 1, 2, 3, 4, 5, 6\}$ so it has order 7, which is prime, then by the theorem every non-identity element of $\mathbb{Z}_7^+$ is a generator of the group (since $\langle 1 \rangle = \mathbb{Z}_7^+$ and so each element is given by $n \cdot 1$). We see this in action below:

1. $\langle 1 \rangle = \{0, 1, 2, 3, 4, 5, 6\}$
2. $\langle 2 \rangle = \{0, 2, 4, 6, 1, 3, 5\}$
3. $\langle 3 \rangle = \{0, 3, 6, 2, 5, 1, 4\}$
4. $\langle 4 \rangle = \{0, 4, 1, 5, 2, 6, 3\}$
5. $\langle 5 \rangle = \{0, 5, 3, 1, 6, 4, 2\}$
6. $\langle 6 \rangle = \{0, 6, 5, 4, 3, 2, 1\}$

In the next theorem, we complete our classification of cyclic groups:

Theorem 9 (fundamental theorem of cyclic groups).

Let $G = \langle a \rangle$ and $H \subseteq G$, then:

1. $H \leq G \iff H$ is cyclic
2. $H \leq G \implies |H| \mid |G|$ (if G is finite)
3. $k \mid |G| \implies \exists! H \leq G (|H| = k)$, and this H is $\langle a^{\frac{|G|}{k}} \rangle$.

Proof. We denote $n = |G| = |\langle a \rangle| = |a|$.

1. $\implies$ Suppose $H \leq G$. If $H = \{e\}$ then H is the trivial subgroup of G and it is cyclic since $\langle e \rangle = \{e\}$, so the proposition holds. If $H \neq \{e\}$, then since H is a subgroup of G it is nonempty and $g \in G$ s.t. $g \neq e$ and $g \in H$, and since H is a group then $g^{-1} \in H$. Since $G = \langle a \rangle$, then $\exists k \in \mathbb{Z}$ s.t. $g = a^k$, so $a^k \in H$ and $a^{-k} \in H$, and also $k \neq 0$ (otherwise $g = a^0 = e$), so $a^{|k|} \in H$, so the subset $A = \{n \in \mathbb{N} \mid a^n \in H \wedge n \neq 0\}$ is a non-empty set of natural numbers, so by the well-ordering principle of $\mathbb{N}$ it has a least element, denote it m , then $a^m \in H$. Since H is closed under multiplication then trivially $\langle a^m \rangle \subseteq H$.

Let $h' \in H$, then $\exists t \in \mathbb{Z}$ s.t. $a^t = h'$. By the division algorithm $\exists! q, r \in \mathbb{Z}, r \in [0, m)$ s.t. $t = qm + r$, so $a^t = a^{qm+r} = a^{qm}a^r$. Since $a^m \in H$ and H is closed under the group operation and under the inverse then also $a^{-qm} \in H$, and by supposition $a^t = a^{qm}a^r \in H$, so $a^{-qm}a^{qm}a^r = a^r \in H$. If $r \neq 0$, then we have that $r \in A$ and also rm , but we denoted the least element of A as m , so it must be that $m \leq r$, a contradiction, so $r = 0$, so $t = qm$, so $a^t = a^{qm} = (a^m)^q$, so $a^t \in \langle a^m \rangle$, so $H \subseteq \langle a^m \rangle$.

By mutual inclusion we conclude $\langle a^m \rangle = H$, so H is a cyclic group with generator a^m .

$\Leftarrow$ By **0.0.14 generated subgroup** we know that if H is a subgroup of G and H is generated by an element then H is a (cyclic) subgroup of G .

2. Suppose $H \leq G$, then by (1) H is cyclic, so $H = \langle a^m \rangle$ for some m . By **Theorem 8**, we have $|a^m| = \frac{|G|}{\gcd(|G|, m)} \in \mathbb{Z}$, so $|G| = k|a^m|$ for some $k \in \mathbb{Z}$, so $|a^m| \mid |G|$. By **11 properties of a cyclic group**, $|a^m| = |\langle a^m \rangle| = |H|$, so $|H| \mid |G|$.

3. **Existence:** Denote $m = \frac{|G|}{k}$. Since $a \in G$ then $\langle a^m \rangle \leq G$ since it is a cyclic group generated by a^m . Also, $(a^m)^k = a^{|G|} = e$ since $|G| = |\langle a \rangle| = |a|$, so $|a^m| \leq k$. For any $t \in \mathbb{N}^+$ s.t. tk , it follows that $0 \leq mt|G|$, and if $(a^m)^t = e$ then $a^{mt} = e$ so $|a| \leq mt|G| = |a|$, a contradiction, so it follows that $\langle a^m \rangle$ is a subgroup of G with order k .

Uniqueness: Suppose $H' \leq G$ s.t. $|H'| = k$. Since G is cyclic with generator a , then by (1) it follows that H' is also cyclic with some generator $a^t \in G$, and by supposition $|H| = |H'| = k$, i.e. $|\langle a^t \rangle| = |\langle a^m \rangle|$. By **Theorem 8**, we know that $|\langle a^t \rangle| = \frac{n}{\gcd(n, t)}$ and similarly for $\langle a^m \rangle$, so $k = \frac{n}{\gcd(n, t)} = \frac{n}{\gcd(n, m)}$, so $\gcd(n, t) = \gcd(n, m)$. Also by **Theorem 8**, $\langle a^m \rangle = \langle a^{\gcd(n, m)} \rangle$, $\langle a^t \rangle = \langle a^{\gcd(n, t)} \rangle$, and as we have shown $\gcd(n, t) = \gcd(n, m)$, so $\langle a^m \rangle = \langle a^t \rangle$, so $H' = H$, so H is the unique subgroup of order k in G .

□

Recall Euler's totient function $\phi : \mathbb{N}^+ \rightarrow \mathbb{N}^+$, where $\phi(x)$ is the amount of numbers which are coprime w.r.t. x and also lesser than x . For example, $\phi(4) = 2$ since out of all positive integers preceding 4 only 1, 3 are coprime w.r.t. 4, so $\phi(4) = |\{1, 3\}| = 2$. Similarly, if x is prime then $\phi(x) = x - 1$. We discuss other explicit formulas and relations regarding Euler's totient function in number theory. Here, we observe the following corollary to the FTG:

Corollary 9.1.

If G is cyclic and d is a positive divisor of $|G|$, then $\phi(d)$ gives the number of distinct elements of order d in G , where ϕ is Euler's totient function.

Proof. Since G is cyclic then it has a generator $a \in G$ s.t. $G = \langle a \rangle$. Denote $|G| = n$, and suppose $d \in \mathbb{N}^+$, $d \mid n$. By proposition (3) of **Theorem 9 fundamental theorem of cyclic groups** it follows that for some $b \in G$, $H = \langle b \rangle$ is the unique subgroup of G with order d (in fact the theorem also tells us which elements generate this subgroup, but that is of no interest to us here). Suppose for some $c \in G$, $|c| = d$, then $|\langle c \rangle| = d$ so by uniqueness, $\langle c \rangle = \langle b \rangle$, so all elements of order d are generators of b .

Since $\langle c \rangle = \langle b \rangle$, then in particular $c \in \langle b \rangle$, so $c = b^k$ for some $k \in \mathbb{N}$, so $\langle b^k \rangle = \langle b \rangle$, which by **8.2** is true iff $\gcd(|b| = d, k) = 1$, i.e. if (k, d) are coprime, and by definition there are exactly $\phi(d)$ positive integers less than d which are coprime w.r.t. d , so $\phi(d)$ is the number of distinct elements of order d in G .

□

Example 9.1.

Let $k \in \mathbb{N}$ and $n = 21k$, and consider $\mathbb{Z}_{21k}^+$ i.e. the cyclic group on modular addition modulus $21k$, then since $|\mathbb{Z}_{21k}^+| = 21k$ and $21|21k = n$, by 9.1 we know that there are exactly $\phi(21) = |\{1, 2, 4, 5, 6, 8, 10, 12, 14, 15, 16, 18, 20\}| = 12$ elements of order 21 in the group, regardless of k . In particular, $\mathbb{Z}_{903}^+$ and $\mathbb{Z}_{21}^+$ both have exactly 12 distinct elements of order 21.

We now present a more general corollary which is true for any group, not just cyclic groups:

Corollary 9.2.

If G is a finite group, then $\forall d \in \mathbb{N}^+$ the number of distinct elements of order d is divisible by $\phi(d)$.

Proof. Recall that $\forall a \in G$, $\langle a \rangle \leq G$ and also $|\langle a \rangle| = |a|$ by 11 properties of a cyclic group. If $\forall a \in G$ ($|a| \neq d$), then G has 0 distinct elements of order d and indeed $\phi(d)|0$. Otherwise, $\exists a \in G$ s.t. $|a| = d$, so $\langle a \rangle = d$, and since $d|d$ then by 9.1 there are exactly $\phi(d)$ distinct elements of order d in $\langle a \rangle$. Now, if $\exists b \in G$ s.t. $b \notin \langle a \rangle$ and $|b| = d$, then we apply the theorem again and count another $\phi(d)$ distinct elements of order d in $\langle b \rangle$, so we now have $2\phi(d)$ distinct elements of order d in G . Since G is finite, eventually we will exhaust all elements of G , and each element either belongs to one of the groups we have already considered, or it doesn't have order d , or it belongs to a disjoint cyclic subgroup we haven't yet considered, which adds another $\phi(d)$ elements of order d , so it must be that the number of distinct elements of d is divisible by $\phi(d)$ since we increment our counter in increments of $\phi(d)$. $\square$

Recall 6.2 generators of a group, in which we considered the cyclic subgroups of $\mathbb{Z}_9^*$. From 9.2 we expect to identify $k\phi(3) = k|\{1, 2\}| = 2k$ elements of order 2 in the group for some $k \in \mathbb{N}$, and indeed the only elements of order 2 in the group were $(4, 7)$, and $2 = 2 \cdot 1$ with $k = 1$.

Using all the properties of cyclic groups we have found so far, we can check if a group is not cyclic:

Recall our example of $\mathbb{Z}_8^*$ (6.3 finite non-cyclic group), which we have demonstrated to be non-cyclic by considering all cyclic subgroups of the group and demonstrating that none of the subgroups is the group itself. In the case of $\mathbb{Z}_8^*$ this isn't too difficult since we only need to check the cyclic subgroups formed by 4 distinct elements and we know that the order of these groups will be at most 4, so the task is manageable. But, given a sufficiently large group, enumerating all cyclic subgroups of the group becomes unmanageable. Instead, we can use results accumulated so far in their contrapositive form to show that a group is non-cyclic.

By 9.1, if G is cyclic and $n = |G|$ and $d|n$ for some positive integer d , then G has exactly $\phi(d)$ distinct elements with order d . Consider the contrapositive: if G is group and $n = |G|$ and $d|n$ for some positive integer d , and G doesn't have exactly $\phi(d)$ distinct elements of order d , then G is non-cyclic. Let $d = 2$ and consider our example of $\mathbb{Z}_8^*$ which has order $\phi(8) = 4$. Clearly $2|4$ so we expect to find exactly 2

distinct elements with order 2, however by checking the cyclic subgroups formed by 3, 5, 7 we find that each element has order 2 so in total we have at least 3 elements of order 2 (in fact we have exactly 3 such elements, since $|1|$ (where 1 is the identity element) is always 1), so G is non-cyclic.

For a more impressive example, consider $\mathbb{Z}_{24}^* = \{1, 5, 7, 11, 13, 17, 19, 23\}$ which demonstrably has order 8. Since $2|8$ we expect that $\mathbb{Z}_{24}^*$ will have exactly 2 elements with order 2, however $\langle 5 \rangle = \{1, 5\}$ (since $5^2 = 25 = 24 + 1$), $\langle 7 \rangle = \{1, 7\}$ (since $7^2 = 49 = 2 \cdot 24 + 1$), $\langle 11 \rangle = \{1, 11\}$ (since $11^2 = 121 = 5 \cdot 24 + 1$), so $\mathbb{Z}_{24}^*$ has at least 3 cyclic subgroups with order 2, so $\mathbb{Z}_{24}^*$ is not cyclic. Note how we only needed to check the first 3 non-trivial subgroups of the group.

Another important result which will become useful to us much later in our discussion of the n^{th} roots of unity is the following, which follows mostly from 9.1 :

Theorem 10 (sufficient condition for finite group to be cyclic).
If G is a finite group and $|G| = n$, and $\forall d \in \mathbb{N}$ s.t. $d | n$ we have that $|\{a \in G | a^d = e\}| \leq d$, then G is cyclic.

Proof. Let G be finite with $|G| = n$. Let $d \in \mathbb{N}$ s.t. $d | n$. From **Theorem 21 Lagrange's theorem** which we see in the future, it follows that $\forall a \in G, |a| | |G|$, hence at least for some $d | n$ we have that $\exists a \in G$ s.t. $|a| = d$.

Let $G_d = \{a \in G | |a| = d\}$ i.e. the subset of G of all elements of order d . There are two cases: either G_d is empty, or it is not. In the case where G_d is not empty, then let $b \in G_d$. Since $|b| = d$, then $\langle b \rangle = d$. Also, since $\forall c \in \langle b \rangle, c^d = e$ (by **Theorem 9 fundamental theorem of cyclic groups** we have $|\langle c \rangle| | |\langle b \rangle| = d$), so $\langle b \rangle \subseteq \{a \in G | a^d = e\}$, but by supposition $|\{a \in G | a^d = e\}| \leq d$, yet $|\langle b \rangle| = d$, so for every $b, c \in G_d$ we have that $\langle b \rangle = \langle c \rangle = \{a \in G | a^d = e\}$, i.e. it is the set of generators of the cyclic subgroup of G of order d . By 9.1, we know that there are exactly $\phi(d)$ such elements, so $|G_d| = \phi(d)$ or $G_d = \emptyset$.

Let $d_1, d_2 \in \mathbb{N}$ s.t. $d_1 | n, d_2 | n$, and also $d_1 \neq d_2$ then $G_{d_1} \cap G_{d_2} = \emptyset$ since otherwise we would have some element in G with order d_1 and also d_2 with the two not being equal, which is nonsense, yet every element of G has order at most n since otherwise we would have a cyclic subgroup of G of order greater than that of G which is also nonsense, so it must be that all these G_d for $d | n$ form a partition on G , so $n = |G| = \sum_{d|n} |G_d| \leq \sum_{d|n} \phi(d)$.

Applying the same argument to a cyclic group of n elements, we find that $\sum_{d|n} \phi(d) = n$ by 9.1, and since $\phi(d)$ is non-negative, it must be that each G_d is nonempty for every $d | n$ and in particular G_n is nonempty (since $n | n$), but G_n is the set of all generators of cyclic subgroups of G with order n , yet $|G| = n$, so G is cyclic.

□

Permutation and Cycles

Definition 0.0.16 (permutation).

A bijection ϕ from a set A to itself is called a permutation of A .

Definition 0.0.17 (symmetric group).

Let A be a set with finite cardinality n , i.e. $|A| = n$. The set of all permutations of A is called the symmetric group of A , and is denoted $\text{Sym}(A)$.

Note that while their names are similar, the symmetry group of a structure and the symmetric group of a set *are not* the same. Recall that the symmetry group is an automorphism (i.e. an isomorphism from the structure to itself), for example the symmetry group of a regular rectangle $\square ABCD$, which is the dihedral group D_8 , does not consist of a mapping ϕ s.t. $A \mapsto A, B \mapsto B, C \mapsto D, D \mapsto C$, since even though it is a bijection it does not preserve the geometric structure of the rectangle (intuitively there is no rigid motion from $\square ABCD$ to $\square ABDC$).

Proposition 12.

$\text{Sym}(A)$ is a group under function composition.

Proof. Recall that the composition of two bijections is also a bijection, so the composition of permutations is a permutation, so permutations are closed under composition. Next, function composition is associative, the identity map $\text{id}_A : a \mapsto a$ is a bijection on A so it is a permutation of A , all bijections are invertible and the inverse of a bijection is also a bijection so $\forall \phi \in \text{Sym}(A), \phi^{-1} \in \text{Sym}(A)$, so the structure $(\text{Sym}(A), \circ)$ has an associative operator and satisfies the identity and inverse properties, so it is a group. (Here we recalled our knowledge of bijections which we gathered in [Introduction to Set Theory](#)). $\square$

Proposition 13.

For a finite A , the order of $\text{Sym}(A)$ is $|A|!$, where $|A|$ is the cardinality of A and $!$ is the factorial defined recursively as $0! = 1, (n+1)! = n! \cdot (n+1)$.

Proof. We prove by a combinatorial argument (the reader may later refine it using induction). Let $\phi \in \text{Sym}(A)$ and consider an ordering $a_1, \dots, a_{|A|}$ on the elements of A (which we suppose is finite). Since ϕ is a permutation then $\phi(a_1) = a_j$ for some $j \in [1, |A|]$, and there are $|A|$ such options (if $|A| = 1$ then we are done). Since ϕ is a bijection, then $\phi(a_2) \neq \phi(a_1)$, so ϕ has $|A| - 1$ options to send a_2 to, so in total we have $|A| \cdot (|A| - 1)$ options so far, and for a_3 we would have $|A| - 2$ etc, until we reach $a_{|A|}$ which will only have one possible element it could map to in all possible permutations we have considered, so we have exactly $\prod_{i=1}^{|A|} i$ possible permutations, which is exactly $|A|!$. Recall that the order of a group is the number of elements in the group, so $|\text{Sym}(A)| = |A|!$. $\square$

Proposition 14 ($\text{Sym}(A)$ non-abelian if $2|A|$).

If $2|A|$, then $\text{Sym}(A)$ is non-abelian.

Proof. Let $a, b, c \in A$ all distinct, then there exists $\sigma, \tau \in \text{Sym}(A)$ s.t. $\sigma(a) = b, \sigma(b) = a, \tau(b) = c, \tau(c) = b$, and all other elements of A are fixed by σ, τ . So, $(\sigma \circ \tau)(c) = \sigma(\tau(c)) = \sigma(b) = a$, yet $(\tau \circ \sigma)(c) = \tau(\sigma(c)) = \tau(c) = b$, but $a \neq b$, so $\sigma \circ \tau \neq \tau \circ \sigma$, so $\text{Sym}(A)$ is non-abelian. $\square$

Definition 0.0.18 (symmetric group of degree n).

The symmetric group of degree n , denoted S_n , is the symmetric group on the set $\{1, 2, \dots, n\}$, i.e. $S_n = \text{Sym}(\{1, 2, \dots, n\})$.

Some writes call S_n the symmetric group on n letters, which appeals to our combinatorial intuition of the structure of S_n .

Proposition 15.

If X is a set with n elements, then $\text{Sym}(X) \cong S_n$.

Proof. Since $|X| = N = |\{1, \dots, n\}|$, we can find a bijection $\phi : X \rightarrow N$. Consider $\sigma \in \text{Sym}(X)$. Since $\sigma : X \rightarrow X$ is a bijection, then $\psi = \phi \circ \sigma \circ \phi^{-1} : N \rightarrow N$ is a bijection as the composition of bijections, so $\psi \in S_n$.

Consider a mapping: $\omega : \text{Sym}(X) \rightarrow S_n, \sigma \mapsto \phi \circ \sigma \circ \phi^{-1}$, then

$$\omega(\sigma_1) \circ \omega(\sigma_2) = (\phi \circ \sigma_1 \circ \phi^{-1}) \circ (\phi \circ \sigma_2 \circ \phi^{-1}) = \phi(\sigma_1 \circ \sigma_2) \circ \phi^{-1} = \omega(\sigma_1 \circ \sigma_2)$$

So ω is a homomorphism. It remains to show it is bijective:

1. (injective) Suppose $\omega(\sigma_1) = \omega(\sigma_2)$, then $\phi \circ \sigma_1 \circ \phi^{-1} = \phi \circ \sigma_2 \circ \phi^{-1}$. Via left-composition with ϕ^{-1} and right-composition with ϕ , we get $\sigma_1 = \sigma_2$.
2. (surjective) Suppose $\psi \in S_n$, so $\phi^{-1} \circ \psi \circ \phi : X \rightarrow X \in \text{Sym}(X)$ (composition of bijections is a bijection). It can be verified that $\omega(\phi^{-1} \circ \psi \circ \phi) = \psi$, so ω is surjective.

We conclude ω is an isomorphism, so $\text{Sym}(X) \cong S_n$. $\square$

We will focus on S_n for now. Later when we discuss group morphisms in depth we will see that all our results trivially extend to any symmetric group isomorphic to S_n as we will find that all “interesting” group properties are preserved under an isomorphism.

Proposition 16.

The dihedral group of order $2n$, D_{2n} , is isomorphic to a subgroup of the symmetric group of degree n , S_n .

Proof. Recall from our discussion on [Rigid motions & symmetry](#) that each element $\phi \in D_{2n}$ is uniquely defined by its action on the n vertices $V_n = \{v_1, \dots, v_n\}$ of a regular n -gon (in fact, each element is uniquely defined by its action on a pair of adjacent vertices, which is a stronger proposition), thus each ϕ can be represented as some permutation on V_n , i.e. in a sense $\phi \in \text{Sym}(V_n)$. More rigorously, for each symmetry ϕ , consider ϕ restricted to V_n , i.e. $\phi|_{V_n}$, then $\phi|_{V_n} \in \text{Sym}(V_n)$, and given a mapping $\eta : D_{2n} \rightarrow \text{Sym}(V_n), \phi \mapsto \phi|_{V_n}$ we identify that η is *injective* (since otherwise $\phi|_{V_n} = \phi'|_{V_n}$ for some $\phi, \phi', \phi \neq \phi'$, but since each symmetry is determined by its action on the vertices, then it must be that $\phi = \phi'$, a contradiction, so η is injective). Consider η' , which is η with the co-restriction to $\text{Im}(\eta) \subseteq \text{Sym}(V_n)$, clearly this η' is surjective as well as injective, so it is a bijection.

Observe that η is trivially a homomorphism (in fact, it is a *monomorphism* since it is an injective homomorphism) since $\forall \phi, \psi \in D_{2n}$ we have $\eta(\phi \circ \psi) = (\phi \circ \psi)|_{V_n} = \phi|_{V_n} \circ \psi|_{V_n} = \eta(\phi) \circ \eta(\psi)$, so η is a group homomorphism, so η' is a group isomorphism, so $D_{2n} \cong \text{Im}(\eta) \leq \text{Sym}(V_n)$, and as we mentioned without proof $\text{Sym}(V_n) \cong S_n$ (the proof is trivial and follows from the natural isomorphism $v_i \mapsto i$), and as we will later see isomorphism is an equivalence relation and subgroups are mapped to subgroups, so $D_{2n} \cong S_D \leq S_n$.

□

Corollary 10.1.

$D_6 \cong S_3$, and it is the only dihedral group with $2n$ s.t. $D_{2n} \cong S_n$.

Proof. By 16 we know that D_6 is isomorphic to a subgroup of S_3 . Trivially group order is preserved under an isomorphism (since an isomorphism is injective & surjective, so both the domain and codomain must have the same cardinality), but we know $|D_6| = 6$ and also $|S_3| = 3! = 6$, and the only subgroup of S_3 of order 6 is S_3 , so $D_6 \cong S_3$. For any other n s.t. $3n$, we have a dihedral group with order $2n$ and a symmetric group of order $n!$, and since $3n$ then $n! = (n-1)!n2n$, so $|D_{2n}| < |S_n|$, so it cannot be that the two groups are isomorphic for n larger than 3.

□

Cycle Notation

Consider a set $N = \{1, \dots, n\}$ and a permutation $\phi \in S_n$. One way to write ϕ is via explicitly specifying the image of every element of the set under ϕ as follows: $\phi : 1 \mapsto 2, 2 \mapsto 3, 3 \mapsto 4, \dots, n-1 \mapsto n, n \mapsto 1$. Identify that this ϕ corresponds to a single rotation $r \in D_{2n}$.

The notation we presented has two major downsides:

1. It is not immediately obvious what the action of ϕ on N is, even though our example is very elementary. Consider, for example, a permutation s.t. it takes every 3 consecutive numbers and exchanges the first and last (i.e. rotates them around the middle number), i.e. if $N = \{1, 2, 3, 4, 5, 6\}$, then $\phi' : 1 \mapsto 3, 2 \mapsto 2, 3 \mapsto 1, 4 \mapsto 6, 5 \mapsto 5, 6 \mapsto 4$. It is even harder to see the pattern here.
2. It has a lot of visual noise coming from all the $\mapsto$ symbols and all the commas.

As an alternative, consider the following notation, called *array notation* or *two-line notation*, applied to the same rotation permutation ϕ :

$$\phi = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{bmatrix}$$

Where ϕ_{1i} is the input and ϕ_{2i} is the output of ϕ for that input, for example $\phi(2)$ is 3, which is the element right under 2. This notation makes the action of the permutation on the group more obvious, however the cyclic nature of ϕ is not immediately obvious (in the sense that it acts on the elements as if they were all vertices of a cycle, and each vertex is sent to the following vertex by ϕ). For our second example, this becomes even more apparent:

$$\phi' = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 2 & 1 & 6 & 5 & 4 \end{bmatrix}$$

It is not immediately obvious from this presentation that ϕ' acts on N as if its elements belong to two *disjoint* cycles, one formed by 123 and one formed by 456. To convince yourself the cycles are disjoint, observe that $(\phi')^2 = \text{id}_N$ and that elements in the subset 1, 2, 3 remain in the same subset under both ϕ' , $(\phi')^2$ and likewise for elements of 4, 5, 6, so the cycles are disjoint.

This observation - that permutations can also be thought of as *cycles*, leads to the introduction of *cycle notations*. In cycle notation, we write permutations as a *composition of cycles*, for example we would write the rotation ϕ mentioned earlier as $\phi = (1234)$, and interpret it as follows: for every element x in the domain, $\phi(x)$ is the element to the right of x in the cycle, or if $x = 4$ then $\phi(x) = 1$, hence the name *cycle*. If an element does not appear in the cycle, then it is fixed by it, for example (123) fixes 4 since $4 \notin (123)$.

Consider another permutation in cycle notation $\psi = (123)(456)$, where (123) and (456) are again *cycles*. We interpret it as follows:

1. To calculate $\psi(i)$, we go from right to left and in each cycle.
2. If we never find i , then $\psi(i)$ fixes i , i.e. $\psi(i) = i$.

3. Otherwise, we find i in some cycle α , and then $\alpha(i)$ is the element to the right of i in α , and if i the rightmost element in α then $\alpha(i)$ is given by the leftmost element of α . More explicitly, if $\alpha = (\alpha_0 \dots \alpha_{n-1})$, then $\alpha(\alpha_i) = \alpha_{(i+1)\%n}$, and we set $i = \alpha(\alpha_i)$.

For example, pick $i = 3$. Going from right to left, we have (456) , which fixes 3 since 3 is not in the cycle, then we have (123) , so $\psi(3) = (123)(3) = 1$. Writing ψ in two-line notation, we get:

$$\psi = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 1 & 5 & 6 & 4 \end{bmatrix}$$

Which has a lot more noise and makes it harder to detect that ψ is in fact composed of two cycles, (123) and (456) .

While this notation takes some time to get used to, it has clear benefits:

1. It makes the cyclic nature of the permutation immediately apparent, especially if ϕ' is expressed as a composition of disjoint cycles (more on that later).
2. It is very concise.

Representing id_N in cycle notation is also trivial: on the one hand, we can explicitly write $\text{id}_N = (1)(2)(3)(4)$ (for $N = \{1, 2, 3, 4\}$) (check that this really is the identity permutation on 4 elements), but we prefer to use a shorthand instead and write $\text{id}_N = ()$ or $\text{id}_N = (1)$.

We can now express ϕ' in cycle notation, as follows: $\phi' = (13)(46)$. In this representation, it is immediately obvious that ϕ' fixes 2, 5 and that 1, 3 and 4, 6 form a cycle.

Definition 0.0.19 (cycle notation).

Let ϕ be a permutation of a set A , then ϕ in cycle notation is given by a composition of cycles $\phi = \alpha \circ \beta \circ \dots \circ \omega$, where $\alpha = (\alpha_0 \dots \alpha_{n_\alpha-1})$, $\beta = (\beta_0 \dots \beta_{n_\beta-1})$, $\dots$, $\omega = (\omega_0 \dots \omega_{n_\omega-1})$, s.t. each cycle c is a permutation of A s.t. $\forall x \in A$, if $x = c_i$ for some c_i in the cycle then $c(x) = c(c_i) = c(c_{(i+1)\%n_c})$, and if x is not in the cycle then it is fixed by c , i.e. $c(x) = x$.

Definition 0.0.20 (disjoint cycle).

Two cycles α, β are disjoint if $\forall x ((\alpha(x) \neq x) \implies (\beta(x) = x))$ as well as the symmetrical statement where α, β are exchanged (i.e. $\forall x ((\beta(x) \neq x) \implies (\alpha(x) = x))$), or informally - two cycles are disjoint if they do not share an element.

We immediately start with a proposition on disjoint cycles, the proof for which is best read after reading the few examples that follow it:

Proposition 17.

If α, β are disjoint cycles, then $\alpha \circ \beta = \beta \circ \alpha$, i.e. disjoint cycles commute.

Proof. Consider some set X and let $x \in X$. There are two cases:

1. x is not fixed by α , so $\alpha(x) = y$ s.t. $y \in X, y \neq x$. Since α, β are disjoint then x must be fixed by β , so $(\alpha \circ \beta)(x) = \alpha(\beta(x)) = \alpha(x) = y$. Similarly, since we suppose $\alpha(x) = y$ so it must be that y is also not fixed by α and belongs to the cycle α , so again β must fix y , so $(\beta \circ \alpha)(x) = \beta(\alpha(x)) = \beta(y) = y$, so $(\alpha \circ \beta)(x) = (\beta \circ \alpha)(x)$.
2. x is fixed by α , so $(\beta \circ \alpha)(x) = \beta(\alpha(x)) = \beta(x)$. If x is also fixed by β then trivially $(\beta \circ \alpha)(x) = \beta(x) = x = \alpha(x) = \alpha(\beta(x)) = (\alpha \circ \beta)(x)$. If x is not fixed by β , then $\beta(x)$ is fixed by α due to the same argument from the previous case, so $(\alpha \circ \beta)(x) = \alpha(\beta(x)) = \beta(x) = (\beta \circ \alpha)(x)$.

So we conclude $\alpha \circ \beta = \beta \circ \alpha$. We note that trivially this argument extends inductively to a composition of n disjoint cycles $\alpha_1 \circ \dots \circ \alpha_n$.

□

In the next example we demonstrate how to calculate the image of an element under a permutation expressed in cycle notation, as well as how to simplify a permutation in cycle notation:

Example 10.1 (using cycle notation).

Consider $\phi = (12)(23)(5612)(23)$. We wish to work out $\phi(1)$. Going from right to left, we see that (23) fixes 1, then (5612) takes 1 to 2, then (23) takes 2 to 3, then (12) again fixes 3, so we conclude $\phi(1) = 3$. Now consider $\phi(3)$, we have (23) maps 3 to 2, then (5612) maps 2 to 5, which remains fixed by the next two cycles, so $\phi(3) = 5$. For $\phi(5)$, the first cycle (from right to left) fixes 5, the second sends 5 to 6, which is then fixed by the rest of the cycle, so $\phi(5) = 6$. For $\phi(6)$, we have that the first cycle fixes 6, the second sends it to 1, the third fixes 1, and finally the fourth sends 1 to 2, so we have $\phi(6) = 2$. Finally for $\phi(2)$, the first cycle sends 2 to 3, the second fixes 3, the third sends 3 to 2 and the fourth sends 2 to 1, so we have $\phi(2) = 1$, note that we have exhausted all values effected by the cycles, and deduced that $\phi : 1 \rightarrow 3, 3 \rightarrow 5, 5 \rightarrow 6, 6 \rightarrow 2, 2 \rightarrow 1$, so again in cycle notation we have $\phi = (13562)$, so ϕ consists of a single cycle, and we have $(12)(23)(5612)(23) = (13562)$.

Note that $(123) = (312) = (231)$ (verify that this is indeed the case!)

[simplifying permutations]

The reader should follow the process described in 10.1 using cycle notation and show that:

1. $(123)(123) = (132)$
2. $(123)(132) = (1) = \text{id}$
3. $(132)(123) = (1) = \text{id}$

And conclude that (123) has order 3.

Proposition 18 (order of a cycle).

If $\alpha = (\alpha_0 \dots \alpha_n)$ where $\alpha_0, \dots, \alpha_n$ are all distinct, then:

1. $\alpha^k(\alpha_i) = \alpha_{(i+k)\%n}$ for $k \in \mathbb{N}$
2. $|\alpha| = n$

Proof. 1. We prove by induction on k . For $k = 0$ we have $\alpha^0 = \text{id}$ and indeed $\text{id}(\alpha_i) = \alpha_i = \alpha_{(i+0)\%n}$. Suppose the proposition is true for some $k \in \mathbb{N}$, so $\alpha^k(\alpha_i) = \alpha_{(i+k)\%n}$. By **Theorem 2 exponent laws (power rules)**, $\alpha^{k+1} = \alpha \circ \alpha^k$, so $\alpha^{k+1}(\alpha_i) = \alpha(\alpha^k(\alpha_i))$, which by the induction hypothesis is $\alpha(\alpha_{(i+k)\%n})$, which by definition is $\alpha_{(i+(k+1))\%n}$, which completes the proof by induction.

2. By (1) we have $\alpha^n(\alpha_i) = \alpha_{(i+n)\%n} = \alpha_i$ for every α_i and also trivially $\alpha^n(x) = x$ for every x not in the cycle, so $|\alpha| \leq n$. If $|\alpha| = kn$, then $\alpha^k(\alpha_i) = \alpha_i$, which by (1) is the same as $\alpha_{(i+k)\%n} = \alpha_i$, so $n|k|$ thus $n \leq k$, but also kn , a contradiction, so $|\alpha| = n$. □

It is not always the case that a permutation in cycle notation can be simplified to a single cycle. For example, $(123)(456)$ is already expressed as two disjoint cycles, and cannot be simplified further. It is however always the case that a permutation can be expressed as the composition of disjoint cycles.

Theorem 11 (disjoint form).

Every permutation ϕ on a finite set can be written as a composition of disjoint cycles.

Proof. We present a constructive proof which generalizes the process described in **10.1 using cycle notation**. Let X be a finite set s.t. $|X| = n$ for some $n \in \mathbb{N}$ and ϕ a permutation on X . Consider the following algorithm:

1. Set $X' = X$, $C = \emptyset$ (where C is a set of cycles. This can also be an array, but here we emphasize that order doesn't matter according to **17** since all cycles in C will be disjoint).
2. While $X' \neq \emptyset$:
 1. Pick some $x \in X'$, and remove it from X' (i.e. set $X' = X' \setminus \{x\}$)
 2. Set $y = \phi(x)$
 3. If $y = x$, then ϕ fixes x , then
 1. go back to 2
 4. Otherwise, set $\alpha = (x)$ (α will represent a cycle, so order of succession matters)
5. While $y \neq x$:
 1. Remove y from X'
 2. Append y to the end of α .
 3. Set $y = \phi(y)$

6. Add α to C
3. C is the set of disjoint cycles whose composition is ϕ

Below is a Python implementation of the algorithm:

```
1 def decompose(domain: list[int], permutation: Callable[[int], int]) ->
  list[list[int]]:
2     cycles: list[list[int]] = []
3     working_set = domain.copy()
4
5     while len(working_set) != 0:
6         x = working_set.pop()
7         y = permutation(x)
8         if x == y:
9             # x is fixed by permutation, skip.
10            continue
11
12        # Start tracking a new cycle.
13        cycle = [x]
14        cycles.append(cycle)
15
16        # Cycles must eventually wrap around assuming this
17        # is a permutation.
18        while y != x:
19            cycle.append(y)
20            # If the permutation is valid, then every element
21            # of the domain should appear exactly once
22            # in the image, so it cannot be that `remove` throws.
23            working_set.remove(y)
24            y = permutation(y)
25
26    return cycles
```

Correctness: Suppose $x \in X$, then $\phi(x) \in X$ as well since ϕ is a permutation. If $\phi(x) = x$ then ϕ fixes x , and in cycle notation we may omit cycles of length 1, so our omission (in step 2.3.1) is justified. Otherwise, $\phi(x) \neq x$, so we start tracking a cycle α , setting $y = \phi(x)$ initially and then setting y to be its successor in each iteration (step 2.7.3). There are 2 cases:

1. If $y \notin \alpha$, i.e. it is not a value already in the cycle, then we add y to the cycle (step 2.7.2) and continue to the next value (step 2.7.3).
2. If $y \in \alpha$, then either $y = x$ or $y \neq x$, but if $y \neq x$ then it must be that y was already the image of some other element in α (since x is the only element of α that is not added due to being the image of another element in α), but then either y has two preimages so ϕ is not injective, but ϕ is a permutation so it is bijective and in particular injective, a contradiction, so if $y \in \alpha$ then $y = x$, which is the condition we check in 2.7.

Since X is finite and each y is either x or not already in α , then 2.7 can repeat at most $|X'|$ times, so it

must eventually terminate. Similarly, 2 must already eventually terminate since it can repeat at most $|X|$ times, so the algorithm always terminates.

It remains to show that the algorithm produces disjoint cycles, and then their composition is indeed ϕ . Suppose the cycles are not disjoint, then we have some $\alpha, \beta \in C$ s.t. $\exists x \in \alpha, x \in \beta$, and suppose w.l.o.g. that α was completed before β , but then x must have been removed from X' either in 2.1 or 2.7.1, so it cannot have been in X' while constructing β , so it cannot be that $x \in \beta$, so it must be that the cycles are disjoint.

Finally, suppose by contradiction that ϕ is not equivalent to the composition of the cycles, which we denote ψ . In other words, $\exists x \in X$ s.t. $\phi(x) \neq \psi(x)$. Since ψ is a composition of disjoint cycles then either $\psi(x) = x$ or $\psi(x) = \alpha(x)$ for some cycle $\alpha \in \psi$, but $\alpha(x)$ is simply $\phi(x)$ by construction, so this case leads to a contradiction, so it must be that $\psi(x) = x$ and also $\phi(x) \neq x$, but again $\psi(x) = x$ if and only if $\phi(x) = x$ (step 2.3.1), so this is also impossible, so we conclude $\phi = \psi$.

□

Transpositions

Definition 0.0.21 (2-cycle (transposition)).

A 2-cycle is a cycle of length 2, i.e. a cycle that consists of 2 elements. It is also called a transposition. Likewise, an n -cycle is a cycle of length n .

Theorem 12.

If $\alpha = (\alpha_1 \alpha_2 \dots \alpha_n)$ is a cycle, then α can be written as a composition of $n - 1$ 2-cycles as $\alpha = (\alpha_1 \alpha_n) \circ (\alpha_1 \alpha_{n-1}) \circ \dots \circ (\alpha_1 \alpha_2)$

Proof. Denote the decomposed form of α into transpositions α' . Let X be a set. We prove by induction on n . If $n=2$ then α is the identity so the theorem is vacuously true (since $(\alpha_1) = \text{id} = ()$ which is a composition of $1 - 1 = 0$ transpositions). If $n = 2$, then α is already written in terms of $2 - 1 = 1$ transpositions.

Suppose by induction the theorem holds for some n , i.e. that $(\alpha_1 \dots \alpha_{n-1} \alpha_n) = (\alpha_1 \alpha_n) \circ (\alpha_1 \alpha_{n-1}) \circ \dots \circ (\alpha_1 \alpha_2) = \alpha'$ and consider $n + 1$. We wish to show by the induction hypothesis that $(\alpha_1 \dots \alpha_n \alpha_{n+1}) = (\alpha_1 \alpha_{n+1}) \circ (\alpha_1 \alpha_n) \circ \dots \circ (\alpha_1 \alpha_2) = (\alpha_1 \alpha_{n+1}) \circ \alpha' = \alpha''$ (the final transition is simply a notation).

Let $x \in X$, there are three cases:

1. $x \notin \alpha$, in which case x is fixed by α and also x not in any of the cycles that compose α'' so it is also fixed by α' , so $\alpha(x) = \alpha''(x)$.
2. $x \in \alpha$, in which case we identify that the difference between the n -cycle and the $n + 1$ -cycle is that

in the $n + 1$ -cycle, $\alpha_n \mapsto \alpha_{n+1}$, $\alpha_{n+1} \mapsto \alpha_1$ whereas in the n -cycle, $\alpha_n \mapsto \alpha_1$ and $\alpha_{n+1} \mapsto \alpha_{n+1}$ (as it is not a part of the cycle), so if $x \notin \{\alpha_n, \alpha_{n+1}\}$ then α and α'' , agree. Suppose then that $x = \alpha_n$, so $(\alpha_1 \alpha_{n+1}) \circ \alpha'(\alpha_n) = (\alpha_1 \alpha_{n+1})(\alpha_1) = \alpha_{n+1}$, so α'' and α agree on $x = \alpha_n$. Similarly, for $x = \alpha_{n+1}$, we have $\alpha'' = (\alpha_1 \alpha_{n+1})(\alpha'(\alpha_{n+1})) = \alpha_1$, so again α'' and α agree.

We conclude that $\alpha = \alpha''$ in the $n + 1$ case, which completes the proof by induction. □

Corollary 12.1 (permutations as a composition of transpositions).

Every permutation can be written as a composition of 2-cycles.

Proof. Immediate from **Theorem 12** given a permutation expressed in cycle notation, which by **Theorem 11 disjoint form** we know that every permutation can be. □

We proceed to a survey of the arithmetic properties of transpositions.

Theorem 13 (transposition arithmetics).

$\forall a, b, c, d$ s.t. all a, b, c, d are different:

1. $(ab) = (ba)$
2. $(ab)(cd) = (cd)(ab)$
3. $(bc)(ab) = (ab)(ac)$
4. $(ac)(ab) = (ab)(bc)$

Proof. 1 is immediate from **0.0.19 cycle notation**. 2 is a special case of **17**.

3. Observe that $\phi = (bc)(ab)$ is not written in disjoint form, so we follow the process described in **Theorem 11 disjoint form** and demonstrated in **10.1 using cycle notation** and observe the following: first, $\phi(c) = (bc)(ab)(c) = (bc)(c) = b$, and also $\phi(b) = (bc)(ab)(b) = (bc)(a) = a$, and finally $\phi(a) = c$ (we know this is the case without direct calculation since a permutation is bijective and we have already found the preimage of the rest of a, b, c), so $\phi = (cba) = (acb)$, which by **Theorem 12** is equivalent to $(ab)(ac)$, so $(bc)(ab) = (ab)(ac)$.

4. Observe that by (1), $(ac)(ab) = (ac)(ba)$, which is simply (3) with the substitutions $b \mapsto a, c \mapsto c$ and $a \mapsto b$, so $(ac)(ba) = (ba)(bc)$, which again by (1) is $(ab)(bc)$. □

Theorem 14.

The identity permutation on $2 \leq n$ elements can be written as a product (composition) of transpositions, and the product always involves an even number of transpositions.

Proof. The first part of the proposition is simply restating **12.1 permutations as a composition of transpositions**. Suppose id is written as a product of m transpositions. If $m = 0$, then the theorem is vacuously true. Otherwise, we have $\text{id} = (a_0 a_1)(a_2 a_3) \dots (a_{2m-1} a_{2m})$ for some $a_0, \dots, a_{2m}$. Denote $\alpha_i = (a_i a_{i+1})$, then $\text{id} = \alpha_0 \alpha_1 \dots \alpha_m$.

Suppose by contradiction that it is possible for m to be 1, i.e. $\text{id} = (a_0 a_1)$, but then clearly it is not the identity since $a_0 \neq a_1$ (otherwise it is not a valid cycle) yet $(a_0 a_1)(a_1) = a_0$, so it cannot be that $m = 1$.

Consider a transposition α_i for some $0 \leq i < m$ (if $m \neq 0$ then by the argument from the previous paragraph we know $1 \leq m$, so there must exist at least one such transposition), then $\text{id} = \alpha_0 \dots \alpha_i \dots$. Consider the product $\alpha_{i-1} \alpha_i$ and $a \in \alpha_i = (ab)$. There are three cases:

1. $\alpha_{i-1} = \alpha_i$, in which case $\alpha_{i-1} \alpha_i = \text{id}$ so the two cancel out
2. $a \in \alpha_{i-1} = (ax)$, in which case by **Theorem 13 transposition arithmetics** we have $(ax)(ab) = (ab)(bx) = \alpha_i \circ (bx)$.
3. $a \notin \alpha_{i-1}$, in which case again by **Theorem 13 transposition arithmetics** we can rewrite $\alpha_{i-1} \alpha_i$ as $\alpha_i \alpha'_{i-1}$ s.t. $a \notin \alpha'_{i-1}$ as well.

Thus in all cases where $\alpha_{i-1} \neq \alpha_i$, we can move α_i to the left while also rewriting the permutation $\dots \alpha_{i-1} \alpha_i \dots$ as $\dots \alpha_i \alpha'_{i-1} \dots$ s.t. $a \in \alpha_i$ and $a \notin \alpha'_{i-1}$. By repeating this process, either α_i is eventually cancelled out, or the permutation becomes $\alpha_i \circ \dots \circ \alpha_{i+1} \circ \dots$, where $\forall k \in [0, i-1], a \notin \alpha_k$.

Pick $i = m$, then by the previous argument either $\exists \alpha_k \in [0, m-1]$ s.t. $\alpha_k = \alpha_m$, or we can rewrite the identity as $\text{id} = \alpha_m \circ \dots$ s.t. α_m is the only transposition with $a \in \alpha_m$, so $\text{id}(a) = \alpha_m(a) = b$ where b is the other element in the transposition α_m , but this is impossible since id is the identity so it must be that $\text{id}(a) = a$, so it must be that $\exists k \neq m$ s.t. $\alpha_k = \alpha_m$. By applying this argument to all permutations (from the rightmost to the leftmost), it is evident that $\forall \alpha_i \exists \alpha_k$ s.t. $\alpha_i = \alpha_k$ and $i \neq k$, so it must be that m is even (since every transposition must be cancelled out by some other transposition, so they must appear in id in pairs).

The proof can trivially be expressed more formally using an inductive argument.

□

Corollary 14.1.

Every permutation ϕ is the product of either an even number of transpositions, or an odd number of transpositions, i.e. ϕ cannot be represented as the product of both an odd and an even number of transpositions.

Proof. Let $\phi \in \text{Sym}(A)$ for some A . Take two representations of ϕ in terms of 2-cycles, not necessarily distinct (we have shown that such representation must exist). So for some $\alpha_1, \dots, \alpha_n$ and $\beta_1, \dots, \beta_m$ we have:

$$\phi = \underbrace{(\alpha_1\alpha_2) \dots (\alpha_{n-1}\alpha_n)}_{\psi} = \underbrace{(\beta_1\beta_2) \dots (\beta_{m-1}\beta_m)}_{\tau}$$

Consider τ . It is the product of $m - 1$ 2-cycles. Each 2-cycle is also a permutation so it is an element of $\text{Sym}(A)$, so by a previous result $\tau^{-1} = ((\beta_1\beta_2) \dots (\beta_{m-1}\beta_m))^{-1} = (\beta_{m-1}\beta_m)^{-1} \dots (\beta_1\beta_2)^{-1} = (*)$. By **18 order of a cycle** the order of every 2-cycle is 2, so each 2-cycle is its own inverse, so $(*) = (\beta_{m-1}\beta_m) \dots (\beta_1\beta_2)$.

Now consider $\tau^{-1} \circ \psi$. Since $\tau = \psi = \phi$ then $\tau^{-1} \circ \psi = \psi^{-1} \circ \psi = \text{id}$. Multiplying the two explicitly according to $(*)$, we get:

$$\tau^{-1} \circ \psi = (\beta_{m-1}\beta_m) \dots (\beta_1\beta_2)(\alpha_1\alpha_2) \dots (\alpha_{n-1}\alpha_n) = \text{id}$$

So we have the identity permutation expressed as a product of 2-cycles, so by **Theorem 14** it must be that there is an even number of 2-cycles in this expression, but the number of 2-cycles in this expression is exactly the sum of the number of two cycles whose product is ψ and the sum of the number of two cycles whose product is τ , so we have two numbers that add up to an even number, which is only true if the parity of both numbers is the same (i.e. either both are even or both are odd), which completes the proof.

□

This results justifies the following definition:

Definition 0.0.22 (parity of a permutation).

A permutation is even if it can be written as the product of an even number of 2-cycles, otherwise it is odd. The sign of a permutation σ , denoted $\text{sgn}(\sigma)$, is defined to be 1 if σ is even and -1 if σ is odd.

Equipped with this result, we can derive the first useful result in our discussion of transpositions:

Theorem 15 (order of product of disjoint cycles (Ruffini)).

If $\phi = \gamma_1\gamma_2 \dots \gamma_n$, where $\gamma_1, \gamma_2, \dots, \gamma_n$ are all disjoint cycles with length $m_1, m_2, \dots, m_n$ respectively, then $|\phi| = \text{lcm}(m_1, m_2, \dots, m_n)$.

Proof. Denote $m = \text{lcm}(m_1, m_2, \dots, m_n)$ and consider $\phi^m = (\gamma_1\gamma_2 \dots \gamma_n)^m$. Since all cycles are disjoint, then by **17** they all commute, so by **Theorem 2 exponent laws (power rules)** we have $(\gamma_1\gamma_2 \dots \gamma_n)^m = \gamma_1^m \gamma_2^m \dots \gamma_n^m$. Since $\forall i \in \mathbb{N} \cap [1, n]$, $m_i | m$ (since m is their least common-multiple), and since each m_i is the length of a cycle γ_i which by **18 order of a cycle** is equal to the order of γ_i , then each terms can be written as $\gamma_i^{z_i m_i} = (\gamma_i^{m_i})^{z_i} = \text{id}^{z_i} = \text{id}$ for some $z_i \in \mathbb{Z}$, so $\phi^m = \text{id}$, so $|\phi| \leq m$.

Suppose $\phi^k = \text{id}$ for some k , so by a similar argument $\prod_{i=1}^n \gamma_i^k = \text{id}$, but then each $\gamma_i^k = \text{id}$ since the cycles are disjoint, so it must be that k is a common-multiple of $m_1, \dots, m_n$, so $m \leq k$, so we conclude $|\phi| = m$.

□

We see the theorem in action in the next example:

Example 15.1 (finding the order of a permutation).

Consider $\phi = (1253)(153)(1475)$. First, we rewrite it as the product of disjoint cycles following the algorithm outlined in **Theorem 11 disjoint form**: Start with 1.

: $\phi(1) = 4, \phi(4) = 7, \phi(7) = 1$, which completes a cycle. Next, pick 5. $\phi(5) = 3, \phi(3) = 2, \phi(2) = 5$, which completes the second cycle. There are no more elements to consider, so we conclude $\phi = (147)(532)$, the order of both cycles is 3, so their least common-multiple is 3, so $|\phi| = 3$. As another example, take $(123)(69)(7337249)$, which is already the product of disjoint cycles with lengths 3, 2, 4, so the order of this permutation is $\text{lcm}(3, 2, 4) = 12$.

We provide a sufficient conditions of S_n the symmetric group on n elements, to be S_n :

Theorem 16.

If p is prime and $H \leq S_p$, and also H consists of a transposition and a p -cycle, then $H = S_p$.

Proof. w.l.o.g suppose the transposition is $\sigma = (12)$, and let τ be the p -cycle, so $\tau^p = (1)$. If $p = 2$ then we are done since σ generates S_p , so suppose $3 \leq p$. By **18 order of a cycle**, $\exists n \in \mathbb{N}$ s.t. $1 \leq np$ and $\tau^n = (12i_3 \dots i_p)$ where $(i_3, \dots, i_p)$ are the other elements of the set of p letters other than 1, 2 (which we also denote $i_1 = 1, i_2 = 2$) in no particular order. The reason such τ^n exists is because, if we look at the distance from 1 to 2 in τ (which both must belong to the cycle τ since it consists of all elements of S_p by supposition) and denote it n , then **18 order of a cycle** tells us that τ^n maps 1 to 2, and primality of p guarantees that this too will be a p -cycle.

Now, note that $\sigma \circ \tau^n = (12) \circ (12i_3 \dots i_p) = (2i_3 \dots i_p)$, which we denote by ψ . Note that $\omega_k = \psi^k \circ (12) \circ \psi^{-k}$ for $k \in \mathbb{N}, 1 \leq k \leq p-2$, behaves on the elements of S_n as follows:

1. Since $1 \notin \psi$, then $1 \notin \psi^k$, so ψ^{-k} fixes 1, then (12) sends 1 to 2, which is then sent by ψ^k to i_{k+2} (since we suppose $1 \leq k \leq p-2$ then all these indices are greater than 3), so $\omega_k(1) = i_{k+2}$.
2. For any other element, there are two cases: either the element is sent to 2 by ψ^{-k} , and then sent to 1 by (12) and fixed by ψ^k , or it is not sent to 2, in which case it is not in (12) since $1 \notin \psi$, so then ω_k on this element reduces to $\psi^k \circ \psi^{-k} = \psi^{k-k} = (1)$ the identity. For i_{k+2} , we have $\psi^{-k}(i_{k+2}) = i_2$ which is 2, so $\omega_k(i_{k+2}) = 1$.

We have $\omega_k = (1i_{k+2})$ for $1 \leq k \leq p-2$ i.e. we have all transpositions of one of the p letters and 1. Since ω is given by a composition of ψ, σ , then $\omega_k \in H$ since H is a subgroup of S_n by supposition. Similarly,

$(1j)(1i)(1j) \in H$, but $(1j)(1i)(1j) = (ij)$ by **Theorem 13 transposition arithmetics**, so we have all transpositions of S_p in H . By **12.1 permutations as a composition of transpositions**, we can write every $\varphi \in S_p$ as a composition of transpositions, so $S_p \subseteq H$, but by supposition $H \leq S_p$, hence $S_p = H$.

□

Alternating Group

Suppose $\sigma, \tau \in S_n$ and both have even parity, then $\sigma \circ \tau$ clearly also has even parity. Also, since n is finite, then S_n is finite, hence by the one-step subgroup test, the set of all even permutations on n elements is a subgroup of S_n . We call this the *alternating group* on n elements:

Definition 0.0.23 (alternating group A_n).

Let $n \in \mathbb{N}$, then the alternating group on n elements, denoted A_n , is the group of all even permutations on n .

As we have argued briefly above, $A_n \leq S_n$.

Lemma 16.1 (every even permutation decomposes to 3-cycles).

If $\sigma \in S_n$ is an even permutation, then σ can be written as a composition of 3-cycles.

Proof. Let σ be an even permutation, then by definition it decomposes to an even number of transpositions. It suffices to show that for σ composed of 2 transpositions, we have that σ is expressible as a 3-cycle, then by induction this fact naturally extends to any even permutation on n letters.

Suppose $\sigma = (ab)(cd)$. If $(ab) = (cd)$ (i.e. $\{a, b\} = \{c, d\}$ in no particular order), then we have $\sigma = (ab)^2 = (1)$ the identity by **18 order of a cycle**, which is exactly the composition of zero 3-cycles so the proposition is vacuously true.

Suppose $(ab) \neq (cd)$. Since $(ab) = (ba)$ and $(cd) = (dc)$, there are really only two cases to consider w.l.o.g: $a = c$ or $a \neq c$.

Suppose $a = c$, then $(ab)(cd) = (ab)(ad)$, in which case by our proof for **Theorem 13 transposition arithmetics** we have that $(ab)(ad) = (adb)$, which is a 3-cycle, so the proposition holds.

Suppose $a \neq c$, then $(ab)(cd)$ is the composition of two disjoint transpositions. Recall that in our proof of **Theorem 16** we have seen that $(ab)(ac)(ab) = (bc)$, hence we can write (cd) as $(cd) = (ac)(ad)(ac)$, so we have $(ab)(cd) = (ab)(ac)(ad)(ac)$, which we can also write as $((ab)(ac)) \circ ((ad)(ac))$, i.e. as a composition of 2 permutations which themselves are the composition of two non-disjoint transpositions. As we have argued in the previous case, each can be written as a single 3-cycle, so $(ab)(cd) = (acb)(acd)$, so the proposition holds.

Completing the proof via induction on the general case is left as an exercise.

□

Corollary 16.1 (A_n is generated by 3-cycles).

A_n is generated by 3-cycles from S_n .

Proof. By 16.1 every even permutation decomposes to 3-cycles and by 0.0.23 alternating group A_n . □

We will collect a few more results on the alternating group on n letters later, once we introduce normal and simple groups. Finally, when we discuss solvability by radicals in the context of Galois theory, we will use these results to demonstrate that there is no general formula for finding the roots of a general quintic over the complex domain (we show a stronger result, but it requires definitions we haven't introduced yet, so for now we state this weakened version as a hint for what's to come).

Group Homomorphism

This section extends greatly on the introduction of the types of group morphisms in [[#^2e2207]]. Let $\phi : G_1 \rightarrow G_2$ be a homomorphism.

Proposition 19.

If $\phi : G_1 \rightarrow G_2$ is a homomorphism, then $\text{Ker}(\phi) \leq G_1$ (where $\text{Ker}(\phi) = \{x \in G_1 \mid \phi(x) = e_{G_2}\}$) and $\text{Im}(\phi) \leq G_2$.

Proof. $\text{Ker}(\phi) \leq G_1$: Recall by **6 group homomorphism preserves identity and commutes with inverse** that ϕ maps the identity of G_1 to the identity of G_2 , so $e_1 \in \text{Ker}(\phi)$, so $\text{Ker}(\phi)$ is a nonempty subset of G_1 . Consider $a, b \in \text{Ker}(\phi)$. By the same proposition, $\phi(b^{-1}) = \phi(b)^{-1}$, and since $b \in \text{Ker}(\phi)$ then $\phi(b)^{-1} = e_2^{-1} = e_2$. Consider ab^{-1} , since ϕ is a homomorphism then $\phi(ab^{-1}) = \phi(a)\phi(b^{-1})$, and $\phi(a) = e_2$ since $a \in \text{Ker}(\phi)$ and we have just shown $\phi(b^{-1}) = e_2$, so $\phi(ab^{-1}) = e_2 \cdot e_2 = e_2$, so $ab^{-1} \in \text{Ker}(\phi)$, so by **Theorem 5 one-step subgroup test** we have $\text{Ker}(\phi) \leq G_1$.

$\text{Im}(\phi) \leq G_2$: Again since $\phi(e_1) = e_2$ then $e_2 \in \text{Im}(\phi)$ so it is a nonempty subset of G_2 . Let $a, b \in \text{Im}(\phi)$, then $\exists x, y \in G_1$ s.t. $\phi(x) = a, \phi(y) = b$. Consider $xy^{-1} \in G_1$, since ϕ is a homomorphism we have $\phi(xy^{-1}) = \phi(x)\phi(y^{-1}) = a\phi(y)^{-1} = ab^{-1} \in \text{Im}(\phi)$, so by **Theorem 5 one-step subgroup test** $\text{Im}(\phi) \leq G_2$.

□

Theorem 17.

Let $\phi : G_1 \rightarrow G_2$ be a homomorphism, then ϕ is injective if and only if $\text{Ker}(\phi) = \{e_1\}$.

Proof. $\implies$ Suppose ϕ is injective. We know $\phi(e_1) = e_2$, so $e_1 \in \text{Ker}(\phi)$. Since ϕ is injective, every image has a single preimage, so $\nexists g \in G_1, g \neq e_1 \wedge \phi(g) = e_2$, so $\text{Ker}(\phi) = \{e_1\}$.

$\impliedby$ Suppose $\text{Ker}(\phi) = \{e_1\}$. Suppose by contradiction ϕ is not injective, i.e. $\exists a, b \in G_1$ s.t. $a \neq b$ and $\phi(a) = \phi(b)$. Observe that $\phi(a), \phi(b) \in G_2$ which is a group, so $\phi(b)^{-1} \in G_2$, so we can multiply both sides by $\phi(b)^{-1}$ and get $\phi(a)\phi(b)^{-1} = e_2$. By **6 group homomorphism preserves identity and commutes with inverse**, $\phi(a)\phi(b)^{-1} = \phi(ab^{-1})$, so $\phi(ab^{-1}) = e_2$, so $ab^{-1} \in \text{Ker}(\phi)$, but then $ab^{-1} = e_1$, so by definition $a^{-1} = b^{-1}$, so $a = b$, a contradiction, so ϕ is injective.

□

Proposition 20 (group homomorphism commutes with group operation).

If $\phi : G_1 \rightarrow G_2$ is a homomorphism and $a \in G_1$, then $\forall z \in \mathbb{Z}, \phi(a^z) = \phi(a)^z$.

Proof. Suppose the proposition holds $\forall 0 \leq z$, then if $z=0$ we recall by **Theorem 2 exponent laws (power rules)** that $a^z = (a^{-z})^{-1}$, and by **6 group homomorphism preserves identity and commutes with inverse** we know that $\phi(a^z) = \phi((a^{-z})^{-1}) = \phi(a^{-z})^{-1}$. Since $z=0$ then $0 = -z$, so for $\phi(a^{-z})$ the proposition holds and we have $\phi(a^{-z})^{-1} = (\phi(a)^{-z})^{-1}$, which by exponents laws is $\phi(a)^{(-z) \cdot (-1)} = \phi(a)^z$, so if the proposition holds $\forall 0 \leq z$ then it holds $\forall z \in \mathbb{Z}$.

We now prove the proposition holds $\forall 0 \leq z$ by induction. For $z = 0$ and $z = 1$ the proposition trivially holds. Suppose it holds for some z , and consider a^{z+1} , which by definition is $a^z a$, so $\phi(a^{z+1}) = \phi(a^z a)$, which since ϕ is a homomorphism, is the same as $\phi(a^z) \phi(a)$. Applying the induction hypothesis, we know $\phi(a^z) = \phi(a)^z$, so $\phi(a^z) \phi(a) = \phi(a)^z \phi(a) = \phi(a)^{z+1}$, which completes the proof by induction.

We conclude the proposition is true. □

Proposition 21 (Group homomorphism preserves subgroups).

If $\phi : G_1 \rightarrow G_2$ is a homomorphism, then:

1. If $H_1 \leq G_1$ then $\phi(H_1) \leq G_2$.
2. If $H_2 \leq G_2$ then $\phi^{-1}(H_2) \leq G_1$.

Proof. 1. Let $\phi : G_1 \rightarrow G_2$ be a group homomorphism and $H_1 \leq G_1$, and consider $\phi|_{H_1}$. Denote $H_2 = \text{Im}(\phi|_{H_1})$, we will prove by **Theorem 5 one-step subgroup test** that $H_2 \leq G_2$. First, since $e_1 \in H_1$ and $\phi(e_1) = e_2$, then $e_2 \in H_2$ so H_2 is nonempty. Consider some $a, b \in H_2$, then $\exists x, y \in H_1$ s.t. $\phi(x) = a, \phi(y) = b$. Consider $xy^{-1} \in H_1$ (it is an element of the set since H_1 is a group), then by **6 group homomorphism preserves identity and commutes with inverse** $\phi(xy^{-1}) = \phi(x)\phi(y)^{-1} = ab^{-1}$, so $ab^{-1} \in \text{Im}(\phi|_{H_1}) = H_2$, so $H_2 \leq G_2$.

2. Suppose $H_2 \leq G_2$, then $e_2 \in H_2$, and by **6 group homomorphism preserves identity and commutes with inverse** $\phi(e_1) = e_2$ so $e_1 \in \phi^{-1}(H_2)$, so $\phi^{-1}(H_2)$ is non-empty. We proceed by the one-step subgroup test. Denote $H_1 = \phi^{-1}(H_2)$ and let $a, b \in H_1$, then $\phi(ab^{-1}) = \phi(a)\phi(b)^{-1}$. Since $a, b \in H_1$ then $\phi(a), \phi(b) \in H_2$, so either $\phi(a) = \phi(b)$, in which case $\phi(a)\phi(b)^{-1} = \phi(a)\phi(a)^{-1} = e \in H_2$, or $\phi(a) \neq \phi(b)$, in which case since H_2 is a group, $\phi(a)\phi(b)^{-1} \in H_2$, in both cases $\phi(ab^{-1}) \in H_2$, so by definition $ab^{-1} \in \phi^{-1}(H_2) = H_1$, so by the one-step subgroup test $\phi^{-1}(H_2) \leq G_1$. □

Group Isomorphism

Isomorphisms are bijective homomorphisms. Intuitively, when two structures are *isomorphic* we understand that they are essentially *the same* under different guises, i.e. they have the same *structure*. A natural question then is, what does it mean for two groups to have the same *structure*? We have

defined certain properties of a group that make groups inherently different from one another, such as whether or not the group is *cyclic* or *abelian*, and the *order* of the group of elements of the group. We have also considered important subgroups, such as the *center* of a group and the *centralizers* of the group. When we say that two groups have the same *structure*, we expect to find a *bijective mapping* that maintains the group structure and keeps these properties as group invariants under the mapping. This is what it means for two groups to be *isomorphic*. All of these follow naturally as a result of supposing that there exists some isomorphism ϕ between the two groups, as we will demonstrate in the next theorem:

Theorem 18 (properties of a group isomorphism).

Let G_1, G_2 be groups and $\phi : G_1 \rightarrow G_2$ be a group isomorphism, then:

1. $|G_1| = |G_2|$.
2. $\phi(Z(G_1)) = Z(G_2)$, i.e. ϕ maps the center of G_1 to the center of G_2 .
3. $\forall a \in G_1, \phi(C(a)) = C(\phi(a))$, i.e. ϕ maps the centralizer of a in G_1 to the centralizer of $\phi(a)$ in G_2 .
4. $\phi(\langle a \rangle) = \langle \phi(a) \rangle$, i.e. ϕ maps cyclic subgroups to cyclic subgroups and generators to generators.
5. $\forall a \in G_1, |a| = |\phi(a)|$, i.e. ϕ preserves order.
6. G_1 is cyclic $\iff G_2$ is cyclic.
7. G_1 is abelian $\iff G_2$ is abelian.
8. Let $k \in \mathbb{Z}$, x a variable in G_1 , and $b \in G_1$, then $x^k = b$ has exactly the same number of solutions as $x^k = \phi(b)$.

Proof. 1. Trivially true since ϕ is a bijection so G_1, G_2 have the same number of elements, so $|G_1| = |G_2|$.

2. Recall **0.0.12 center of a group** - $Z(G_1) = \{g \in G_1 | \forall x \in G_1 (xg = gx)\}$. Let $y \in G_2$. Since ϕ is an isomorphism it is bijective and in particular surjective, i.e. $\forall y \in G_2 \exists x \in G_1$ s.t. $\phi(x) = y$. Let $g \in Z(G_1)$, so $xg = gx$, so $\phi(xg) = \phi(gx)$, and since ϕ is a group isomorphism we have $\phi(xg) = \phi(x)\phi(g) = y \cdot \phi(g)$ and $\phi(gx) = \phi(g)\phi(x) = \phi(g) \circ y$, which is true $\forall y \in G_2 \forall g \in Z(G_1)$, so $\phi(g) \in Z(G_2)$, so $\phi(Z(G_1)) \subseteq Z(G_2)$. Since ϕ^{-1} is also an isomorphism by [\[\[#^7c54e6\]\]](#) then by an identical argument we have $\phi^{-1}(Z(G_2)) \subseteq Z(G_1)$, so $Z(G_2) \subseteq \phi(Z(G_1))$, so by mutual inclusion $\phi(Z(G_1)) = Z(G_2)$.

3. Let $a \in G_1$. Recall **0.0.13 centralizer of an element** - $C(a) = \{g \in G_1 | ag = ga\}$. Take some $g \in C(a)$, then $\phi(a)\phi(g) = \phi(ag) = \phi(ga) = \phi(g)\phi(a)$, so $\phi(g) \in C(\phi(a))$, so $\phi(C(a)) \subseteq C(\phi(a))$. Symmetrically for ϕ^{-1} , we find that $\phi^{-1}(C(\phi(a))) \subseteq C(a)$, so we conclude $\phi(C(a)) = C(\phi(a))$.

4. Let $a \in G_1$ and consider a^k for some $k \in \mathbb{Z}$, then by **20 group homomorphism commutes with group operation** it follows that $\phi(a^k) = \phi(a)^k$, and since $\langle a \rangle = \{a^k | k \in \mathbb{Z}\}$ then $\phi(\langle a \rangle) = \{\phi(a)^k | k \in \mathbb{Z}\}$,

which is exactly $\langle \phi(x) \rangle$.

5. By (1), (4) it follows that $|\langle a \rangle| = |\langle \phi(a) \rangle|$, and by **11 properties of a cyclic group** $|a| = |\langle a \rangle|$, $|\phi(a)| = |\langle \phi(a) \rangle|$, so $|a| = |\phi(a)|$.

6. Immediately from (4).

7. $\implies$ Suppose G_1 is abelian, and let $a, b \in G_2$. Since ϕ is an isomorphism it is in particular surjective so $\exists x, y \in G_1$ s.t. $\phi(x) = a, \phi(y) = b$. Since G_1 is abelian, $xy = yx$, so $\phi(xy) = \phi(yx)$, and since ϕ is a homomorphism $ab = \phi(x)\phi(y) = \phi(xy) = \phi(yx) = \phi(y)\phi(x) = ba$, so G_2 is abelian. $\longleftarrow$ Follows the same argument with $\phi^{-1} : G_2 \rightarrow G_1$.

8. Suppose $x_0^k = b$ for some x_0 , then $\phi(x_0^k) = \phi(b)$, but $\phi(x_0^k) = \phi(x_0)^k$ by **20 group homomorphism commutes with group operation**, so $\phi(x_0)^k = \phi(b)$, so $\phi(x_0)$ is a solution for $x^k = \phi(b)$. Conversely, suppose $y_0^k = \phi(b)$ for some y_0 , then $\phi^{-1}(y_0^k) = \phi^{-1}(\phi(b))$, so $\phi^{-1}(y_0)^k = b$, so $\phi^{-1}(y_0)$ is a solution for $x^k = b$. Thus, every solution for $x^k = b$ is also a solution for $x^k = \phi(b)$, so the number of solutions to both equations is the same.

□

Since so many of the group properties are invariant under isomorphism, then we have many ways of determining that two groups are not isomorphism.

Example 18.1.

We will show $(\mathbb{R}, \times) \not\cong (\mathbb{C}, \times)$ (it is left as an exercise for the reader to verify the two structure are groups, with the usual definitions for multiplication in either field). Consider the equation $x^4 = 1$ in $\mathbb{R}$, then we have $x^4 - 1 = (x^2 - 1)(x^2 + 1) = (x - 1)(x + 1)(x^2 + 1) = 0$, and the only solutions to this equation in $\mathbb{R}$ are ± 1 , i.e. there are 2 solutions. If $(\mathbb{R}, \times) \cong (\mathbb{C}, \times)$, then for every isomorphism $\phi : \mathbb{R} \rightarrow \mathbb{C}$, $1 \mapsto 1$, so by the final proposition of **Theorem 18 properties of a group isomorphism** we expect to find 2 solutions in $\mathbb{C}$ as well, however in $\mathbb{C}$ the expression can be written as $(x - 1)(x + 1)(x - i)(x + i) = 0$, and the solutions to this equation are $\pm 1, \pm i$, i.e. there are 4 solutions compared to 2 in $\mathbb{R}$, so $(\mathbb{R}, \times) \not\cong (\mathbb{C}, \times)$.

We now turn our attention to automorphisms, which are isomorphisms from a structure to itself. Let G be a group, then we denote the set of automorphisms of G as $\text{Aut}(G)$.

Proposition 22.

$$\text{Aut}(G) \leq \text{Sym}(G)$$

Proof. id_G is an automorphism so $\text{Aut}(G)$ is nonempty. We proceed according to the one-step subgroup test. Let $\phi, \psi \in \text{Aut}(G)$, then $\phi \circ \psi^{-1} : G \rightarrow G$ is also an isomorphism since the inverse of an isomorphism is also an isomorphism ([#^7c54e6]) and isomorphisms form an equivalence relation

($[[\#^1\text{ea7eb}]]$), and clearly it is an isomorphism from G to itself, so $\phi \circ \psi^{-1} \in \text{Aut}(G)$, so $\text{Aut}(G) \leq \text{Sym}(G)$ by the one-step subgroup test. $\square$

We turn our attention to a particular subgroup of automorphisms. We begin with an example. Let $M \in \text{GL}(n, \mathbb{R})$, then M^{-1} exists. Now let $X \in \text{GL}(n, \mathbb{R})$ and consider a mapping $\phi : X \mapsto M^{-1}XM$. Recall from linear algebra that we call this *conjugation* by the matrix M . Now observe that $\det(M^{-1}XM) = \det(M^{-1})\det(X)\det(M) = \det(M)^{-1}\det(M)\det(X) = \det(X) \neq 0$, so $M^{-1}XM \in \text{GL}(n, \mathbb{R})$, so ϕ a mapping from $\text{GL}(n, \mathbb{R})$ to itself. Likewise, $\phi(MXM^{-1}) = X$, so ϕ is surjective and also if $\phi(X) = \phi(Y)$ then trivially $X = Y$ so it is also injective, so ϕ is bijective, and $\phi(XY) = M^{-1}XYM = M^{-1}XMM^{-1}YM = \phi(X)\phi(Y)$ so it is a homomorphism, so we conclude ϕ is an automorphism. We call automorphisms of this form *inner automorphisms*.

Definition 0.0.1 (inner automorphism).

Let G be a group. An inner automorphism $\phi_g : G \rightarrow G$ is an automorphism of the form $\phi_g : x \mapsto gxg^{-1}$ for some $g \in G$. We call ϕ_g conjugation by g . The set of all inner automorphisms of a group G is denoted $\text{Inn}(G)$.

Proposition 23.

Let ϕ_g be an inner automorphism $x \mapsto gxg^{-1}$, then $\phi_g^{-1} = \phi_{g^{-1}}$, i.e. $\phi_g^{-1}(x) = g^{-1}xg$.

Proof. Since ϕ_g is an automorphism then it is an isomorphism so it is a bijection so its inverse exists and is unique. Observe via direct substitution that $g(g^{-1}xg)g^{-1} = (gg^{-1})x(gg^{-1})$ from associativity, which is just $e \cdot x \cdot e = x = \text{id}(x)$, so $\phi_g^{-1}(x) = g^{-1}xg$, which is just $\phi_{g^{-1}}$. $\square$

Proposition 24.

$$\text{Inn}(G) \leq \text{Aut}(G)$$

Proof. Observe that $\text{id}_G(x) = e \cdot x \cdot e^{-1}$ so $\text{id}_G \in \text{Inn}(G)$ so it is nonempty. We proceed by the one-step subgroup test. Let $\phi, \psi \in \text{Inn}(G)$, then $\exists g, h \in G$ s.t. $\phi(x) = gxg^{-1}$ and $\psi(x) = h x h^{-1}$. By 23, $\psi^{-1}(x) = h^{-1}xh$. Consider $(\phi \circ \psi^{-1})(x) = \phi(\psi^{-1}(x)) = g(h^{-1}xh)g^{-1} = (gh^{-1})x(hg^{-1})$. Observe that $(gh^{-1})^{-1} = hg^{-1}$, so $(\phi \circ \psi^{-1})(x) = (gh^{-1})x(gh^{-1})^{-1}$, which is exactly the inner automorphism of gh^{-1} , so $\phi \circ \psi^{-1} \in \text{Inn}(G)$, so $\text{Inn}(G) \leq \text{Aut}(G)$ by the subgroup test $\square$

Consider $\mathbb{Z}_n^+$, i.e. the group of modular addition $\pmod n$. Since the group is cyclic (with generator 1), then every subgroup of it is also cyclic, and we know by a previous result that for an element $k \in \mathbb{Z}_n$ to generate it, it must satisfy $\gcd(n, k) = 1$, i.e. (n, k) must be coprime, and the amount of such elements is also exactly $|\mathbb{Z}_n^*|$, i.e. the order of the group of modular multiplication $\pmod n$. This observation motivates the following theorem:

Theorem 19.

$$\text{Aut}(\mathbb{Z}_n^+) \cong \mathbb{Z}_n^*$$

Proof. Let $\psi \in \text{Aut}(\mathbb{Z}_n^+)$ and let $k \in \mathbb{Z}_n^+$, then $\psi(k) = \psi(k \cdot 1) = k\psi(1)$ (here we used **20 group homomorphism commutes with group operation** in additive notation), so ψ is uniquely determined by where it maps $1 \in \mathbb{Z}_n^+$. By **Theorem 18 properties of a group isomorphism**, since $\langle 1 \rangle$ is a cyclic subgroup of $\mathbb{Z}_n^+$ (in fact, it is the same group), then ψ maps 1 to a generator of a cyclic subgroup in $\mathbb{Z}_n^+$ of same order, i.e. it must map 1 to another generator of the group. As we recalled above, the only elements that satisfy this property are those which are coprime w.r.t. n , the amount of which is exactly $\phi(n)$ where ϕ is Euler's totient function evaluated at n , and also $|\mathbb{Z}_n^+|$.

Consider a mapping $\tau : \text{Aut}(\mathbb{Z}_n^+) \rightarrow \mathbb{Z}_n^*$ given by $\tau(\psi) = \psi(1)$. As we reasoned above, $\psi(1)$ is coprime w.r.t. n , and the elements of $\mathbb{Z}_n^*$ are exactly the numbers which are coprime w.r.t. n (and less than n), so the codomain is valid, and trivially the function is a bijection. Consider $\psi, \eta \in \text{Aut}(\mathbb{Z}_n^+)$. By **22** we know that $\text{Aut}(\mathbb{Z}_n^+)$ is a group, so $\tau(\psi \circ \eta)$ is well-defined, but also $(\psi \circ \eta)(1) = \psi(\eta(1))$, and since ψ is an automorphism and $\eta(1) \in \mathbb{Z}_n^+$ then by the same reasoning as above $\psi(\eta(1)) = \psi(1 \cdot \eta(1)) = \psi(1)\eta(1)$, and by definition of τ , $\tau(\psi)\tau(\eta) = \psi(1)\eta(1)$, so we have $\tau(\psi \circ \eta) = \tau(\psi)\tau(\eta)$, so τ is a homomorphism and also bijective, so it is an isomorphism, so $\text{Aut}(\mathbb{Z}_n^+) \cong \mathbb{Z}_n^*$.

□

Intuitively speaking, the theorem implies that we can think of the automorphisms of $\mathbb{Z}_n^+$ as elements of $\mathbb{Z}_n^*$.

Cosets

Definition 0.0.1 (coset).

Let $H \subseteq G$ where G is a group with multiplicative notation, and let $a \in G$.

1. $\{ah|h \in H\}$ is called the left coset of H with representative a , denoted aH .
2. $\{ha|h \in H\}$ is called the right coset of H with representative a , denoted Ha .

Cosets let us perform a group operation where one of the operands is fixed on the entire set. For example, $2\mathbb{Z}$ is the set of all even integers (which we also know to be a group), and $1 + \mathbb{R}$ is simply the real line again, and also $1 + \mathbb{R} = \mathbb{R} + 1$. We see that sometimes the left and right cosets with same representative are the same, and sometimes they are equal to the set itself, i.e. there is a natural bijection from H to aH or Ha in that case. Also of note is that sometimes the representative is not a member of the coset, for example $2 + 3\mathbb{N}^+$ is a left coset of $3\mathbb{N}^+$ with is a subset (but not a subgroup!) of $(\mathbb{Z}, +)$, yet 2 is not a member of that coset since $2 = 2 + 3x$ has no solutions in $\mathbb{N}^+$. When $H \leq G$, then trivially a is a member of all a -cosets:

Proposition 25.

Let G be a group with $a \in G$. If $H \leq G$, then $a \in aH$ and $a \in Ha$.

Proof. Since $H \leq G$ then $e \in H$, so $a = ae \in aH$ and $a = ea \in Ha$. □

We now suppose that $H \leq G$ and show that all “nice” properties we expect are preserved.

Theorem 20 (properties of a coset).

Let $H \leq G$ and $a, b \in G$, then:

1. $aH = H \iff a \in H$
2. $aH = bH$ or $aH \cap bH = \emptyset$
3. $aH = bH \iff a^{-1}b \in H$
4. $|aH| = |bH|$ and $|Ha| = |Hb|$.
5. $aH = Ha \iff H = aHa^{-1}$
6. $aH \leq G \iff a \in H$.

Proof. 1. $\implies$ Suppose $aH = H$. By 25, $H \leq G \implies a \in aH$, so $a \in H$. $\impliedby$ Suppose $a \in H$, so $a^{-1} \in H$ since H is a group. Let $h \in H$, then $a^{-1}h \in H$, so $a(a^{-1}h) = h$ and also $a(a^{-1}h) \in aH$, so $H \subseteq aH$. Now let $ah \in aH$, since $a \in H$ and $h \in H$ and H is a group then trivially $ah \in H$, so $aH \subseteq H$, so $H = aH$.

2. Suppose $aH \neq bH$ and suppose by contradiction $aH \cap bH \neq \emptyset$, i.e. $\exists k$ s.t. $k = ah$ and $k = bg$ for some $h, g \in H$. Let $bt \in bH$ for some $t \in H$, so $g^{-1}t \in H$, so $a(hg^{-1}t) \in aH$, but $a(hg^{-1}t) =$

$(ah)(g^{-1}t) = b(gg^{-1})t = bt$, so $bt \in aH$, so $bH \subseteq aH$. By a symmetrical argument we find $aH \subseteq bH$, thus $aH = bH$, a contradiction, so either $aH = bH$ or $aH \cap bH = \emptyset$.

3. $\implies$ Suppose $aH = bH$. $b \in bH$ by **25**, so $b \in aH$, so $b = ah$ for some $h \in H$, and the only solution to the equation is $h = a^{-1}b$, so $a^{-1}b \in H$. $\Leftarrow$ Suppose $a^{-1}b \in H$, so $a(a^{-1}b) = b \in aH$, and by **25** $b \in bH$, so $b \in aH \cap bH$, so by (2), $aH = bH$.

4. By definition $aH = \{ah|h \in H\}$. Consider a mapping $\phi : H \rightarrow aH, h \mapsto ah$. Evidently it is surjective. Suppose $\phi(h) = \phi(g)$, then $ah = ag$, then since $a \in G$ and G is a group, by **Theorem 1 cancellation law** $h = g$, so ϕ is injective, so it is a bijection, so $|H| = |aH|$. Since aH is any left coset of H , then also $|H| = |bH|$, so $|aH| = |bH|$. By a similar argument we find that $\phi : H \rightarrow Ha, h \mapsto ha$ is a bijection, so we have $|aH| = |Ha|$ and $|Ha| = |Hb|$.

5. First, notice that, in general aHb is well defined even without parenthesis (i.e. $(aH)b = a(Hb)$, so the expression aHb is well defined), this follows directly from the associativity of G .

$\implies$ Suppose $aH = Ha$. Let $aha^{-1} \in aHa^{-1}$. Since $aH = Ha$ then $ah = ga$ for some $g \in H$, so $aha^{-1} = gaa^{-1} = g \in H$, so $aHa^{-1} \subseteq H$. Conversely, let $h \in H$. Solving $h = aga^{-1}$ in G , we have $g = a^{-1}ha$, and again since $Ha = aH$ then $\exists k \in H$ s.t. $ha = ak$, so $a^{-1}ha = a^{-1}ak$, so $g = k \in H$, so $aga^{-1} \in aHa^{-1}$, but $aga^{-1} = h$, so $H \subseteq aHa^{-1}$, so $H = aHa^{-1}$. $\Leftarrow$ Suppose $H = aHa^{-1}$, then by definition $aHa^{-1} = \{aha^{-1}|h \in H\}$ and also $Ha = \{ha|h \in H\}$, and since $aHa^{-1} = H$ we can write $h \in H$ as aga^{-1} for some $g \in H$, so $Ha = \{(aga^{-1})a|g \in H\} = \{ag|g \in H\} = aH$ by definition, so $Ha = aH$.

6. $\implies$ Suppose $aH \leq G$, then $e \in aH$, so $ah = e$ for some H , so $h = a^{-1}$, so $a^{-1} \in H$, and since $H \leq G$ it follows that $a \in H$. $\Leftarrow$ Suppose $a \in H$, then by (1), $aH = H$, and by supposition $H \leq G$, so $aH \leq G$.

□

Theorem 21 (Lagrange's theorem).

Let G be a finite group and let $H \leq G$, then $|H||G|$.

Proof. Suppose $H \leq G$. If $H = G$ then the theorem is trivially true, so suppose $H \neq G$, then $G \setminus H$ is nonempty, so take some $a_1 \in G \setminus H$, and consider a_1H . By **Theorem 20 properties of a coset**, $|a_1H| = |eH| = |H|$. Now consider $G \setminus a_1H$, it is also nonempty since $|a_1H| = |H| < |G|$, so let $a_2 \in G \setminus a_1H$. Consider a_2H , by the previous theorem we also know $|a_2H| = |a_1H| = |H|$, and also since $a_2 \notin a_1H$ and yet $a_2 = a_2e \in a_2H$ it follows that $a_1H \neq a_2H$, so by the theorem they are disjoint. Now, either $a_1H \cup a_2H = G$, or the union is a proper subset of G , in which case we consider some $a_3 \in G \setminus \cup_{i=1}^2 a_iH$. Again $|a_3H| = |H|$. If $G \setminus \cup_{i=1}^3 a_iH = \emptyset$, then since all left cosets are disjoint, we have $|G| = \sum_{i=1}^3 |a_iH| = 3|H|$, so $|H||G|$. Otherwise, we proceed as above. Since G is finite and all cosets are disjoint, eventually this process will terminate, and it is always the case that

$$\sum_{i=1}^n |a_i H| = n|H|, \text{ so } |H| \mid |G|.$$

□

Note that **Theorem 21 Lagrange's theorem** generalizes a result we have already proven for cyclic groups in **Theorem 9 fundamental theorem of cyclic groups**.

Corollary 21.1.

If $H \leq G$ and G is a finite group, then G has exactly $\frac{|G|}{|H|}$ distinct left H -cosets.

Proof. In the proof of Lagrange's theorem we have seen that for some finite sequence $(a_1, \dots, a_n)$, we have $G = \cup_{i=1}^n a_i H$, such that $\forall a_i, a_j (i \neq j \implies (a_i H \cap a_j H = \emptyset))$. Consider some $b \in G$ s.t. $b \notin \{a_1, \dots, a_n\}$, and consider bH . For every i , either $a_i H$ and bH are disjoint, or they are the same coset (by **Theorem 20 properties of a coset**), and since G is covered by $\mathcal{H} = \{a_i H \mid a_i \in \{a_1, \dots, a_n\}\}$ then for some a_i it must be that $a_i H = bH$, and only for that a_i since for every other a_j we have $a_i H \cap a_j H = \emptyset$, so every left H -coset is an element of $\mathcal{H}$, so G has exactly $|\mathcal{H}| = n = \frac{|G|}{|H|}$ distinct left H -cosets. □

Lagrange's theorem motivates the following definition:

Definition 0.0.2 (index of a subgroup).

Let $H \leq G$, then the index of H in G , denoted $[G : H]$, is the number of disjoint left H -cosets in G .

By **Theorem 21 Lagrange's theorem**, the index of a subgroup is well defined.

Corollary 21.2.

If G is a group of prime order, then it is cyclic.

Proof. Let G be a group of prime order, then the only divisors of $|G|$ are itself and 1, which is the order of the trivial cyclic subgroup of G . Let $a \in G$ s.t. $a \neq e$ (there must exist such element since otherwise $|G| = 1$ which is not prime), and consider $\langle a \rangle$, it is a subgroup of G so by **Theorem 21 Lagrange's theorem**, $|\langle a \rangle| \mid |G|$, so $|\langle a \rangle|$ is either 1 or $|G|$, but by choice $a \neq e$ so $|\langle a \rangle| \neq 1$, so $|\langle a \rangle| = |G|$, so $\langle a \rangle = G$, so G is cyclic. □

Proposition 26.

If G is a finite group and $a \in G$, then $a^{|G|} = e_G$.

Proof. Consider $\langle a \rangle \leq G$. By **Theorem 21 Lagrange's theorem**, $|a| = |\langle a \rangle|$ divides $|G|$, so $|G| = n|a|$, so $a^{|G|} = a^{n|a|} = (a^{|a|})^n = e_G^n = e_G$. □

Theorem 22 (every group of composite order has proper nontrivial cyclic subgroup).

If G is a group and $|G|$ is composite (i.e. not prime nor 1), then $\exists H \leq G$ s.t. H is proper nontrivial, i.e. $H \neq G$ and also $H \neq \{e\}$. Furthermore, we can find such H that is cyclic.

Proof. Consider $a \in G$ s.t. a is not the identity of G . If $\langle a \rangle \neq G$ then since $\langle a \rangle \leq G$ we are done. Otherwise, write $G = nd$ where (n, d) are coprime and consider $\langle a \rangle, \langle a^d \rangle$. By supposition $\langle a \rangle = G$, so $|a| = nd$. By 8.2, $\langle a \rangle = \langle a^d \rangle$ if and only if $(|a|, d)$ are coprime, but by supposition $|a| = nd$ and clearly d is a common divisor of $|a|, d$, so $\langle a \rangle \neq \langle a^d \rangle$, so it follows that $\langle a^d \rangle$ is a proper nontrivial subgroup of G . $\square$

Normal Subgroups and Factor Groups

We build our way towards a reduction of group to smaller building blocks. To do that, we introduce the notion of a normal subgroup.

Definition 0.0.3 (normal subgroup).

If G is a group and $H \leq G$ is a subgroup of G s.t. $\forall a \in G, aH = Ha$, then we say that H is a normal subgroup of H , denoted $H \trianglelefteq G$ (normality).

Proposition 27.

Let G be a group, then $\{e\}, Z(G)$ and G are always normal subgroups of G .

Proof. Let $g \in G$.

1. $\{e\} = \langle e \rangle$ which is the trivial subgroup of G , and also $eg = ge$, so $g\langle e \rangle = \langle e \rangle g$, so $\{e\} \trianglelefteq G$.
2. Consider $Z(G)$. By definition, $gZ(G) = \{gz | z \in Z(G)\}$, but since $z \in Z(G)$ then $gz = zg$, so $\{gz | z \in Z(G)\} = \{zg | z \in Z(G)\} = Z(G)g$, so $Z(G) \trianglelefteq G$.
3. Consider G , since $g \in G$ then by Theorem 20 properties of a coset $gG = Gg = G$, so $G \trianglelefteq G$.

$\square$

Proposition 28 (subgroup with index 2 is normal subgroup).

Let G be a group and $H \leq G$ s.t. $[H : G] = 2$, then $H \trianglelefteq G$.

Proof. Suppose G is a group with subgroup H , and that the order of H in G is 2, then G is exactly H and aH for some $a \notin H$. Since $a \notin H$, then $Ha \neq H$, hence $Ha = aH$, hence H is normal by definition, hence $H \trianglelefteq G$. $\square$

Definition 0.0.4 (simple group).

A group G is simple if $\forall H \trianglelefteq G$, either $H = \{e\}$ or $H = G$, i.e. it has no non-trivial proper normal subgroups.

Example 22.1 (simple group).

Consider $\mathbb{Z}_p^+$ for some prime p which is a cyclic group (either by identifying it has prime order, or by identifying that 1 is a generator of the group) then by 8.4 it follows that every cyclic subgroup of $\mathbb{Z}_p^+$ (aside from the identity) is the same as $\mathbb{Z}_p$, and by *Theorem 9 fundamental theorem of cyclic groups* we know that all subgroups of $\mathbb{Z}_p^+$ are cyclic, so if $H \leq \mathbb{Z}_p^+$ then $H = \{0\}$ or $H = \mathbb{Z}_p^+$, so $\mathbb{Z}_p^+$ has no proper non-trivial subgroups, so it is also the case that it has no non-trivial proper normal subgroups, so $\mathbb{Z}_p^+$ is simple.

We will later show that normal subgroups and simplicity of a group are preserved under an isomorphism.

Recall *Theorem 20 properties of a coset*. We have shown that $aH = Ha \iff H = aHa^{-1}$. We can now express this relation in terms of normality, and write $H \trianglelefteq G \iff H = aHa^{-1}$. The direction in the backwards direction ($(H = aHa^{-1}) \implies (H \trianglelefteq G)$) can be further relaxed:

Proposition 29.

If $H \leq G$ and $\forall a \in G, aHa^{-1} \subseteq H$, then $H \trianglelefteq G$.

Proof. Left as an exercise (can be shown by the same argument as the one in the respective proposition of *Theorem 20 properties of a coset*). □

We collect a result on normal subgroups of A_n for use later in our discussion of the insolubility of the quintic:

Theorem 23.

If $N \trianglelefteq A_n$ and $\exists \sigma \in N$ s.t. σ is a 3-cycle, then $N = A_n$ (for $3 \leq n$).

Proof. Suppose $N \trianglelefteq A_n$ and $\sigma \in N$ is a 3-cycle (abc) , then by *Theorem 20 properties of a coset* $\forall \tau \in A_n, \tau \circ \sigma \circ \tau^{-1} \in N$. Also, recall that A_n is generated by 3-cycles on S_n (*16.1 A_n is generated by 3-cycles*), so it is sufficient to show that if $\tau \in A_n$ is a 3-cycle, then $\tau \in N$, from which it would follow that $N = A_n$.

w.l.o.g. let $\sigma = (123)$, and let $\tau = (abc)$ be a general 3-cycle in A_n . We will show $(abc) \in N$ via gradual substitution of 1, 2, 3 with a, b, c .

Consider conjugation of σ by $(12a) \in A_n$ (since it is a 3-cycle it is in A_n), then $(12a)^{-1}(123)(12a) \in N$ since N is normal in A_n . Calculating the composition explicitly, we have that it produces the 3-cycle $(13a)$, so $(13a) \in N$. As a formal formula in FOL, we can write this as

$$(((123) \in N) \wedge ((12a) \in A_n)) \implies ((13a) \in N) \quad (1)$$

We label the above formula (1). The formula above lets us substitute a letter in our 3-cycle with an arbitrary letter and get a new 3-cycle also in N .

We repeat the same process for $(13a) \in N$ and $(13b) \in A_n$, then by substitution $1 \mapsto 1, 2 \mapsto 3, 3 \mapsto a, a \mapsto b$ we get from (1) above that $(1ab) = (ab1) \in N$. Since N is a subgroup then $(ab1)^2 \in N$, and by **18 order of a cycle** and primality of 3 we find that $(ab1)^2 = (a1b)$, so $(a1b) \in N$.

Finally, repeating the same process one final time with $(a1b) \in N, (a1c) \in A_n$, then again by (1) above and substitution, we find $(abc) \in N$, so N consists of all 3-cycles on n letters, so by **16.1 A_n is generated by 3-cycles**, $N = A_n$.

□

The key to the insolubility of the quintic lies in the next theorem, the proof of which we skip as it is rather technical, although it definitely can be read and understood at this point:

Lemma 23.1.

If $5 \leq n$, and $N \trianglelefteq A_n$ and also $N \neq \{e\}$, then N contains a 3-cycle.

Theorem 24 ($A_{5 \leq n}$ is simple).

A_n is simple for $5 \leq n$.

Proof. Suppose $N \trianglelefteq A_n$ and also $5 \leq n$. Suppose $N \neq \{e\}$. By **23.1**, $\exists \sigma \in N$ s.t. σ is a 3-cycle. By **Theorem 23**, $N = A_n$, so A_n has no proper normal subgroup, so A_n is simple by **0.0.4 simple group**.

□

By **Theorem 21 Lagrange's theorem**, we know that H divides G into $\frac{|G|}{|H|}$ distinct cosets, denote them $G/H = \{aH | a \in G\}$. This is a quotient set under the relation $aRb \iff aH = bH$, which trivially is an equivalence relation. If H

is a normal subgroup of G , we equip this quotient set (also called a factor set) with a binary operator to form a group, which we call the quotient group of G under H (also sometimes denoted $G \bmod H$):

Definition 0.0.5 (factor group).

Let $H \trianglelefteq G$, then the structure $(G/H, \cdot)$, where:

1. $G/H = \{aH | a \in G\}$, i.e. it is the quotient set of G under H .
2. $\cdot : (G/H) \times (G/H) \rightarrow G/H$ is given by $aH \cdot bH = abH$. We call this operation coset multiplication.

Is called the factor group of G under H .

It remains to show that:

1. The binary operator is well-defined.
2. The structure satisfies the group axioms.

Proposition 30.

G/H for $H \trianglelefteq G$ is a group under coset multiplication.

Proof. First, we show the operation is well-defined. Suppose $H \trianglelefteq G$ and consider G/H , and consider two representatives of two elements of the quotient set, $a_1H = a_2H$ and $b_1H = b_2H$. It remains to show that $a_1H \cdot b_1H = a_2H \cdot b_2H$, i.e. that $a_1b_1H = a_2b_2H$. Since H is a normal subgroup of G , then $b_1H = Hb_1$, so $a_1b_1H = a_1Hb_1$. Now, since $a_1H = a_2H$, then $a_1Hb_1 = a_2Hb_1$, and since $b_1H = b_2H$ and H is normal then $Hb_1 = Hb_2$, so $a_2Hb_1 = a_2Hb_2$, and finally $a_2Hb_2 = a_2b_2H$ since $Hb_2 = b_2H$, so we have $a_1b_1H = a_2b_2H$, so coset multiplication is well defined on normal subgroups. Also, since $\forall a, b \in G, ab \in G$ then abH is a left coset of H , so it is an element of G/H by 21.1, so the domain of the mapping is the set, so it is indeed a binary operator on G/H .

It remains to show that $(G/H, \cdot)$ satisfies the group axioms. Note that we cannot use a subgroup test since $G/H \not\subseteq H$.

1. (left identity) since eH is a left coset of H (where e is the identity of H so it is also the identity of G since $H \leq G$), then by definition $\forall a \in G (ea = a)$, so by definition of coset multiplication $\forall aH \in G/H (eH \cdot aH = aH)$, so $eH = H$ is the (left) identity of G/H . By the same process we can demonstrate that it is also a right identity, but that is not necessary (see 0.1).
2. (left inverse) let $a \in G$, so $a^{-1}H, aH \in G/H$, and by definition $a^{-1}H \cdot aH = a^{-1}aH = eH = H$, which is the identity of G/H , so every element has a left inverse.
3. (associativity) suppose $aH, bH, cH \in G/H$, then $aH \cdot (bH \cdot cH) = aH \cdot (bcH) = a(bc)H$, and since $a, b, c \in G$ then by associativity of G , $a(bc)H = (ab)cH = (aH \cdot bH) \cdot cH$, so the operator is associative on G/H .

We conclude that $(G/H, \cdot)$ is a group. Note that normality of H was necessary for the operator to be well-defined.

□

Proposition 31 (natural homomorphism from group to factor group).

Let $H \trianglelefteq G$, then $\phi : G \rightarrow G/H, a \mapsto aH$ is a group homomorphism. We call this ϕ the natural homomorphism of $G \rightarrow G/H$. This homomorphism is also trivially surjective.

Proof. We prove this ϕ is a homomorphism. Let $a, b \in G$, then $\phi(ab) = (ab)H = aH \cdot bH = \phi(a)\phi(b)$.

□

Proposition 32 (order of element in the factor group).

Let $N \trianglelefteq G$ and G/N be a factor group, then $\forall aN \in G/N$, the order of aN is the smallest $k \in \mathbb{N}$ s.t. $a^k \in N$.

Proof. By definition of coset multiplication, $(aN)^k = a^k N$. The order of aN is the smallest positive k s.t. $(aN)^k = \text{id}_{G/N} = eN = N$. Clearly if we take $(aN)^{|a|}$, where $|a|$ is the order of a in G , then $(aN)^{|a|} = a^{|a|} N = eN = N$, so by the well-ordering principle the order of aN exists and is well-defined since $|a|$ exists, so we have $a^k N = N$, and by **Theorem 20 properties of a coset** this is true if and only if $a^k \in N$. $\square$

Proposition 33 (order of a factor group).

Let $N \trianglelefteq G$ and G/N be a factor group, then $|G/N| = [G : N]$, where $[G : N]$ is the index of N in G .

Proof. By **0.0.2 index of a subgroup**, $[G : N]$ is the number of distinct left N -cosets of G . In **0.0.5 factor group** we define G/N to be the group whose elements are the distinct left N -cosets of G , so by definition $|G/N| = [G : N]$. $\square$

Normal subgroup homomorphism & first isomorphism theorem

Here we demonstrate that normality of a subgroup and simplicity of a group are preserved under an isomorphism, and then present the first isomorphism theorem and show a few examples of the theorem.

Theorem 25 (homomorphism preserves normality).

If $\phi : G_1 \rightarrow G_2$ is a homomorphism, then:

1. If ϕ is surjective and $H_1 \trianglelefteq G_1$, then $\phi(H_1) \trianglelefteq G_2$ (the image of a normal subgroup is a normal subgroup of the codomain)
2. If $H_2 \trianglelefteq G_2$, then $\phi^{-1}(H_2) \trianglelefteq G_1$ (the pre-image of a normal subgroup is a normal subgroup of the domain)

Proof. Let $\phi : G_1 \rightarrow G_2$.

1. Suppose ϕ is surjective (note: this condition can be removed entirely since we know that $\phi : G_1 \rightarrow \phi(G_1)$ is surjective, and since G_1 is a group and ϕ is a homomorphism then the image of subgroups of G_1 is a subgroup of the domain so trivially $\phi(G_1)$ is also a group by **21 Group homomorphism preserves subgroups**). Suppose $H_1 \trianglelefteq G_1$, so $H_1 \leq G_1$, so by **21 Group homomorphism preserves subgroups**

$\phi(H_1) \leq G_2$, so it remains to show that $\phi(H_1)$ is normal. Let $y \in G_2$, since ϕ is surjective, $\exists x \in G_1$ s.t. $\phi(x) = y$. Since H_1 is normal, then $xH_1 = H_1x$, i.e. $\forall h \in H_1 (xh = hx)$. Let $h' \in \phi(H_1)$, then $\exists h \in H_1$ s.t. $\phi(h) = h'$, and since $h \in H_1$ then $xh = hx$, so $\phi(xh) = \phi(hx)$, now since ϕ is a homomorphism $\phi(xh) = \phi(x)\phi(h) = yh'$ and $\phi(hx) = \phi(h)\phi(x) = h'y$, and this is true $\forall y \in G_2 \forall h' \in \phi(H_1)$, so by definition $y\phi(H_1) = \phi(H_1)y$, so $\phi(H_1) \trianglelefteq G_2$.

2. Suppose $H_2 \trianglelefteq G_2$, so $H_2 \leq G_2$ and by **21 Group homomorphism preserves subgroups** $\phi^{-1}(H_2) \leq G_1$. It remains to show $\phi^{-1}(H_2)$ is normal. Since H_2 is normal, then $\forall g_2 \in G_2, g_2H_2 = H_2g_2$, so by **Theorem 20 properties of a coset** $g_2H_2g_2^{-1} = H_2$.

Let $g_1 \in G_1$, then $\phi(g_1) \in G_2$ since ϕ is a mapping from G_1 to G_2 , so g_1 is a pre-image of some $g_2 \in G_2$. Denote $H_1 = \phi^{-1}(H_2)$ and let $h_1 \in H_1$, then it is the preimage of some $h_2 \in H_2$, so we have $\phi(h_1) = h_2$ and $\phi(g_1) = g_2$. Consider $g_1h_1g_1^{-1} \in G_1$, since ϕ is a homomorphism then $\phi(g_1h_1g_1^{-1}) = \phi(g_1)\phi(h_1)\phi(g_1)^{-1} = g_2h_2g_2^{-1} \in g_2H_2g_2^{-1}$, which as we argued last paragraph is equivalent to H_2 , so $g_1h_1g_1^{-1} \in \phi^{-1}(H_2) = H_1$, and this holds $\forall g_1 \in G_1 \forall h_1 \in H_1$, so by **29**, $H_1 \trianglelefteq G_1$.

□

Corollary 25.1.

If $\phi : G_1 \rightarrow G_2$ is a homomorphism, then $\text{Ker}(\phi) \trianglelefteq G_1$.

Proof. By **27** we know that $\{e_2\} \trianglelefteq G_2$, and by definition $\phi^{-1}(\{e_2\}) = \text{Ker}(\phi)$, so by **Theorem 25 homomorphism preserves normality** $\text{Ker}(\phi) \trianglelefteq G_1$.

□

Corollary 25.2.

If $\phi : G_1 \rightarrow G_2$ is a homomorphism, then:

1. The factor group $G_1/\text{Ker}(\phi)$ is well-defined.
2. If $\text{Ker}(\phi) = \{e_1\}$, then $G_1/\text{Ker}(\phi) \cong G_1$.

Proof. 1. By **25.1**, $\text{Ker}(\phi) \trianglelefteq G_1$. By **30**, if $H_1 \trianglelefteq G_1$ then the factor group G_1/H_1 is well-defined, so $G_1/\text{Ker}(\phi)$ is well-defined.

2. Suppose $\text{Ker}(\phi) = \{e_1\}$ and consider the natural homomorphism $\psi : G_1 \rightarrow G_1/\text{Ker}(\phi), a \mapsto a\text{Ker}(\phi)$ (see **31 natural homomorphism from group to factor group**).

The identity of $G_1/\text{Ker}(\phi)$ is $\text{Ker}(\phi)$. Suppose $\psi(a) = \text{Ker}(\phi)$ for some $a \in G_1$, so $a = \text{Ker}(\phi)$, but also by definition $a = a\text{Ker}(\phi)$, so $a\text{Ker}(\phi) = \text{Ker}(\phi)$. By **Theorem 20 properties of a coset**, this is true if and only if $a \in \text{Ker}(\phi)$, but we supposed $\text{Ker}(\phi) = \{e_1\}$, so $a = e_1$, so $\text{Ker}(\psi) = \{e_1\}$, so by **Theorem 17**, ψ is injective. Also, ψ is surjective, so ψ is an isomorphism, so $G_1 \cong G_1/\text{Ker}(\phi)$, and since isomorphism is an equivalence relation, it is symmetric, so $G_1/\text{Ker}(\phi) \cong G_1$.

□

Proposition 34 (isomorphism preserves simplicity).

If $\phi : G_1 \rightarrow G_2$ is an isomorphism then G_1 is simple if and only if G_2 is simple.

Proof. Suppose G_1 is simple and G_2 is not, then G_2 has some normal subgroup $H_2 \trianglelefteq G_2$ s.t. $H_2 \notin \{\{e_2\}, G_2\}$. By **Theorem 25 homomorphism preserves normality**, $\phi^{-1}(H_2) \trianglelefteq G_1$. By **Theorem 18 properties of a group isomorphism**, $|H_2| = |\phi^{-1}(H_2)|$, and since $|G_1| = |G_2|$ (since ϕ is bijective) then it must also be that $\phi^{-1}(H_2) \notin \{\{e_1\}, G_1\}$, but then $\phi^{-1}(H_2)$ is a normal non-trivial subgroup of G_1 , a contradiction. Following the same argument, it is shown that if G_2 is simple then G_1 must also be simple. □

We now prove a lemma towards the first isomorphism theorem (some writers call this lemma the first isomorphism theorem, however we reserve this name for a more general theorem)

Lemma 25.1.

If $\phi : G_1 \rightarrow G_2$ is a homomorphism, then $\psi : G_1/\text{Ker}(\phi) \rightarrow \phi(G_1), x\text{Ker}(\phi) \mapsto \phi(x)$ is an isomorphism, and $G_1/\text{Ker}(\phi) \cong \phi(G_1)$.

Proof. $G_1/\text{Ker}(\phi)$ is a factor group as per **25.1**. First, we show ψ is well-defined. Suppose $a\text{Ker}(\phi) = b\text{Ker}(\phi)$ for $a, b \in G_1$, then by definition $\psi(a\text{Ker}(\phi)) = \phi(a)$ and $\psi(b\text{Ker}(\phi)) = \phi(b)$. For ψ to be well-defined it must be single-valued, i.e. it must be that $\phi(a) = \phi(b)$, and indeed since $a\text{Ker}(\phi) = b\text{Ker}(\phi)$ then by **Theorem 20 properties of a coset** $ab^{-1} \in \text{Ker}(\phi)$, so $\phi(ab^{-1}) = e_2$, and since ϕ is a homomorphism then $\phi(ab^{-1}) = \phi(a)\phi(b)^{-1} = e_2$, so $\phi(b)^{-1} = \phi(a)^{-1}$, so $\phi(a) = \phi(b)$, so ψ is single-valued and so it is well-defined.

Next, we show ψ is a group homomorphism: Indeed, $\psi(a\text{Ker}(\phi) \cdot b\text{Ker}(\phi)) = \psi(ab\text{Ker}(\phi)) = \phi(ab) = \phi(a)\phi(b) = \psi(a\text{Ker}(\phi))\psi(b\text{Ker}(\phi))$, so ψ is a group homomorphism.

Let e_2 be the identity on $\phi(G_1)$, then $\psi(e_1\text{Ker}(\phi)) = \phi(e_1) = e_2$, so $e_1\text{Ker}(\phi) \in \text{Ker}(\psi)$, so it is non-empty. Now let $x \in \text{Ker}(\psi)$, then $x = a\text{Ker}(\phi)$ for some $a \in G_1$, and $\psi(x) = \phi(a)$, and also $\psi(x) = e_2$ since $x \in \text{Ker}(\psi)$, so $\phi(a) = e_2$, so $a \in \text{Ker}(\phi)$ by definition, but $\forall a \in \text{Ker}(\phi) (a\text{Ker}(\phi) = \text{Ker}(\phi))$ by **Theorem 20 properties of a coset**, so in fact all choices for a are congruent $\pmod{\text{Ker}(\phi)}$, i.e. they all map to the same element, so $\text{Ker}\psi = \{a\text{Ker}(\phi) | a \in \text{Ker}(\phi)\} = \{\text{Ker}(\phi)\}$, which is the identity on $G_1/\text{Ker}(\phi)$, so by **Theorem 17** ψ is injective, and clearly ψ is surjective (let $y \in \phi(G_1)$, then $\exists x \in G_1$ s.t. $\phi(x) = y$, and also $x\text{Ker}(\phi) \in G/\text{Ker}(\phi)$, so $\psi(x\text{Ker}(\phi)) = \phi(x) = y$, so ψ is surjective), so it is a bijection, so ψ is an isomorphism, so $G_1/\text{Ker}(\phi) \cong \phi(G_1)$. □

Theorem 26 (first isomorphism theorem).

If $\phi : G_1 \rightarrow G_2$ is a group homomorphism, then:

1. $\text{Ker}(\phi) \trianglelefteq G_1$
2. $\phi(G_1) \leq G_2$
3. $G_1 / \text{Ker}(\phi) \cong \phi(G_1)$

Proof. (1) is proven in 25.1, (2) is proven in 21 Group homomorphism preserves subgroups, and (3) is proven in 25.1. $\square$

Corollary 26.1.

If $\phi : G_1 \rightarrow G_2$ is a group homomorphism and G_1 is finite, then $|G_1| = |\text{Ker}(\phi)| \cdot |\text{Im}(\phi)|$.

Proof. By Theorem 26 first isomorphism theorem $G_1 / \text{Ker}(\phi) \cong \text{Im}(\phi)$, by 21.1 since $\text{Ker}(\phi)$ is a normal subgroup of G_1 and G_1 is finite, we know $[G_1 : \text{Ker}(\phi)] = \frac{|G_1|}{|\text{Ker}(\phi)|}$. By 33 order of a factor group, $|G_1 / \text{Ker}(\phi)| = [G_1 : \text{Ker}(\phi)]$, so $|G_1 / \text{Ker}(\phi)| = \frac{|G_1|}{|\text{Ker}(\phi)|}$. Since the two groups are isomorphic, then $|G_1 / \text{Ker}(\phi)| = |\text{Im}(\phi)|$, so $|G_1| = |\text{Ker}(\phi)| \cdot |\text{Im}(\phi)|$. $\square$

Proposition 35 (order of element under homomorphism).

If $\phi : G_1 \rightarrow G_2$ is a group homomorphism, and $a \in G_1$, then the order of $\phi(a)$ in G_2 divides the order of a in G_1 , i.e. $|\phi(a)|_{G_2} \mid |a|_{G_1}$.

Proof. Consider ϕ restricted to $\langle a \rangle$, then $\phi(a^k) = \phi(a)^k \in \langle \phi(a) \rangle$, so $\psi = \phi|_{\langle a \rangle}$ is a group homomorphism $\langle a \rangle \rightarrow \langle \phi(a) \rangle$. Suppose $\psi(a^k) = e$, then $\psi(a^k) = \psi(a)^k = \phi(a)^k = e$, so $|\phi(a)|$ must divide k by previous results on cyclic groups. Since $\phi(e) = e$ (by 6 group homomorphism preserves identity and commutes with inverse), and $a^{|a|} = e$, then $\phi(a^{|a|}) = \phi(a)^{|a|} = e$, so $|\phi(a)| \mid |a|$. $\square$

We proceed with some examples where we use the first isomorphism theorem.

Example 26.1.

Recall $GL(n, \mathbb{R})$ is a group, and consider $\det : GL(n, \mathbb{R}) \rightarrow \mathbb{R}^\times$ (where $\mathbb{R}^\times$ is the group of real numbers with usual multiplication), then $e_{\mathbb{R}^\times}$ is the multiplicative identity of $\mathbb{R}$ which is 1, so by definition $\text{Ker}(\det) = \{A \in GL(n, \mathbb{R}) \mid \det(A) = 1\}$, which is just the definition of $SL(n, \mathbb{R})$ (note: this also immediately proves that $SL(n, \mathbb{R})$ is a group under matrix multiplication once we show that $\det$ is a group homomorphism). Next, let $A, B \in GL(n, \mathbb{R})$, then by properties of the determinant $\det(AB) = \det(A) \det(B)$, so $\det$ is a group homomorphism, and it is also surjective (let $r \in \mathbb{R}$, then the uniform scaling matrix rI has determinant r , so $\det$ is surjective) so by Theorem 26 first isomorphism theorem we have that $GL(n, \mathbb{R}) / SL(n, \mathbb{R}) \cong \mathbb{R}^*$.

We present the next example as a theorem:

Theorem 27 (cyclic groups are isomorphic to modular addition groups of integers of same order).

If $G = \langle a \rangle$, then:

1. If $|G| = \infty$, $G \cong \mathbb{Z}^+$ (which is just the group of integers under addition)
2. If $|G| = n$ for some $n \in \mathbb{N}^+$, $G \cong \mathbb{Z}_n^+$ (the group of integers under modular addition mod n).

Proof. 1. Let $G = \langle a \rangle$ and suppose $|G| = \infty$, and consider the mapping $\phi : \mathbb{Z}^+ \rightarrow G, z \mapsto a^z$. Observe that $\phi(z + n) = a^{z+n} = a^z a^n = \phi(z)\phi(n)$, so it is a group homomorphism from $\mathbb{Z}^+$ to G . Let $z \in \text{Ker}(\phi)$, then $\phi(z) = e$ and also $\phi(z) = a^z$ (by definition of ϕ), so $a^z = e$, which is true if and only if $z = 0$ since G is a non-finite cyclic group, so $\text{Ker}(\phi) = \{0\}$, which is also the identity of $\mathbb{Z}^+$ (this is to be expected since the identity of the domain is taken to the identity of the codomain by a group homomorphism), so by **Theorem 26 first isomorphism theorem** $\mathbb{Z}^+ / \{0\} \cong G$. By 25.2, we have $\mathbb{Z}^+ \cong \mathbb{Z}^+ / \{0\}$, so since isomorphism is an equivalence relation we have $G \cong \mathbb{Z}^+$.

2. Let $G = \langle a \rangle$ and suppose $|G| = n$, and similarly consider the mapping $\phi : \mathbb{Z}^+ \rightarrow G$. By the same reasoning as before ϕ is a group homomorphism. Now let $z \in \text{Ker}(\phi)$, then as before $a^z = e$, which is true if and only if $n|z$ by 3.1, so $z \in \text{Ker}(\phi) \iff (z = nk \wedge k \in \mathbb{Z}^+)$, so by definition of the right coset, $\text{Ker}(\phi) = n\mathbb{Z}^+$. By **Theorem 26 first isomorphism theorem**, $\mathbb{Z}^+ / n\mathbb{Z}^+ \cong G$, but $\mathbb{Z}^+ / n\mathbb{Z} = \mathbb{Z}_n^+$, so $G \cong \mathbb{Z}_n^+$.

□

Example 27.1.

Consider $\mathbb{R}^+$ (the group of real numbers under addition) and $\mathbb{C}^\times$ (the group of complex numbers under complex multiplication), and define $\phi : x \mapsto e^{ix}$, then $\phi(x + y) = e^{i(x+y)} = e^{ix+iy} = e^{ix}e^{iy} = \phi(x)\phi(y)$, so it is a group homomorphism. Furthermore, $\text{id}(\mathbb{C}^\times) = 1$, so $\forall x \in \text{Ker}(\phi) (\phi(x) = 1)$, i.e. $e^{ix} = 1$. Recall from Euler's formula that $e^{ix} = \cos x + i \sin x$, so $1 = \cos x + i \sin x$ if and only if $\sin x = 0$ and $\cos x = 1$, which is true if and only if $x = 2\pi k$ for $k \in \mathbb{Z}$, so $\text{Ker}(\phi) = 2\pi\mathbb{Z}$, so by **Theorem 26 first isomorphism theorem** $\mathbb{R}^+ / 2\pi\mathbb{Z} \cong \phi(\mathbb{R}^+)$. Recall the geometric interpretation of $\cos x + i \sin x$; it is a point in the complex unit circle which we denote S_1 , i.e. $\text{Im}(\phi) \subseteq S_1$. Let $s \in S_1$, then $s = \cos x + i \sin x$ for some x , so $x \in \mathbb{R}^+$, so $\phi(x) = s$, so $S_1 \subseteq \text{Im}(\phi)$, so $\text{Im}(\phi) = S_1$, so in fact $\mathbb{R}^+ / 2\pi\mathbb{Z} \cong S_1$. Geometrically we interpret it as follows: take the real line and slice it into intervals of the form $I_k = [2\pi k, 2\pi(k + 1)]$ for $k \in \mathbb{Z}$, then by wrapping I_k we can construct a unit circle, and also addition of points in I_k has the same affect as multiplying those points in S_1 .

Theorem 28 ($S_n/A_n \cong \mathbb{Z}_2$).

Let $4 \leq n$. The alternating group on n letters A_n is a normal subgroup of the symmetric group on n letters S_n , and also $S_n/A_n \cong \mathbb{Z}_2$.

Proof. Consider the mapping $\phi : S_n \rightarrow \mathbb{Z}_2, \sigma \mapsto \text{sgn}(\sigma)$, where $\text{sgn}(\sigma)$ is the parity of the cycle σ as defined in 0.0.22 parity of a permutation, only that we let $\text{sgn}(\sigma)$ be 0 when the permutation is even, and let it be 1 otherwise. Since the parity of a permutation is well-defined, so is ϕ . Clearly, ϕ is surjective for any $4 \leq n$ since $(12)(13)(14) \in S_n$ which has odd parity so $\phi((12)(13)(14)) = 1$, while $\text{id} \in S_n$ has even parity so $\phi(\text{id}) = 0$, hence ϕ is surjective, i.e. $\text{Im}(\phi) = \mathbb{Z}_2$.

Next, we show ϕ is a group homomorphism: let $\sigma, \tau \in S_n$, then there are 3 cases:

1. σ, τ are both even, $\sigma \circ \tau$ is even, so $\phi(\sigma \circ \tau) = 0 = 0 + 0 = \phi(\sigma) + \phi(\tau)$
2. σ, τ are both odd, so $\sigma \circ \tau$ is even, so $\phi(\sigma \circ \tau) = 0 \equiv_2 2 = 1 + 1 = \phi(\sigma) + \phi(\tau)$
3. w.l.o.g σ is even and τ is odd, so $\sigma \circ \tau$ is odd, so $\phi(\sigma \circ \tau) = 1 = 0 + 1 = \phi(\sigma) + \phi(\tau)$

So, ϕ is a surjective group homomorphism. Note that by definition of ϕ , $\text{Ker}(\phi)$ is the set of even permutations on n letters, i.e. it is exactly A_n , hence $\text{Ker}(\phi) = A_n$, so by Theorem 26 first isomorphism theorem:

1. $A_n \trianglelefteq S_n$
2. $S_n/A_n \cong \mathbb{Z}_2$.

□

Corollary 28.1 ($[S_n : A_n] = 2$).

For $4 \leq n$, $[S_n : A_n] = 2$.

Proof. By 26.1 and Theorem 28 $S_n/A_n \cong \mathbb{Z}_2$, $|S_n| = |A_n| \cdot |\mathbb{Z}_2| = 2|A_n|$, so by 0.0.2 index of a subgroup, $[S_n : A_n] = 2$. □

Rank-Nullity Theorem

Recall the rank-nullity theorem from linear algebra:

Theorem 29 (rank-nullity theorem).

If T is a linear transformation between vector spaces and $\text{Domain}(T)$ finite dimensional, then

$$\dim(\text{Domain}(T)) = \dim(\text{Ker}(T)) + \dim(\text{Im}(T))$$

In this section we will briefly demonstrate that the first isomorphism theorem is a generalization of the rank-nullity theorem. First, we provide a quick refresher of relevant terminology from linear algebra (although usually in undergraduate linear algebra courses, these definitions appear before group theory has been properly introduced, so we offer a revised, more compact version of the same definitions, using definitions from group theory.)

Definition 0.0.6 (vector space).

The structure $(V, +, \cdot, \mathbb{F})$ where:

1. $\mathbb{F}$ is a field
2. V is a nonempty set
3. $+: V \times V \rightarrow V$ is a binary operator on V (vector addition)
4. $\cdot: \mathbb{F} \times V \rightarrow V$ is a binary operator (scalar multiplication)

Is called a vector space, if the following properties are satisfied $\forall u, v \in V \forall a, b \in \mathbb{F}$:

1. $(V, +)$ is an abelian group
2. $a(bv) = (ab)v$ (scalar multiplication is compatible with field multiplication)
3. If $1 \in \mathbb{F}$ is the multiplicative identity of $\mathbb{F}$, then $1v = v$.
4. $a(u + v) = au + av$ (scalar multiplication is distributive over vector addition)
5. $(a + b)v = av + bv$ (scalar multiplication is distributive over field addition)

We note that 2, 3 can be abbreviated to saying that $\mathbb{F}$ acts on V via $\cdot$, but we have not yet presented group or field actions so we ignore that for now.

Definition 0.0.7 (quotient vector space).

If V is a vector space and $U \subseteq V$ is a vector subspace of V , then the factor group V/U is called the quotient vector space of V under U , where vector addition and scalar multiplication are defined as follows:

1. $[u] + [w] = [u + w]$
2. $\alpha[u] = [\alpha u]$

Since a group generalizes vector spaces (vector spaces satisfy all group axioms and have more structure), then all our previous results apply, namely that vector addition in V/U is well-defined and that the natural homomorphism $V \rightarrow V/U, v \mapsto v + U$ is a homomorphism from V to its quotient space under U . More importantly, the first isomorphism theorem also holds. It is left to the reader to demonstrate that $\alpha[u] \mapsto [\alpha u]$ is also well defined.

We now show that the V/U is indeed a vector space under the same field $\mathbb{F}$ as V .

Proposition 36.

Let V be a vector space over a field $\mathbb{F}$ and U be a vector subspace of U , then V/U is also a vector space over $\mathbb{F}$.

Proof. We already know from group theory that V/U is a group, and it is trivial to demonstrate that if V is abelian and $U \trianglelefteq V$ then V/U is also abelian. We proceed to demonstrate the 4 other axioms are satisfied. Let $[u], [w] \in V/U$ and $a, b \in \mathbb{F}$, then:

1. $a(b[u]) = a([bu]) = [abu] = (ab)[u]$
2. $1[u] = [1u] = [u]$
3. $a([u] + [w]) = a([u + w]) = [a(u + w)] = [au + aw] = [au] + [aw] = a[u] + a[w]$
4. $(a + b)[u] = [(a + b)u] = [au + bu] = [au] + [bu] = a[u] + b[u]$

Thus we conclude that V/U with the operations defined in **0.0.7 quotient vector space** is a vector space. □

Since V/U is a vector space, then we can discuss its *dimension*.

Proposition 37.

If V has finite dimension, then $\dim(V/U) = \dim(V) - \dim(U)$

Proof. Let b_U be a basis of U , then it is a set of $\dim(U)$ linearly independent vectors in V and $\dim(U) \leq \dim(V)$ since U is a subspace of V , so we can complete b_U to a basis b_V of V by extending b_U with $\dim(V) - \dim(U)$ new linearly independent vectors, so now we have $|b_V| = |\dim(V)|$. Note that this extension is only possible since V is *finite*.

Let $[v] \in V/U$, then $[v] = x + U$ for some $x \in V$. Let $\phi : V \rightarrow V/U$ be the natural homomorphism $x \mapsto x + U$, then by **31 natural homomorphism from group to factor group** it is trivially a surjection, so since b_V spans V then trivially $\phi(b_V)$ spans V/U .

By **Theorem 20 properties of a coset** we know that $\forall x \in U, x + U = U$, and if $x \in U$ then it can be expressed as a linear combination of b_U . Let $x \in V$, then we can write x as $x = x_U + x_V$, where x_U is x projected to U and x_V is x projected to $V \setminus U$, so $x + U = x_U + x_V + U$ and since V is abelian then $x_U + x_V + U = x_V + x_U + U = x_V + (x_U + U)$ and as we argued $x_U + U = U$ since $x_U \in U$, so we have that every $[v] \in V/U$ can be written as $x_V + U$ where x_V is a linear combination of the basis vectors of $b_V \setminus b_U$, so $\dim(V/U) \leq |b_V \setminus b_U| = \dim(V) - \dim(U)$. Since $b_V \setminus b_U$ is a set of linearly independent vectors, then it is trivial to show that it cannot be that $\dim(V/U) < \dim(V) - \dim(U)$ (suppose it is, then it must be that at least two of the basis vectors which we denote $b_1, b_2 \in b_V \setminus b_U$

map to the same coset, but then it must be that $b_1 - b_2 \in U$ by **Theorem 20 properties of a coset**, so $\{b_1, b_2\} \cup b_U$ is not a set of linearly independent vectors, a contradiction).

Thus we conclude $\dim(V/U) = \dim(V) - \dim(U)$. □

Lemma 29.1.

If $T : V \rightarrow U$ is a bijective linear map and V be finite dimensional, then $\dim(V) = \dim(U)$.

Proof. Let $v \in V$ and $B_V = \{b_1, \dots, b_n\}$ where $n = \dim V$, then $v = \sum_{i=1}^n \alpha_i b_i$ for some set of scalars. Let $u \in U$, since T is bijective then $\exists v \in V$ s.t. $T(v) = u$. Substituting v for a linear combination of basis vectors, we have $T(v) = \sum_{i=1}^n \alpha_i T(b_i) = u$ from linearity of T , so $T(B_V)$ spans U , so $\dim U \leq \dim V$. Similarly, by considering the inverse bijection T^{-1} (it is left as an exercise to the reader to demonstrate that T^{-1} is also linear) and a basis B_U of U , we find that $T^{-1}(B_U)$ spans V , so $\dim V \leq \dim U$. So we conclude $\dim V = \dim U$. □

We can now prove the rank-nullity theorem (**Theorem 29 rank-nullity theorem**) using the first isomorphism theorem:

Proof. Denote $\text{Domain}(T) = V$, thus the theorem can be restated as $V = \dim(\text{Ker}(T)) + \dim(\text{Im}(T))$.

Since T is a linear transformation then $\forall v, w \in V$, $T(u + w) = T(u) + T(w)$, so it is a group homomorphism, so by **Theorem 26 first isomorphism theorem** $V/\text{Ker}(T) \cong \text{Im}(T)$.

It is to show that $\psi : v\text{Ker}(T) \mapsto T(v)$ is a linear map ($(\alpha v + u)\text{Ker}(T) \mapsto T(\alpha v + u) = \alpha T(v) + T(u)$ since T is linear, and also $\alpha v\text{Ker}(T) \mapsto T(\alpha v) = \alpha T(v)$, so $\psi((\alpha v + u)\text{Ker}(T)) = \alpha\psi(v\text{Ker}(T)) + \psi(u\text{Ker}(T))$, so ψ is linear), and ψ is the mapping under which $V/\text{Ker}(T) \cong \text{Im}(T)$ (as per the proof of **Theorem 26 first isomorphism theorem**), then it is by **29.1**, $\dim(V/\text{Ker}(T)) = \dim(\text{Im}(T))$.

By **37**, $\dim(V/\text{Ker}(T)) = \dim(V) - \dim(\text{Ker}(T))$, so $\dim(V) - \dim(\text{Ker}(T)) = \dim(\text{Im}(T))$. Rearranging the final expression, we get $\dim(V) = \dim(\text{Ker}(T)) + \dim(\text{Im}(T))$. □

Direct Product

Let $(G, \cdot), (H, \times)$ be groups and consider their cartesian product $G \times H$, whose elements are the tuples (g, h) where $g \in G, h \in H$. It is natural to define a binary operator on $G \times H$ as follows: $\star : (G \times H)^2 \rightarrow G \times H, (g_1, h_1) \star (g_2, h_2) \mapsto (g_1 \cdot g_2, h_1 \times h_2)$, i.e. the operation on the product set is realized by applying the respective group operation on each component. We can verify that the resulting structure $(G \times H, \star)$ is indeed a group whose identity element is (e_G, e_H) . We call this group the *external direct product* of G and H . We present a more generalized definition below, and then proceed to show it is indeed a group:

Definition 0.0.1 (external direct product).

Suppose $\mathcal{G} = \{G_1, \dots, G_n\}$ is a finite collection of groups, then the structure $(\prod_{i=1}^n G_i, \cdot)$, where:

1. $\prod_{i=1}^n G_i = G_1 \times \dots \times G_n$, i.e. it is the cartesian product of all underlying sets
2. $\cdot : (\prod_{i=1}^n G_i)^2 \rightarrow \prod_{i=1}^n G_i$ is a binary operation on the product set, given by $(g_1, \dots, g_n) \cdot (h_1, \dots, h_n) = (g_1 h_1, \dots, g_n h_n)$, where each $g_i h_i$ term is the result of applying the group operation of G_i on $g_i, h_i \in G_i$.

Is called the external direct product on $\mathcal{G}$. We denote it simply as $G_1 \times \dots \times G_n$ or times $\prod_{i=1}^n G_i$, via abuse of notation.

Proposition 38.

1. The external direct product is a group
2. The identity of the external direct product is the n -tuple whose i -th element is the identity of the i -th group in the product.
3. The inverse of an element of the external direct product a is the n -tuple $a^{-1} = (a_1^{-1}, \dots, a_n^{-1})$.

Proof. We show the structure defined in **0.0.1 external direct product** satisfies the group axioms $\forall a, b, c \in \times_{i=1}^n G_i$:

1. (associativity) by definition $(a(bc))_i = a_i(b_i c_i)$. By associativity of the group operation of G_i , $a_i(b_i c_i) = (a_i b_i) c_i$, which is exactly $((ab)c)_i$, so $a(bc) = (ab)c$.
2. (left identity) let $e = (e_1, \dots, e_n)$ i.e. the n -tuple of identity elements of the underlying groups of the product, then by definition $(ea)_i = e_i a_i$, but since e_i is the identity of G_i then $e_i a_i = a_i$, so $ea = a$, so e is the identity of the product.
3. (left inverse) let $a^{-1} = (a_1^{-1}, \dots, a_n^{-1})$, then by definition $(a^{-1}a)_i = a_i^{-1} a_i = e_i$, so $a^{-1}a = e$, so a^{-1} is indeed the left inverse of a .

We conclude that the external direct product is a group and that sentences (2) and (3) of the proposition are true.

□

If $H, K \leq G$, then we can consider the subset $HK = \{hk \in G | h \in H, k \in K\}$ under the group operation of G . Trivially we find that since $e \in H \cap K$ then $e \in HK$ (identity) and given $h_1k_1, h_2k_2, h_3k_3 \in HK$, then $(h_1k_1h_2k_2)h_3k_3 = h_1k_1(h_2k_2h_3k_3)$ (associativity) since $HK \subseteq G$ and the group operation is associative on G . However when we attempt to show that the resulting structure $(HK, \cdot)$ where $\cdot$ is the group operation of G has the inverse property, we run into an issue: we want to find an inverse of hk in HK . We know that $(hk)^{-1} = k^{-1}h^{-1}$, so we want $k^{-1}h^{-1} \in HK$ so we wish to find $h' \in H, k' \in K$ s.t. $h'k' = k^{-1}h^{-1}$, but this equation is not always solvable. However, if $H, K \trianglelefteq G$, then we have $HK = KH$ and clearly since $k^{-1} \in K, h^{-1} \in H$, we find that $k^{-1}h^{-1} \in KH = HK$. This discussion motivates the following definition:

Definition 0.0.2 (independent subgroups).

Let $H_1, \dots, H_n \leq G$ where $n \in \mathbb{N}, 2 \leq n$, then $H_1, \dots, H_n$ are independent if $\forall i \in \mathbb{N} \cap [1, n-1], (\prod_{j=1}^i H_j) \cap H_{i+1} = \{e\}$ (we say that the intersection is trivial), where $\prod_{j=1}^i H_j = \{\prod_{j=1}^i h_j | h_j \in H_j\}$.

Definition 0.0.3 (internal direct product).

Let $H_1, \dots, H_n \trianglelefteq G$ be a finite set of independent normal subgroups, then the structure $(H_1 \dots H_n, \cdot)$ where $H_1 \dots H_n = \{h_1 \dots h_n | h_i \in H_i\}$ and $\cdot$ is the group operation of G is called the internal direct product of $H_1, \dots, H_n$ and is denoted $H_1 \oplus \dots \oplus H_n$, or $\oplus_{i=1}^n H_i$.

We have motivated why we are only interested in *normal* subgroups. The motivation behind the restriction that the subgroups be *independent* will become clearer later, when we will show that the internal direct product and the external direct product of the same subgroups are isomorphic.

Proposition 39.

The internal direct product is a group.

Proof. The proof is identical to the proof presented for the case where $n = 2$ in the discussion a few paragraphs above. Filling out the detail of the group for a general finite n is left to the reader. □

We now prove a lemma towards the isomorphism theorem for external and internal direct product:

Lemma 29.2.

Suppose $H_1, \dots, H_n \trianglelefteq G$ and the intersection of any two $H_i, H_j, i \neq j$ is the identity (equivalently, suppose $\oplus_{i=1}^n H_i$ can be defined), then for all $a = \prod_{i=1}^n a_i$ and $b = \prod_{i=1}^n b_i$, where $a_i, b_i \in H_i$ for all $i \in \mathbb{N} \cap [1, n]$:

1. If $a = b$, then for all such i , $a_i = b_i$.
2. $\prod_{i=1}^n a_i = \prod_{i=n}^1 a_i$ (i.e. for $n = 3$, we have $a_1a_2a_3 = a_3a_2a_1$)

Proof. We prove both propositions by induction on n . For the direct product to be defined we must choose n at least 2, so our base case will be $n = 2$.

1. Suppose $n = 2$, then $a_1a_2 = b_1b_2$ where $a_1, b_1 \in H_1, a_2b_2 \in H_2$. Right-multiplying by a_1^{-1} and left-multiplying by b_2^{-1} , we get

$$\begin{aligned} a_1^{-1}(a_1a_2)b_2^{-1} &= a_1^{-1}(b_1b_2)b_2^{-1} \\ (a_1^{-1}a_1)a_2b_2^{-1} &= a_1^{-1}b_1(b_2b_2^{-1}) \\ \underbrace{a_2b_2^{-1}}_{\in H_2} &= \underbrace{a_1^{-1}b_1}_{\in H_1} \end{aligned}$$

So we have $a_2b_2^{-1} \in H_1 \cap H_2$, and by supposition $H_1 \cap H_2 = \{e\}$, so $a_2b_2^{-1} = e$, so $a_2 = b_2$. Similarly since we have $a_1^{-1}b_1 \in H_1 \cap H_2$, we have $a_1 = b_1$, which completes the proof for the case where $n = 2$.

Suppose the proposition holds for some n , and consider $n + 1$, then we have

$$\begin{aligned} \prod_{i=1}^{n+1} a_i &= \prod_{i=1}^{n+1} b_i \\ \left(\prod_{i=1}^n a_i\right)a_{n+1} &= \left(\prod_{i=1}^n b_i\right)b_{n+1} \end{aligned}$$

Note that $\prod_{i=1}^n a_i, \prod_{i=1}^n b_i \in \oplus_{i=1}^n H_i$, so by letting $H = \prod_{i=1}^n H_i, K = H_{n+1}$ we have $H, K \trianglelefteq G$ and since all subgroups are independent, then by **0.0.2 independent subgroups** $\oplus_{i=1}^n H_i \cap H_{n+1} = \{e\}$, so $H \cap K = \{e\}$, so denoting $h_1 = \prod_{i=1}^n a_i, g_1 = \prod_{i=1}^n b_i \in H$ and $h_2 = a_{n+1}, g_2 = b_{n+1} \in K$, we have the $n = 2$ case which we have already proven, so it follows that $h_1 = h_2$ and $g_1 = g_2$, i.e. $a_{n+1} = b_{n+1}$.

Consider $h_1 = h_2$ and recall that this is just a change of notation for $\prod_{i=1}^n a_i = \prod_{i=1}^n b_i$, so by the induction hypothesis $a_i = b_i$ for every $i \in \mathbb{N} \cap [1, n]$. Combining the two results together, we have that the proposition holds for $n + 1$, so by induction the proposition is true.

Note that we didn't use normality here. This result applies to every set of independent subgroups.

3. For $n = 2$, let $a_1 \in H_1, a_2 \in H_2$. Since $a_1 \in H_2 \subseteq G$, and H_1 is a normal subgroup, then by **Theorem 20 properties of a coset** $a_2H_1a_2^{-1} = H_1$, so $a_2a_1^{-1}a_2^{-1} \in H_1$, so $a_1(a_2a_1^{-1}a_2^{-1}) \in H_1$ (by closure of H_1 under the group operation). Similarly, since $H_2 \trianglelefteq G$ then $a_1H_1a_1^{-1} = H_2$, so $a_1a_2a_1^{-1} \in H_2$, so $(a_1a_2a_1^{-1})a_2^{-1} \in H_2$ (since $a_2 \in H_2 \therefore a_2^{-1} \in H_2$). Now by associativity, $a_1(a_2a_1^{-1}a_2^{-1}) = (a_1a_2a_1^{-1}a_2^{-1})$, and as we shown this is an element of $H_1 \cap H_2$, so by independency of the subgroups, $a_1a_2a_1^{-1}a_2^{-1} = e$, so $a_1a_2 = a_2a_1$.

Suppose the proposition holds for some n and consider $n + 1$, again we observe that $\prod_{i=1}^n a_i \in \oplus_{i=1}^n H_i \trianglelefteq \oplus_{i=1}^{n+1} H_i$, so by denoting $h_1 = \prod_{i=1}^n a_i, h_2 = a_{n+1}$, this case is again reduced to the case

where $n = 2$ which we have already proven, so $h_1 h_2 = h_2 h_1$, and by the induction hypothesis $\prod_{i=1}^n a_i = \prod_{i=n}^n a_i$, so we have $\prod_{i=1}^{n+1} a_i = \prod_{i=n+1}^1 a_i$, which concludes the proof by induction. $\square$

Theorem 30 (direct products are isomorphic).

Let $H_1, \dots, H_n \trianglelefteq G$ all independent (i.e. the internal direct product of all subgroups is defined), then their internal direct product is isomorphic to their external direct product:

$$\times_{i=1}^n H_i \cong \oplus_{i=1}^n H_i$$

Proof. Denote $H_{\times} = \times_{i=1}^n H_i$, $H_{\oplus} = \oplus_{i=1}^n H_i$.

We define a natural homomorphism from the external inner product to the internal inner product as follows:

$$\phi : H_{\times} \rightarrow H_{\oplus}, (h_1, \dots, h_n) \mapsto \prod_{i=1}^n h_i$$

Let $\prod_{i=1}^n h_i \in H_{\oplus}$, then by definition of ϕ , $\phi(h_1, \dots, h_n) = \prod_{i=1}^n h_i$, and $(h_1, \dots, h_n) \in H_{\times}$ as $h_i \in H_i$ for every h_i , so ϕ is surjective.

Let $a = (a_1, \dots, a_n), b = (b_1, \dots, b_n) \in H_{\times}$. If $\phi(a) = \phi(b)$, then $\prod_{i=1}^n a_i = \prod_{i=1}^n b_i$, so by (1) in 29.2 $a_i = b_i$ for every $a_i \in H_i$, so $a = b$, so ϕ is injective. Since ϕ is also surjective, then it is a bijection. It remains to show it is a homomorphism.

To show ϕ is a homomorphism, we prove the following lemma by induction: if $H_{\oplus}$ is defined and $a = (a_1, \dots, a_n), b = (b_1, \dots, b_n) \in H_{\times}$, then $(\prod_{i=1}^n a_i)(\prod_{i=1}^n b_i) = \prod_{i=1}^n (a_i b_i)$.

For $n = 2$ we have by (2) in 29.2 that $a_1 a_2 = a_2 a_1$ and $b_1 b_2 = b_2 b_1$, so $a_1 a_2 b_1 b_2 = a_1 a_2 b_2 b_1$, and now we have $(a_2 b_2) b_1 \in H_2 \oplus H_1$ so again we have $(a_2 b_2) b_1 = b_1 (a_2 b_2)$, so $a_1 a_2 b_2 b_1 = a_1 b_1 a_2 b_2$, so we conclude $a_1 a_2 b_1 b_2 = a_1 b_1 a_2 b_2$, so the proposition holds for $n = 2$.

Suppose by induction that it holds for some n , and consider $n + 1$, then we follow the usual trick of

observing that by rewriting $\prod_{i=1}^{n+1} a_i$ as $\underbrace{\left(\prod_{i=1}^n a_i\right)}_{\in \oplus_{i=1}^n H_i \trianglelefteq G} \cdot \underbrace{a_{n+1}}_{\in H_{n+1} \trianglelefteq G}$, then the case reduces to $n = 2$, which

we have already shown, so $\prod_{i=1}^{n+1} a_i \prod_{i=1}^{n+1} b_i = (\prod_{i=1}^n a_i \prod_{i=1}^n b_i)(a_{n+1} b_{n+1})$, and by the induction hypothesis $\prod_{i=1}^n a_i \prod_{i=1}^n b_i = \prod_{i=1}^n (a_i b_i)$, so we have $\prod_{i=1}^{n+1} a_i \prod_{i=1}^{n+1} b_i = \prod_{i=1}^{n+1} (a_i b_i)$, which completes the proof by induction.

Consider $\phi(ab) = \phi(a_1 b_1, \dots, a_n b_n)$. By definition, $\phi(ab) = \prod_{i=1}^n (a_i b_i)$. By the lemma we have just proven, $\prod_{i=1}^n (a_i b_i) = \prod_{i=1}^n a_i \prod_{i=1}^n b_i = \phi(a)\phi(b)$, so ϕ is a homomorphism.

We find that ϕ is a bijective homomorphism, so it is an isomorphism, so $H_{\times} \cong H_{\oplus}$.

□

Consider the external direct product of H_1, H_2, H_3 , given by $H_1 \times H_2 \times H_3 = \{(h_1, h_2, h_3) | h_1 \in H_1, h_2 \in H_2, h_3 \in H_3\}$. If we think of the cartesian product as an n -ary operator, then trivially it is not associative since $H_1 \times (H_2 \times H_3) = \{(h_1, (h_2, h_3)) | h_i \in H_i\}$ whereas $(H_1 \times H_2) \times H_3 = \{((h_1, h_2), h_3) | h_i \in H_i\}$, and clearly the intersection of this two sets is the empty set. However when we consider the group structure of each group, we find that they are isomorphic to each other and to $H_1 \times H_2 \times H_3$, so in the group theoretic sense we can think of the external direct product as associative. We show this partly in the following theorem:

Theorem 31 (associativity of external direct product).

$$H_1 \times H_2 \times H_3 \cong (H_1 \times H_2) \times H_3 \cong H_1 \times (H_2 \times H_3)$$

Proof. First we show $H_1 \times H_2 \times H_3 \cong (H_1 \times H_2) \times H_3$. Consider a mapping $\phi : (a, b, c) \mapsto ((a, b), c)$, it is easy to see it is a bijection, and also

$$\begin{aligned} \phi(a_1, b_1, c_1) \cdot \phi(a_2, b_2, c_2) &= \\ ((a_1, b_1), c_1) \cdot ((a_2, b_2), c_2) &= \\ ((a_1 a_2, b_1 b_2), c_1 c_2) &= \\ \phi((a_1, b_1, c_1) \cdot (a_2, b_2, c_2)) \end{aligned}$$

So ϕ is a homomorphism, so it is an isomorphism, so $H_1 \times H_2 \times H_3 \cong (H_1, H_2) \times H_3$.

Similarly it can be shown that $H_1 \times H_2 \times H_3 \cong H_1 \times (H_2, \times H_3)$ via $\phi : (a, b, c) \mapsto (a, (b, c))$, and since isomorphism is an equivalence relation then by transitivity we retrieve the original proposition.

□

Theorem 32 (external direct product preserves isomorphism).

If $H \cong K$, then $H \times G \cong K \times G$, and also $G \times H \cong G \times K$.

Proof. Let $\phi : H \rightarrow K$ be an isomorphism, then it is easy to verify that $\psi : H \times G \rightarrow K \times G, (h, g) \mapsto (\phi(h), g)$ is an isomorphism. Similarly we can show that $G \times H \cong G \times K$ by considering $\tau : G \times H \rightarrow G \times K, (g, h) \mapsto (g, \phi(h))$ and showing τ is an isomorphism.

□

Theorem 33.

If $(a_1, \dots, a_n) \in \times_{i=1}^n H_i$, and the order of each $a_i \in H_i$ is finite, then $|(a_1, \dots, a_n)| = \text{lcm}(|a_1|, \dots, |a_n|)$.

Proof. Let $m = \text{lcm}(|a_1|, \dots, |a_n|)$, then $(a_1, \dots, a_n)^m = (a_1^m, \dots, a_n^m)$. Since $|a_i| \mid m$ for every a_i , then $a_i^m = e_i$, so $(a_1^m, \dots, a_n^m) = (e_1, \dots, e_n)$, which by 38 is the identity of $\times_{i=1}^n H_i$, so $|(a_1, \dots, a_n)| \leq m$. Suppose $(a_1, \dots, a_n)^k = (e_1, \dots, e_n)$ for some k , then $a_i^k = e_i$ for every i , so $|a_i| \mid k$, so k is a common divisor of $(|a_1|, \dots, |a_n|)$, so $m \leq k$, so we conclude $|(a_1, \dots, a_n)| = m$. $\square$

Theorem 34.

If $\times_{i=1}^n H_i$ is defined and $H_1, \dots, H_n$ are finite cyclic groups, then $\times_{i=1}^n H_i$ is cyclic if and only if the orders of the cyclic subgroups are pairwise coprime.

Proof. If $H_1, \dots, H_n$ are cyclic, then for generators $h_1 \in H_1, \dots, h_n \in H_n$, we have that every $a \in H_i$ can be written as $h_i^{k_i}$ for some $k_i \in \mathbb{N}$. Consider $\langle (h_1, \dots, h_n) \rangle \leq \times_{i=1}^n H_i$, it is a cyclic subgroup of the external direct product. Note that $|\langle (h_1, \dots, h_n) \rangle| = |(h_1, \dots, h_n)|$ by 11 properties of a cyclic group, and $|(h_1, \dots, h_n)| = \text{lcm}(|h_1|, \dots, |h_n|)$ by Theorem 33, and also $|\langle (h_1, \dots, h_n) \rangle| \leq |\times_{i=1}^n H_i| \iff \langle (h_1, \dots, h_n) \rangle = \times_{i=1}^n H_i$ since it is a subgroup of that group.

Finally by counting we note that for $a \in \times_{i=1}^n H_i$, there are $|H_i| = |h_i|$ distinct choices for each a_i , so in total $|\times_{i=1}^n H_i| = \prod_{i=1}^n |h_i|$.

$\implies$ We prove by induction on n .

For $n = 2$, suppose $H_1 \times H_2$ is cyclic, so $\langle (a_1, a_2) \rangle = H_1 \times H_2$ for some $a_1 \in H_1, a_2 \in H_2$, so $|(a_1, a_2)| = |H_1 \times H_2| = |H_1| \cdot |H_2|$.

Denote $m = |H_1| \cdot |H_2|$, and consider $d = \gcd(|H_1|, |H_2|)$, then $\frac{|H_i|}{d} \in \mathbb{N}$ and so trivially $\frac{m}{d} \in \mathbb{N}$, so $(a_1, a_2)^{\frac{m}{d}}$ is defined, so:

$$(a_1, a_2)^{\frac{m}{d}} = (a_1^{\frac{m}{d}}, a_2^{\frac{m}{d}}) = ((a_1^{|H_1|})^{\frac{|H_2|}{d}}, (a_2^{|H_2|})^{\frac{|H_1|}{d}}) = (*)$$

By Theorem 8, we know that $|a_i| \mid |H_i|$, so $a_i^{|H_i|} = e_i$, so

$$(*) = (e_1^{\frac{|H_2|}{d}}, e_2^{\frac{|H_1|}{d}}) = (e_1, e_2) = e_{H_1 \times H_2}$$

So it follows that $|(a_1, a_2)| \leq \frac{m}{d}$, but also $|(a_1, a_2)| = m$ by supposition, so $\frac{m}{d} \leq m$, so $d = 1$, so $\gcd(|H_1|, |H_2|) = 1$, so $|H_1|, |H_2|$ are coprime.

Suppose by induction the proposition is true for some n (i.e. that for this n , $\gcd(|H_i|, |H_j|) = 1$ for any $H_i \neq H_j$), and consider $n + 1$.

By definition, $\times_{i=1}^{n+1} H_i = \times_{i=1}^n H_i \times H_{n+1}$. Again we use our usual trick of reducing this case to the case where $n = 2$: by **Theorem 31 associativity of external direct product** we have $\times_{i=1}^n H_i \times H_{n+1} \cong (\times_{i=1}^n H_i) \times H_{n+1}$. By **Theorem 18 properties of a group isomorphism**, we know that $\times_{i=1}^n H_i \times H_{n+1}$ is cyclic if and only if $(\times_{i=1}^n H_i) \times H_{n+1}$ is cyclic, but $(\times_{i=1}^n H_i) \times H_{n+1}$ is simply the direct product of two groups so it reduces to the case where $n = 2$ which we have already covered, so $\gcd(|\times_{i=1}^n H_i|, |H_{n+1}|) = 1$, but by the counting argument $|\times_{i=1}^n H_i| = \prod_{i=1}^n |H_i|$, so $\gcd(\prod_{i=1}^n |H_i|, |H_{n+1}|) = 1$, so it must be that $\forall i \in \mathbb{N} \cap [1, n] (\gcd(|H_i|, |H_{n+1}|) = 1)$ (suppose not and denote the non-unit divisor d , then d must also be the divisor of $\prod_{i=1}^n |H_i|$, so $\gcd(\prod_{i=1}^n |H_i|, |H_{n+1}|) \neq 1$, a contradiction), so $|H_{n+1}|$ is coprime w.r.t. any the order of any other cyclic subgroup H_i . For any pair of the form $(|H_i|, |H_j|)$ where $i, j \neq n + 1$, we know that it is a pair of coprimes by the induction hypothesis, thus we complete the proof by induction.

$\Leftarrow$ Suppose $|H_1|, \dots, |H_n|$ are all pairwise coprime, then $|h_1|, \dots, |h_n|$ are coprime, so $\text{lcm}(|h_1|, \dots, |h_n|) = \prod_{i=1}^n |h_i|$, so $|\langle (h_1, \dots, h_n) \rangle| = \prod_{i=1}^n |h_i|$, so we have $|\times_{i=1}^n H_i| = |\langle (h_1, \dots, h_n) \rangle|$, so $\times_{i=1}^n H_i$ is a cyclic group with generator $(h_1, \dots, h_n)$.

□

Corollary 34.1 (decomposition of modular addition group).

Let $k = \prod_{i=1}^n m_i$ (where each $m_i \in \mathbb{N}$), then $\mathbb{Z}_k^+ \cong \times_{i=1}^n \mathbb{Z}_{m_i}^+$ if and only if all elements of $m_1, \dots, m_n$ are pairwise relatively prime.

Proof. $\Rightarrow$ Suppose $\mathbb{Z}_k^+ \cong \times_{i=1}^n \mathbb{Z}_{m_i}^+$, then since $\mathbb{Z}_k^+$ is cyclic (with generator 1) then by **Theorem 18 properties of a group isomorphism** $\times_{i=1}^n \mathbb{Z}_{m_i}^+$ is also cyclic, and all of $\mathbb{Z}_{m_i}$ are finite since all $m_i \in \mathbb{N}$, so by **Theorem 34** it follows that all $m_1, \dots, m_n$ are relatively prime.

$\Leftarrow$ Suppose $m_1, \dots, m_n$ are pairwise coprime, then by **Theorem 34** $\times_{i=1}^n \mathbb{Z}_{m_i}^+$ is cyclic, and by counting $|\times_{i=1}^n \mathbb{Z}_{m_i}^+| = \prod_{i=1}^n |\mathbb{Z}_{m_i}| = \prod_{i=1}^n m_i = k$, so by **Theorem 27 cyclic groups are isomorphic to modular addition groups of integers of same order** it follows that $\mathbb{Z}_k^+ \cong \times_{i=1}^n \mathbb{Z}_{m_i}^+$.

□

We use this result, along with **Theorem 33**, to count the number of subgroups of a finite cyclic group with order k , via the process demonstrated in the following example:

Example 34.1 (subgroups of order 3 in cyclic group with order 138).

First denote $138 = 3 \cdot 46$, and 3, 46 are relatively prime, so if G is cyclic and $|G| = 138$ then by [34.1 decomposition of modular addition group](#), $G \cong \mathbb{Z}_3^+ \times \mathbb{Z}_{46}^+$. Let ϕ be a group isomorphism between the isomorphic groups and let $H \leq G$, then since G is cyclic by [Theorem 9 fundamental theorem of cyclic groups](#) H is cyclic, then $a \in H$ is a generator of H , i.e. $H = \langle a \rangle$, and by [Theorem 18 properties of a group isomorphism](#) $\phi(\langle a \rangle)$ is the generator of a subgroup with the same order in $\mathbb{Z}_3^+ \times \mathbb{Z}_{46}^+$. Denote this generator (q, p) , where $q \in \mathbb{Z}_3^+, p \in \mathbb{Z}_{46}^+$. By [Theorem 33](#) we know that $|(q, p)| = \text{lcm}(|q|, |p|)$. Since $H \cong \langle (q, p) \rangle$, then $|H| = \text{lcm}(|q|, |p|)$. We are looking for subgroups of G with order 3, so we can convert the problem to counting the total number of distinct elements (q, p) s.t. $\text{lcm}(|q|, |p|) = 3$, which greatly simplifies the problem.

Suppose then that $\text{lcm}(|q|, |p|) = 3$, and consider the following cases:

1. $q = e$, then $|q| = 1$, then $\text{lcm}(|q|, |p|) = 3 \iff |p| = 3$, but since $p \in \mathbb{Z}_{46}^+$ and $|\mathbb{Z}_{46}^+| = 46$ and by a previous result $|p|$ must divide $|\mathbb{Z}_{46}^+|$ and yet (3, 46) are coprime, then there are no such elements.
2. $p = e$, then $|p| = 1$, then $\text{lcm}(|q|, |p|) = 1 \iff |q| = 3$, and since $q \in \mathbb{Z}_3^+$ which has primer order, by [8.4](#) all non-identity elements of $\mathbb{Z}_3^+$ generate the group and so have order 3, so this case contributes 2 total generators.
3. $p \neq e$ and $q \neq e$, in which case it must be that both $|q|, |p|$ are multiples of 3 and also greater than 1, but 3 and 46 are coprime so it is impossible that $3|p|$ and also $3||p|$, so there are no such cases.

We conclude that if G is cyclic with order 138, it has exactly two subgroups of order 3.

It can further be shown that all groups under modular multiplication (i.e. $\mathbb{Z}_n^\times$) can be expressed as the external direct product of finite cyclic groups. We show this in [RSA Encryption](#).

Second Isomorphism Theorem

In the previous section we discussed the internal and external direct products, and we defined the internal direct product such that, given independent normal subgroups of G , their internal direct product is also a subgroup of G . We mentioned that the requirement that the subgroups be independent is not needed for the product to be a subgroup. Let us relax our definition further and examine the case where $H_1, H_2 \leq G$ but neither are necessarily normal subgroups of G .

Definition 0.0.4 (product of two subgroups).

If $H_1, H_2 \leq G$, then their product is the structure $(H_1H_2, \cdot)$ where $H_1H_2 = \{a \cdot b \mid a \in H_1, b \in H_2\}$ and $\cdot$ is the group operation of G .

This is similar to the definition of the internal direct product but without the condition that H_1, H_2 be normal subgroups nor independent.

Definition 0.0.5 (normalizer of a subgroup).

If $H \leq G$, then the normalizer of H in G , denoted $N_G(H)$, is given by $N_G(H) = \{g \in G \mid gHg^{-1} = H\}$.

The motivation behind the name is that the normalizer *normalizes* H as a normal subgroup of itself, as we show in the following:

Theorem 35 (properties of the normalizer).

1. $N_G(H) \leq G$
2. $H \trianglelefteq N_G(H)$
3. If $H \trianglelefteq G$, then $N_G(H) = G$
4. $N_G(H)$ is the maximal (largest) subgroup of G s.t. $H \trianglelefteq N_G(H)$

Proof. 1. First observe $e \in N_G(H)$ since $eHe^{-1} = H$, so $N_G(H)$ is a nonempty subset of G . We proceed by the one-step subgroup test: let $a, b \in N_G(H)$ and consider ab^{-1} . Consider that $(ab^{-1})H(ab^{-1})^{-1} = (ab^{-1})H(ba^{-1}) = a(b^{-1}Hb)a^{-1}$, now since $b \in N_G(H)$ then trivially also $b^{-1}Hb = H$, so we have $a(b^{-1}Hb)a^{-1} = aHa^{-1}$, and since $a \in N_G(H)$ then $aHa^{-1} = H$, so $ab^{-1} \in N_G(H)$, so by the one-step subgroup test $N_G(H) \leq G$.

2. Let $h \in H$, then $hH = H$ by **Theorem 20 properties of a coset** and also $Hh^{-1} = H$, so $hHh^{-1} = H$, so $h \in N_G(H)$, and also $H, N_G(H)$ are groups with the same operation, so $H \leq N_G(H)$.

To show H is a normal subgroup of $N_G(H)$, let $n \in N_G(H)$. By **29** it suffices to show $nHn^{-1} \subseteq H$ to show $H \trianglelefteq N_G(H)$. Consider nhn^{-1} , then since $n \in N_G(H)$ by **0.0.5 normalizer of a subgroup** it follows that $nhn^{-1} = h' \in H$, so we conclude $H \trianglelefteq N_G(H)$.

3. Suppose $H \trianglelefteq G$, then $\forall g \in G (gH = Hg)$, so $\forall g \in G (gHg^{-1} = H)$, so $N_G(H) = G$ by the membership test of $N_G(H)$.

4. Suppose by contradiction $N_G(H)$ is not maximal, then $\exists K \leq G$ s.t. $H \trianglelefteq K$ and $|N_G(H)| < |K|$. From normality of K , $\forall k \in K$ it follows that $kHk^{-1} = H$, but then by **0.0.5 normalizer of a subgroup** $k \in N_G(H)$, so $K \subseteq N_G(H)$, but then $|K| \leq |N_G(H)|$, a contradiction, so $N_G(H)$ is maximal.

□

Via this properties of the normalizer, we can prove sufficient conditions for the product of two subgroups, as defined in [0.0.4 product of two subgroups](#), to be a group:

Theorem 36.

If:

1. G is a group
2. $H_1, H_2 \leq G$
3. $H_1 \leq N_G(H_2)$

Then $H_1H_2 \leq G$ (here H_1H_2 is the product as defined in [0.0.4 product of two subgroups](#))

Proof. Suppose all conditions are satisfied. First, note that $e \in H_1, H_2$ so $ee = e \in H_1H_2$, and also $H_1H_2 \subseteq G$. We proceed via the one-step subgroup test: let $a_1b_1, a_2b_2 \in H_1H_2$ s.t. $a_i \in H_1, b_i \in H_2$, then

$$(a_1b_1)(a_2b_2)^{-1} = (a_1b_1)(b_2^{-1}a_2^{-1}) = a_1 \underbrace{(b_1b_2^{-1})}_{\in H_2, \text{denote } b_{12}} a_2^{-1} = a_1b_{12}a_2^{-1} = (*)$$

Since $H_1 \leq N_G(H_2)$, then by definition $a_1b_{12}a_1^{-1} = b_3$ for some $b_3 \in H_2$. Post-multiplying both sides by a_1 , we get $a_1b_{12} = b_3a_1$. Substituting for $(*)$, we get

$$(*) = b_3 \cdot \underbrace{a_1a_2^{-1}}_{\in H_1, \text{denote } a_{12}}$$

By the same argument we have $a_{12}^{-1}b_3a_{12} = b_4$ for some $b_4 \in H_2$, so $b_3a_{12} = a_{12}b_4$, which is clearly in H_1H_2 by definition, so $(a_1b_1)(a_2b_2)^{-1} \in H_1H_2$, so $H_1H_2 \leq G$ by the one-step subgroup test. □

We proceed with a couple of corollaries to the theorem:

Corollary 36.1.

If all conditions of [Theorem 36](#) are satisfied, then:

1. $H_2 \leq H_1H_2$
2. $H_1 \cap H_2 \leq H_1$

Proof. 1. $H_2 \leq H_1H_2$ is immediate from H_1H_2, H_2 being groups and from the definition of the product group, so it remains to show $(ab)H_2(ab)^{-1} \in H_2$ for every $a \in H_1, b \in H_2$. Let $h \in H_2$, then

$(ab)h(ab)^{-1} = (ab)h(b^{-1}a^{-1}) = a(bhb^{-1})a^{-1}$. Since $b, h \in H_2$ then from closure of H_2 , $bhb^{-1} \in H_2$, now since $bhb^{-1} \in H_2$ and $b \in H_1 \leq N_G(H_2)$, it follows that $a(bhb^{-1})a^{-1} \in H_2$, so we conclude that $H_2 \trianglelefteq H_1H_2$.

2. By **9 intersection of subgroups is a subgroup** we know that $H_1 \cap H_2 \leq H_1$, so it remains to show it is a normal subgroup. We need to show $a(H_1 \cap H_2)a^{-1} \subseteq H_1 \cap H_2$ for every $a \in H_1$. Let $b \in H_1 \cap H_2$, then $b \in H_1$, so $aba^{-1} \in H_1$ from closure. Also, since $a \in H_1 \leq N_G(H_2)$, and $b \in H_1 \cap H_2 \therefore b \in H_2$, then $aba^{-1} \in H_2$. We conclude $aba^{-1} \in H_1 \cap H_2$, so $H_1 \cap H_2 \trianglelefteq H_1$.

□

This leads us directly to the second isomorphism theorem:

Theorem 37 (second isomorphism theorem).

If:

1. G is a group
2. $H_1, H_2 \leq G$
3. $H_1 \leq N_G(H_2)$

Then:

1. $H_1H_2 \leq G$
2. $H_2 \trianglelefteq H_1H_2$
3. $H_1 \cap H_2 \trianglelefteq H_1$
4. $H_1H_2/H_2 \cong H_1/(H_1 \cap H_2)$

Proof. The first proposition in the theorem is covered by **Theorem 36**, and propositions 2, 3 are exactly **36.1**. For 4, consider a mapping $\phi : H_1 \rightarrow H_1H_2/H_2$, given by $\phi : h_1 \mapsto h_1H_2$. Note that since $H_2 \trianglelefteq H_1H_2$, the codomain is well-defined.

Let $a, b \in H_1$, then $\phi(ab) = abH_2 = (aH_2)(bH_2) = \phi(a)\phi(b)$ so ϕ is a homomorphism (note that $abH_2 = (aH_2)(bH_2)$ since this is how we defined the group operation of a factor group).

We show ϕ is surjective. Let $(h_1h_2)H_2 \in H_1H_2/H_2$. Since $h_2 \in H_2$, then $(h_1h_2)H_2 = h_1H_2$ where $h_1 \in H_1$, so for $\phi(h_1)$ we get $\phi(h_1) = h_1H_2$, so ϕ is surjective, so $\text{Im}(\phi) = H_1H_2/H_2$.

Consider $\text{Ker}(\phi)$. By definition $x \in \text{Ker}(\phi) \iff \phi(x) = H_2$ (which is the identity on the codomain), but also $\phi(x) = xH_2$, so we require $xH_2 = H_2$. This is true if and only if $x \in H_2$, but $x \in H_1$ since x is taken from the domain of ϕ , so this is true if and only if $x \in H_1 \cap H_2$, so $\text{Ker}(\phi) = H_1 \cap H_2$.

Thus by **Theorem 26 first isomorphism theorem** since ϕ is surjective and $\text{Ker}(\phi) = H_1 \cap H_2$, we have $H_1/(H_1 \cap H_2) \cong H_1H_2/H_2$, which completes the proof.

Note that we did not use proposition (3) in the theorem in the proof of (4), so by 25.1 and (4), we immediately get (3). □

Corollary 37.1 (product formula for normal subgroups).

If $H_1, H_2 \trianglelefteq G$ are finite, then $|H_1 H_2| = \frac{|H_1||H_2|}{|H_1 \cap H_2|}$. (in fact, it suffices that $H_1, H_2 \leq G$ and $H_1 \leq N_G(H_2)$).

Proof. By the isomorphism from the last proposition in Theorem 37 second isomorphism theorem, and

Theorem 18 properties of a group isomorphism. □

Using this theorem, we can prove the following result:

Theorem 38.

If G is a finite abelian group with order divisible by a prime p , then $\exists a \in G$ s.t. $|a| = p$.

Proof. Suppose G has order divisible by a prime p , so $|G| = np$ for some $n \in \mathbb{N}$. Since a group is nonempty then $n \neq 0$. We proceed by induction on n .

If $n = 1$, then $|G| = p$, so by 21.2 it follows that G is cyclic, so $G = \langle a \rangle$ for some $a \in G$, so $|G| = |\langle a \rangle| = |a| = p$.

Suppose the proposition holds up to some n , and consider the case where $|G| = (n+1)p$. Since G is composite, then by Theorem 22 every group of composite order has proper nontrivial cyclic subgroup G admits a proper subgroup $H \leq G$ s.t. $|H||G| = (n+1)p$. We choose a maximal such H (i.e. H is the proper subgroup of G of greatest order, which must exist since G is finite).

Consider H , there are two cases:

1. $p \mid |H|$, in which case by the induction hypothesis $\exists a \in H$ s.t. $|a| = p$, and also $a \in G$ since $H \leq G$, which completes the proof in this case.

2. $p \nmid |H|$, in which case consider some $x \in G \setminus H$ (one must exist since H is proper). This x generates a cyclic group $\langle x \rangle \leq G$, and this cyclic group is also proper nontrivial since $H \neq \emptyset$ and $e \in H$ so $x \neq e$. Consider $h \in H$. Consider the internal direct product $\langle a \rangle \oplus H$. Since G is abelian, then $H, \langle a \rangle \trianglelefteq G$ so by 37.1 product formula for normal subgroups it follows that $|H\langle a \rangle| = \frac{|H||a|}{|H \cap \langle a \rangle|}$. Since by assumption $a \notin H$ then $H \cap \langle a \rangle = \{e\}$ so $|H\langle a \rangle| = |H||a|$ (note - since $H, \langle a \rangle$ are independent we could have also used Theorem 30 direct products are isomorphic to derive this relation). Consider two cases:

1. $H\langle a \rangle$ is proper nontrivial. Since $|H| = \frac{|H\langle a \rangle|}{|a|}$ and $|H|$ is maximal, it must be that $|a| = 1$ (otherwise $H\langle a \rangle$ would violate the maximality of H), then $|H| = |H\langle a \rangle|$, but then $a \in H$, a contradiction, so $H\langle a \rangle$ cannot be a proper subgroup.

2. $H\langle a \rangle = G$, in which case $|G| = |H||a|$, so $|H||a| = (n+1)p$. By supposition, $p \nmid |H|$, so by Euclid's lemma $p \mid |a|$, so by the induction hypothesis since $\langle a \rangle$ is a proper nontrivial subset of G it follows that $\exists s \in \mathbb{N}$ s.t. $|a^s| = p$, which completes the proof.

□

We present an application to the second isomorphism theorem below. Consider the following definitions, most of which you should already be familiar with:

1. $\text{GL}_n(V)$ the *general linear group* of degree n over the vector space V , which is the group of invertible $n \times n$ matrices whose entries are elements in V . We have demonstrated earlier that $\text{GL}_n(V)$ is a group.
2. $\text{SL}_n(V)$ the *special linear group* of degree n over the vector space V , which is the group of $n \times n$ matrices with entries in V that have determinant 1. By **Theorem 4 two-step subgroup test** it is trivial to see that $\text{SL}_n(V) \leq \text{GL}_n(V)$.
3. $\text{Z}_n(V)$ the group of *scalar transformations* over V , i.e. the set of all nonzero scalar $n \times n$ matrices of the form $Z = vI$ for some $v \in V$. It is trivial to see that $\text{Z}_n(V) \leq \text{GL}_n(V)$.
4. $\text{SZ}_n(V)$ the group of *special scalar transformations* over V , i.e. the set of all $n \times n$ scale matrices with determinant 1.

Then, for $V = \mathbb{C}$:

1. $\text{Z}_n(\mathbb{C}) = \{cI \mid c \in \mathbb{C} \setminus \{0\}\}$
2. $\text{SZ}_n(\mathbb{C}) = \{\pm I\}$
3. $\text{SL}_n(\mathbb{C}) \cap \text{Z}_n(\mathbb{C}) = \{\pm I\} = \text{SZ}_n$
4. The internal direct product $\text{SL}_n(\mathbb{C})\text{Z}_n(\mathbb{C})$ is exactly $\text{GL}_n(\mathbb{C})$
5. $\text{Z}_n(\mathbb{C}) \trianglelefteq \text{GL}_n(\mathbb{C})$, since from linear algebra for every $cI \in \text{Z}_n(\mathbb{C})$, $A \in \text{GL}_n(V)$, we have $A^{-1}(cI)A = (A^{-1}c)IA = cA^{-1}A = cI$, so $\text{Z}_n(\mathbb{C})$ is normal in $\text{GL}_n(V)$.
6. $\text{SL}_n(\mathbb{C}) \leq \text{N}_G(\text{Z}_n(\mathbb{C}))$, since $\text{Z}_n(\mathbb{C})$ is normal in $\text{GL}_n(\mathbb{C})$ so by **Theorem 35 properties of the normalizer**, the normalizer of $\text{Z}_n(\mathbb{C})$ in $\text{GL}_n(\mathbb{C})$ is the entire group, and when recalling the definitions we noted that $\text{SL}_n(\mathbb{C}) \leq \text{GL}_n(\mathbb{C})$.

So, letting $H_2 = \text{Z}_n(\mathbb{C})$, $H_1 = \text{SL}_n(\mathbb{C})$, $G = \text{GL}_n(\mathbb{C})$, all conditions of the second isomorphism theorem hold and we get:

$$\begin{aligned} (\text{SL}_n(\mathbb{C})\text{Z}_n(\mathbb{C})) / \text{Z}_n(\mathbb{C}) &\cong \text{SL}_n(V) / (\text{SL}_n(\mathbb{C}) \cap \text{Z}_n(\mathbb{C})) \\ \text{GL}_n(\mathbb{C}) / \text{Z}_n(\mathbb{C}) &\cong \text{SL}_n(\mathbb{C}) / \text{SL}_n(\mathbb{C}) \end{aligned}$$

In projective geometry, we define the *complex projective linear group* $\text{PGL}_n(\mathbb{C})$, as $\text{GL}_n(\mathbb{C}) / \text{Z}_n(\mathbb{C})$, and the *complex projective special linear group* $\text{PSL}_n(\mathbb{C})$, as $\text{SL}_n(\mathbb{C}) / \text{SL}_n(\mathbb{C})$. The second isomorphism theorem provides the following results:

1. Both definitions are well-defined.
2. Both definitions are equivalent in the sense that the groups they define are isomorphic.

Third Isomorphism Theorem

The third isomorphism theorem tells us that factoring a group further does not teach us anything new about its structure:

Theorem 39 (third isomorphism theorem).

If G is a group and $K \trianglelefteq H \trianglelefteq G$, then

$$(G/K)/(H/K) \cong G/H$$

Proof. Consider a mapping $\phi : G/K \rightarrow G/H$, given by $\phi(gK) \mapsto gH$. First, we show ϕ is single-valued: let $g_1K = g_2K$, then $\phi(g_1K) = g_1H$, $\phi(g_2K) = g_2H$. Since $g_1K = g_2K$ then $g_1g_2^{-1} \in K$, and since $K \trianglelefteq H$ then $g_1g_2^{-1} \in H$, so $g_1H = g_2H$ (we use [Theorem 20 properties of a coset](#)), so $\phi(g_1K) = \phi(g_2K)$ for any representatives g_1K, g_2K of a congruence class of G/K , so ϕ is single-valued, so it is a function.

Let $gH \in G/H$, then $\phi(gK) = gH$ by definition of ϕ , so ϕ is surjective. Furthermore, via the definition of the group operation of the factor group, $\phi(g_1K)\phi(g_2K) = (g_1H)(g_2H) = (g_1g_2)H = \phi((g_1g_2)K) = \phi((g_1K)(g_2K))$, so ϕ is a homomorphism.

Finally, let $xK \in \text{Ker}(\phi)$, then $\phi(xK) = H$ and also by definition $\phi(xK) = xH$, so $xH = H$, which is true if and only if $x \in H$, so $\text{Ker}(\phi) = H/K$ (which also immediately gives us that $H/K \trianglelefteq G/K$ by [25.1](#)), so by [Theorem 26 first isomorphism theorem](#) $(G/K)/(H/K) \cong G/H$.

□

Isomorphism Theorems as Commutative Diagrams

The first three isomorphism theorems provide a good setting for the introduction of commutative diagrams. One of the objects studied in category theory, commutative diagrams are a way to represent the relation between objects under morphisms. A commutative diagram is drawn as a drawing of a directed graph, where the objects correspond to vertices and the morphisms correspond to edges.

A commutative diagram *commutes* in the sense that every two paths that share the same start and finish, start and end with the same result. We will not formally discuss commutative diagrams here, but only touch on them briefly as they provide a nice way of visualizing the isomorphism theorems for groups.

First, a warm up. Suppose G, H, K are groups s.t. $\phi : G \rightarrow H$ is an isomorphism and $\psi : H \rightarrow K$ is a homomorphism, then $\tau : G \rightarrow K$ given by $\psi \circ \phi$ is also a homomorphism. This result can be expressed in the following commutative diagram:

$$\begin{array}{ccc} G & \xrightarrow{\tilde{\phi}} & H \\ & \searrow \phi \circ \psi & \downarrow \psi \\ & & K \end{array}$$

We read the diagram as follows: a straight arrow ($\rightarrow$) represents a morphism. In our case we know what the morphisms are so we denote them via their symbol over the arrow. If the morphism is an isomorphism, we indicate this with a tilde ($\sim$) over the arrow representing the morphism. The diagram says that if we follow along edges from G to K , we get the same result no matter which path we take - indeed, by following the edge ϕ to H and then following the edge ψ , we arrive at K (via a homomorphism), and we arrive at the same homomorphism via following the arrow from G to K which represents the composition $\phi \circ \psi$ - this is pretty much the definition of a composition of functions.

First Isomorphism Theorem

We now turn our attention to a more exciting example, the first isomorphism theorem. Recall that the first isomorphism theorem states the following:

If $\phi : G_1 \rightarrow G_2$ is a group homomorphism, then $\text{Ker}(\phi) \trianglelefteq G_1$, $\phi(G_1) \leq G_2$, and $G_1/\text{Ker}(\phi) \cong \phi(G_1)$.

To represent the theorem in a commutative diagram, we need to introduce a few new arrow types. In total, we use the following arrow:

1. $A \xrightarrow{\phi} B$ means that A is homomorphic to B via $\phi : A \rightarrow B$, and if $A \cong B$ we denote $A \xrightarrow{\sim} B$.
2. $A \twoheadrightarrow B$ means that there is a surjective homomorphism from A to B . Here we use this to denote that B is a factor group of A over some normal subgroup.
3. $A \hookrightarrow B$ means that $\phi : A \rightarrow B$ is an injective homomorphism (monomorphism), we usually use this to indicate $A \leq B$ (since then id_A is an injective homomorphism from A to B)

We can then state the first homomorphism theorem as follows:

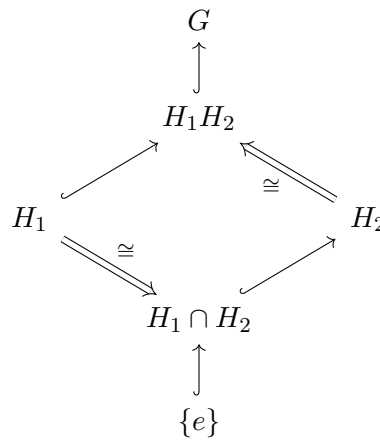
If $\phi : G_1 \rightarrow G_2$ is a group homomorphism. then the following graph commutes:

$$\begin{array}{ccc} & & G_2 \\ & \nearrow & \\ G_1 & \xrightarrow{\phi} & \phi(G_1) \\ \downarrow & \nearrow \sim & \\ G_1/\text{Ker}(\phi) & & \end{array}$$

We read the diagram as follows: if $\phi : G_1 \rightarrow G_2$ is a homomorphism from G_1 to $\phi(G_1)$ which is a subgroup of G_2 , we can also go from G_1 to the quotient $G_1/\text{Ker}(\phi)$, which will be isomorphic to $\phi(G_1)$.

Second Isomorphism Theorem

Recall **Theorem 37 second isomorphism theorem**. It can be restated by the following commutative diagram (although we should still mention all conditions of the theorem, since for example $H_1 \leq N_G(H_2)$ is not captured in this diagram), where we introduce a new two-lined arrow $A \twoheadrightarrow B$ to denote normality (A is a normal subgroup of B):



We read the diagram as follows:

1. $H_1H_2 \leq G$
2. $H_2 \trianglelefteq H_1H_2$
3. $H_1 \cap H_2 \trianglelefteq H_1$
4. $H_1 / (H_1 \cap H_2) \cong H_1H_2 / H_2$ (this isomorphism is indicated by the $\cong$ symbol on the parallel edges $H_1 \twoheadrightarrow H_1 \cap H_2, H_2 \twoheadrightarrow H_1H_2$)

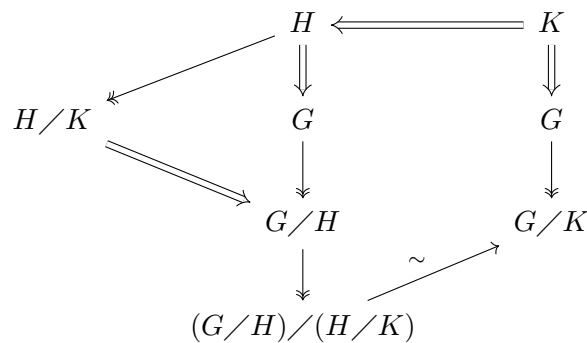
Which is exactly what the second isomorphism theorem states.

Now that we have seen the second isomorphism theorem as a commutative diagram, it should come as no surprise that it is also called the *diamond theorem* or the *parallelogram theorem*.

Third Isomorphism Theorem

Recall **Theorem 39 third isomorphism theorem**. We can restate it as follows:

*If all conditions of **Theorem 39 third isomorphism theorem** are satisfied, the following diagram commutes:*



Interpreting this diagram is left as an exercise to the reader.

Group Action

Consider the dihedral group D_{2n} , which is the group of symmetries of a regular n -gon. We have seen in [Rigid Motions & Symmetry](#) and earlier on in this discussion that $D_{2n} = \{\langle r, f \rangle \mid |r| = n, |f| = 2, frf = r^{-1}\}$. Using this representation, we understand the dihedral group of order $2n$ by its group structure in the abstract sense, i.e. we forget the geometric motivation for D_{2n} as the group of symmetries of a regular n -gon, and instead consider some abstract multiplicative group with elements r, f and some rules (or relations) that govern the group (namely, the order of r , the order of f , and the relation $frf = r^{-1}$).

Recall that in our geometric motivation for D_{2n} , we thought of r as a rotation by $\frac{2\pi}{n}$ radians, and as f as a reflection through an axis of effective symmetry. We have shown that these are in fact symmetries of the regular n -gon, and furthermore that their compositions are symmetries, and that we can realize all symmetries of the shape by the composition of only one reflection f and one rotation r (either CW or CCW).

More formally, what we did was show that for any $\sigma \in D_{2n}$, if V_n is the set of vertices of the n -gon, then $\sigma(V_n)$ acts on the vertices in a way that preserves the geometric structure of the n -gon, i.e. it produces a *symmetry*.

Note the distinction here: D_{2n} is a *group* which we define in the abstract sense without any geometric meaning, and then we *define* how elements of the group *act* on a set, in this case *the vertices of a regular n -gon*, which formalizes the connection between the geometric motivation and the abstract algebraic structure.

What we have just demonstrated is a *group action* on the set V_n by the group D_{2n} .

Definition 0.0.6 (group action).

If G is a group and S is a set, then a mapping $\star : G \times S \rightarrow S$ is a group action if $\forall x \in S$:

1. $\forall a, b \in G, a \star (b \star x) = (ab) \star x$ (where ab is group multiplication in G)
2. $e \star x = x$, where e is the identity of G .

Example 39.1 (symmetric define a group action).

Let S be a finite set and G be a subgroup of $\text{Sym}(S)$ (i.e. the group of permutations, or the symmetric group, of S), then $\star : (\sigma, s) \mapsto \sigma(s)$ defines a group action:

1. Clearly $\star$ is a mapping $G \times S \rightarrow S$.
2. Take $\sigma_a, \sigma_b \in G$, then $(\sigma_a \sigma_b) \star (x) = (\sigma_a \circ \sigma_b)(x) = \sigma_a(\sigma_b(x)) = \sigma_a \star (\sigma_b \star x)$
3. $\text{id}_S \star x = \text{id}_S(x) = x$

Example 39.2 (group multiplication defines group action).

Let $H \leq G$, then $\star : H \times G \rightarrow G, h \star x \mapsto hx$ defines a group action:

1. Let $h, g \in H$, then $hg \star x = hgx = h(gx) = h \star (gx) = h \star (g \star x)$.
2. Let $e \in H$, then e is the identity on G , so $e \star x = ex = x$.

The next theorem demonstrates that group action $G \times S \rightarrow S$ is really just another way of looking at a group homomorphism from a group G to $\text{Sym}(S)$ of some set S .

Theorem 40 (group actions are equivalent to group homomorphism to symmetric group).

Let G be a group and S be a set, then:

1. If $\phi : G \rightarrow \text{Sym}(S)$ is a group homomorphism, then $a \star x \mapsto \phi(a)(x)$ is a group action
2. If $\star : G \times S \rightarrow S$ is a group action, then $\star$ arises from a homomorphism $\phi : G \rightarrow \text{Sym}(S)$ as described in (1).

Proof. 1. Let $\phi : G \rightarrow \text{Sym}(S)$ be a group homomorphism, then we show $\star : G \times S \rightarrow S, a \star x \mapsto \phi(a)(x)$ is a group action:

1. First we verify the domain and co-domain are indeed correct. Since $\text{Im}(\phi) \leq \text{Sym}(S)$ and $\text{Sym}(S)$ is the group of bijections $S \rightarrow S$, then $a \star x$ as given above is a mapping $G \times S \rightarrow S$.
2. Let $a, b \in G$, then $\phi(ab) = \phi(a)\phi(b)$ since ϕ is a group homomorphism. By definition, $a \star (b \star x) = a \star (\phi(b)(x)) = \phi(a)\phi(b)(x) = \phi(ab)(x) = (ab) \star x$.
3. Since ϕ is a group homomorphism then $\phi(e_G) = \text{id}_S$, so $e_G \star x = \text{id}_S(x) = x$.

We conclude that ϕ realizes a group action.

2. Suppose $\star : G \times S \rightarrow S$ is a group action, and define $\phi : a \mapsto \star_a$ where $\star_a$ is the partial mapping of $\star$ where the first argument ($a \in G$) is fixed. We wish to show that:

1. ϕ is a group homomorphism
2. ϕ is a mapping from G to $\text{Sym}(S)$

Let $a, b \in G$ and let $x \in S$, then $\phi(ab) = \star_{ab}$, and since $\star$ is a group action then $\phi(ab)(x) = \star_{ab}x = (ab) \star x$, which by the properties of a group action is the same as $a \star (b \star x)$, which is the same as $\star_x(\star_b x)$, which by definition of ϕ is the same as $(\phi(a) \circ \phi(b))(x)$, so we have $\phi(ab) = \phi(a)\phi(b)$, so ϕ is a group homomorphism.

We show (2) by definition: $\phi(a) \in \text{Sym}(S)$ if and only if $\phi(a)$ is a bijective mapping from S to itself. Let $a \in G$ and consider $\phi(a)$, then since $\phi(a)$ describes the action of a on S , $\phi(a)$ is a mapping $S \rightarrow S$. Let $y \in S$.

1. (Surjective) Since $\star$ is a group action, then $e \star y = y$, but $e = aa^{-1}$ so $e \star y = (aa^{-1}) \star y$ which by the properties of group action is equivalent to $a \star (a^{-1} \star y)$, but since $a^{-1} \star y \in S$, then we have $\phi(a)(a^{-1} \star y) = \star_a(a^{-1} \star y) = y$, so $\phi(a)(x) = y$ has a solution $a^{-1} \star y$, so $\phi(a)$ is surjective.
2. (Injective) Suppose $\phi(a)(y) = \phi(a)(x)$, so $\phi(a^{-1})\phi(a)(y) = \phi(a^{-1})\phi(a)(x)$, so $\phi(a^{-1}a)(y) = \phi(a^{-1}a)(x)$, so $\text{id}_S(y) = \text{id}_S(x)$, so $y = x$, so $\phi(a)$ is injective.

We conclude $\phi(a) \in \text{Sym}(S)$, so $\phi : G \rightarrow \text{Sym}(S)$ and also ϕ is a group homomorphism, and by construction the group action $\star$ arises from ϕ .

□

Example 40.1 (group action of dihedral group).

Consider the dihedral group D_{2n} and set of n vertices V_n , then the homomorphism $\phi : D_{2n} \rightarrow \text{Sym}(V_n)$, which we showed in 16, realizes a group action. Conversely, any group action of the dihedral group on a set S is realized via a homomorphism $\phi : D_{2n} \rightarrow \text{Sym}(S)$ as per the theorem.

The theorem we have shown trivially implies Cayley's theorem:

Theorem 41 (Cayley's theorem).

If G is a group then G is isomorphic to a subgroup of the symmetric group $\text{Sym}(G)$.

Proof. Consider a group action $\star : G \times G \rightarrow G$, then by Theorem 40 group actions are equivalent to group homomorphism to symmetric group the group action arises from a homomorphism $\phi : G \rightarrow \text{Sym}(G)$. Also, the only element for which $a \star g = g$ is e , so $\text{Ker}(\phi) = \{e\}$, so by Theorem 26 first isomorphism theorem and 25.2 it follows that $G \cong \phi(G) \leq \text{Sym}(G)$. □

Orbits and Stabilizers

Definition 0.0.7 (orbit).

If G acts on a set S , then the orbit of x under G , denoted $\text{orb}_G(x)$, for some $x \in G$ is given by the image of x under the action of G :

$$\text{orb}_G(x) = \{s \in S \mid \exists g \in G (s = g \star x)\}$$

We sometimes denote the orbit of x under G as $O(x)$.

Definition 0.0.8 (stabilizer).

If G acts on a set S , then the stabilizer of x under G , denoted $\text{stab}_G(x)$, for some $x \in G$, is given by the elements of G that fix x under the group action:

$$\text{stab}_G(x) = \{g \in G \mid \forall x \in S (g \star x = x)\}$$

We sometimes denote the stabilizer of x under G as $S(x)$.

Example 41.1 (orbit and stabilizer of group multiplication).

Let G be a group and $H \leq G$. Recall **39.2 group multiplication defines group action**. Let $x \in G$. The orbit of g in G is the set of all elements of G s.t. $h \star x = g$ for some $h \in H$, but $h \star x = hx$, so the orbit is the set of all $g \in G$ s.t. $hx = g$, which by definition is the right H -coset of x , so $\text{orb}_H(x) = Hx$.

To find the stabilizer of x , we wish to find all $h \in H$ s.t. $h \star x = hx = x$. By definition such h is the identity, so $\text{stab}_H(x) = \{e\}$.

Example 41.2 (orbit and stabilizer of scalar multiplication).

Consider a vector space V over a field of scalars $\mathbb{F}$, then scalar multiplication defines a group action on V , $\star : \mathbb{F} \times V \rightarrow V$, $a \star v \mapsto av$:

1. $a \star (b \star v) = a(bv) = (ab)v = (ab) \star v$
2. $1 \star v = v$

Let $v \in V$, then:

1. $\text{stab}_{\mathbb{F}}(v) = \{a \in \mathbb{F} \mid av = v\} = \{1\}$.
2. $\text{orb}_{\mathbb{F}}(v) = \{av \mid a \in \mathbb{F}\}$, i.e. it is the line in direction v .

Example 41.3 (stabilizer of conjugation).

Let G be a group and consider $\star : G \times G \rightarrow G, g \star x \mapsto gxg^{-1}$, then:

1. $g \star (h \star x) = g \star (h x h^{-1}) = g(h x h^{-1})g^{-1} = (gx)h(gh)^{-1} = (gh) \star x$
2. $e \star x = exe^{-1} = x$

So $\star$ is the group action defined by conjugation.

Consider $\text{stab}_G(x)$, it is the set of all $g \in G$ s.t. $g \star x = x$, but $g \star x = gxg^{-1}$, so $gxg^{-1} = x$. Post-multiplying both sides by g we have $gx = xg$, so $\text{stab}_G(x) = \{g \in G \mid gx = xg\}$, which is exactly the centralizer of x in G as defined in 0.0.13 *centralizer of an element*, so we have $\text{stab}_G(x) = C(G)$.

In all three examples, we have seen that the stabilizer of an element in a group is a subgroup of the group acting on the element. This is true in general.

Theorem 42.

If G acts on a set S , and $x \in S$, then $\text{stab}_G(x) \leq G$.

Proof. We prove by the two-step subgroup test. First, note $e \star x = x$ always, so $e \in \text{stab}_G(x)$, so it is non-empty. For closure, suppose $a, b \in \text{stab}_G(x)$, so $ab \star x = a \star (b \star x) = a \star x = x$, so $ab \in \text{stab}_G(x)$, so it is closed under the group operation. For inverse, let $a \in \text{stab}_G(x)$, then $(a^{-1}a) \star x = a^{-1} \star (a \star x) = a^{-1} \star x$, but also $a^{-1}a = e$ and $e \star x = x$, so $a^{-1} \star x = x$, so $a^{-1} \in \text{stab}_G(x)$, so $\text{stab}_G(x) \leq G$ by the two-step subgroup test. $\square$

We now define an equivalence relation which will assist in proving the next theorem:

Lemma 42.1 (orbit defines an equivalence relation).

If G acts on S , then the relation $(x \sim y) \iff (x \in \text{orb}_G(y))$ defined on S is an equivalence relation.

Proof. We show $\sim$ satisfies all axioms of an equivalence relation:

1. (reflexive) $x = e \star x$ so $x \in \text{orb}_G(x)$ so $x \sim x$.
2. (symmetric) suppose $x \sim y$ so $x \in \text{orb}_G(y)$ so $x = g \star y$ for some $g \in G$, and also $g^{-1} \star (g \star y) = (g^{-1}g) \star y = e \star y = y$, but $g \star y = x$, so $g^{-1} \star x = y$, so $y \in \text{orb}_G(x)$, so $y \sim x$.
3. (transitive) suppose $x \sim y$ and $y \sim z$, then $x \in \text{orb}_G(y)$ and $y \in \text{orb}_G(z)$, so for some g_1, g_2 we have $g_1 \star y = x$, $g_2 \star z = y$, so $x = g_1 \star y = g_1 \star (g_2 \star z) = (g_1 g_2) \star z$, so $x \in \text{orb}_G(z)$, so $x \sim z$.

$\square$

Theorem 43 (orbits partition the set).

If G acts on a set S , then the orbits of the elements in S under the group action of G partition S (i.e. the union of all orbits is S).

Proof. Recall the equivalence relation $\sim$ defined in 42.1 orbit defines an equivalence relation. Since $\sim$ is an equivalence relation, then it partitions S , so the theorem follows immediately. $\square$

Corollary 43.1 (size of set expressed via orbits).

If a finite G acts on a finite set S , and S/G is the quotient of S under the equivalence relation defined by the orbits of S under the group action of G , then

$$|S| = \sum_{[x] \in S/G} |\text{orb}_G(x)|$$

The proof is immediate from Theorem 43 orbits partition the set and counting.

Theorem 44 (size of orbit of an element is the index of the stabilizer of the element).

If G acts on a set S , and S is finite, and $x \in S$, then:

1. There is a one-to-one correspondence between $\text{orb}_G(x)$ and the left-cosets of $\text{stab}_G(x)$ in G .
2. $|\text{orb}_G(x)| = [G : \text{stab}_G(x)]$, i.e. the size of $\text{orb}_G(x)$ is the index of the stabilizer of x in G (where the index is defined as in 0.0.2 index of a subgroup).

Proof. Let $G/\text{stab}_G(x)$ be the set of left-cosets of $\text{stab}_G(x)$ in H (by Theorem 42 the stabilizer of G is a subgroup of G , so by 21.1 $\text{stab}_G(x)$ partitions G). To show there is a one-to-one correspondence, we show a bijective mapping between the two sets.

Consider $\phi : \text{orb}_G(x) \rightarrow G/\text{stab}_G(x)$ defined as follows: for $y \in \text{orb}_G(x)$, there exists $g_y \in G$ s.t. $g_y \star x = y$ (by definition of the orbit of x under G). We choose one such g_y (if G is infinite this requires the axiom of choice, if G is finite then $H = \{g \in G | g \star x = y\} \subseteq G$ is also finite and also non-empty as we have just argued so by definition $\exists \psi : H \rightarrow [1, n]$ for some $n \in \mathbb{N}$ s.t. ψ is bijective, so we can declare $g_y = \psi(1)$). Now, we define $\phi(y) = g_y \text{stab}_G(x)$, i.e. ϕ maps y to the left coset of $\text{stab}_G(x)$ by g_y .

The choice of g_y is in fact irrelevant: given g_{y_1}, g_{y_2} s.t. $g_{y_1} \star x = g_{y_2} \star x = y$, we have $g_{y_1}^{-1} g_{y_2} \star x = x$, so $g_{y_1}^{-1} g_{y_2} \in \text{stab}_G(x)$, so by Theorem 20 properties of a coset $g_{y_1} \text{stab}_G(x) = g_{y_2} \text{stab}_G(x)$.

Clearly, $\phi(y) \in G/\text{stab}_G(x)$ since $g_y \text{stab}_G(x)$ is a left-coset of $\text{stab}_G(x)$, so ϕ is well-defined.

Now, suppose $\phi(z) = \phi(y)$, so $g_y \text{stab}_G(x) = g_z \text{stab}_G(x)$, so by Theorem 20 properties of a coset $g_y^{-1} g_z \in \text{stab}_G(x)$, so $g_y^{-1} g_z \star x = x$, so $g_y^{-1} \star (g_z \star x) = x$, acting on both sides by g_y we get

$g_y \star (g_y^{-1} \star (g_z \star x)) = g_y \star x$, simplifying we get $g_z \star x = g_y \star x$, but $g_z \star x = z$ and $g_y \star x = y$ by construction, so $z = y$, so ϕ is injective.

To show ϕ is surjective, let $g \text{stab}_G(x) \in G/\text{stab}_G(x)$. Since $g \star x \in \text{orb}_G(x)$ by definition, then $g \star x \in \text{Domain}(\phi)$. As we argued above, it must be that $\phi(g \star x) = g \text{stab}_G(x)$, so ϕ is surjective.

We conclude ϕ is a bijection, which proves part 1 of the theorem. For part 2, since ϕ is a bijection then trivially $|\text{orb}_G(x)| = |G/\text{stab}_G(x)|$, where the later is by definition $[G : \text{stab}_G(x)]$.

□

Corollary 44.1.

If G is finite, then $\forall x \in G$ the number of unique conjugates of x (i.e. distinct elements of the form $gxg^{-1} \in G$ for some $g \in G$) is the index of its centralizer $C(x)$.

Proof. By **41.3 stabilizer of conjugation**, $\text{stab}_G(x) = C(x)$. By **Theorem 44 size of orbit of an element is the index of the stabilizer of the element**, $|\text{orb}_G(x)| = [G : \text{stab}_G(x)] = [G : C(x)]$. Since $\text{orb}_G(x) = \{gxg^{-1} | g \in G\}$ (since we consider the group action defined by conjugation), the theorem follows.

□

We use the previous theorem to prove the orbit-stabilizer theorem:

Theorem 45 (orbit-stabilizer theorem).

If a finite group G acts on a set S , then $\forall x \in S$:

1. $|\text{orb}_G(x)| \mid |G|$.
2. $|G| = |\text{orb}_G(x)| \cdot |\text{stab}_G(x)|$

Proof. We prove (1) and (2) together since (2) immediately implies (1). By **Theorem 44 size of orbit of an element is the index of the stabilizer of the element**, $|\text{orb}_G(x)| = [G : \text{stab}_G(x)]$. By **21.1**, $[G : \text{stab}_G(x)] = \frac{|G|}{|\text{stab}_G(x)|}$, so $|G| = |\text{orb}_G(x)| \cdot |\text{stab}_G(x)|$.

□

The orbit-stabilizer theorem, and group actions in general, have useful application in counting, as we will show in the next couple of examples, the first of which we state as a proposition:

Proposition 40 (order of the dihedral group).

$$|D_{2n}| = 2n$$

Proof. Consider the dihedral group D_{2n} acting on a set V_n with n elements as discussed in the introduction to [[#Group Action]]. In [Rigid Motions & Symmetry](#), we argue that for every vertex of an n -gon, it can be rotated to any other vertex, i.e. the orbit of some $v \in V_n$ under D_{2n} is in fact V_n , and it is preserved either by the identity or by a reflection where v coincides with the line of reflexive symmetry, so the stabilizer of v is id_{V_n}, r . By [Theorem 45 orbit-stabilizer theorem](#), $|D_{2n}| = |\text{orb}_{D_{2n}}(v)| \cdot |\text{stab}_{D_{2n}}(v)| = n \cdot 2 = 2n$, so the dihedral group has order $2n$. $\square$

Example 45.1 ((informally) counting the rotational symmetries of a cube).

Consider the rotational symmetries of a cube. A cube has 6 faces, and we can act on the cube via rotation to map any face to any other face, so the size of the orbit of any face is 6. The face remains fixed by rotations whose axis is perpendicular to the face. There are 4 such rotational symmetries (those that map the vertices of the face to themselves and not to vertices of other faces), so every face has stabilizer of order 4, so by the orbit-stabilizer theorem the cube has 24 symmetries.

We proceed with more applications of the orbit-stabilizer theorem.

Definition 0.0.9 (conjugacy class).

If G is a group and $x \in G$, then the conjugacy class of x in G , denoted x^G or $\text{Cl}(s)$, is given by the set $\{gxg^{-1} | g \in G\}$.

Observe that the conjugacy class of x in G is exactly the orbit of x in G under the action of conjugation, and by [42.1 orbit defines an equivalence relation](#) we can trivially define an equivalence relation based on conjugation which partitions the set acted on by the group into equivalence classes, so the use of *class* in the definition is properly motivated.

[44.1](#) can be restated using conjugacy classes as follows:

Proposition 41.

$$|x^G| = [G : C(x)]$$

This result gives rise to the conjugacy class equation, which we define in the following theorem:

Theorem 46 (conjugacy class equation).

If G is a group, then the class equation of G is given by

$$|G| = |Z(G)| + \sum_i [G : C(x_i)] = |Z(G)| + \sum_i |x_i^G|$$

Where $Z(G)$ is the center of G , and the sum is over representatives of the conjugacy classes of $x \in G \setminus Z(G)$ (i.e. the conjugacy classes of elements of G not in its center).

Proof. Recall that **Theorem 43** orbits partition the set and the conjugacy class of x is the orbit of x under conjugation by **41**, so the sum of the order of all distinct conjugacy classes is exactly $|G|$ (since G acts on itself by conjugation), so by considering only the conjugacy classes of $G \setminus Z(G)$, we have that the sum of their orders is $|G| - |Z(G)|$. Adding $|Z(G)|$ to both sides produces the class equation of G . $\square$

Lemma 46.1.

If G is a group then $|x^G| = 1 \iff x \in Z(G)$ ($Z(G)$ is the center of G).

Proof. $\Leftarrow$ Suppose $x \in Z(G)$, then $\forall g \in G (gx = xg)$, so $gxg^{-1} = g(xg^{-1}) = g(g^{-1}x) = (gg^{-1})x = ex = x$, so $\forall g \in G (gxg^{-1} = x)$ in this case, so $x^G = \{x\}$, so $|x^G| = 1$.

$\Rightarrow$ Suppose $|x^G| = 1$. First observe that for $e \in G$, we have $exe^{-1} = x$, so $x \in x^G$ always, so if $|x^G| = 1$ then it must be that $\forall g \in G, gxg^{-1} = x$, which is equivalent to $\forall g \in G (gx = xg)$, which is by definition equivalent to $x \in Z(G)$. $\square$

We use the lemma to prove Cauchy's theorem:

Theorem 47 (Cauchy's theorem).

If G is a finite group and p is a prime s.t. $p \mid |G|$, then $\exists a \in G$ s.t. $|a| = p$.

Proof. Suppose $p \mid |G|$. We prove by induction on $|G|$. If $|G| = 2$ (smallest prime), then $G \cong \mathbb{Z}_2^+$ and $|1| = 2$ in $\mathbb{Z}_2^+$. 2 is the only prime that divides itself, so the theorem holds for $|G| = 2$.

Suppose by induction the theorem holds for some n and consider the case where $|G| = n + 1$. Let $x \in G$, then by **41** $|x^G| = [G : C(x)] = \frac{|G|}{|C(x)|}$.

If $Z(G) = G$, then $\forall g \forall x \in G (gx = xg)$ i.e. G is abelian, then by **Theorem 38** it follows that since p divides G then $\exists a \in G$ s.t. $|a| = p$, which completes the proof.

If $Z(G) \neq G$, then $\exists x \in G \setminus Z(G)$. By **46.1**, if $x \notin Z(G)$ then $1 \neq |x^G|$, so $|C(x)| \mid |G|$. If $p \mid |C(x)|$, in which case by the induction hypothesis $\exists a \in C(x) \leq G$ s.t. $|a| = p$. Consider the case where $C(x)$ is not divisible by p , then since $|x^G| \mid |C(x)| = |G|$ and $|G|$ is divisible by p , then by Euclid's lemma it must be that $p \mid |x^G|$.

Recall by **Theorem 46** conjugacy class equation that $|G| = |Z(G)| + \sum_i |x_i^G|$ where x_i is a representative element of a conjugacy class and also $x_i \in G \setminus Z(G)$. Since the argument from the last paragraphs holds for every such x_i , then it follows that $\sum_i |x_i^G|$ is divisible by p , so since $|Z(G)| = |G| - \sum_i |x_i^G|$ is the difference of two multiples of p then p also divides $|Z(G)|$. Since $Z(G)$ is abelian and also $p \mid |Z(G)|$, then again by **Theorem 38** it follows that $\exists a \in Z(G) \leq G$ s.t. $|a| = p$, which completes the proof.

□

This theorem is a weaker version of a more general theorem, known as Sylow's first theorem. We state the first and third of Sylow's theorems without proof below. Proving the theorems relies on the Orbit-Stabilizer theorem, Cauchy's theorem, induction and the conjugacy class equation. Sylow's theorem are a powerful tool in the classification of subgroups of finite group, and provide a connection between the prime factors of the order of the group, and subgroups of the group. Thus, it reduces the group-theoretic problem of classifying the subgroups of some group, to a number-theoretic problem of finding prime factors of the order of the group and their multiplicity.

Definition 0.0.10 (p -group).

If G has order p^n for some $p \in \mathbb{G}$, then G is called a p -group.

Theorem 48 (Sylow's first theorem).

If G is a finite group and p is a prime factor with multiplicity n s.t. $p^n \mid |G|$, then $\exists H \leq G$ s.t. $|H| = p^n$.

Sylow's first theorem guarantees the existence of a p -subgroup for every prime factor p of $|G|$, which is why they are sometimes called Sylow p -subgroups. Sylow's third theorem teaches us about the structure of these p -groups:

Theorem 49 (Sylow's third theorem).

If G is a finite group, and $p, q \in \mathbb{N}$ s.t. p is prime, $\gcd(p, q) = 1$, and $|G| = p^n q$ for some $n \in \mathbb{N}$, then:

1. The index of the Sylow p -subgroups of G is q .
2. Let n_p be the number of Sylow p -subgroups of G , then $n_p \mid q$.
3. $n_p \equiv 1 \pmod{p}$
4. $n_p = [G : N_G(P)]$ where P is any Sylow p -subgroup of G , and $N_G(P)$ is the normalizer of P in G .

The next theorem shows another result in the same spirit of the conjugacy class equation:

Theorem 50 (Burnside's lemma).

If G is a finite group acting on a finite set X , and X^g is the set of elements in X fixed by $g \in G$ under the group action (i.e. $X^g = \{x \mid g \star x = x\} \subseteq X$, not to be confused with the stabilizer of x in G , which is a subset of G s.t. its elements fix x under the group action, nor with x^G (note the lowercase x) which is the conjugacy class of some $x \in G$ under the group action of conjugation), then the number of (distinct) orbits, denoted $|X/G|$, is given by

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|$$

Proof. First we note that by definition of X^g , $\text{stab}_G(x)$, and counting, we get:

$$\sum_{g \in G} |X^g| = |\{(g, x) \in G \times X \mid g \star x = x\}| = \sum_{x \in X} |\text{stab}_G(x)|$$

By **Theorem 45 orbit-stabilizer theorem** $|\text{stab}_G(x)| = \frac{|G|}{|\text{orb}_G(x)|}$. Via summation over all $x \in X$, we get:

$$\sum_{g \in G} |X^g| = \sum_{x \in X} |\text{stab}_G(x)| = \sum_{x \in X} \frac{|G|}{|\text{orb}_G(x)|} = |G| \sum_{x \in X} \frac{1}{|\text{orb}_G(x)|}$$

Where the first transition is justified by the relation we developed in the beginning of the proof.

Consider $\sum_{x \in X} \frac{1}{|\text{orb}_G(x)|}$. Since the orbits of X partition the set, we can express X directly as the disjoint union of representatives of X/G , and write

$$\sum_{x \in X} \frac{1}{|\text{orb}_G(x)|} = \sum_{\text{orb}_G(x) \in X/G} \sum_{x' \in \text{orb}_G(x)} \frac{1}{|\text{orb}_G(x)|} = |X/G|$$

Putting it all together, we get

$$\begin{aligned} |G| \sum_{x \in X} \frac{1}{|\text{orb}_G(x)|} &= \\ |G| |X/G| &= \\ \sum_{g \in G} |X^g| &= \\ \therefore & \\ |X/G| &= \frac{1}{|G|} \sum_{g \in G} |X^g| \end{aligned}$$

□

We use Burnside's lemma in the next example, adapted from Wikipedia:

Example 50.1 (colorings of a cube).

Consider a cube. We wish to count the rotationally distinct colorings of faces of the cube using 3 colors, that is - we wish to count how many ways there are to paint the faces of a cube using 3 colors such that one coloring χ cannot be mapped to another coloring χ' via rotation (this is what it means for a coloring to be rotationally distinct).

Note that this is not the same question as counting the rotational symmetries of a cube, which we did in 45.1 (informally) counting the rotational symmetries of a cube (and arrived at the number 24), however only by acting on the coloring via a rotational symmetry of the cube can we map one colored face to another via rotation, so the question becomes counting the distinct orbits of possible colorings under the rotational symmetry group of the faces of the cube.

Denote G to be the group of rotational symmetries of the faces of the cube. We know $|G| = 24$ by 45.1 (informally) counting the rotational symmetries of a cube. Given n colors, there are n^6 possible different colorings of the faces of the cube. Let X be the set of colorings of the cube, then $|X| = n^6$. The question is now finding $|X/G|$, which by Burnside's lemma is exactly $\frac{1}{|G|} \sum_{g \in G} |X^g| = \frac{1}{24} \sum_{g \in G} |X^g|$.

Consider X^g for some $g \in G$ - it is the set of colorings that are fixed by the group action of g , i.e. fixed under the rotation represented by g . Consider the action of the rotational symmetries of the cube on the coloring. By considering all rotations (we have not classified these yet, but you can convince yourself that the enumeration is true by considering the action of each described rotation on the cube, and recalling that we did show that in total there should be 24 such rotations), we find that:

1. The identity rotation fixes all colorings, so we have $|X^e| = n^6$
2. All six 90 rotations of each face each fix n^3 colorings, so in total they contribute $6n^3$ to the count
3. All three 180 rotations of each face fix n^4 colorings, so they contribute $3n^4$
4. All eight 120 vertex rotations fix n^2 colorings, so they contribute $8n^2$
5. All six 180 edge rotations fix n^3 colorings, so they contribute $6n^3$

Note that we have accounted for all 24 rotations which we know represent all rotational symmetries of the cube, so in total, we have $\sum_{g \in G} |X^g| = n^6 + 6n^3 + 3n^4 + 8n^2 + 6n^3$. Substituting 3 for n , we get that this sum is 1368. Dividing by $|G| = 24$, we get $|X/G| = 57$, so we conclude that there are 57 rotationally distinct colorings of the faces of the cube using 3 colors.

Ring & Field Theory

Rings are an algebraic structure which, while having group structure, also have other properties. Fields, in turn, are rings with additional structure. Field and group theory were brought together by Evariste Galois in Galois theory, which reduces problems concerning fields to problems concerning groups, thus making them easier. A very famous application of Galois theory is proving that the quintic does not have a general solution by radicals. We will not discuss Galois theory here, but instead build a base of knowledge in ring & field theory, which can later be used to develop Galois theory.

Basic Definitions

We begin with a survey of definitions. Once we have collected enough definitions to properly examine interesting examples, we show a few examples.

Definition 0.0.1 (ring).

A structure $(R, +, \cdot)$ where:

1. R is a nonempty set
2. $+$ is a binary operator on R called addition
3. $\cdot$ is a binary operator on R called multiplication

Is called a ring, if it satisfies the following properties $\forall a, b, c \in R$:

1. $(R, +)$ is an abelian group
2. Multiplication is associative, i.e. $a \cdot (b \cdot c) = (a \cdot b) \cdot c$
3. Multiplication is distributive over addition, i.e. $a \cdot (b + c) = a \cdot b + a \cdot c$ (right distributive), and also $(a + b) \cdot c = a \cdot c + b \cdot c$ (left distributive)

Note that our definition of a ring is often referred to as a *pseudo-ring* or *non-unital ring*, or *rng*, and *ring* is reserved for what we call a *ring with unity*. I follow Gallian's definitions which call what is more commonly called a *rng* a *ring*, and a *ring* a *unital ring*.

Definition 0.0.2 (commutative ring).

A ring is commutative if multiplication in the ring commutes, i.e. $\forall a, b \in R, a \cdot b = b \cdot a$.

Definition 0.0.3 (unital ring).

If $\exists x \in R$ s.t. $\forall r \in R, xr = r = rx$, then we denote $x = 1$ and say that R has unity 1, and say that R is a unital ring. The unity is the multiplicative identity of the ring.

Alternatively, a *unital ring* is a ring s.t. $(R, \cdot)$ is a **0.0.1 monoid** where the identity of that structure is denoted 1. Some writers call a *unital ring* (or a *ring with unity*) a *ring with identity*. We try to reserve

the term *identity* for the additive identity of the ring, and use the term *unity* for the multiplicative identity.

Now that we have the definition of a ring with unity, the name *rng* should perhaps be more obvious - it is a *ring* but without *i*(dentity).

Unital rings can succinctly be defined (without referring to **0.0.1 ring**) as a structure $(R, +, \cdot)$ s.t.:

1. $(R, +)$ is an abelian group
2. $(R, \cdot)$ is a monoid
3. Multiplication is (left & right) distributive over addition

The notion of *factor*, *divisor* and *multiple* generalize naturally to rings:

Definition 0.0.4 (factor, divisor, multiple).

If R is a ring and for some $a, b \in R$ exists $k \in R$ s.t. $a = kb$, we say that b divides a , denoted $b|a$. We also say that a is a multiple of b , and that b is a factor of a .

Definition 0.0.5 (unit of a ring).

If R has unity 1 and for $a \in R$ there exists $b \in R$ s.t. $ab = 1$, we say that a is a unit and write $b = a^{-1}$ and call b the multiplicative inverse of a .

Theorem 51.

1. If $1 \in R$ (i.e. if R has unity 1), then it is unique.
2. If $a^{-1} \in R$, then it is unique.
3. If $ab = 1$ then both a, b are units of R .

Proof. For (1), refer to **1 uniqueness of the identity**. For (2), (3), refer to **3 properties of the inverse** $\square$

Definition 0.0.6 (zero-divisors).

If R is a ring and $ab = 0$ for some $a, b \in R$ both nonzero (where 0 is the additive identity of the ring), then we call a, b zero-divisors.

Given $a \in R, z \in \mathbb{Z}$, we define an *action* of $\mathbb{Z}$ on R along the same fashion of **0.0.4 power notation** with **0.0.5 additive notation**. Note that now we can say something more interesting than we did in those definitions - we can identify this as a *group action*. Once we have defined this group action, we can define the ring characteristic:

Definition 0.0.7 (ring characteristic).

If R is a ring, then the characteristic of R , denoted $\text{char}(R)$, is the least $n \in \mathbb{Z}^+$ for which $\forall x \in R, nx = 0$ (i.e. n acting on any element of R results in the additive identity of R). If no such n exists, then we say $\text{char}(R) = 0$.

Definition 0.0.8 (integral domain).

A ring R is an integral domain if R is a commutative ring with unity (equivalently, a commutative monoid with respect to multiplication), that has no zero-divisors.

Definition 0.0.9 (field).

A ring R is a field if R is a commutative ring where every nonzero (where 0 is the additive identity of the field) element is a unit, i.e. it has an inverse w.r.t multiplication.

Equivalently, a field can be defined as a structure $(F, +, \cdot)$ where F is a nonempty set, $+$, $\cdot$ are binary operators on F called *addition* and *multiplication* respectively, s.t. $(F, +)$ is a commutative group, $(F, \cdot)$ is a commutative monoid, multiplication is distributive over addition, and $\forall a \in F, a \neq 1 \implies a^{-1} \in F$.

Definition 0.0.10 (group of units).

The set of units of a field R is the set of all units in R , denoted $U(R)$. If $U(R)$ is nonempty then the group of units is the structure $(U(R), \cdot)$ where $\cdot$ is multiplication in the group.

Proposition 42.

$U(R)$ is a group under ring multiplication (if $U(R)$ is nonempty).

Proof. $(U(R), \cdot)$ is a commutative monoid, so it remains to show that the inverse property is satisfied, and indeed if $a \in U(R)$ then a has an inverse $a^{-1} \in R$, and $(a^{-1})^{-1} = a$ so a^{-1} is also a unit so $a^{-1} \in U(R)$, thus $(U(R), \cdot)$ satisfies all group axioms, thus it is properly named. $\square$

We now survey a few important examples.

Example 51.1 (the integers).

Consider $\mathbb{Z}$ with the usual addition and multiplication operations. By the properties of the integers we already know that $\mathbb{Z}$ satisfies all ring axioms, so it is a ring. Furthermore, multiplication on integers commutes, so $\mathbb{Z}$ is a commutative ring.

$1 \in \mathbb{Z}$ is the unity of the ring (i.e. multiplicative identity), so $\mathbb{Z}$ is a commutative ring with unity, and also $ab = 0$ implies either $a = 0$ or $b = 0$, so it has no zero-divisors, so it is an integral domain by definition. However, it is not a field - take $2 \in \mathbb{Z}$, we know the multiplicative inverse of 2 is $\frac{1}{2}$, but $\frac{1}{2} \notin \mathbb{Z}$, so 2 is not a unit of $\mathbb{Z}$, so $\mathbb{Z}$ is not a field. In fact, it is trivial to show that $U(\mathbb{Z}) = \{-1, 1\}$.

Finally, the characteristic of $\mathbb{Z}$ is 0 since we know $nx = 0$ for $x \neq 0$ implies $n = 0$.

Example 51.2 (integers under modular arithmetic).

Consider $\mathbb{Z}_n$ i.e. the quotient set of the integers under a congruence defined by some modulus n , equipped with modular addition and multiplication. We know (from [Elementary Number Theory](#)) that this structure satisfies all ring axioms and also that multiplication is commutative so it is a commutative ring. Furthermore, we know that the only elements of $\mathbb{Z}_n$ that have a multiplicative inverse also in $\mathbb{Z}_n$, i.e. the units of $\mathbb{Z}_n$, are numbers which are relatively coprime w.r.t. n , and they form the group we denoted mostly by $\mathbb{Z}_n^\times$ before.

Since it is always true that $1 \in \mathbb{Z}_n$ then these are commutative rings with unity, however for $a, b \in \mathbb{Z}_n$ s.t. $a \cdot b$ calculated in $\mathbb{Z}$ is a multiple of n , then $ab \equiv_n 0$ by definition of modular congruence, for example with $n = 9$ and $a, b = 3$, we get $3 \not\equiv_n 0$ and yet $3 \cdot 3 = 9 \equiv_9 0$, so 3 is a zero divisor in $\mathbb{Z}_9$, so it is not true in general that $\mathbb{Z}_n$ is an integral domain.

Finally, since $\forall x \in \mathbb{Z}_n, nx \equiv_n 0$, then $\text{char}(\mathbb{Z}_n) \leq n$. By a standard division algorithm argument we can show that $\text{char}(\mathbb{Z}_n) = n$.

Example 51.3 ($\mathbb{Q}$ is a field).

We know from previous knowledge that the rationals, the reals and the complex numbers satisfy all field axioms under the usual addition and multiplication operations defined for these sets, so $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are all examples of fields.

We turn our attention to a more exciting example - *polynomials over a commutative ring*. We will formally define this structure later, so the following example is not entirely accurate and appeals to previous math knowledge, but nonetheless it is worth considering and we will build upon it in later examples before we formally introduce the underlying structure discussed in this example.

First, a (slightly informal) definition:

Definition 0.0.11 (univariate polynomial).

A polynomial in one indeterminate (one variable) is called a univariate polynomial.

We tend to use *univariate polynomial* and *polynomial* interchangeably.

Definition 0.0.12 (polynomial over a commutative ring).

If R is a commutative ring then a polynomial over R is a polynomial with coefficients from R . The set of all such univariate polynomials is denoted $R[x]$.

We will later see that if R is a commutative ring then $R[x]$ is also a commutative ring, and that if R is an integral domain then so is $R[x]$. For now, we present an example based on said previous knowledge:

Example 51.4 ($\mathbb{Z}[n]$ is a ring).

Recall from previous knowledge that a univariate polynomial can be written as a finite sum $\sum_{i=0}^n a_i x^i$ where x is a variable and $a_0, \dots, a_n$ are coefficients. For some $p(x) \in \mathbb{Z}[n]$, we have $p(x) = \sum_{i=0}^n a_i x^i$ for some n , s.t. all coefficients are in $\mathbb{Z}$. We know from previous knowledge that these polynomials satisfy all ring axioms, and furthermore they commute under multiplication ($(gf)(x) = (fg)(x)$) and $o(x) = 1$ is the unity of $\mathbb{Z}[n]$ and if $(gf)(x) = 0$ then either $g(x) = 0$ or $f(x) = 0$, so $\mathbb{Z}[n]$ forms an integral domain.

We define an external direct product on rings similarly to our definition on groups:

Definition 0.0.13 (product ring).

If $R_1, \dots, R_n$ are rings, then $R_1 \times \dots \times R_n$ (the Cartesian product of all elements of the rings) equipped with component-wise addition and multiplication (i.e. if $s, r \in R_1 \times \dots \times R_n$, then $s + r = (s_1 + r_1, \dots, s_n + r_n)$ and $s \cdot r = (s_1 \cdot r_1, \dots, s_n \cdot r_n)$ where each component-wise operation is the respective operation of the ring the component belongs to) is called the product ring of $R_1, \dots, R_n$.

Proposition 43.

The product ring is a ring.

Proof. Since the proof is simple and we have a lot of similar proofs in the past, we only sketch it and leave filling out the details to the reader. By 38 we know that the product ring forms a commutative group with respect to addition (in truth, the proposition shows that it forms a group w.r.t. addition, but it is trivial to see that if each component-wise addition commutes, then addition on the product commutes). Closure of multiplication as defined above is also obvious, and associativity of multiplication follows from the associativity of component-wise multiplication operations. The distributive property is also shown in a similar fashion. $\square$

This last definition provides one way of forming new rings from existing rings. Note that the polynomial ring is also another such method as we will see later. Finally, consider matrices over a ring.

Definition 0.0.14 (matrix ring).

If R is a ring, then the set of all $n \times n$ matrices with elements from R (i.e. the set of all matrices s.t. $\forall i, j \in \mathbb{N} \cap [1, n](A_{ij} \in R)$), denoted $R^{n \times n}$ or $M_n(R)$, equipped with usual matrix addition (which is carried component-wise) and multiplication ($(A \cdot B)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$), is called the matrix ring of R .

Proposition 44.

The matrix ring forms a ring under matrix addition and multiplication, and if R has unity 1 then matrix ring of R has unity I where I is an $n \times n$ matrix where $I_{ij} = \begin{cases} i & i = j \\ 0 & \text{otherwise} \end{cases}$, i.e. it is a diagonal matrix where all entries along the diagonal are 1.

Proof. Let R be a ring. First, we show $(M_n(R), +)$ (where $+$ is matrix addition) forms a commutative group. Let $A, B, C \in M_n(R)$, then:

1. (closure under addition) By definition, $(A + B)_{ij} = (A_{ij} + B_{ij})$ where the latter is addition in R , and since R is a ring then elements of R are closed under addition, so every ij^{th} cell of $A + B$ is in R , so the $n \times n$ matrices whose elements are $(A + B)_{ij}$ is also a matrix where all elements belong to R , so it belongs to $M_n(R)$, hence $M_n(R)$ is closed under addition.
2. (associativity under addition) Since R is a ring then addition in R is associative, so $A_{ij} + (B_{ij} + C_{ij}) = (A_{ij} + B_{ij}) + C_{ij}$, the former is $(A + (B + C))_{ij}$ by definition and the latter is $((A + B) + C)_{ij}$, so $(A + B) + C = A + (B + C)$, so matrix addition is associative.
3. (additive identity) Consider $0 \in R$ the additive identity of R , and let 0_M be an $n \times n$ matrix s.t. all its elements are $0 \in R$, clearly $0_M \in M_n(R)$, and also $(A + 0_M)_{ij} = A_{ij} + 0_{M,ij} = A_{ij} + 0 = A_{ij}$, so $A + 0_M = A$, so 0_M is the additive identity of $(M_n(R), +)$ (we have only shown right identity but by an earlier theorem from group theory we know that it suffices to show the structure is closed under the operation, associative, and has identity and inverse in one side, to imply that the identity and inverse also exists for the other side and are in fact equal in both sides).
4. (additive inverse) Since R is a ring then $-A_{ij} \in R$ as well. Consider a matrix D given by $D_{ij} = -A_{ij}$, then $(A + D)_{ij} = A_{ij} + D_{ij} = A_{ij} - A_{ij} = 0$, so $A + D = 0_M$, so D is the additive inverse of A .
5. (addition commutes) Since R is a ring then addition is commutative on R , so by an identical argument to those presented above we find that $A + B = B + A$.

We conclude $(M_n(R), +)$ is a commutative group. We now consider $(M_n(R), +, \cdot)$ where $\cdot$ is matrix multiplication. Again let $A, B, C \in M_n(R)$, then:

1. (closure under multiplication) By definition $(A \cdot B)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$. The RHS of the equation is a sum of product of elements of R . Since R is a ring it is closed under these operations, so the RHS also belongs to R , so $AB \in M_n(R)$.
2. (associativity under multiplication) By definition,

$$(A(BC))_{ij} = \sum_{k=1}^n (A_{ik} (BC)_{kj}) = \sum_{k=1}^n (A_{ik} \sum_{k'=1}^n (B_{kk'} C_{k'j})) = (*)$$

Since R is a ring, multiplication in R is distributive over addition in R , so

$$(*) = \sum_{k=1}^n \left(\sum_{k'=1}^n (A_{ik} (B_{kk'} C_{k'j})) \right) \stackrel{R \text{ mult is associative}}{=} \sum_{k=1}^n \left(\sum_{k'=1}^n ((A_{ik} B_{kk'}) C_{k'j}) \right)$$

Exchanging the symbols for k, k' , the above expression becomes:

$$\sum_{k'=1}^n \sum_{k=1}^n ((A_{ik'} B_{k'k}) C_{kj}) = (*)$$

On the other hand, again by definition:

$$((AB)C)_{ij} = \sum_{k=1}^n (AB)_{ik} C_{kj} = \sum_{k=1}^n \left(\sum_{k'=1}^n A_{ik'} B_{k'k} \right) C_{kj}$$

Since ring addition commutes, we can exchange the order of summation, which gives us $(**)$, which is $(A(BC))_{ij}$, so $A(BC) = (AB)C$, so matrix multiplication is associative.

1. (distributive property) we only show right distributivity

$$\begin{aligned} (AB + AC)_{ij} &= \\ \sum_{k=1}^n (A_{ik} B_{kj} + A_{ik} C_{kj}) &= \\ \sum_{k=1}^n A_{ik} (B_{kj} + C_{kj}) &= \\ \sum_{k=1}^n A_{ik} (B + C)_{kj} &= \\ (A(B + C))_{ij} \end{aligned}$$

So $A(B + C) = AB + AC$.

The matrix ring satisfies all ring axioms, so we conclude it is indeed a ring.

Next, suppose R has unity 1, and let I be a diagonal $n \times n$ matrix where all entries along the diagonal are 1, then $\forall A \in M_n(R)$:

$$(IA)_{ij} = \sum_{k=1}^n I_{ik} A_{kj} = 0 \left(\sum_{k=1}^{i-1} I_{ik} A_{kj} \right) + I_{ii} A_{ij} + 0 \left(\sum_{k=i+1}^n I_{ik} A_{kj} \right) = A_{ij}$$

Similarly we have $(AI)_{ij} = A_{ij}$, so $IA = AI = A$, so I is indeed the unity of $M_n(R)$.

□

It can be further shown that if R is unital, then the group of units of $M_n(R)$ is the set $\{A \in M_n(R) \mid \det A \in U(R)\}$ where $U(R)$ is the group of units of R .

Note that this result can be applied recursively - since a matrix ring can be defined over an arbitrary ring, we can consider the following: consider $\mathbb{Z}$ the ring of integers, then $M_n(\mathbb{Z})$ the set of $n \times n$ matrices with elements from $\mathbb{Z}$ forms a ring under matrix addition and multiplication by 44, but then $M_n(M_n(\mathbb{Z}))$ the set of $n \times n$ matrices whose elements are themselves $n \times n$ matrices with elements from $\mathbb{Z}$ is also a ring by the same proposition. A similar argument can be applied to polynomial rings, i.e. the coefficients of the polynomials can themselves be polynomials and the structure would still form a ring.

Another immediate result is that every field is a vector space over itself.

Theorem 52 (F is a vector space over itself with dimension 1).

If F is a vector space, then F is a vector space over F , and $\dim(F) = 1$.

Proof. Recall that a vector space is a set of vectors V and some scalar field $\mathbb{F}$ satisfying a set of axioms. By setting $V = \mathbb{F}$, we find that the axioms of a vector space coincide with those of a field:

1. F is an abelian group under addition
2. Since we define $V = F$ and also F to be the field of scalars, then obviously $\forall a, b \in F, u, v \in V = F$, we have $a(bv) = (ab)v$ (F is a monoid w.r.t. multiplication), $1v = v$ (F is unital), $a(u + v) = au + av$ and $(a + b)v = av + bv$ (distributive property of multiplication over addition in a ring).

So, F is a vector space over itself. Furthermore, let $1 \in F$ and $r \in F$, then since $r1 = r$ for every $r \in F$, it follows that $\{1\}$ is a basis of F and $\dim(F) = 1$.

□

We define *subrings* in an analogous manner to *subgroups*:

Definition 0.0.15 (subring).

If R is a ring and $S \subseteq R$ is a nonempty set, then we say that S is a subring of R if it is a ring with respect to the ring operations of R .

Theorem 53 (subring test).

If $S \subseteq R$ is nonempty, then S is a ring w.r.t the ring operations on R , if:

1. *It is closed under subtraction, i.e. $\forall a, b \in S, a - b \in S$*
2. *It is closed under multiplication, i.e. $\forall a, b \in S, ab \in S$.*

Proof. Observe that (1) is simply the one-step subgroup test for $(R, +)$, thus if (1) is satisfied then $(S, +) \leq (R, +)$, and since $(R, +)$ is abelian then $(S, +)$ is also abelian. Suppose (2) holds, then S is closed under multiplication. Associativity of multiplication immediately follows from associativity of

multiplication in R , and also right and left distributivity behave the same since the operators are the same, so S is a subring of R . $\square$

The next example is a cautionary tale:

Example 53.1 (unities of subring and unity of ring aren't necessarily identical).
 Consider $\mathbb{Z}_6$ whose elements are $\{0, 1, 2, 3, 4, 5\}$, and consider the coset $2\mathbb{Z}_6 = \{0, 2, 4, 6, 8, 10\} = \{0, 2, 4, 0, 2, 4\} = \{0, 2, 4\}$. We can verify that $2\mathbb{Z}_6$ forms a (commutative) ring under modular addition and multiplication $\pmod 6$, however note that 1 is the unity of $\mathbb{Z}_6$, whereas $4 \cdot 0 = 0$, $4 \cdot 2 = 8 \equiv_6 2$, $4 \cdot 4 = 16 \equiv_6 4$, so 4 is the unity of $2\mathbb{Z}_6$ even though it is a subring of $\mathbb{Z}_6$ and $\mathbb{Z}_6$ is also a ring with unity.

Of course, the additive identity is the same across all subrings since all subrings are in particular subgroups under addition.

Elementary Results

The next theorem collect basic arithmetic properties of rings:

Theorem 54 (ring arithmetic).
 If R is a ring then $\forall a, b, c \in R$:

1. $a0 = 0a = 0$ (here we talk about $0 \in R$, and not $0 \in \mathbb{Z}$)
2. $a(-b) = (-a)b = -(ab)$
3. $(-a)(-b) = ab$
4. $a(b - c) = ab - ac$ and $(b - c)a = ba - ca$
5. If R has unity 1, then $(-1)a = -a$ and $(-1)(-1) = 1$
6. (cancellation law) R has no zero divisors if and only if $a \neq 0$ and $ab = ac$, then $b = c$

Proof. 1. Observe that $0 = 0 + 0$ by the group structure of $(R, +)$, so $a0 = a(0 + 0) = a0 + a0$ by right-distributivity, so $a0 = a0 + a0$, so $a0 = 0$. Similarly $0a = (0 + 0)a = 0a + 0a$ hence $0a = 0$ (here we used left-distributivity), so $0a = a0 = 0$.

2. Note $0 = b - b$. By 1 we know $a0 = 0$, so $a0 = a(b - b) = ab + a(-b) = 0$, so $a(-b) = -(ab)$ by uniqueness of the additive inverse. Similarly, $0b = (a - a)b = ab + (-a)b = 0$, so $(-a)b = -(ab)$, so in conclusion $a(-b) = (-a)b = -(ab)$.

3. Denote $d = -b$, so by 2 we have $(-a)(-b) = (-a)d = -(ad)$. Now, $a(-b) = -(ab)$ also by 2, so we have $(-a)(-b) = -(-(ab))$, so $(-a)(-b) = ab$ since inverse is an involution.

4. Via right-distributivity $a(b - c) = a(b + (-c)) = ab + a(-c) = ab - ac$ (the last transition is justified by (2)), similarly we find that $(b - c)a = ba + (-c)a = ba - ca$ via left-distributivity and (2).

5. Suppose R has unity 1, then $1a = a$ and also $1 - 1 = 0$, so $(1 - 1)a = 0a = 0$ (by (1)). By (4), $(1 - 1)a = 1a + (-1)a = a + (-1)a = 0$, so $(-1)a = -a$. Choosing $a = 1$, we get $(-1)(-1) = 1$.

6. $\implies$ Suppose R has no zero divisors and suppose $a \neq 0$ and also $ab = ac$. Consider $a(b - c)$, by (4) we have $a(b - c) = ab - ac$. By supposition $ab = ac$ so $-(ab) = ac$, so $ab - ac = 0$, so $a(b - c) = 0$, so since R has no zero divisors then either $a = 0$ or $b - c = 0$, but by supposition $a \neq 0$ so $b - c = 0$, so $b = c$.

$\Leftarrow$ Suppose $ab = 0$. If $a = 0$ then we're done, otherwise let $c \in R$, then $\exists d \in R$ s.t. $d = b + c$, so $b = d - c$, so $ab = a(d - c)$, by (4) we have $a(d - c) = ad - ac = 0$, adding ac to both sides we get $ad = ac$. Since $a \neq 0$, then it follows that $d = c$ by supposition, so $b = d - d = 0$. We conclude that R has no zero divisors.

□

Proposition 45.

If R is a commutative ring, then if a is a zero-divisor and $ab \neq 0$ for some $b \in R$, then ab is a zero-divisor.

Proof. Suppose a is a zero-divisor and $ab \neq 0$. Since a is a zero-divisor, then $ac = 0$ for some $c \neq 0$. Since R is commutative, then $(ab)c = (ac)b = 0b = 0$ by **Theorem 54 ring arithmetic**, but $ab \neq 0$ and also $c \neq 0$, so ab is a zero-divisor.

□

Proposition 46 (a finite integral domain is a field).

If R is a finite integral domain, then R is a field.

Proof. Since R is assumed to be a finite integral domain, by definition we have that R is a commutative ring with unity 1 and no zero-divisors. It remains to show that the nonzero elements of R are units. We follow an identical argument to the one outlined in **7 order of elements in finite group is finite**.

Let $x \in R$. Consider the set $X = \{x^n \in R | n \in \mathbb{N}\}$ (since R is a ring it is closed under multiplication and so x^n as defined in **0.0.4 power notation** is also in R for every $n \in \mathbb{N}$). Since R is finite and $X \subseteq R$ then X is also finite, so $\forall n \in \mathbb{N} \exists m \in \mathbb{N}$ s.t. $x^m = x^n$. Suppose w.l.o.g. $m > n$, then $x^{m-n} = x^m x^{-n} = x^n x^{-n} = x^{n-n} = x^0 = 1$, so $x^{m-n-1}x = x^{m-n} = 1$, so $x^{m-n-1} \in R$ is the multiplicative inverse of $x \in R$, in other words x is a unit, so we conclude R is a field.

□

Theorem 55.

If R is a ring with unity 1 and no zero-divisors, the additive identity 0 and the multiplicative identity 1 are distinct elements.

Proof. Suppose by contradiction $1 = 0$, then also $1 \cdot 1 = 0$ (here $\cdot$ denotes group multiplication), but then 1 is a zero-divisor, a contradiction, so $1 \neq 0$. $\square$

Proposition 47.

If F is a field, then:

1. $\forall a, b \in F, (ab)^{-1} = b^{-1}a^{-1}$ (socks-shoes property is valid for fields)
2. F has no zero-divisors.

Proof. (1) is proven using an identical argument to one for groups, presented in **3 properties of the inverse**. For (2), let F be a field and suppose by contradiction F has zero-divisors $a, b \in F, a, b \neq 0$, so $ab = 0$. Since by supposition $a \neq 0$ then $a^{-1} \in F$ by field axioms, so $a^{-1}ab = a^{-1}0$. By **Theorem 54 ring arithmetic**, $a^{-1}0 = 0$, and by associativity of field multiplication $a^{-1}ab = (a^{-1}a)b = 1b = b$, so $b = 0$, a contradiction, so F has no zero divisors. $\square$

Corollary 55.1 (field are integral domains).

If F is a field then F is an integral domain.

Proof. By **47**, if F is a field then F has no zero divisors, and also by field axioms F is a commutative ring with unity 1, so F satisfies all integral domain axioms. $\square$

Lemma 55.1.

If R is a ring with unity 1, then $\forall n \in \mathbb{Z} \forall x \in R, n \cdot x = (n \cdot 1)x$, where juxtaposition denotes ring multiplication, and $\cdot$ denotes the action of $\mathbb{Z}$ on R .

Proof. Suppose R is a ring with unity 1. Let $n \in \mathbb{Z}^+$ and let $x \in R$, and consider $n \cdot x$, which we defined as $n \cdot x = \sum_{i=1}^n x$ (where $\sum$ is a summation over ring addition). Since $x = 1x$ by **Theorem 54 ring arithmetic**, it follows that $\sum_{i=1}^n x = \sum_{i=1}^n 1x$. By the distributive property, $\sum_{i=1}^n 1x = (\sum_{i=1}^n 1)x$. By definition of the action of n on R , we have $\sum_{i=1}^n 1 = n \cdot 1$, so $(\sum_{i=1}^n 1)x = (n \cdot 1)x$. In conclusion, $n \cdot x = (n \cdot 1)x$. $\square$

Theorem 56 (characteristic of a ring with unity).

If R is a ring with unity 1 then:

1. If $|1| = \infty$ ($|1|$ is the order of 1 in $(R, +)$) then $\text{char}(R) = 0$
2. If $|1| = n$ for some finite n , then $\text{char}(R) = n$

Proof. Let $x \in R$. By **55.1** $n \cdot x = (n \cdot 1)x$.

1. We prove the contrapositive - if $\text{char}(R) = n$ then $|1|$ is finite. Suppose $\text{char}(R) \neq 0$, then $n \cdot x = 0$ for some positive integer n , but then $(n \cdot 1)x = 0$ for the same n . By **Theorem 54 ring arithmetic**, $n \cdot 1 = 0$. By definition, $n \cdot 1 = \sum_{i=1}^n i$, so $|i| \leq n$. It follows that if $|1| = \infty$, $\text{char}(R) = 0$ via contraposition.
2. Suppose $|1| = n$, then $n \cdot 1 = 0$, then $n \cdot x = 0x = 0$, so $\text{char}(R) \leq n$. Suppose $\text{char}(R) = mn$, then for $m \cdot x = 0$, i.e. $m \cdot 1 = 0$, but then $|1| \leq mn$, a contradiction, so $\text{char}(R) = n$.

□

Corollary 56.1 (unital subrings preserve characteristic).

If R is a unital ring and S is a unital subring of R (i.e. a subring of R with the same unity as R), then $\text{char}(R) = \text{char}(S)$.

Proof. By the theorem, the order of the unity of the ring uniquely determines the characteristic of the ring. Since S is a subring of R then it is a subgroup of R and so order of shared elements is preserved, and since they share the same unity we have $\text{char}(R) = \text{char}(S)$.

□

Corollary 56.2 (characteristic of integral domain).

If R is an integral domain, then $\text{char}(R)$ is either 0 or a prime.

Proof. Let R be an integral domain. Again by **55.1**, $n \cdot x = (n \cdot 1)x$. By **Theorem 56 characteristic of a ring with unity**, if $|1| = \infty$ then $\text{char}(R) = 0$. Suppose then that $|1| = n$ for some $n \in \mathbb{N}$, then by the theorem $\text{char}(R) = n$, so it remains to show that if R is an integral domain and $|1|$ is finite, then $|1|$ is prime.

Suppose $\text{char}(R) = 1$, then $|1| = 1$ then $1 = 0$, and by **Theorem 55** this is impossible, so $n \neq 1$.

Since $n \in \mathbb{N}$ and also $n \neq 1$ as we have argued above, then by the fundamental theorem of arithmetic we can write $n = kp$ for p prime and $k \in \mathbb{N}$, so $n \cdot 1 = (kp) \cdot 1 = k \cdot (p \cdot 1)$ (the last transition is due to the properties of a group action). Again we refer to **55.1** and get $k \cdot (p \cdot 1) = (k \cdot 1)(p \cdot 1)$ (where the RHS is multiplication in the ring), which is 0 by choice of n . Since R is an integral domain, it has no zero divisors, and so $(k \cdot 1)(p \cdot 1) = 0$ implies that $k \cdot 1 = 0$ or $p \cdot 1 = 0$ (or both).

Since n is minimal s.t. $n \cdot 1 = 0$, and also $n = kp$ and $(k \cdot 1)(p \cdot 1) = 0$, there are two cases:

1. $k \cdot 1 = 0$, but then since n is minimal then it must be that $k = n$, but then $p = 1$ while 1 is not a prime, a contradiction.
2. $p \cdot 1 = 0$, then again $n = p$ and $k = 1$.

Since only the second case is possible, we conclude $|1| = \text{char}(R)$ is prime.

□

Proposition 48 ($\mathbb{Z}_p$ is a field).

If p is prime then $\mathbb{Z}_p$, equipped with modular addition and multiplication, is a field.

Proof. The elements of $\mathbb{Z}_p$ are identified with the numbers $0, \dots, p-1$. Note that $\mathbb{Z}_p$ equipped with modular addition is always a group. Also by properties of modular arithmetic, $(\mathbb{Z}_p, +_p, \times_p)$ is a *commutative ring* with unity 1. Consider the group of units of this ring, denoted $U(p)$ (previously we also used the notation $\mathbb{Z}_p^\times$). We know from number theory that the elements of this group are all positive natural numbers that are less than p and are coprime w.r.t p . Since p is prime, then this is exactly the set $1, \dots, p-1$, so $U(p) = \mathbb{Z}_p \setminus \{0\}$, which by definition means that $\mathbb{Z}_p$ is a field. $\square$

Ideals

Ideals are to rings as normal subgroup are to groups, as we will see in this section.

Definition 0.0.16 (ideal).

If R is a ring then a subring S is called an ideal of R if $\forall r \in R \forall s \in S, sr \in S$ and $rs \in S$.

Example 56.1 ($n\mathbb{Z}$ is an ideal of the integers).

For $n \in \mathbb{Z}$, the left coset $n\mathbb{Z} = \{nz | z \in \mathbb{Z}\}$ is a subring. Since $\mathbb{Z}$ is a commutative ring, then $n\mathbb{Z}$ is also commutative. so $nz = zn$. Let $nz \in n\mathbb{Z}$ and $q \in \mathbb{Z}$, then $(nz)q = n(zq)$ and $zq \in \mathbb{Z}$ so by definition $(nz)q \in n\mathbb{Z}$, and also $q(nz) = n(qz)$ by commutativity of multiplication on $\mathbb{Z}$, so $q(nz) \in n\mathbb{Z}$, so $n\mathbb{Z}$ is an ideal of $\mathbb{Z}$ by definition.

Proposition 49 (ideals are normal subgroups).

If R is a ring and P is an ideal of R , then P is a normal subgroup of $(R, +)$.

Proof. Suppose P is an ideal, then P is a subring so it is a subgroup, it remains to show it is normal i.e. that $\forall r \in R, r + P - r \subseteq P$. Let $p \in P$ and $r + p - r \in P$, since $(R, +)$ is abelian then $r + p - r = r - r + p = p$, so $r + p - r = p \in P$, so $r + P - r \subseteq P$, so $P \trianglelefteq (R, +)$. Note that we have already seen this result - every subgroup of an abelian group is normal. $\square$

Theorem 57.

If R is a ring, then:

1. $\{0\}$ is an ideal of R
2. R is an ideal of R

We call such ideals trivial ideals. If S is an ideal of R that is non trivial, we say it is proper.

Proof. 1. From **Theorem 54 ring arithmetic** we know that $\forall r \in R, r0 = 0r = 0$, so $r\{0\} = \{0\}r = \{0\}$, so $\{0\}$ is an ideal of R .

2. Since R is closed under multiplication then $\forall x, y \in R$ we have $xy \in R$ and $yx \in R$ so R is an (improper) ideal of R . □

Theorem 58 (ideal test).

If $S \subseteq R$ and $S \neq \emptyset$, then S is an ideal of R if:

1. S is closed under subtraction
2. If $s \in S$ and $r \in R$, then $sr \in S$ and $rs \in S$

Proof. Suppose $S \subseteq R$ and $S \neq \emptyset$. If we pick $r \in R \cap S$ then by (2) we have that S is closed under multiplication so by **Theorem 53 subring test** S is a subring of R . The second part of (2) is simply stating the condition for a subring to be an ideal as per **0.0.16 ideal**. □

Definition 0.0.17 (principal ideal).

If R is a commutative ring with unity and $a \in R$, then $\langle a \rangle = \{ar \mid r \in R\}$ is the principal ideal generated by a .

Note that a principal ideal of some $a \in \mathbb{Z}_n$ is simply the cyclic subgroup $\langle a \rangle = \{na\}$ in additive notation, equipped with modular multiplication.

Proposition 50.

Principal ideal is an ideal.

Proof. Consider $\langle a \rangle$ under the ring operations. Clearly $a0 = 0 \in \langle a \rangle$ so it is nonempty. We proceed by **Theorem 58 ideal test**. Let $ar_1, ar_2 \in \langle a \rangle$, then $ar_1 - ar_2 = a(r_1 - r_2)$ by the distributive property. Since R is a ring then $\exists r \in R$ s.t. $r = r_1 - r_2$, so $ar = a(r_1 - r_2)$, so $ar \in \langle a \rangle$ by definition, so $\langle a \rangle$ is closed under subtraction. Next, let $ar_1 \in \langle a \rangle$ and $r_2 \in R$, then $(ar_1)r_2 = a(r_1r_2)$ from associativity, and also $r_1r_2 \in R$ from closure of R , so $a(r_1r_2) \in \langle a \rangle$ by definition. Consider $r_2(ar_1)$, again from associativity $r_2(ar_1) = (r_2a)r_1$. Since R is a commutative ring (see **0.0.17 principal ideal**) then $r_2a = ar_2$, so $(r_2a)r_1 = (ar_2)r_1 = a(r_2r_1)$. Again $r_2r_1 \in R$ so $a(r_2r_1) = r_2(ar_1) \in \langle a \rangle$. We conclude by **Theorem 58 ideal test** that $\langle a \rangle$ is an ideal. □

Proposition 51 (principal ideal of a unit).

If R is a commutative ring with unity and $a \in U(R)$ (i.e. a is a unit of R), then $\langle a \rangle = R$.

Proof. $\langle a \rangle \subseteq R$ is immediate. To prove the other direction, let $r \in R$ and note that since $a \in U(R)$ then $a^{-1} \in R$, so $a^{-1}r \in R$, so $a(a^{-1}r) \in \langle a \rangle$ by definition, but this is simply r , so $R \subseteq \langle a \rangle$, so $\langle a \rangle = R$. $\square$

Theorem 59 (ideal is a superset of the principal ideals of its elements).

If I is an ideal in a commutative ring R , and $d \in I$, then $\langle d \rangle \subseteq I$.

Proof. Suppose I is an ideal in a commutative ring R and $d \in I$ for some $d \in R$, then by definition $\forall s \in R, ds, rd \in I$. Since R is commutative, $dr \in I \iff rd \in I$. By definition, $\langle d \rangle$ is the set of all elements of the form dr for $r \in R$, so $\langle d \rangle \subseteq I$. $\square$

Definition 0.0.18 (principal ideal domain).

If D is an integral domain, and every ideal of D is principal, then D is called a principal ideal domain, or PID for short.

Theorem 60 (fields are PIDs).

If F is a field, then it is a PID.

Proof. Suppose F is a field and I is an ideal of F . If $I = \{0\}$, then $I = \langle 0 \rangle$, otherwise $a \in I$ s.t. $a \neq 0$, so $a^{-1} \in F$, so $a^{-1}a \in I$ by definition of an ideal, but $a^{-1}a = 1$, so $1 \in I$. Let $f \in F$, then $f \cdot 1 = f \in I$, so $I = F$, so a field has no nontrivial ideals, and also note that $\langle a \rangle = F$ for every non-zero a , and we have shown that the only ideals of F are $\langle 0 \rangle, F$ and we have shown that F is principal and every non-zero element in F is a generator of F , so F is a PID by definition. $\square$

The notion of a principal ideal can be generalized to any finite number of elements.

Definition 0.0.19 (ideal generated by multiple elements).

If R is a commutative ring with unity and $a_1, \dots, a_n \in R$, then $\langle a_1, \dots, a_n \rangle = \{ \sum_{i=1}^n a_i r_i \mid r_i \in R \}$ is called the ideal generated by $a_1, \dots, a_n$.

Proposition 52.

The ideal generated by $a_1, \dots, a_n$ is an ideal.

Proof. Denote $A = \langle a_1, \dots, a_n \rangle$. For an n -tuple $(r_1, \dots, r_n) = (0, \dots, 0)$ we have $\sum_{i=1}^n a_i r_i = \sum_{i=1}^n a_i 0 = 0$ so $0 \in A$ so A is nonempty. We proceed by **Theorem 58 ideal test**. Let $a, b \in A$, then $\exists r_a, r_b \in R^n$ s.t. $a = \sum_{i=1}^n a_i r_{a,i}$ and $b = \sum_{i=1}^n a_i r_{b,i}$, then $a - b = \sum_{i=1}^n a_i (r_{a,i} - r_{b,i})$. Since each $r_{a,i} - r_{b,i} \in R$, then $r_{a-b} = (r_{a,1} - r_{b,1}, \dots, r_{a,n} - r_{b,n}) \in R^n$, so $a - b \in A$ (here we take a to be $(a_1, \dots, a_n)$ and $\cdot$ to be the dot product of the two n -tuples), so A is closed under subtraction.

Let $a \in A$ where $a = \sum_{i=1}^n a_i r_{a,i}$ for some $r_a \in R^n$, and let $r \in R$, then:

$$ar = \left(\sum_{i=1}^n a_i r_{a,i} \right) r \stackrel{\text{left distributive, associativity}}{=} \sum_{i=1}^n a_i \underbrace{(r_{a,i} r)}_{\in R}$$

So $ar \in A$ by definition. Similarly, from commutativity of the ring, $ar = ra$, so $ra \in A$ as well, so by **Theorem 58 ideal test**, A is an ideal. □

Example 60.1.

Consider the polynomial ring $\mathbb{R}[x]$, and consider the polynomial $x \in \mathbb{R}[x]$. The ideal generated by x is, by definition, $\{xp(x) | p(x) \in \mathbb{R}[x]\}$, clearly all polynomials $f(x)$ in $\langle x \rangle$ satisfy $f(0) = 0$ (since $x = 0 \therefore f(x) = xp(x) = 0$). Consider a polynomial $h(x)$ s.t. $h(0) = 0$, then either $h(x) = 0$ or $h(x) = xh'(x)$ where $h'(x) \in \mathbb{R}[x]$, but since $0 = 0x$ then in both cases we have $h(x) \in \langle x \rangle$, so via mutual inclusion we can characterize the ideal generated by $\langle x \rangle$ in $\mathbb{R}[x]$ as the set of all polynomials $p(x)$ s.t. $p(0) = 0$. This example relies on previous knowledge, but we will later discuss polynomials in depth and develop the appropriate machinery to derive this result without referring to external knowledge.

By now it should become quite apparent that ideals generated by elements are conceptually very similar to the span of vectors in linear algebra.

Factor Rings

We define factor rings in a similar manner to how we defined factor groups.

Definition 0.0.20 (factor ring).

If R is a ring and A is a subring of R , then if the following operators are well-defined on the quotient set R/A (which is always well-defined since $A \trianglelefteq (R, +)$ since $(R, +)$ is an abelian group and from group theory we know that A partitions R so R/A is well-defined):

$$1. + : (R/A)^2 \rightarrow R/A, (s + A) + (t + A) \mapsto (s + t) + A$$

$$1. \cdot : (R/A)^2 \rightarrow R/A, (s + A) \cdot (t + A) \mapsto st + A$$

Then we say that $(R/A, +, \cdot)$ is the factor ring of R over A .

Theorem 61 (factor ring is a ring).

The factor ring is a ring if and only if A is an ideal. Furthermore, the additive identity 0 of R/A is A , and if R has unity 1 then $1 + A$ is the unity (multiplicative identity) of the factor ring.

Proof. $\Leftarrow$ Suppose A is an ideal. Since $A \trianglelefteq (R, +)$ as argument in the definition of a factor ring, then addition as defined on the structure is well-defined, so it remains to show multiplication is also well-defined. Choose two representatives from each equivalence class $s_1 + A = s_2 + A, t_1 + A = t_2 + A$, it remains to show $s_1 t_1 + A = s_2 t_2 + A$.

Recall that $r + A$ for some $r \in R$ is a *left coset* of A (in additive notation), so by **Theorem 20 properties of a coset** (converted to additive notation), $s_1 + A = s_2 + A \iff s_1 - s_2 \in A$, and similarly $t_1 + A = t_2 + A \iff t_1 - t_2 \in A$, and also $s_1 t_1 + A = s_2 t_2 + A \iff s_1 t_1 - s_2 t_2 \in A$. We will show $s_1 t_1 - s_2 t_2 \in A$.

Since $s_1 - s_2 \in A$, then $\exists a_1 \in A$ s.t. $s_1 - s_2 = a_1$, i.e. $s_1 = s_2 + a_1$, similarly $\exists a_2 \in A$ s.t. $t_1 = t_2 + a_2$. Taking the product $s_1 t_1$, we get $s_1 t_1 = (s_2 + a_1)(t_2 + a_2)$, which by ring axioms (specifically the distributive property and commutativity of addition) expands to $s_1 t_1 = s_2 t_2 + a_1 t_2 + (s_2 + a_1) a_2$. Solving for $s_1 t_1 - s_2 t_2$, we get $s_1 t_1 - s_2 t_2 = a_1 t_2 + (s_2 + a_1) a_2$. Since $t_2 \in R$ then since A is an ideal, $a_1 t_2 \in A$, and similarly since $s_2, a_1 \in R$ then by closure under addition $s_2 + a_1 \in R$ and since A is an ideal $(s_2 + a_1) a_2 \in A$, finally since A is a subring then it is closed under addition so $a_1 t_2 + (s_2 + a_1) a_2 \in A$, and we conclude $s_1 t_1 - s_2 t_2 \in A$, so $s_1 t_1 + A = s_2 t_2 + A$, so multiplication is well-defined.

It remains to show that R/A with these operations satisfies all ring axioms. Note that R/A being an abelian group under addition follows from A being a normal subgroup of R/A under addition, so we only need to show:

1. multiplication is associative - let $a, b, c \in R$, then $a(bc) = (ab)c$ from associativity of multiplication in the ring R , then by definition $(a + A)(bc + A) = a(bc) + A = (ab)c + A = (ab + A)(c + A)$
2. distributive property - let $a, b, c \in R$, then $a(b + c) = ab + ac$, and by definition $(a + A)((b + A) + (c + A)) = (a + A)(b + c + A) = a(b + c) + A = ab + ac + A = (ab + A)(ac + A)$, which proves right distributivity. Left distributivity can be shown via an identical argument.

Thus we conclude R/A is a *ring* under the operations defined. Furthermore, the additive identity of R/A is A by known results from group theory. If R has unity 1, then let $r \in R$, then by definition and by **Theorem 54 ring arithmetic** $(1 + A)(r + A) = 1r + A = 1 + A = r1 + A = (r + A)(1 + A)$, so $1 + A$ is the unity of R/A .

$\Rightarrow$ We prove by contraposition - if A is not an ideal, then R/A is not a ring. It suffices to show that multiplication as defined in **0.0.20 factor ring** is not well defined.

Suppose A is not an ideal but is a subring, then for some $r \in R$ and $a \in A$, either $ra \notin A$ or $ar \notin A$. Consider that by **Theorem 20 properties of a coset** $\forall a \in A (a + A = 0 + A)$. Consider $a \in A, r \in R$ s.t. not both $ar \in A \wedge ra \in A$, then if multiplication on the quotient set is well defined then since $a + A = 0 + A$ we expect $(a + A)(r + A) = (0 + A)(r + A)$, but $(a + A)(r + A) = ar + A$ while $(0 + A)(r + A) = 0r + A = A$ (by **Theorem 54 ring arithmetic**). If $ar + A = A$, then $ar \in A$. If not, then the operation is not well-defined for the a, r we chose so R/A has no hope of being a ring,

and we are done. Suppose then $ar \in A$, then by supposition $ra \notin A$, so $ra + A \neq A$, but then $(r + A)(a + A) \neq (r + A)(0 + A)$, so multiplication is again not well defined, so R/A can't be a ring under these operations, which completes the proof.

□

Example 61.1 (identifying the complex numbers in factor ring of polynomials).

Consider the polynomial ring $\mathbb{R}[x]$. Consider the factor ring $\mathbb{R}[x]/\langle x^2 + 1 \rangle$, which per **Theorem 61** factor ring is a ring forms a ring. Denote this factor ring $A = \mathbb{R}[x]/\langle x^2 + 1 \rangle$. We begin with a warm up to help the reader get acquainted to ring arithmetic in the factor ring:

Suppose $f(x), g(x) \in \mathbb{R}[x]$ are in the same equivalence class, i.e. $f(x) + \langle x^2 + 1 \rangle = g(x) + \langle x^2 + 1 \rangle$, then by **Theorem 20 properties of a coset** it follows that $f(x) - g(x) \in \langle x^2 + 1 \rangle$. By **0.0.17 principal ideal**, this means $f(x) - g(x) = (x^2 + 1)h(x)$ for some $h(x) \in \mathbb{R}[x]$, or equivalently $g(x) = f(x) - (x^2 + 1)h(x)$. Consider the case where $h(x) = 1$ and $f(x) = x^2 + 2x + 2$, then $g(x) = x^2 + 2x + 2 - x^2 - 1 = 2x + 1$, so $2x + 1 + \langle x^2 + 1 \rangle = x^2 + 2x + 2 + \langle x^2 + 1 \rangle$.

Now we consider the equivalence class represented by $x^2 + 1$. By **Theorem 20 properties of a coset**, we find that in this factor ring, $(x^2 + 1) + \langle x^2 + 1 \rangle = \langle x^2 + 1 \rangle$. Subtracting $-1 + \langle x^2 + 1 \rangle$ from both sides, we get by ring arithmetic in the factor ring that $x^2 + \langle x^2 + 1 \rangle = -1 + \langle x^2 + 1 \rangle$. Using $a \equiv b \pmod{x^2 + 1}$ to denote $a + \langle x^2 + 1 \rangle = b + \langle x^2 + 1 \rangle$, we find that $x^2 \equiv -1 \pmod{x^2 + 1}$.

Using $a \equiv b \pmod{x^2 + 1}$ to denote $a + \langle x^2 + 1 \rangle = b + \langle x^2 + 1 \rangle$, we can rewrite the above as $x^2 \equiv -1 \pmod{x^2 + 1}$,

Using this last relation, we can write every $f(x) = \sum_{i=0}^n a_i x^i \in \mathbb{R}[x]$ as follows mod $x^2 + 1$:

1. If i is even, then $x^i = -1$
2. If i is odd, then $x^i = x$

Then by commutativity of addition in the ring $\mathbb{R}[x]$, we can write $f(x) \equiv_{x^2+1} a_0 + a_1x - a_2 + a_3x - a_4x + \dots$, i.e. for some $b_0, b_1 \in \mathbb{R}$ we have $f(x) \equiv b_0 + b_1x \pmod{x^2+1}$ for every polynomial in $\mathbb{R}[x]$, so $\mathbb{R}[x]/\langle x^2 + 1 \rangle = \{b_0 + b_1x + \langle x^2 + 1 \rangle \mid b_0, b_1 \in \mathbb{R}\}$.

Consider multiplication in this factor ring, choose two representatives $a + bx, c + dx$, then $(a + bx)(c + dx) \equiv_{x^2+1} ac + adx + bcx + bdx^2$ (we used the fact that $\mathbb{R}[x]$ is a commutative ring to get $bx \cdot c = bcx$), and by the aforementioned result that $x^2 \equiv -1 \pmod{x^2 + 1}$, this expression becomes $(ac - bd) + (ad + bc)x$, i.e. in terms of coefficients, multiplication can be expressed as $(a, b)(c, d) = (ac - bd, ad + bc)$. Recall the definition of multiplication of complex numbers: $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$. These are in fact the same. In conclusion, we identified a structure similar to the field $\mathbb{C}$ in $\mathbb{R}[x]/\langle x^2 + 1 \rangle$ (of course, this exploration does not yet show that they are in fact the isomorphic, but it is trivial to show that as well once we have presented the first isomorphism theorem for rings, which is analogous to that of groups).

Definition 0.0.21 (prime ideal).

If R is a commutative ring, then an ideal P of R is prime if it is a proper nontrivial ideal of R , s.t. $\forall a, b \in R (ab \in P \implies (a \in P \vee b \in P))$.

This definition is similar to the statement of Euclid's lemma, which states that if some prime $p \in \mathbb{N}$ divides the product of two integers, then it must also divide one of the integers. The following examples explores the connection between the two:

Example 61.2 ($p\mathbb{Z}$ is prime ideal of integer ring for p prime).

Consider $\mathbb{Z}$. Let $p \in \mathbb{Z}$ be prime, and consider the left coset $p\mathbb{Z}$. By 56.1 $n\mathbb{Z}$ is an ideal of the integers we know $p\mathbb{Z}$ is an ideal. Let $a, b \in \mathbb{Z}$ and suppose $ab \in p\mathbb{Z}$, then $ab = pz$ for some $z \in \mathbb{Z}$, so $p|ab$. By Euclid's lemma, it follows that either $p|a$ or $p|b$. In the former case we have $a \in p\mathbb{Z}$ and in the later we have $b \in p\mathbb{Z}$, so it follows that $p\mathbb{Z}$ is a prime ideal of $\mathbb{Z}$.

Definition 0.0.22 (maximal ideal).

An ideal P of a commutative ring R is maximal if P is proper and also there does not exist another proper ideal A s.t. $P \subseteq A$.

Example 61.3 ($\langle x^2 + 1 \rangle$ is a maximal ideal of $\mathbb{R}[x]$).

Suppose by contradicition it is not maximal, then $\exists A$ ideal of $\mathbb{R}[x]$ s.t. $\langle x^2 + 1 \rangle$ is a proper subset of A and also $A \neq \mathbb{R}[x]$. Take $f(x) \in A \setminus \langle x^2 + 1 \rangle$, then by polynomial division we can write $f(x) = q(x)(x^2 + 1) + r(x)$ for polynomials $q(x), r(x)$ over the reals such that $r(x) = ax + b$ for some $a, b \in \mathbb{R}$, and also $r(x) \neq 0$ (otherwise $f(x) \in \langle x^2 + 1 \rangle$ which by supposition is not the case).

Since $r(x) = f(x) - q(x)(x^2 + 1)$, and $f(x) \in A$ and also $q(x)(x^2 + 1) \in \langle x^2 + 1 \rangle$ by definition and by supposition $\langle x^2 + 1 \rangle \subset A$, then $r(x) \in A$ since A being an ideal means it is closed under subtraction (since it is a subring of $\mathbb{R}[x]$). Take $ax - b \in \mathbb{R}[x]$ (with the same a, b), then since A is an ideal and $r(x) = ax + b \in A$, then $(ax - b)(ax + b) = a^2x^2 - b^2 \in A$, and since $x^2 + 1 \in \langle x^2 + 1 \rangle$ and $a^2 \in \mathbb{R}[x]$ then since $\langle x^2 + 1 \rangle$ is an ideal we have $(x^2 + 1)a^2 = x^2a^2 + a^2 \in \langle x^2 + 1 \rangle$ so it is also in A . Again from closure under subtraction of A we get $(a^2x^2 + a^2) - (a^2x^2 - b^2) = a^2 + b^2 \in A$.

Since $r(x) = ax + b \neq 0$ then at least one of $a, b \neq 0$ so $a^2 + b^2 \neq 0$. Since $\mathbb{R}$ is a field and $a^2 + b^2 \neq 0$, then $\exists (a^2 + b^2)^{-1} \in \mathbb{R}$ s.t. $(a^2 + b^2)^{-1}(a^2 + b^2) = 1$, and since $a^2 + b^2 \in A$ and A is an ideal then this product is also in A , so $1 \in A$.

Let $p(x) \in \mathbb{R}[x]$, then $1p(x) \in A$, but $1p(x) = p(x)$, so $\mathbb{R}[x] \subseteq A$ and also $A \subseteq \mathbb{R}[x]$ by supposition, so $A = \mathbb{R}[x]$, but we supposed $A \neq \mathbb{R}[x]$, a contradiction, so $\langle x^2 + 1 \rangle$ is a maximal ideal of $\mathbb{R}[x]$.

Definition 0.0.23 (comaximal ideals).

If R is a ring with ideals I, P s.t. I, P are maximal and $I + P = R$, then we say I, P are comaximal.

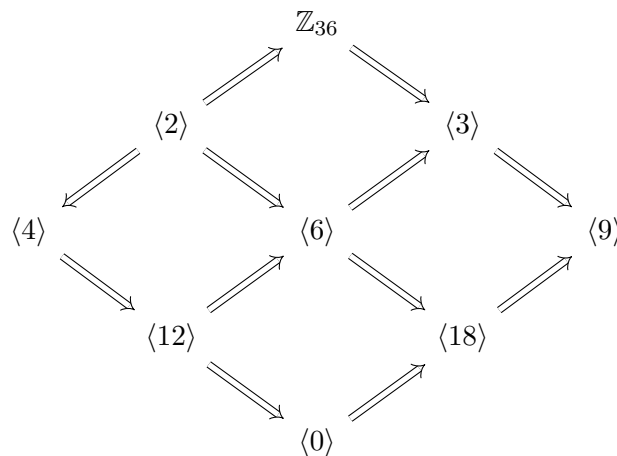
The notion of comaximal ideals somewhat corresponds to the idea of coprimes, as we demonstrate in the next example:

Example 61.4 (coprime numbers span the integers).

Let a, b be coprime, i.e. $\gcd(a, b) = 1$, and consider $\langle a, b \rangle$ the ideal generated by a, b . By definition, $\forall x, y \in \mathbb{Z}, xa + yb \in \langle a, b \rangle$. By Bezout's identity, $\exists x, y \in \mathbb{Z}$ s.t. $xa + yb = \gcd(a, b) = 1$, so $1 \in \langle a, b \rangle$. Let $z \in \mathbb{Z}$, by definition of an ideal $\forall s \in \langle a, b \rangle, sz \in \langle a, b \rangle$, so for $s = 1$ we find $z \in \langle a, b \rangle$, so $\langle a, b \rangle = \mathbb{Z}$.

It can be shown that in $\mathbb{Z}$, prime ideals are also maximal, and any pair of primes are coprime, so every pair of prime ideals in $\mathbb{Z}$ is also comaximal by the above.

As another example, consider the lattice diagram of the ring $\mathbb{Z}_{36}$:



Identify that $\mathbb{Z}_{36}/\langle 2 \rangle \cong \mathbb{Z}_2$, and $\mathbb{Z}_{36}/\langle 3 \rangle \cong \mathbb{Z}_3$, which are both fields by 48 $\mathbb{Z}_p$ is a field since 2, 3 are both prime. Also, by the lattice diagram it is clear that $\langle 2 \rangle, \langle 3 \rangle$ are maximal ideals of $\mathbb{Z}_{36}$, and they are in fact comaximal. This is not a coincidence, as we will soon see.

Theorem 62.

If R is a commutative ring with unity 1, and P is an ideal of R , then R/P is an integral domain if and only if P is prime.

Proof. $\implies$ Suppose R/P is an integral domain and let $a, b \in R$ s.t. $ab \in P$ (there has to be at least one such pair, for example the case where $a = b = 0$). Since $ab \in P$ then $ab + P = P$, but

by **Theorem 61 factor ring is a ring** we know that the 0 of R/P is P and by definition of factor ring multiplication $(a + P)(b + P) = ab + P$, so we have $ab \equiv_P 0$. Since R/P is an integral domain, then either $a \equiv_P 0$ or $b \equiv_P 0$, so either $a + P = P$ or $b + P = P$. By **Theorem 20 properties of a coset** this means that either $a \in P$ or $b \in P$, so by definition P is prime.

$\Leftarrow$ Suppose P is prime and let $ab \in P$, then $ab + P = P$, again by definition $ab + P = (a + P)(b + P)$, so by the same argument either $a + P = P$ or $b + P = P$, so by definition R/P is an integral domain.

□

Theorem 63.

If R is a commutative ring with unity 1, and P is an ideal of R , then R/P is a field if and only if P is maximal.

Proof. $\Rightarrow$ Suppose R/P is a field and suppose by contradiction P is not maximal, i.e. there exists some $P \subset A \subset R$ s.t. A is an ideal of R . Pick $a \in A \setminus P$, then $a + P \neq P$, however since R/P is a field then $\exists b \in R$ s.t. $(a + P)(b + P) = ab + P = 1 + P$. By **Theorem 20 properties of a coset**, it follows that $1 - ab \in P$. Since $P \subset A$ then $1 - ab \in A$ as well. Finally, since $a \in A$ and $b \in R$, and A is an ideal, then $ab \in A$, so $ab + (1 - ab) \in A$ due to closure of A as a subring of R , but $ab + (1 - ab) = 1$, so $1 \in A$. Let $r \in R$, then $r = 1 \cdot r$ by **Theorem 54 ring arithmetic**, but then by definition since A is an ideal $r \in A$, so $R = A$, but we supposed $A \subset R$, a contradiction, so P is maximal.

$\Leftarrow$ Let $a \in R$ and consider the coset $aR + P = \{ar + p \mid r \in R, p \in P\}$. We will show $aR + P$ is an ideal by **Theorem 58 ideal test**:

1. (closure under subtraction) $(ar_1 + p_1) - (ar_2 + p_2) = a(r_1 - r_2) + (p_1 - p_2)$ by **Theorem 54 ring arithmetic** and the commutative property of ring addition. Since R, P are rings then $r_1 - r_2 \in R, p_1 - p_2 \in P$, so $aR + P$ is closed under subtraction.
2. (closure under multiplication by $r \in R$) let $r \in R$ and let $ar_0 + p_0 \in aR + P$, then $r(ar_0 + p_0) = rar_0 + rp_0$ by the distributive property, and since R is a commutative ring then $r(ar_0 + p_0) = (ar_0 + p_0)r$ so it suffices to show $r(ar_0 + p_0) \in aR + P$. Since P is an ideal and $r \in R, p_0 \in P$, then $rp_0 \in P$. Consider rar_0 , since R is commutative then $rar_0 = arr_0$, and $rr_0 \in R$ from closure under ring multiplication, so for $r' = rr_0$ and $p' = rp_0$ we have $r(ar_0 + p_0) = ar' + p' \in aR + P$.

We conclude $aR + P$ is an ideal. Now, since $0 \in P, 1 \in R$ (since R is unital by supposition) then $a1 + 0 = a \in aR + P$, and also by fixing $r = 0$ we get $ar + p = a0 + p = p$, so $P \subseteq aR + P$.

Consider two cases:

1. $a \notin P$, then since $P \subseteq aR + P$ and also $a \in aR + P$, and P is maximal, it follows that $aR + P = P$. Since $1 \in R$ and $aR + P = P$ then $ar + p = 1$ for some $r \in R, p \in P$. Since $p \in P$ then by coset properties $(ar + p) + P = ar + P$, and since $ar + p = 1$ then $(ar + p) + P = 1 + P$, so $ar + P = 1 + P$, which is the unity of R/P . By definition of factor ring multiplication, $(a + P)(r + P) = ar + P = 1 + P$,

so $r + P$ is the multiplicative inverse of $a + P$.

2. $a \in P$, then $a + P = 0 + P$, so $a + P$ is the zero element of R/P .

By the above argument it follows that every non zero element of R/P is a unit. It remains to show that R/P is a commutative ring: let $a + P, b + P \in R/P$ for some $a, b \in R$, then $ab = ba$ since R is commutative by supposition, then $ab + P = ba + P$, then $(a + P)(b + P) = (b + P)(a + P)$, then R/P is a commutative ring.

We find that R/P is a commutative ring where every nonzero element is a unit, hence by 0.0.9 field, it is a field.

□

Example 63.1 ($\mathbb{R}[x]/\langle x^2 + 1 \rangle$ is a field).

By 61.3 $\langle x^2 + 1 \rangle$ is a maximal ideal of $\mathbb{R}[x]$ and by Theorem 63, $\mathbb{R}[x]/\langle x^2 + 1 \rangle$ is a field.

Ring Homomorphism

A ring homomorphism is a mapping that preserves ring structure. In this section, we give precise definition for a ring homomorphism, consider some examples, and prove that all properties we expect to be invariant under a ring homomorphism are in fact invariant.

Definition 0.0.24 (ring homomorphism).

If R, S are rings then a function $\phi : R \rightarrow S$ is a ring homomorphism if it satisfies the following properties $\forall a, b \in R$:

1. $\phi(a + b) = \phi(a) + \phi(b)$ (i.e. it is a group homomorphism with respect to the group structure of R, S under the respective addition operations)
2. $\phi(ab) = \phi(a)\phi(b)$ (ϕ preserves multiplication)
3. If R, S are unital, then $\phi(1_R) = 1_S$. (ϕ preserve unity)

If we remove (3) from the above definition, we have what is commonly known as a *rng homomorphism*.

Proposition 53 (rng homomorphism and unity).

If $\phi : R \rightarrow S$ is a rng homomorphism and R, S are unital, then:

1. If R is an integral domain, either $\phi(1_R) = 1_S$ or $\phi(1_R) = 0_S$.
2. If ϕ is surjective, then $\phi(1_R) = 1_S$.

Proof. 1. If ϕ is a rng homomorphism then we have $\phi(1_R) = \phi(1_R 1_R) = \phi(1_R)\phi(1_R)$, then $\phi(1_R)(1_S - \phi(1_R)) = 0_S$ (here we used Theorem 54 ring arithmetic to justify the transition).

If R is an integral domain, then $\phi(1_R)(1_S - \phi(1_R)) = 0_S$ implies then either $\phi(1_R) = 0_S$ or $1_S - \phi(1_R) = 0_S \therefore \phi(1_R) = 1_S$.

2. If ϕ is surjective then $\forall s \in S \exists r \in R$ s.t. $\phi(r) = s$. By **Theorem 54 ring arithmetic** $1_R r = r$, so $\phi(1_R r) = \phi(r) = s$, but also $\phi(1_R r) = \phi(1_R)\phi(r)$ since ϕ preserves multiplication, so $\phi(1_R)\phi(r) = \phi(r)$, and symmetrically $\phi(r1_R) = \phi(r)\phi(1_R) = \phi(r)$, so by definition $\phi(1_R) = 1_S$. □

Definition 0.0.25 (ring isomorphism).

A ring isomorphism is a bijective ring homomorphism. If $\phi : R \rightarrow S$ is a ring isomorphism. we write $R \cong S$.

As mentioned earlier, all properties we expect a ring homomorphism to preserve are preserved:

Theorem 64 (properties of ring homomorphism).

If $\phi : R \rightarrow S$ is a ring homomorphism, then:

1. $\forall r \in R \forall n \in \mathbb{Z}, \phi(n \cdot r) = n\phi(r)$ and $\phi(r^n) = \phi(r)^n$, where $n \cdot r$ is the additive action defined by $\mathbb{Z}$ on R and r^n is the multiplicative action defined by $\mathbb{Z}$ or R .
2. $\text{Im}(\phi)$ is a subring of S
3. ϕ preserves ideals, i.e. if P is an ideal of R then $\phi(P)$ is an ideal of $\text{Im}(\phi)$.
4. The preimage of an ideal is an ideal of the domain, i.e. if B is an ideal of S , then $\phi^{-1}(B)$ is an ideal of R .
5. If R is commutative, then $\text{Im}(\phi)$ is commutative.
6. $\text{Ker}(\phi)$ is an ideal of R .
7. ϕ is a ring isomorphism if and only if ϕ is surjective and also $\text{Ker}(\phi) = \{0_R\}$.
8. If ϕ is a ring isomorphism then ϕ^{-1} is also a ring isomorphism.

Proof. Most proofs are analogous to the proof of the analogous proposition stated for groups, where ideals are replaced by normal subgroups.

1. $\phi(n \cdot r) = n\phi(r)$ by **20 group homomorphism commutes with group operation** (in additive notation).

By inspecting the proof we find that we have not used the inverse property of the group operation, so the same proof also holds for ring multiplication, so $\phi(r^n) = \phi(r)^n$.

2. A ring homomorphism is also a group homomorphism so $\text{Im}(\phi) \leq (S, +_S)$. It remains to show it is closed under multiplication, then by **Theorem 53 subring test** we will have $\text{Im}(\phi)$ is a subring of S . Let $y_1, y_2 \in \text{Im}(\phi)$, then $y_1 y_2 = \phi(x_1)\phi(x_2)$ for some $x_1, x_2 \in R$. Since ϕ is a ring homomorphism, then by definition $\phi(x_1 x_2) = \phi(x_1)\phi(x_2) = y_1 y_2$, so $y_1 y_2 \in \text{Im}(\phi)$, so it is a subring of R .

3. From group theory we know ϕ preserves normal subgroups. Suppose P is an ideal of R , then it is a normal subgroup of R under addition, then $\phi(P)$ is a normal subgroup of $\text{Im}(\phi)$ (which is a subring of S by (2)) under addition. We proceed by **Theorem 58 ideal test**, it remains to show that

$\forall x \in \phi(P) \forall y \in \text{Im}(\phi), xy, yx \in \phi(P)$. Since $x \in \phi(P), \exists a \in P$ s.t. $\phi(a) = x$, similarly $\exists b \in R$ s.t. $\phi(b) = y$, then $\phi(ab) = \phi(a)\phi(b) = xy$ and also $\phi(ba) = \phi(b)\phi(a) = yx$. Since P is an ideal, then both $ab, ba \in P$, so it follows that $xy, yx \in \phi(P)$, so it is an ideal of $\text{Im}(\phi)$.

4. Again from group theory we know $\phi^{-1}(B)$ is a subgroup of $(R, +_R)$. We proceed by **Theorem 58 ideal test**. Let $a \in \phi^{-1}(B), b \in S$, then $\phi(ab) = \phi(a)\phi(b)$ where $\phi(a) \in B$ and $\phi(b) \in S$, since B is an ideal of S then $\phi(a)\phi(b) \in B$, so $ab \in \phi^{-1}(B)$. Similarly, we find $ba \in \phi^{-1}(B)$. We conclude $\phi^{-1}(B)$ is an ideal of R .

5. Suppose R is commutative and let $x, y \in \text{Im}(\phi)$, then $\exists a, b \in R$ s.t. $\phi(a) = x, \phi(b) = y$. Since R is commutative, $ab = ba$, so $\phi(ab) = \phi(ba)$, so $\phi(a)\phi(b) = \phi(b)\phi(a)$, so $xy = yx$, so $\text{Im}(\phi)$ is commutative.

6. We already know $\text{Ker}(\phi)$ is a normal subgroup of R . To show it is an ideal, let $a \in \text{Ker}(\phi), b \in R$, then $\phi(ab) = \phi(a)\phi(b) = 0_S\phi(b) = 0_S$, so $ab \in \text{Ker}(\phi)$. Similarly, $\phi(ba) = \phi(b)0_S = 0_S$ so $ba \in \text{Ker}(\phi)$, so $\text{Ker}(\phi)$ is an ideal of R .

7. Since a ring isomorphism is also a group isomorphism and this proposition is exactly the same as a known result from group theory, no new argument is needed.

8. We know from group theory that if ϕ is a group isomorphism then ϕ^{-1} is a group isomorphism, so it remains to show ϕ^{-1} preserves multiplication and unity if the rings are unital. Let $x, y \in S$, then $\exists a, b \in S$ s.t. $\phi(a) = x, \phi(b) = y$, and also $\phi(ab) = xy$ since ϕ is a ring homomorphism, so $\phi^{-1}(xy) = ab = \phi^{-1}(x)\phi^{-1}(y)$, so ϕ^{-1} preserves multiplication. Suppose R, S are unital, then $\phi^{-1}(\phi(1_R)) = \phi^{-1}(1_S) = 1_R$. We conclude ϕ^{-1} is a ring isomorphism.

□

Proposition 54 (ring homomorphism and multiplicative inverse associate).

If $\phi : R \rightarrow S$ is a ring homomorphism and $r \in R$ is a unit, then $\phi(r^{-1}) = \phi(r)^{-1}$.

Proof. Suppose $r \in R$ is a unit, then $\phi(r^{-1}r) = \phi(r^{-1})\phi(r)$ and also $\phi(r^{-1}r) = \phi(1_R) = 1_S$ by definition of ring homomorphism, so $\phi(r^{-1})\phi(r) = 1_S$, so $\phi(r), \phi(r^{-1})$ belong to $U(S)$ the group of units of S , and $\phi(r^{-1}) = \phi(r)^{-1}$. □

Corollary 64.1 (ring homomorphism is a group homomorphism between group of unit).

If $\phi : R \rightarrow S$ is a ring homomorphism, and $U(R), U(S)$ are nonempty, then $\phi|_{U(R)}$ is a group homomorphism $U(R) \rightarrow U(S)$.

Proof. Suppose $r \in U(R)$ then $r^{-1} \in U(R)$, so $\phi(r^{-1}) = \phi(r)^{-1}$, so $\phi(r) \in U(S)$, so the domain and range of ϕ are correct. Since ϕ is a ring homomorphism it also satisfies all group homomorphism

properties, and since $U(R), U(S)$ are groups, then $\phi|_{U(R)}$ is a group homomorphism $U(R) \rightarrow U(S)$. $\square$

Proposition 55 (ring isomorphism preserves characteristic).

If $\phi : R \rightarrow S$ is a ring isomorphism, then $\text{char}(R) = \text{char}(S)$.

Proof. Suppose $\text{char}(R) = 0$ and $\text{char}(S) \neq 0$, then $\exists n \in \mathbb{N}$ s.t. $\forall s \in S, n \cdot s = 0_S$. Let $a \in R$ s.t. $\phi(a) = s \in \phi(R)$, then $n \cdot s = s + \dots + s = \phi(a) + \dots + \phi(a) = n \cdot \phi(a)$, and also $n \cdot \phi(a) = 0_S = \phi(0_R)$ by supposition. By **Theorem 64 properties of ring homomorphism**, $n \cdot \phi(a) = \phi(n \cdot a) = \phi(0_R)$. Since ϕ is an isomorphism, it is injective, so $n \cdot a = 0_R$ for every $a \in R$, but by supposition $\text{char}(R) = 0$, a contradiction, so it must be that $\text{char}(R) = 0$ implies $\text{char}(S) = 0$.

Suppose now that $\text{char}(R) = n$ and let $s \in S$. Since ϕ is surjective, $\exists r \in R$ s.t. $\phi(r) = s$, so $n \cdot \phi(r) = n \cdot s$, and also $n \cdot \phi(r) = \phi(n \cdot r) = \phi(0_R) = 0_S$, so $n \cdot s = 0_S$, so $\text{char}(S) \leq n$. Suppose $m = \text{char}(S)n$, then by the same argument applied to the ring isomorphism ϕ^{-1} we find that $\text{char}(R) \leq mn$, but by supposition $\text{char}(R) = n$, a contradiction, so we conclude $\text{char}(R) = \text{char}(S)$. $\square$

Proposition 56.

If there exists a ring homomorphism from R to S , and R, S are unital with finite characteristic, then $\text{char}(S) | \text{char}(R)$.

Proof. By **Theorem 56 characteristic of a ring with unity**, $|1_R| = \text{char}(R)$ and $|1_S| = \text{char}(S)$ since the characteristics are finite by supposition. By **35 order of element under homomorphism**, we have $\text{char}(S) = |1_S| |1_R| = \text{char}(R)$. $\square$

The first isomorphism theorem also holds for rings using the appropriate definitions:

Theorem 65 (first isomorphism theorem for rings).

If $\phi : R \rightarrow S$ is a ring homomorphism, then:

1. $\text{Ker}(\phi)$ is an ideal of R
2. $\phi(R)$ is an ideal of S
3. $R/\text{Ker}(\phi) \cong \phi(R)$ (via $\psi : r + \text{Ker}(\phi) \rightarrow \phi(r)$)

Proof. First, (1) and (2) were already proven in **Theorem 64 properties of ring homomorphism**. For (3), we know that ψ is well defined and a group isomorphism by the proof for **Theorem 26 first isomorphism theorem**, so it remains to show ψ is a ring homomorphism, then by **Theorem 64 properties of ring homomorphism** (specifically, proposition (7)) it will follow that ψ is a ring isomorphism.

1. (multiplication is preserved) let $r_1 + \text{Ker}(\phi), r_2 + \text{Ker}(\phi) \in R/\text{Ker}(\phi)$, then $(r_1 + \text{Ker}(\phi))(r_2 + \text{Ker}(\phi)) = r_1 r_2 + \text{Ker}(\phi)$. By definition, $\psi(r_1 r_2 + \text{Ker}(\phi)) = \phi(r_1 r_2)$. Since ϕ is a ring homomorphism, $\phi(r_1 r_2) = \phi(r_1)\phi(r_2)$. By definition, $\phi(r_1)\phi(r_2) = \psi(r_1 + \text{Ker}(\phi))\psi(r_2 + \text{Ker}(\phi))$, so ψ preserves multiplication.
2. (unity is preserved) Suppose R, S are unital, then $\psi(1_R + \text{Ker}(\phi)) = \phi(1_R) = 1_S$ by definition.

We conclude ψ is a ring homomorphism, and it is also bijective, so $R/\text{Ker}(\phi) \cong \phi(R)$.

□

We proceed with a short survey of examples of ring homomorphisms:

Example 65.1 (conjugation map).

Let $\phi : \mathbb{C} \rightarrow \mathbb{C}, c \mapsto c^*$ where c^* is the complex conjugate of c , i.e. if $c = a + bi$, then $c^* = a - bi$. The reader can verify via the definition of complex number multiplication and addition that ϕ indeed preserves multiplication and addition, and also since $1_{\mathbb{C}} = 1 + 0i = 1$, we have $\phi(1 + 0i) = 1 - 0i = 1 + 0i$ so ϕ preserves unity, so ϕ is a ring homomorphism. Furthermore, $(\phi \circ \phi)(a + bi) = \phi(a - bi) = a + bi$, so ϕ is its own inverse (i.e. complex conjugation is an involution), so it is a bijective homomorphism. so it is a ring isomorphism, so $\phi \in \text{Aut}(\mathbb{C})$ and also $|\phi| = 2$ (since $\phi^2 = \text{id}$ as we have shown).

Furthermore, note that $\phi(c) = c \iff \text{Im}(c) = 0$, so the real numbers can be characterized as the set of elements in $\mathbb{C}$ for which ϕ acts as the identity.

Example 65.2 (evaluation map).

Consider the evaluation map of a polynomial $\phi_a : R[x] \rightarrow R$ given by $\phi_a(f(x)) = f(a)$ for some $a \in R$, i.e. ϕ_a evaluates a polynomial over R given some value a (i.e. x in the polynomial expression is replaced with a , and then the expression is evaluated in R). Since $f(x) = \sum_{i=0}^n \alpha_i x^i$ where $\alpha_i \in R$, then $f(a) = \sum_{i=0}^n \alpha_i a^i \in R$ due to closure of a ring under addition and multiplication, so the codomain is well defined.

$R[x]$ has unity if and only if R has unity, and $1_{R[x]} = 1_R$. By definition, $\phi_a(1_{R[x]}) = 1_R$, so ϕ_a preserves unity. Furthermore, $\phi_a((f + g)(x)) = (f + g)(a) = f(a) + g(a) = \phi_a(f(x)) + \phi_a(g(x))$, so ϕ_a preserves multiplication (here we rely on previous knowledge of polynomials), similarly we find that ϕ_a preserves multiplication, so the evaluation map is a ring homomorphism.

In group theory, once we have presented the idea of a group isomorphism, we spent a great amount of time understanding the structure of groups in terms of well-studied subgroups which we identify (via an isomorphism, or a direct product, or both) in the groups. We begin to give rings the same treatment.

Theorem 66 (group action of integers on a unital ring defines a ring homomorphism).
If R is a unital ring with unity 1 , then $\phi : \mathbb{Z} \rightarrow R, n \mapsto n \cdot 1$ is a ring homomorphism.

Proof. First, we show ϕ preserves unity: observe $\phi(1) = 1 \cdot 1$ which by definition is 1 (of R). Next, we show ϕ preserves multiplication:

For $n = 0$, trivially $\forall m \in \mathbb{Z} \phi(nm) = \phi(0) = 0_R = 0_R \phi(m) = \phi(n)\phi(m)$, so ϕ preserves multiplication. Consider $mn \in \mathbb{Z}$.

For n positive, we defined $n \cdot 1$ as $\sum_{i=1}^n 1$ (where summation is via the addition of R), so for positive m , $mn \cdot 1 = \sum_{i=1}^{mn} 1$. By the associative property of addition and distributive property of multiplication over addition in R , we have $\sum_{i=1}^{mn} 1 = \sum_{i=1}^m \sum_{i=1}^n 1$. By **Theorem 54 ring arithmetic** $r = 1r$, so $\sum_{i=1}^m \sum_{i=1}^n 1 = \sum_{i=1}^m 1 \sum_{i=1}^n 1 = (m \cdot 1)(n \cdot 1)$, so $\phi(mn) = \phi(m)\phi(n)$ if $0m, n$ we have that ϕ preserves multiplication.

For n negative, we defined $n \cdot 1 = \sum_{i=1}^n -1$ which by ring arithmetic is equal to $-\sum_{i=1}^{|n|} 1$. Let m be negative, then mn is positive by properties of multiplication of integers (and also by **Theorem 54 ring arithmetic**), then by the same argument as before $\sum_{i=1}^{mn} 1 = \sum_{i=1}^{|m|} 1 \sum_{i=1}^{|n|} 1$. By ring arithmetic, we have $ab = (-a)(-b)$, so $\sum_{i=1}^{|m|} \sum_{i=1}^{|n|} 1 = (-\sum_{i=1}^{|m|} 1)(-\sum_{i=1}^{|n|} 1) = \phi(m)\phi(n)$, so if $m, n < 0$ we have that ϕ preserves multiplication.

Since $\mathbb{Z}$ is commutative, then $\phi(nm) = \phi(mn)$, so it suffices to show that for n negative and m positive, ϕ preserves multiplication. Suppose that is the case, then $mn < 0$ so by definition $\phi(mn) = \sum_{i=1}^{|mn|} -1$. By the same argument we used before, $\sum_{i=1}^{|mn|} -1 = \sum_{i=1}^{|m|} \sum_{i=1}^{|n|} -1 = \sum_{i=1}^m 1 \sum_{i=1}^{|n|} -1 = \phi(m)\phi(n)$.

We conclude ϕ preserves multiplication.

Next, we show ϕ preserves addition. Consider $m + n \in \mathbb{Z}$. If n is positive and m is positive, then $\phi(m + n) = (m + n) \cdot 1 = \sum_{i=1}^{m+n} 1 = \sum_{i=1}^m 1 + \sum_{i=1}^n 1 = \phi(m) + \phi(n)$ by associativity of addition in R . A similar argument holds for the case where m, n are both negative. If n is zero then $\phi(m + n) = \phi(m + 0) = \phi(m) = \phi(m) + 0 = \phi(m) + \phi(0) = \phi(m + n)$.

If m is positive and n is negative (since addition in $\mathbb{Z}$ commutes it suffices to check only this case and not also the case where m is negative and n is positive), then there are two cases:

1. $|m| \geq |n|$, then $\phi(m + n) = \sum_{i=1}^{m+n} 1 = \sum_{i=1}^{m-|n|} 1 + \sum_{i=1}^{|n|} 1 - 1 = \phi(m) + \phi(n)$.
2. $|m| < |n|$, then $\phi(m + n) = \sum_{i=1}^{m+n} -1 = \sum_{i=1}^{|n|-m} -1 + \sum_{i=1}^m -1 + \sum_{i=1}^{|n|} -1 = \phi(m) + \phi(n)$

We conclude that ϕ satisfies all the properties of a ring homomorphism.

□

Corollary 66.1.

If R is a ring with unity, then:

1. If $\text{char}(R) = n$, then R contains a subring isomorphic to $\mathbb{Z}_n$.
2. If $\text{char}(R) = 0$, then R contains a subring isomorphic to $\mathbb{Z}$.

Proof. 1. Suppose $\text{char}(R) = n$, then by **Theorem 56 characteristic of a ring with unity** it follows $|1| = n$. Consider the homomorphism described in **Theorem 66 group action of integers on a unital ring defines a ring homomorphism**: $\phi : \mathbb{Z} \rightarrow R, z \mapsto z \cdot 1$ defines a ring homomorphism. Let $z \in \text{Ker}(\phi)$, then $\phi(z) = 0$, and also $\phi(z) = z \cdot 1$. By the division algorithm, $z = qn + r$ for some $q, r \in \mathbb{Z}, r \in [0, n)$, so $\phi(z) = \phi(qn + r)$. By the theorem, ϕ is a ring homomorphism, so $\phi(qn + r) = \phi(q)\phi(n) + \phi(r) = (q \cdot 1)(n \cdot 1) + r \cdot 1$. By supposition $n \cdot 1 = 0$, so $\phi(z) = (q \cdot 1)0 + r \cdot 1 = r \cdot 1$. If r is a positive integer, and also by the division algorithm r less than n , then it cannot be that $r \cdot 1 = 0$ since otherwise $|1| \leq rn$, a contradiction, so it follows $r = 0$, so $z = qn$, so $\text{Ker}(\phi) = n\mathbb{Z}$.

By **Theorem 65 first isomorphism theorem for rings**, $\mathbb{Z}/\text{Ker}(\phi) = \mathbb{Z}/n\mathbb{Z} = \mathbb{Z}_n \cong \phi(\mathbb{Z})$. By **Theorem 64 properties of ring homomorphism**, $\phi(\mathbb{Z})$ is a subring of R , so a subring of R is isomorphic to $\mathbb{Z}_n$, which completes the proof.

2. Suppose $\text{char}(R) = 0$, then $|1| = \infty$ by **Theorem 56 characteristic of a ring with unity**, then $\phi(z) = 0$ has no solution except for the trivial solution $z = 0$, then $\text{Ker}(\phi) = 0$, then $\mathbb{Z}/\{0\} \cong \phi(\mathbb{Z})$ which is a subring of R . By an analogous theorem to **25.2** stated for rings, we have $\mathbb{Z}/\{0\} \cong \mathbb{Z}$, since isomorphism is an equivalence relation it follows $\mathbb{Z}$ is isomorphic to a subring of R .

□

Corollary 66.2.

The mapping $\phi : \mathbb{Z} \rightarrow \mathbb{Z}_n, z \mapsto [z]_n$ is a ring homomorphism.

Proof. Observe that in $\mathbb{Z}$, $z = z \cdot 1$, so $[z]_n = [z \cdot 1]_n = z \cdot [1]_n$, so it is simply a special case of the isomorphism from **Theorem 66 group action of integers on a unital ring defines a ring homomorphism**.

□

Corollary 66.3 (Steinitz, 1910).

If $\mathbb{F}$ is a field, then:

1. If $\text{char}(\mathbb{F}) = p$, then $\mathbb{F}$ contains a subfield isomorphic to $\mathbb{Z}_p$. (here, p is prime)
2. Otherwise (by **56.2 characteristic of integral domain**, the only other case is $\text{char}(\mathbb{F}) = 0$), $\mathbb{F}$ contains a subfield isomorphic to $\mathbb{Q}$, the rationals.

1. By **66.1**, $\text{char}(\mathbb{F}) = p$ implies that $\mathbb{F}$ contains a subring isomorphic to $\mathbb{Z}_p$. By **48 $\mathbb{Z}_p$ is a field**, it follows that the subring isomorphic to $\mathbb{Z}_p$ must also be a field, so it is a subfield.

2. Suppose $\text{char}(\mathbb{F}) = 0$, then by 66.1 $\mathbb{F}$ has a subring isomorphic to $\mathbb{Z}$ via an isomorphism $\phi : \mathbb{Z} \rightarrow \mathbb{F}$. Note that $\mathbb{Q} = \{ab^{-1} | a, b \in \mathbb{Z}, b \neq 0\}$. Since $\forall a, b \in \mathbb{Z}, \phi(a), \phi(b) \in \mathbb{F}$, we find that if $\phi(b) \neq 0$, then $\phi(a)\phi(b)^{-1} \in \mathbb{F}$, i.e. $Q = \{\phi(a)\phi(b)^{-1} | a, b \in \mathbb{Z}, b \neq 0\}$ is a subset of $\mathbb{F}$. Notice that Q is an isomorphic copy of $\mathbb{Q}$ (for a more formal argument, one can recall (as we will soon discuss at length) that $\mathbb{Q}$ is constructed from $\mathbb{Z}$ via equivalence relations and addition and multiplication which arise from the definitions of addition and multiplication in $\mathbb{Z}$, and following the same construction we can create a field from any integral domain. If two integral domains are isomorphic, then we will find that their isomorphism naturally extends to an isomorphism between these structures. Here we simply saw a special case of this, rather hand-wavely).

Definition 0.0.26 (prime subfield).

If $\mathbb{F}$ is a field and $\mathbb{P}$ is a subfield of $\mathbb{F}$ that is also contained in all other subfields of $\mathbb{F}$, we call $\mathbb{P}$ the prime subfield of $\mathbb{F}$.

Proposition 57 (prime subfield is unique).

The prime subfield of a field is unique.

Proof. Suppose not, then $\mathbb{P}, \mathbb{Q}$ are both prime subfields of $\mathbb{F}$, so by definition $\mathbb{P}$ and $\mathbb{Q}$ are contained in each other, so $\mathbb{P} = \mathbb{Q}$. □

Proposition 58 (corollary to Steinitz).

If $\mathbb{F}$ is a field, then:

1. *If $\text{char}(\mathbb{F}) = p$ for some prime p , then the prime subfield of $\mathbb{F}$ is isomorphic to $\mathbb{Z}_p$.*
2. *Otherwise, the prime subfield of $\mathbb{F}$ is isomorphic to $\mathbb{Z}$.*

Proof. Suppose $\mathbb{S}$ is a subfield of $\mathbb{F}$, then by 56.1 **unital subrings preserve characteristic** $\text{char}(\mathbb{S}) = \text{char}(\mathbb{F})$, so by 66.3 **Steinitz, 1910** if $\text{char}(\mathbb{F})$ is prime then $\mathbb{S}$ has a subring isomorphic to $\mathbb{Z}_p$, otherwise it has a subring isomorphic to $\mathbb{Z}$. Since the field characteristic is preserved under subfields, we can always find such subfields, so the one matching the field characteristic must be the prime subfield of $\mathbb{F}$. □

Field of Fractions

In [Introduction to Set Theory](#), we presented a formal construction of the rationals $\mathbb{Q}$ from the integers $\mathbb{Z}$. In our construction, we defined the rationals via the quotient of an equivalence relation on $\mathbb{Z} \times \mathbb{Z}^+$ (here we use $\mathbb{Z}^+$ to denote the positive subset of $\mathbb{Z}$):

Definition 0.0.27 (set theoretic rational number).

The rational numbers $\mathbb{Q}$ are given by $\mathbb{Q} = \mathbb{Z} \times \mathbb{Z}^+ / \sim$, where $\sim$ is the following equivalence relation: $(a, b) \sim (c, d) \iff ad = bc$, where ad, bc are multiplications in $\mathbb{Z}$.

The motivation behind this definition is that what we really want is to satisfy the known property of fractions, that $\frac{a}{b} = \frac{c}{d} \iff b, d \neq 0 \wedge ad = bc$. Equivalently, we could have simply specified that $\mathbb{Q}$ is the quotient of $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$.

In our construction, we have shown that $\sim$ is indeed an equivalence relation, so $\mathbb{Q}$ as a quotient group is well defined. We later defined *addition* and *multiplication* in $\mathbb{Q}$ as follows:

Definition 0.0.28 (rational number arithmetic).

We equip $\mathbb{Q}$ with the following operations:

1. $+$: $(a, b) + (c, d) = (ad + bc, bd)$ where $+$ in the RHS is addition in $\mathbb{Z}$ and juxtaposition is multiplication in $\mathbb{Z}$.
2. $\times$: $(a, b) \times (c, d) = (ac, bd)$ where juxtaposition in the RHS is multiplication in $\mathbb{Z}$.

We have shown that this operations are well defined, and further seen that $1_{\mathbb{Q}} = (1, 1)$ and $0_{\mathbb{Q}} = (0, 1)$. We have shown that $(\mathbb{Q}, +, \times)$ is a *field*, and shown that the mapping $\phi : \mathbb{Z} \rightarrow \mathbb{Q}, z \mapsto (z, 1)$ is the natural injective ring homomorphism between the two structures (since $\phi(ab) = (ab, 1) = (a, 1)(b, 1) = \phi(a)\phi(b)$, $\phi(a+b) = (a+b, 1) = (a, 1) + (b, 1) = \phi(a) + \phi(b)$ and finally $\phi(1) = (1, 1)$ so ϕ satisfies all properties of a ring homomorphism, injectivity follows from observing that $(a, 1) = (b, 1)$ in $\mathbb{Q}$ if and only if $a1 = 1b$, therefore if and only if $a = b$, so ϕ is injective).

The reader can verify that, in that construction and in the verification of all results relevant to showing that addition and multiplication on $\mathbb{Q}$ are well defined and that $\mathbb{Q}$ equipped with those operations is a field, we only used the properties of $\mathbb{Z}$ as an integral domain (i.e. a commutative ring with unity and no zero divisors). Using only these properties, we were able to construct a new structure $(\mathbb{Q}, +, \times)$ that is a *field* and also found a natural *ring homomorphism* from $\mathbb{Z}$ to this field. The field behaves the same way we expect fractions to behave, so we called its elements *fractions*.

All of this is to say, that the exact same process, with the exact same proofs, can be carried for every integral domain D . If the reader is not convinced, the reader should refer to the aforementioned construction and follow the same arguments for a general integral domain D instead of $\mathbb{Z}$, and consider $\mathbb{Q}$ as a quotient set of $D \times (D \setminus \{0_D\})$ under the equivalence relation defined the same way as in **0.0.27 set theoretic rational number**. Substituting the domain and range of ϕ with the appropriate once for this new construction, we find a ring homomorphism from D to the field constructed from D , which, since it acts in a similar manner to fractions, we can the *field of fractions* of the integral domain D .

Definition 0.0.29 (field of fractions).

If D is an integral domain, then the field F constructed as above is called the field of fractions of D . In this field, we denote the ordered pairs belonging to the equivalence classes used in the construction of the field by $(a, b) = \frac{a}{b}$, then we have that addition in the field is given by $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$ and multiplication is given by $\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$.

Theorem 67 (every integral domain is isomorphic to a subring of its field of fractions).

If D is an integral domain and F is its field of fractions, then F has a subring isomorphic to D .

Proof. As argued above, $\phi : D \rightarrow F, a \mapsto \frac{a}{1}$ is an injective ring homomorphism. By **Theorem 64 properties of ring homomorphism**, $\phi(D)$ is a subring of F . Since ϕ is injective, then $\phi : D \rightarrow \phi(D)$ is a bijection, so it is an isomorphism, so D is isomorphic to a subring of its field of fractions F . $\square$

Theorem 68 (fraction fields of isomorphic integral domains are isomorphic).

If D_1, D_2 are integral domains and F_1, F_2 are their respective fields of fraction, and $D_1 \cong D_2$, then $F_1 \cong F_2$.

Proof. Suppose D_1, D_2 are integral domain isomorphic via $\phi : D_1 \rightarrow D_2$. ϕ naturally gives rise to another mapping $\psi : F_1 \rightarrow F_2$ given by $\psi : (a, b) \mapsto (\phi(a), \phi(b))$. We will show that ψ is a ring homomorphism.

First, we will show ψ is well-defined. Suppose $(a, b) \sim (a', b')$, then by definition (**0.0.27 set theoretic rational number**) $ab' = ba'$, so $\phi(ab') = \phi(ba')$, and since ϕ is an isomorphism $\phi(ab') = \phi(a)\phi(b')$ and $\phi(ba') = \phi(b)\phi(a')$, which by definition implies $\psi(a, b) = (\phi(a), \phi(b)) \sim (\phi(a'), \phi(b')) = \psi(a', b')$, so ψ is well-defined.

Next, we show it is a homomorphism.

Let $(a, b), (c, d) \in F_1$, then $(a, b) + (c, d) = (ad + bc, bd)$. Observe that

$$\begin{aligned} \psi(ad + bc, bd) &= \\ (\phi(ad + bc), \phi(bd)) &= \\ (\phi(a)\phi(d) + \phi(b)\phi(c), \phi(b)\phi(d)) &= \\ (\phi(a), \phi(b)) + (\phi(c), \phi(d)) \end{aligned}$$

Where the second to last transition is justified since ϕ is a ring homomorphism, and the last transition is justified by the definition of addition in a field of fractions. For multiplication, by definition $(a, b)(c, d) = (ac, bd)$, so

$$\psi(ac, bd) = (\phi(ac), \phi(bd)) = (\phi(a)\phi(c), \phi(b), \phi(d)) = \phi(a, b)\phi(c, d)$$

So, ψ preserves addition. Finally, $\psi(1_{D_1}, 1_{D_1}) = (\phi(1_{D_1}), \phi(1_{D_1})) = (1_{D_2}, 1_{D_2})$ again since ϕ is a ring homomorphism.

So, we conclude, ψ is a field homomorphism. Let $(a, b) \in \text{Ker}(\psi)$, then $\psi(a, b) = 0$, so $(\phi(a), \phi(b)) = (0, r)$ for some $r \in D_2$, so $\phi(a) = 0$, so $a = 0_{D_1}$ since ϕ is an isomorphism so in particular it is injective, so ψ is injective. Next, suppose $(c, d) \in D_2$, then since ϕ is an isomorphism it is surjective and $\exists a, b \in D_1$ s.t. $\phi(a) = c, \phi(b) = d$, so $\psi(a, b) = (c, d)$ by definition, so ψ is surjective.

We conclude that $\psi : F_1 \rightarrow F_2$ is a bijective field homomorphism so it is a field isomorphism, so $F_1 \cong F_2$.

□

Note that the above theorem immediately gives a more satisfying proof for [66.3 Steinitz, 1910](#).

Polynomial Ring

Motivation

In analysis, we think of polynomials as function (potentially multivariable) whose rule is given explicitly (i.e., given a vector of variables X and a polynomial f , an explicit rule is given for calculating $f(X)$). Furthermore, for f to be a polynomial, the function rule has to be a mathematical expression consisting of indeterminates (which in analysis is synonymous with variables) and coefficients (which are taken from a field but more generally, from some commutative ring R), involving only addition (and subtraction), multiplication, and exponentiation to positive integer powers. Finally, we want f to have *finitely many nonzero terms*. The following definition somewhat formalizes this notion, but it is still informal:

Definition 0.0.30 (polynomial function (informal)).

A polynomial function is a function which can be described explicitly in the form of a mathematical expression consisting of indeterminates (variables) and coefficients, involving only basic arithmetical operations, that is - addition (and subtraction), multiplication, and exponentiation to positive integer powers, consisting of finitely many nonzero terms.

As a concrete example, consider $f(x) = x^2 + 5x - x^5$. The expression consists only of coefficients from $\mathbb{Z}$ (the nonzero coefficients are 1, 5, -1 and they are the coefficients of x^2, x, x^5 respectively), and consists of finitely many terms, so it is a polynomial.

On the other hand, consider a convergent power series, i.e. an expression of the form $f(x) = \sum_{n=0}^{\infty} a_n x^n$. Suppose all coefficients a_n are elements of a ring. Is $f(x)$ a polynomial function over the ring? No, because although it is given explicitly by an expression involving only addition, multiplication, and exponentiation of coefficients and a variable x , it involves infinitely many terms, and it can be that $\forall n \in \mathbb{N} \exists m \in \mathbb{N}$ s.t. mn and also $a_m \neq 0$, i.e. it potentially involves infinitely many nonzero terms. so by definition a power series, even if it is convergent, is not necessarily a polynomial. On the other hand, if from some $n \in \mathbb{N}$, all $a_m = 0$ for every mn , then it is a polynomial function.

As an example of the other extreme, consider $f(x) = 0$. It is a polynomial, since it is a power series that consists of finitely many nonzero terms (0 such terms, in fact). We call it the *zero polynomial*.

Consider $f(x) = 1 + \sin(x)$, is $f(x)$ a polynomial function over the reals? No, because although the Taylor expansion of $\sin(x)$ is convergent everywhere in $\mathbb{R}$, i.e. $\sin(x)$ can be expressed in terms of a power series, it is:

1. Pedantically, $1 + \sin(x)$ is not a power series unless we specifically allow substitution of $\sin(x)$ with its power series expansion in $\mathbb{R}$, and more importantly
2. The power series of $\sin(x)$ consists of infinitely many nonzero terms, so any expression involving $\sin(x)$ cannot express a polynomial function, as argued above.

Of course, polynomial functions need not always be considered in $\mathbb{R}$ or $\mathbb{C}$. Consider the following motivating example: consider $f(x) = x^3 - x$ in $\mathbb{Z}_3$, whose elements are $\{0, 1, 2\}$. Note that the nonzero coefficients of the polynomial are -1 and 1 , and $-1 \equiv 2 \pmod{3}$, so $f(x)$ is a polynomial function over $\mathbb{Z}_3$ (and also we can write $f(x) = x^3 + 2x$). Formally speaking, a function is a mapping such that every element in the domain maps to a single element in the codomain, so we can also describe $f(x)$ in terms of where it sends the elements of $\mathbb{Z}_3$. We find $f(0) = 0^3 - 0 = 0$, $f(1) = 1^3 - 1 = 0$, $f(2) = 2^3 - 2 = 8 - 2 = 6 \equiv_3 0$, so we find that $f(x) = 0$ in $\mathbb{Z}_3$, so we have that the polynomial $f(x) = x^3 - x$ and the zero polynomial, when considered as functions, are equivalent in $\mathbb{Z}_3$, so the two polynomial *functions* are the same, yet the two polynomial *expressions* are different, in the sense that different powers of x have different coefficients. This example motivates defining a *polynomial expression* more carefully.

Furthermore, consider the polynomial expression $2x^3 + 1$. The same expression can describe a real-valued polynomial, where x is a variable representing some real number, or a matrix polynomial where x is some matrix. Indeed, we can replace the indeterminates of a polynomial by any element on which we define *scalar multiplication* and *exponentiation by positive integers*. This means that in our formalism, we should not think of x as a variable that stands for some element in the commutative ring from which we take the coefficients, but instead think of x as some *constant* the we *substitute* when we *evaluate* the polynomial at some value (this is precisely what we have shown in [65.2 evaluation map](#) although at that point we haven't introduced the proper formalism for polynomial rings).

What follows is a formal construction of polynomial rings in the abstract algebra sense.

Construction

Our end-goal here is to provide a framework that, given a ring R and a polynomial expression $\sum_{i=0}^n a_i x^i$ where $a_i \in R$, identifies the polynomial with some finite sequence $(a_0, \dots, a_n)$, and equip these sequences with addition and multiplication s.t. these operations behave as we would expect addition and multiplication of polynomial *functions* to behave, i.e. we wish to build a *model* for polynomials within abstract algebra.

For the sake of discussion, we let R be a commutative ring, and we take our coefficients from R . In our construction, we will build the *polynomial ring* of R .

We start with a model of a power series which as we have seen in our introductory discussion, generalizes the notion of a polynomial expression.

Definition 0.0.31 (formal power series over a ring).

A formal power series of a commutative ring R is a sequence $\phi : \mathbb{N} \rightarrow R, n \mapsto a_n$, where each a_i is called a coefficient of the power series, and a_i is called the i^{th} coefficient of the power series.

We call these *formal power series* since we do not require that the power series *converge*. At first it may seem strange to think of a power series, which we are conditioned to think of as $\sum_{i=0}^{\infty} a_i x^i$, in terms of a sequence. We will make that connection later when we formally define what we mean by x , but for now consider that intuitively the expression is determined by its coefficient, so we don't lose any information when we represent a power series in terms of its coefficients.

Definition 0.0.32 (polynomial over a ring).

A polynomial over a ring R is a formal power series s.t. $\exists n \in \mathbb{N}$ s.t. $\forall m \in \mathbb{N} (mn \implies a_m = 0)$ where 0 is the additive identity of R . Equivalently, it is a formal power series such that there are finitely many nonzero coefficients.

Example 68.1.

Given $R = \mathbb{Z}_3$, the sequence $(1, 0, 2, 1, 0, \dots)$ is a polynomial, yet $(1, 0, 2, 1, 1, \dots)$ is not (where $\dots$ means that we repeat the previous element ad infinitum).

We introduce the familiar of a *degree*, and *leading* and *free* coefficients.

Definition 0.0.33 (zero polynomial).

The polynomial $(0, \dots)$ (i.e. the polynomial whose formal power series is $\phi : n \mapsto 0$) is called the zero polynomial.

Definition 0.0.34 (degree of a polynomial).

The degree of a polynomial f is the smallest $n \in \mathbb{N}$ after which, all coefficients are zero. If f is the zero polynomial, the degree of the polynomial is undefined. We denote the degree of a polynomial f by $\deg(f)$.

Example 68.2.

If $f = (c, \dots)$ for some nonzero $c \in R$, then we say that f is a constant polynomial, and by definition $\deg(f) = 0$. If $f = (0, 1, 0, 0, 0, 3, 1, 0, \dots)$, then by definition $\deg(f) = 6$.

Definition 0.0.35 (leading and free coefficients).

If f is a nonzero polynomial, then its free coefficient is a_0 (i.e. $\phi(0)$ as defined in 0.0.31 formal power series over a ring), and its leading coefficient is $a_{\deg(f)}$.

Definition 0.0.36 (monic polynomial).

If f is a nonzero polynomial over a unital ring R and its leading coefficient is the unity of R , then f is monic.

Since polynomials have finitely many nonzero coefficients, a simpler notation for f would be $f = (a_0, \dots, a_{\deg(f)})$ as opposed to $(a_0, \dots, a_{\deg(f)}, 0, \dots)$. In this notation, we can think of f as some finite sequence of coefficients.

Consider two polynomials over a ring R . We equip it with addition and multiplication as defined below:

Definition 0.0.37 (polynomial addition and multiplication).

If f, g are polynomials over R , and the coefficients of f are denoted a_i and those of g are denoted b_i , then:

1. $f + g = (a_0 + b_0, \dots)$ (i.e. we add the sequences component wise)
2. $f \cdot g = (s_0, \dots)$ where $s_i = \sum_{k=0}^i a_k b_{i-k}$ (readers may identify that this is the discrete convolution)

The reader should note that these definitions agree with polynomial multiplication as studied in high-school math.

Proposition 59.

The above operations are well-defined.

Proof. We need to show that $+$, $\cdot$ as defined in 0.0.37 polynomial addition and multiplication are indeed binary operators on the polynomials over R .

For addition, observe that since we perform addition component-wise, it is the case that $\deg(f + g) \leq \max\{\deg(f), \deg(g)\}$, so $f + g$ is clearly a polynomial over R (since it has finitely many nonzero terms).

For multiplication, we can see that $\deg(f \cdot g) \leq \deg(f) + \deg(g)$, so again $f \cdot g$ is a polynomial over R , which would complete the proof. Denote $\deg(f) = m$, $\deg(g) = n$, and denote $f = (f_0, \dots, f_m)$, $g = (g_0, \dots, g_n)$, then $\forall i \leq m+n$, $(fg)_i = \sum_{k=0}^i f_k g_{i-k}$, if $k > m$ then $f_k = 0$, so this sum is the same as $\sum_{k=0}^m f_k g_{i-k}$. Now, both $m+n$ and k are non-negative, so $i-k \leq n$, so $g_{i-k} = 0$, so $\forall i > m+n$, $(fg)_i = 0$, so $\deg(f \cdot g) \leq \deg(f) + \deg(g)$. $\square$

Since R is a ring, then addition on R commutes, so instead of writing $\sum_{k=0}^i a_k b_{i-k}$ we can rearrange the term any way we'd like, which motivates the following notation: instead of writing $s_i = \sum_{k=0}^i a_k b_{i-k}$, we write $s_i = \sum_{s+t=i} a_s b_t$, where $\sum_{s+t=i}$ is shorthand for a summation over all unique s, t s.t. $s+t=i$ and s, t are non-negative integers. We show the two are equivalent in the next proposition:

Proposition 60.

As defined above,

$$\sum_{k=0}^i a_k b_{i-k} = \sum_{s+t=i} a_s b_t$$

Proof. Consider $s+t=i$, equivalently we can write $t=i-s$, and by the definition from the paragraph above it is implied that $s, t \geq 0$, so we have

$$\sum_{s+t=i} a_s b_t = \sum_{s+(i-s)=i} a_s b_{i-s} = a_0 b_i + a_1 b_{i-1} + a_2 b_{i-2} + \dots + a_i b_0 = \sum_{k=0}^i a_k b_{i-k}$$

$\square$

Aside from being more compact, the above notation plays nicely with multiplication, as we will see later on.

Example 68.3.

Consider the polynomials $f = (1, -7, 3)$ and $g = (5, 9, -1)$ over $\mathbb{Z}$. We calculate $f + g$ and $f \cdot g$ from definition:

1. $f + g = (6, 2, 2)$
2. We calculate $f \cdot g$ by definition. By 59, we can stop once we have calculated the coefficient at index $\deg(f) + \deg(g) = 2 + 2 = 4$.

1. $(f \cdot g)_0 = f_0 g_0 = 1 \cdot 5 = 5$.
2. $(f \cdot g)_1 = f_0 g_1 + f_1 g_0 = 1 \cdot 9 + (-7) \cdot 5 = 9 - 35 = -26$
3. $(f \cdot g)_2 = f_0 g_2 + f_1 g_1 + f_2 g_0 = 1 \cdot (-1) + (-7) \cdot 9 + 3 \cdot 5 = -1 - 63 + 15 = -49$
4. $(f \cdot g)_3 = f_0 g_3 + f_1 g_2 + f_2 g_1 + f_3 g_0 = 0 + 7 + 27 + 0 = 34$
5. $(f \cdot g)_4 = f_0 g_4 + f_1 g_3 + f_2 g_2 + f_3 g_1 + f_4 g_0 = 3 \cdot (-1) = -3$

We conclude $f \cdot g = (5, -26, -49, 34, -3)$.

We will eventually want to show that f is a faithful representative of the polynomial expression whose coefficients are the elements of f , i.e. that f represents $1 - 7x + 3x^2$, and similarly g represents $5 + 9x - x^2$. Indeed, by carrying out the calculations as thought in school, we find $(1 - 7x + 3x^2) + (5 + 9x - x^2) = 6 + 2x + 2x^2$, and similarly $(1 - 7x + 3x^2)(5 + 9x - x^2) = 5 - 26x - 49x^2 + 34x^3 - 3x^4$.

Consider all polynomials over a ring R . Equipped with the operations defined above, it is the *polynomial ring* of R .

Definition 0.0.38 (polynomial ring).

If R is a commutative ring and $R[x]$ is the set of all polynomials over R as defined in 0.0.32 *polynomial over a ring*, then $(R[x], +, \cdot)$ (as defined in 0.0.37 *polynomial addition and multiplication*) is called the polynomial ring of R .

Theorem 69 (polynomial ring is a commutative ring).

$R[x]$ is a commutative ring over polynomial addition and multiplication.

Proof. Let R be a commutative ring. First, we show $(R[x], +)$ is an abelian group.

Let $f, g, h \in R[x]$ and let $n = \max\{\deg(f), \deg(g), \deg(h)\}$, and denote $f = (a_0, \dots, a_n)$, $g = (b_0, \dots, b_n)$ and $h = (c_0, \dots, c_n)$, then:

1. (associative) $(f + (g + h))_i = a_i + (b_i + c_i) \underset{R \text{ is a ring}}{=} (a_i + b_i) + c_i = ((f + g) + h)_i$
2. (identity) $(f + 0)_i = a_i + 0 = a_i = f$, where 0 is 0.0.33 *zero polynomial*.
3. (inverse) let $f' = (-a_0, \dots, -a_n)$, then $(f + f')_i = a_i - a_i = 0$, so $f + f' = 0$, so $f' = -f$.
4. (commutative) $(f + g)_i = a_i + b_i \underset{R \text{ is a ring}}{=} b_i + a_i = (g + f)_i$

We conclude $(R[x], +)$ is an abelian group.

Next, we show $R[x]$ associates and commutes under multiplication.

1. (associative) by 60 ,

$$\begin{aligned}
 (f \cdot (g \cdot h))_i &= \\
 \sum_{s+t=i} (f_s (g \cdot h)_t) &= \\
 \sum_{s+t=i} f_s \sum_{x+y=t} g_x h_y &\stackrel{\text{distributivity in } R}{=} \\
 \sum_{s+t=i} \sum_{x+y=t} f_s g_x h_y &\stackrel{\text{associativity in } R, \text{ counting}}{=} \\
 \sum_{s+x+y=i} f_s g_x h_y
 \end{aligned}$$

Similarly we show the same for $((f \cdot g) \cdot h)_i$, and conclude associativity.

1. (commutative) $(f \cdot g)_i = \sum_{k=0}^i a_k b_{i-k} = \sum_{k=0}^i b_{i-k} a_k$ from commutativity of R . Note that the latter expands to $b_i a_0 + b_{i-1} a_1 + b_{i-2} a_2 + \dots + b_0 a_i$. Since R is a ring then addition in R also commutes, so this expression is the same as $b_0 a_i + b_1 a_{i-1} + \dots + b_i a_0 = \sum_{k=0}^i b_k a_{i-k}$, which by definition is the i^{th} coefficient of $g \cdot f$, so multiplication commutes.

Finally, we show multiplication is distributive w.r.t addition:

1. (left distributive) $(fg + fh)_i = \sum_{k=0}^i a_k b_{i-k} + \sum_{k=0}^i a_k c_{i-k} = \sum_{k=0}^i a_k (b_{i-k} + c_{i-k}) = (f(g+h))_i$
2. (right distributive) Since $fg + fh = f(g+h)$ from left distributivity and multiplication in $R[x]$ commutes by the above, $f(g+h) = (g+h)f = fg + fh = gf + hf$, so we have right distributivity.

We conclude $R[x]$ is a commutative ring. It is worth noting that the same proof, with the omission of commutativity, shows that if R is a (not necessarily commutative) ring then $R[x]$, defined as in 0.0.38 polynomial ring, is a ring.

□

From now on, we also assume that R is also unital with unity 1.

Proposition 61 (unity of $R[x]$).

R is unital if and only if $R[x]$ is unital, furthermore $1_{R[x]} = (1_R, 0_R, \dots)$, i.e. it is a constant polynomial whose only coefficient is 1_R .

Proof. $\implies$ Suppose R is unital and consider $p = (1, 0, \dots) \in R[x]$. Let $f = (a_0, \dots, a_n) \in R[x]$. By definition $(p \cdot f)_i = \sum_{k=0}^i p_k a_{i-k}$, where $p_i = 1$ if $i = 0$, otherwise $p_i = 0$, then $\sum_{k=0}^i p_k a_{i-k} = p_0 a_{i-0} + 0 \sum_{k=1}^i p_k a_{i-k} = a_i$, so $p \cdot f = f$. Since $R[x]$ is commutative, $f \cdot p = f$ (we could have also shown this by calculating $f \cdot p$ again by definition), so p is the unity of $R[x]$.

$\Leftarrow$ Suppose $R[x]$ is unital with unity $(1, 0, \dots)$, and let $c \in R$. Consider $(c, 0, \dots) \in R[x]$, since $(1, 0, \dots)$ is the unity of $R[x]$ then $(1, 0, \dots) \cdot (c, 0, \dots) = (c, 0, \dots) \cdot (1, 0, \dots) = (c, 0, \dots)$, so we only need to calculate the first coefficient of the product polynomial. By definition, $((1, 0, \dots)(c, 0, \dots))_0 = 1 \cdot c$, so $1 \cdot c = 1$ and also $c \cdot 1 = 1$, so 1 is the unity of R . $\square$

Recall our original motivation - we sought a formal model which we can identify with polynomial expressions. Consider some polynomial $f \in R[x]$. Intuitively, we understand that $f = (1, 5, -3, 0, 4)$ corresponds to the univariate polynomial expression $1 + 5x - 3x^2 + 4x^4$.

One way to formalize this connection, is to *embed* R into $R[x]$ as follows: $\phi : R \rightarrow R[x], r \mapsto (r, 0)$ (one can show that this is a ring homomorphism trivially). Next, we introduce an *indeterminate* x into $R[x]$, with the following operations:

1. *multiplication with itself (exponentiation)*: we define $x^0 = 1_R$ and define $xx^n = x^n x = x^{n+1}$ for every integer n .

2. *multiplication with elements of R* : we define $r \cdot x^n$ as an $n + 1$ -tuple (representing a polynomial of degree $n + 1$) s.t. $(r \cdot x^n)_i = \begin{cases} r & i = n + 1 \\ 0 & \text{otherwise} \end{cases}$ for $r \neq 0$, and for 0 we define $r \cdot x^n = (0)$ the zero polynomial. Furthermore, we want $r \cdot x^n = x^n \cdot r$, in other words we want the operation to be commutative.

Once we have defined these operations, if we are willing to identify $r \in R$ with its corresponding image $\phi(r)$, then the following expression

$$\sum_{i=0}^n a_i x^i$$

is well-defined in $R[x]$ via the definitions provided above and **0.0.37 polynomial addition and multiplication**, and clearly

$$\sum_{i=0}^n a_i x^i = (a_0, 0) + (0, a_1) + (0, 0, a_2) + \dots (0, \dots, a_n) = (a_0, a_1, a_2, \dots, a_n)$$

So the sequence of coefficients faithfully represents the polynomial expression.

If R is a ring with unity, we can model such x via a concrete element of $R[x]$, as shown below:

Definition 0.0.39 (x in $R[x]$ over a unital R).

We define $x = (0, 1, 0, \dots)$ where 1 is the unity in R . We define exponentiation of x the same way we define exponentiation in any ring or group - as a shorthand for consecutive multiplications of the same element with itself.

Proposition 62 (x models an indeterminate).

x as defined above models an indeterminate, namely:

1. $x^2 = (0, 0, 1, 0, \dots)$
2. $\forall n \in \mathbb{N}, x^n = e_n$ where e_n is an $n+1$ -tuple s.t. $e_{n,i} = 0$ if $i \neq n$ and $e_{n,n} = 1$.
3. $(c, 0) \cdot x^n = \vec{v}$ where $\vec{v}_i = c$ if $i = n+1$, otherwise $\vec{v}_i = 0$.

Proof. 1. We know $\deg(x^2) \leq \deg(x) + \deg(x) = 2$, so it suffices to calculate $(x^2)_0, (x^2)_1, (x^2)_2$. By definition, $(x^2)_0 = x_0x_0 = 0$, and also $(x^2)_1 = x_0x_1 + x_1x_0 = 0 + 0 = 0$, finally $(x^2)_2 = x_0x_2 + x_1x_1 + x_2x_0 = 0 + 1 \cdot 1 + 0 = 1$, so $x^2 = (0, 0, 1)$.

2. We prove by induction. $x^0 = 1_{R[x]} = (1, \dots)$ by definition, for x the proposition holds also by definition of x , and we have shown the proposition holds for x^2 in (1). Suppose by induction the proposition holds up to some n , and consider x^{n+1} . By definition, $x^{n+1} = x^n x$. Consider $(x^n x)_i = \sum_{s+t=i} (x^n)_s x_t$. We know that $x_t \neq 0 \iff t = 1$, so $(x^n x)_0 = 0$ since it involves no terms with $t = 1$, and $\forall i > 0, (x^n x)_i = \sum_{s+1=i} (x^n)_s$. By the induction hypothesis, $(x^n)_s \neq 0 \iff s = n$, so $(x^n x)_i \neq 0 \iff n+1 = i$, in which case we have $(x^n x)_i = 1$, so $x^{n+1} = e_{n+1}$, which completes the proof by induction.

3. Denote $(c, 0) = \vec{c}$. By definition, $(\vec{c} \cdot x^n)_i = \sum_{k=0}^i \vec{c}_k (x^n)_{i-k}$. Since $\vec{c}_k$ is nonzero only for $k = 0$, in which case it is equal to c , we have that the sum is simply $c(x^n)_i$. From (2), we know $(x^n)_i$ is nonzero (and equal to 1) only for $i = n+1$, so we conclude $(\vec{c} \cdot x^n)_i$ is 0 if $i \neq n+1$, otherwise it is c . □

Once these results have been established, the correspondence between a polynomial expression and a polynomial sequence quickly follows.

Theorem 70 (polynomial sequence and polynomial expression correspondence).

Let R be a unital ring of coefficients and $f \in R[x]$ s.t. $f = (a_0, \dots, a_n)$, then $f = \sum_{i=0}^n a_i x^i$, where x^i defined as in 0.0.39 x in $R[x]$ over a unital R .

Proof. Follows trivially from 62 x models an indeterminate as argued informally a few paragraphs earlier. □

We now have a working formal model for polynomial expressions. Now that we got all the formalism out of the way, the reader should gain a newfound appreciation for 65.2 evaluation map - the evaluation map connects a polynomial expression to a polynomial function with variables from some domain.

We further find that if R is an integral domain, then so is $R[x]$:

Corollary 70.1 (polynomial ring of integral domain is an integral domain).

If R is an integral domain, then $R[x]$ is an integral domain.

Proof. Suppose R is an integral domain, then R is a commutative ring with unity and no zero-divisors. From **Theorem 69 polynomial ring is a commutative ring** and **61 unity of $R[x]$** we know that $R[x]$ is a commutative ring with unity, so it remains to show that $\forall f, g \in R[x], f \cdot g = 0$ implies $f = 0$ or $g = 0$, where 0 is the zero polynomial. If $f = 0$ then we are done, otherwise we proceed by induction on $\deg(f)$ to prove that $f \neq 0$ implies $g = 0$.

Since $f \cdot g = 0$ then $\forall i \in \mathbb{N}, (f \cdot g)_i = 0$. If $\deg(f) = 0$, then $f = (c, 0, \dots)$ for some $c \in R$, so $(f \cdot g)_i = cg_i$, so $cg_i = 0$ for every i . If $c = 0$, then $f = 0$, which we supposed is not the case, so $c \neq 0$, but then since R is an integral domain then it must be that $g_i = 0$. This holds for every g_i , so $g = 0$.

Suppose the proposition holds for all $\deg(f) \leq n \in \mathbb{N}$, and consider the case where $\deg(f) = n + 1$, then $f = (a_0, \dots, a_n, a_{n+1})$ where $a_i \in R, a_{n+1} \neq 0$. By definition of addition in $R[x]$, we have $f = (a_0, \dots, a_n) + (0, \dots, a_{n+1})$, so $fg = (a_0, \dots, a_n)g + (0, \dots, a_{n+1})g = 0 = (0, \dots, 0)$. Let $g = (b_1, \dots, b_m)$ where $m = \deg(g)$. Observe that, for $i \leq n + 1$ in particular, we have by definition $(f \cdot g)_i = a_{n+1}g_{i-(n+1)}$, which must be 0 since $(0)_i = 0$ (since all coefficients of the zero polynomial are zero). By the same argument as the previous case, it follows that either $a_{n+1} = 0$, or $g_{i-(n+1)} = 0$. So, there are two cases: either $a_{n+1} = 0$, or $g = 0$ i.e. g is the zero polynomial. If $g = 0$, then we are done, otherwise $g \neq 0$, so $a_{n+1} = 0$ and we have $fg = (a_0, \dots, a_n)g + (0, \dots, a_{n+1})g = (a_0, \dots, a_n)g = 0$, and now by the induction hypothesis since $g \neq 0$, $(a_0, \dots, a_n) = (0, \dots, 0)$, and since $a_{n+1} = 0$ as well then $f = 0$, but we supposed $f \neq 0$, a contradiction, so $g = 0$.

We conclude by induction that if $f \cdot g = 0$ and $f \neq 0$, then $g = 0$, so $R[x]$ is an integral domain. $\square$

Proposition 63 (degree equation for polynomial ring of integer domain).

If D is an integer domain, then $\forall f, g \in D[x], f, g \neq 0, \deg(fg) = \deg(f) + \deg(g)$.

Proof. Suppose D is an integral domain. By **59**, $\deg(fg) \leq \deg(f) + \deg(g)$. Denote $\deg(f) = n, \deg(g) = m$, and consider $i = n + m$, then by definition $(fg)_{n+m} = \sum_{k=0}^{n+m} f_k g_{n+m-k}$. w.l.o.g suppose nm (an integral domain is a commutative ring so $fg = gf$), then

$$\begin{aligned}
 \sum_{k=0}^{n+m} f_k g_{n+m-k} &\stackrel{nk \Rightarrow f_{k=0}}{=} \\
 \sum_{k=0}^n f_k g_{n+m-k} &= \\
 f_0 g_{n+m} + f_1 g_{n+m-1} + \cdots + f_n g_m &\stackrel{mi \Rightarrow g_i=0}{=} \\
 f_n g_m
 \end{aligned}$$

Suppose $f_n g_m = 0$, then since D is an integral domain either $f_n = 0$ or $g_m = 0$, but we know this is not the case since $\deg(f) = n$ means that $f_n \neq 0$, and similarly $g_m \neq 0$, so $f_n g_m \neq 0$, so $\deg(fg) = \deg(f) + \deg(g)$. □

It is worthwhile to note here that if D is an integral domain then the units of $D[x]$ arise naturally from $U(D)$ the units of D .

Corollary 70.2 (group of units of polynomial over integral domain).

If D is an integral domain, then $U(D[x]) = \phi(U(D))$ where ϕ is the natural homomorphism $\phi : D \rightarrow D[x], c \mapsto (c, 0)$.

Proof. Suppose $f(x) \in U(D[x])$, then $\exists g(x) \in U(D[x])$ s.t. $f(x)g(x) = 1$ and also both a nonzero polynomials so their degree is well-defined. By **63 degree equation for polynomial ring of integer domain**, $\deg(f(x)g(x)) = \deg(f(x)) + \deg(g(x))$. Since the degree of a polynomial is non-negative and their sum must be $\deg(1) = 0$, then $\deg(f(x)) = \deg(g(x)) = 0$, so $f(x) = (c, 0)$ for some $c \in D$, and $g(x) = (d, 0)$ for some $d \in D$. By definition of polynomial multiplication, $f(x)g(x) = (cd, 0)$, which by supposition is equal to the unity of $D[x]$, i.e. $cd = 1$, i.e. $c, d \in U(D)$. Conversely, if $c \in U(D)$ then $c^{-1} \in D$ and so $\phi(c)\phi(c^{-1}) = 1$, so $\phi(c) \in U(D[x])$. □

Proposition 64 (coefficient ring homomorphism induces polynomial ring homomorphism).

If $\phi : R \rightarrow S$ is a ring homomorphism, then $\psi : R[x] \rightarrow S[x], (r_0, \dots, r_n) \mapsto (\phi(r_0), \dots, \phi(r_n))$ is a homomorphism between the polynomial rings of the two rings. We call such ψ the natural polynomial homomorphism induced by the coefficient rings homomorphism.

Proof. Verify that ψ satisfies all ring homomorphism properties. Left as an exercise. □

Zeros and Division

Theorem 71 (polynomial division algorithm).

If F is a field, and $f(x), g(x) \in F[x]$, then $\exists! q(x), r(x) \in R[x]$ s.t. $\deg(r(x)) < \deg(g(x))$, and $f(x) = q(x)g(x) + r(x)$, where $q(x)$ is called the quotient of the division of $f(x)$ by $g(x)$, and $r(x)$ is called the remainder of the division.

Proof. Existence: We prove by induction on $\deg(f(x))$. If $\deg(f(x)) < \deg(g(x))$, then choosing $q(x) = 0$ and $r(x) = f(x)$ we have $f(x) = 0g(x) + f(x) = f(x)$. Consider the case where $\deg(f(x)) \geq \deg(g(x))$, and suppose the proposition holds up to some $n = \deg(f(x))$. Denote $f = (a_0, \dots, a_n)$ and $\deg(g(x)) = m$ and $g = (b_0, \dots, b_m)$. By definition of the degree of a polynomial, $a_n, b_m \neq 0$. Since F is a field, b_m has a multiplicative inverse in F and so $a_n b_m^{-1} \in F$, so $a_n b_m^{-1} x^{n-m} \in F[x]$ (since by supposition $m \leq n$, so $n - m \in \mathbb{N}$). Now, consider $a_n b_m^{-1} x^{n-m} g(x)$, by definition of polynomial multiplication we have for $i = n$:

$$\begin{aligned} (a_n b_m^{-1} x^{n-m} g(x))_n &= \\ \sum_{k=0}^n (a_n b_m^{-1} x^{n-m})_k b_{n-k} &= \\ a_n b_m^{-1} b_{n-(n-m)} &= \\ a_n b_m^{-1} b_m &= \\ a_n \end{aligned}$$

Consider $f'(x) = f(x) - a_n b_m^{-1} x^{n-m} g(x)$. Also by definition, $f'(x)_n = a_n - a_n = 0$, and by **63 degree equation for polynomial ring of integer domain** $\deg(a_n b_m^{-1} x^{n-m} g(x)) = (n - m) + \deg(g(x)) = n - m + m = n$, so it follows that $\deg(f'(x)) \leq n - 1$ since $f'(x)_i = 0$ for every $n \leq i$, so by the induction hypothesis $\exists q(x), r(x)$ s.t. $f'(x) = q(x)g(x) + r(x)$, so

$$\begin{aligned} f(x) - a_n b_m^{-1} x^{n-m} g(x) &= f'(x) = q(x)g(x) + r(x) \\ f(x) &= (q(x) + a_n b_m^{-1} x^{n-m})g(x) + r(x) \end{aligned}$$

Note that $r(x)$ is the same remainder term as before so it satisfies $\deg(r(x)) < \deg(g(x))$, and the expression is of the form $f(x) = q'(x)g(x) + r(x)$. This completes the existence part of the proof via induction.

Uniqueness: We prove uniqueness similarly to the proof of Euclidean division algorithm for integers. Let $q(x), p(x), r(x), s(x) \in R[x]$ s.t. $\deg(r(x)) < \deg(g(x))$ and also $\deg(s(x)) < \deg(g(x))$, and $f(x) = q(x)g(x) + r(x)$ and also $f(x) = p(x)g(x) + s(x)$, then

$$\begin{aligned} q(x)g(x) + r(x) &= p(x)g(x) + s(x) \\ (q(x) - p(x))g(x) &= s(x) - r(x) \end{aligned}$$

If $q(x) = p(x)$, then $q(x) - p(x) = 0$, so the above becomes $0 = s(x) - r(x)$, therefore $s(x) = r(x)$, and we are done. Otherwise, $q(x) \neq p(x)$, then by **63 degree equation for polynomial ring of integer domain** we have $\deg(q(x) - p(x))g(x) = \deg(q(x) - p(x)) + \deg(g(x)) \geq \deg(g(x))$, so $\deg(g(x)) \leq \deg(s(x) - r(x))$, but as we have argued in **59** $\deg(s(x) - r(x))$ is at most $\max\{\deg(s(x)), \deg(r(x))\}$ which as we have argued above must be less than $\deg(g(x))$, a contradiction, so it must be that $q(x) = p(x)$, and then also $s(x) = r(x)$, so the quotient and remainder of the division are unique. $\square$

Next, we use the fact that polynomial evaluation is a ring homomorphism to prove a useful corollary to the theorem:

Corollary 71.1.

*If F is a field and $a \in F$, $f(x) \in F[x]$, then $f(a)$ (i.e. f evaluated at $x = a$ as per **65.2 evaluation map**), then written as a polynomial via the natural homomorphism) is the remainder of the division of $f(x)$ by $x - a$.*

Proof. By the division algorithm, $\exists! q(x), r(x) \in F[x]$ s.t. $f(x) = q(x)(x - a) + r(x)$. Now, $f(a) = \phi_a(f(x)) = \phi_a(q(x)(x - a) + r(x))$. Since the evaluation map is a ring homomorphism, $\phi_a(q(x)(x - a) + r(x)) = q(a)(a - a) + r(a) = q(a)0 + r(a) = r(a)$, so $f(a) = r(a)$.

Now, since $\deg(x - a) = 1$, then either $r(x) = 0$, or $\deg(r(x)) = 0$ (since it must be less than $\deg(x - a)$ by definition). By the embedding of F in $F[x]$, it follows that $r(x)$ is given by some constant $r \in R$, equivalently $\forall a, b \in R (r(a) = r(b))$. As we found above, $r(a) = f(a)$, so $r(x) = f(a)$, so $f(a)$ is indeed the remainder of the division. $\square$

Definition 0.0.40 (zero of a polynomial).

If $f(x) \in R[x]$ and for some $a \in R$, $f(a) = 0$, we say that a is a zero of the polynomial $f(x)$.

Proposition 65.

The following are equivalent:

1. a is a zero of $f(x)$
2. $x - a$ is a factor of $f(x)$, where by factor we mean a polynomial that divides $f(x)$ s.t. the remainder term is zero.

Proof. $1 \implies 2$: By **71.1** we have that $f(x) = q(x)(x - a) + f(a)$. By **0.0.40 zero of a polynomial**, $f(a) = 0$, so $f(x) = q(x)(x - a)$.

$2 \implies 1$: Suppose $f(x) = q(x)(x - a)$, via evaluation $f(a) = q(a)(a - a) = 0$, so a is a zero of $f(x)$ by definition.

□

This equivalence lets us define the *multiplicity of a zero*:

Definition 0.0.41 (multiplicity of a zero).

If $f(x) \in R[x]$ and a is a zero of $f(x)$, then the multiplicity of a is the least $n \in \mathbb{N}$ s.t. the polynomial $(x - a)^{n+1}$ is not a factor of $f(x)$.

When we say that a polynomial has n zeros, we mean that we are also counting multiplicities. For example, $f(x) = (x - 2)^2$ in $\mathbb{R}[x]$ has *two* zeros, where 2 is counted twice since it has multiplicity 2.

Proposition 66 (a polynomial of degree n has at most n zeros).

If F is a field and $f(x) \in F(x)$, then $f(x)$ has at most $\deg(f(x))$ zeros, counting multiplicity.

Proof. We prove by contradiction. Let $\deg(f(x)) = n$. Suppose $f(x)$ has more than n zeros, then each i^{th} zero generates some factor (not necessarily distinct) $x - a_i$, so $\prod_{i=1}^n (x - a_i) | f(x)$. By **63 degree equation for polynomial ring of integer domain**, $\deg(\prod_{i=1}^n (x - a_i)) = \sum_{i=1}^n 1 = n$.

Now, suppose $f(x)$ has more than n zeros, so there must be at least another factor $x - a_{n+1}$, so $\prod_{i=1}^{n+1} (x - a_{n+1}) | f(x)$, but then the product of these polynomials has degree $n + 1$ which is greater than the degree of $f(x)$, but $f(x) = q(x) \prod_{i=1}^{n+1} (x - a_{n+1})$ for some nonzero $q(x)$, so $\deg(f(x))$ must be at least $n + 1$, a contradiction, so $f(x)$ has at most n zeros.

□

Recall that a PID (**0.0.18 principal ideal domain**) is an integral domain where all ideals are principal. The next theorem draws a connection between F being a field and its polynomial ring $F[x]$ being a PID.

Theorem 72 (polynomial ring over a field is a PID).

If F is a field, then $F[x]$ is a PID.

Proof. Suppose F is a field, then F is an integral domain, so by **70.1 polynomial ring of integral domain is an integral domain** we know that $F[x]$ is an integral domain. It remains to show that all ideals of $F[x]$ are principal.

Let $I \in F$ be an ideal of F . If $I = \{0\}$, then $I = \langle 0 \rangle$ by **Theorem 57**. Otherwise, suppose $I \neq \{0\}$, and consider the set $I \setminus \{0\}$ of all nonzero polynomials in I . Since by definition the degree of a nonzero polynomial is a natural number, by the well-ordering principle $I \setminus \{0\}$ has a polynomial of least degree (not necessarily unique), denote this polynomial $g(x)$. Let $f(x) \in I$ and suppose $f(x) \neq 0$, then by

Theorem 71 polynomial division algorithm $f(x) = q(x)g(x) + r(x)$ for unique $q(x), r(x) \in F[x]$, where $\deg(r(x)) < \deg(g(x))$ or $r(x) = 0$.

By supposition $f(x) \in I$ and also $g(x) \in I$. Since $g(x) \in I$ and I is an ideal, $q(x)g(x) \in I$. Since $q(x)g(x), f(x) \in I$ then from closure under subtraction, $f(x) - q(x)g(x) \in I$, but $f(x) - q(x)g(x) = r(x)$. If $r(x)$ is nonzero then we have found a polynomial in $r(x)$ with degree less than $g(x)$, violating the minimality of the degree of $g(x)$, so it must be that $r(x) = 0$, so $f(x) = q(x)g(x)$, so $f(x) \in \langle g(x) \rangle$ by definition. Since this is true for every $f(x) \in I$, and since trivially $\langle g(x) \rangle \subseteq I$, then $\langle g(x) \rangle = I$, so $F[x]$ is a PID. □

Corollary 72.1.

If F is a field and I is a nonzero ideal of $F[x]$, and $g(x) \in I$, then $I = \langle g(x) \rangle$ if and only if $g(x)$ is a nonzero polynomial of least degree in I .

Proof. The above proof is a constructive proof wherein we showed exactly this. □

Corollary 72.2 (polynomial field generator).

If F is a field and $p(x) \in F[x]$, then $\langle p(x) \rangle = F[x]$ if and only if $\deg(p(x)) = 0$.

Proof. Suppose F is a field and $p(x) \in F[x]$.

$\implies$ Suppose $\langle p(x) \rangle = F[x]$, then $\forall f(x) \in F[x] \exists h(x) \in F[x]$ s.t. $f(x) = p(x)h(x)$. Suppose $f(x)$ is constant nonzero, then $\deg(f(x)) = 0$ and also $p(x), h(x) \neq 0$ (if $p(x) = 0$ then $\langle p(x) \rangle$ is the trivial zero ideal so it cannot be $F[x]$, if $h(x) = 0$ then $p(x)h(x) = 0$ but we suppose $f(x) \neq 0$). By the degree equation, $\deg(f(x)) = \deg(p(x)) + \deg(h(x))$, where both operands on the RHS are non-negative and the LHS is 0 by supposition, so it must be that $\deg(p(x)) = 0$ and $\deg(h(x)) = 0$, so in particular $\deg(p(x)) = 0$.

$\impliedby$ Suppose $\deg(p(x)) = 0$ (which also implies $p(x) \neq 0$), then $p(x) = c$ for some nonzero constant $c \in F$. Let $f(x) \in F[x]$, since F is a field and $c \neq 0$ then $c^{-1} \in F$, so $h(x) = c^{-1} \in F[x]$, and from closure $h(x)f(x) \in F[x]$. Consider $p(x)(h(x)f(x)) \in \langle p(x) \rangle$, by commutativity of multiplication in $F[x]$ we have $p(x)(h(x)f(x)) = (p(x)h(x))f(x) = (cc^{-1})f(x) = f(x)$, so $f(x) \in \langle p(x) \rangle$, so $\langle p(x) \rangle = F[x]$. □

We now get back to our example of $\mathbb{R}[x]/\langle x^2 + 1 \rangle$. We saw, in 61.1 identifying the complex numbers in factor ring of polynomials, that the structure of this quotient ring seems suspiciously similar to that of $\mathbb{C}$. We later saw that the principle ideal generated by $x^2 + 1$ is maximal in 61.3 $\langle x^2 + 1 \rangle$ is a maximal

ideal of $\mathbb{R}[x]$, which by **Theorem 63** makes $\mathbb{R}[x]/\langle x^2 + 1 \rangle$ maximal. Now we will show formally that this structure is indeed isomorphic to $\mathbb{C}$.

Example 72.1 ($\mathbb{R}[x]/\langle x^2 + 1 \rangle \cong \mathbb{C}$).

Consider $\phi : \mathbb{R}[x] \rightarrow \mathbb{C}$, $f(x) \mapsto f(i)$, then:

1. $\phi(1) = 1$
2. $\phi(f(x)) \cdot \phi(g(x)) = f(i)g(i) = (f \circ g)(i) = \phi((f \circ g)(x))$
3. $\phi(f(x)) + \phi(g(x)) = f(i) + g(i) = (f + g)(i) = \phi((f + g)(x))$

So ϕ is a ring homomorphism, and also $\phi(f(x)) = 0 \iff f(i) = 0$. Take such nonzero $f(x) \in \text{Ker}(\phi)$. If $\deg(f(x)) = 0$, then $f(x) = c$ for some nonzero $c \in \mathbb{R}$, and clearly $c \neq 0$, so there is no such $f(x)$ with degree 0. If $\deg(f(x)) = 1$, then $f(x) = mx + b$ for some $m, b \in \mathbb{R}$, then $f(i) = mi + b$, so $f(i) = 0 \therefore b = -mi$, but $-mi$ is purely imaginary whereas b is real, so this is also impossible, so $\deg(f(x))$ is at least 2.

Consider $f(x) = x^2 + 1$, since $i^2 = -1$ (this is the defining property of i in $\mathbb{C}$) then $f(i) = i^2 + 1 = -1 + 1 = 0$, so $f(i) = 0$, so $x^2 + 1 \in \text{Ker}(\phi)$. By **Theorem 64 properties of ring homomorphism**, $\text{Ker}(\phi)$ is an ideal of $\mathbb{R}[x]$, so by

72.1 $\langle x^2 + 1 \rangle = \text{Ker}(\phi)$.

By **Theorem 65 first isomorphism theorem for rings**, $\mathbb{R}[x]/\langle x^2 + 1 \rangle \cong \text{Im}(\phi)$. Let $a + bi \in \mathbb{C}$, then given $f(x) = a + bx$, we have $\phi(f(x)) = f(i) = a + bi$, so ϕ is surjective, so its image is $\mathbb{C}$, so we have that $\mathbb{R}[x]/\langle x^2 + 1 \rangle \cong \mathbb{C}$.

Polynomial Factorization

From high school algebra we have some basic informal understanding of polynomial factorization. For example, we know that $f(x) = x^2 - 4$ can be written as $f(x) = (x - 2)(x + 2)$, or that $h(x) = 5x^3 + 15$ can be written as $h(x) = 5(x^3 + 3)$. In both cases, we took a polynomial and expressed it as the product of two other polynomials, which we call *factors*. However, context is important - consider $g(x) = x^2 - 1$. We know that in $\mathbb{C}$, $g(x) = (x - i)(x + i)$, however if we consider the polynomial in $\mathbb{R}[x]$ then we have no such coefficient $i \in \mathbb{R}$, so $g(x)$ in $\mathbb{R}[x]$ cannot be factorized further. In this section, we discuss how we factor polynomials (mostly using the term *reduction* instead of *factorization*).

Definition 0.0.42 (irreducible polynomial).

If:

1. D is an integral domain
2. $f(x) \in D[x]$ is nonzero polynomial and not a unit in $D[x]$, and
3. $\forall g(x), h(x) \in D[x], g(x), h(x) \neq 1$, if $f(x) = g(x)h(x)$ then one of $g(x), h(x)$ is a unit in $D[x]$,

then we say that $f(x)$ is irreducible over D . Otherwise, we say $f(x)$ is reducible over D .

The idea we attempt to capture in the above definition is that, a polynomial is *reducible* if it has a non-trivial factorization. Of course, *non-trivial* is relatively ill-defined, hence the definition.

Proposition 67 (zero implies reducible).

If D is an integral domain and $f(x) \in D[x]$ a polynomial of degree at least 2 has a zero in D , then $f(x)$ is reducible in D .

Proof. Suppose $f(x) \in D[x]$ has a zero $a \in D$, then for $g(x) = x - a$, $\exists h(x)$ s.t. $g(x)h(x) = f(x)$ per 65. By 70.2 group of units of polynomial over integral domain, $g(x)$ cannot be a unit of $D[x]$ since it has nonzero degree. By 63 degree equation for polynomial ring of integer domain, $\deg(f(x)) = \deg(g(x)) + \deg(h(x)) = 1 + \deg(h(x))$. By supposition, $\deg(f(x))$ is at least 2, so it follows that $\deg(h(x))$ is at least 1, so again by 70.2 group of units of polynomial over integral domain, $h(x) \notin U(D[x])$, so we have found some $g(x), h(x) \in D[x]$ s.t. $g(x), h(x) \notin U(D[x])$ and also $f(x) = g(x)h(x)$, so by 0.0.42 irreducible polynomial we say that $f(x)$ is reducible in D . $\square$

Note that the converse is not always true, as illustrated in the next example:

Example 72.2.

Consider $2x^2 + 2 \in \mathbb{Z}$, clearly $2 \notin U(\mathbb{Z})$ so $2 \notin U(\mathbb{Z}[x])$ by 70.2 group of units of polynomial over integral domain, and also $x^2 + 1$ has degree 2 so it too is not a unit of $\mathbb{Z}[x]$, and also $2x^2 + 2 = 2(x^2 + 1)$, so $2x^2 + 2$ is reducible in $\mathbb{Z}$, however $x^2 + 1$ has no zero in $\mathbb{Z}[x]$ (since we know that the roots of polynomial are in fact $\pm i$), so the converse of 67 zero implies reducible is not true.

It is worth noting that $2x^2 + 2$, while reducible in $\mathbb{Z}$, is not reducible in $\mathbb{Q}$, since $\mathbb{Q}$ is a field and so $2 \in U(\mathbb{Q})$ so $2 \in U(\mathbb{Q}[x])$, so $2(x^2 + 1)$ no longer demonstrated that $2x^2 + 2$ is reducible since 2 is a unit in $\mathbb{Q}[x]$. In this case, we have not disproven the converse of the previous proposition.

The second part of the example hints at the possibility that, at least in some cases, the converse of **67 zero implies reducible** does hold for fields (the main difference between $\mathbb{Z}$ and $\mathbb{Q}$ is that $\mathbb{Q}$ is a field whereas $\mathbb{Z}$ is an integral domain). The next theorem proves that this is indeed the case for polynomials of degrees 2 and 3:

Theorem 73.

If F is a field and $f(x) \in F[x]$ is a polynomial of degree 2 or 3, then $f(x)$ is reducible in F if and only if it has a zero in F .

Proof. Let F be a field and $f(x) \in F[x]$ be a polynomial of degree 2 or 3.

$\implies$ Suppose $f(x)$ is reducible in F , so $\exists g(x), h(x) \in F[x]$ non-units and nonzero so they have degree at least 1 (by **70.2 group of units of polynomial over integral domain**) s.t. $f(x) = g(x)h(x)$, so $\deg(f(x)) = \deg(g(x)) + \deg(h(x))$.

There are two cases: $\deg(f(x)) = 2$, in which case the degrees of both $g(x)$ and $h(x)$ are 1, or $\deg(f(x)) = 3$, in which case the degree of one of the polynomials factors is 1 while the other is 2, so in both cases one of the polynomials has degree 1. Suppose w.l.o.g that this is $g(x)$. Since $g(x)$ has degree 1 it is of the form $ax + b$ for some $a, b \in F$. Since F is a field, then $-ba^{-1} \in F$, and clearly $g(-ba^{-1}) = 0$, so $g(x)$ has a zero in F , and since $g(x)$ is a factor of $f(x)$ then $-ba^{-1}$ is also a zero of $f(x)$, so $f(x)$ has a zero in F .

$\Leftarrow$ Demonstrated in **67 zero implies reducible**.

□

Note that the theorem fails for degrees greater than 3. As an example, consider a polynomial of degree 4, $x^4 + 4x^2 + 4 = (x^2 + 2)^2$ in $\mathbb{Q}$. Clearly it is reducible in $\mathbb{Q}$ since it has $x^2 + 2 \in \mathbb{R}[x]$ as a factor, however $(x^2 + 2)^2$ has no zeros in $\mathbb{Q}$.

Content of a Polynomial and Primitive Polynomials

In this section, we initially limit our discussion to $\mathbb{Z}[x]$.

To motivate our next definition, consider $f(x) = 2x^2 + 6x + 12$. We can factor out 2 and get $f(x) = 2(x^2 + 3x + 6)$. Suppose we wish to write $f(x) = nh(x)$ for some $n \in \mathbb{Z}$ (here we finally stop the ceremonious mentioning that formally, this n is actually an element of $\mathbb{Z}[x]$ induced via the natural homomorphism $\mathbb{Z} \rightarrow \mathbb{Z}[x]$, and refer to n simply as an element of $\mathbb{Z}$), $h(x) \in \mathbb{Z}[x]$. If $h(x) \in \mathbb{Z}[x]$ then all its coefficients must also be integers, so any such n must be a common divisor of all coefficients of $f(x)$ (because $f_i = (nh(x))_i = nh_i$, so $n|f_i$ must). From number theory, we know that such maximal n exists, and it is the greatest common divisor of all coefficients of $f(x)$. Since $\gcd(2, 6, 12) = 2$,

then the factor we produced is maximal. In the context of polynomials with integer coefficients, we call the gcd of these coefficients the *content* of the polynomial, and we call polynomials for which $\gcd(f_0, \dots, f_n) = 1$ *primitive polynomials*.

Definition 0.0.43 (content of a polynomial).

If $f(x) \in \mathbb{Z}[x]$ and $f(x) = \sum_{i=0}^n a_i x^i$, then the greatest common divisor of its coefficients $\gcd(a_0, \dots, a_n)$ is called the content of $f(x)$. If the content of $f(x)$ is 1, then we say that $f(x)$ is primitive.

Clearly, every polynomial where one of the nonzero coefficients is 1 is primitive, so in particular monic polynomials (in $\mathbb{Z}[x]$) are primitive. The content captures the natural factorization of $f(x)$ in $\mathbb{Z}[x]$ (since the gcd of all coefficients can trivially be factored out), which then produces a primitive polynomial, as shown in the next proposition:

Proposition 68 (primitive representative).

If $f(x) \in \mathbb{Z}[x]$ is non-zero and c is its content and $c \neq 1$, then $f(x) = ch(x)$ where $h(x) \in \mathbb{Z}[x]$ is primitive, and $h(x)$ is unique. We call this $h(x)$ the primitive representative of $f(x)$.

Proof. By definition c is a divisor of all coefficients of $f(x)$ so clearly the constant polynomial $g(x) = c$ divides $f(x)$, so by **Theorem 71 polynomial division algorithm**, there exists a unique $h(x) \in \mathbb{Z}[x]$ s.t. $f(x) = ch(x)$. $\square$

Theorem 74 (Gauss' lemma (primitivity)).

If $f(x), g(x) \in \mathbb{Z}[x]$ are primitive, then their product $f(x)g(x)$ is primitive.

Proof. Suppose by contradiction $f(x), g(x) \in \mathbb{Z}[x]$ are primitive while $f(x)g(x)$ is not. Since $f(x)g(x)$ is non-primitive, then its content is not 1 and hence divisible by some prime p , so $f(x)g(x) = p \cdot h(x)$ for some $h(x) \in \mathbb{Z}[x]$.

Via the natural homomorphism $\phi : \mathbb{Z} \rightarrow \mathbb{Z}_p, z \mapsto [z]_p$, we induce a ring homomorphism $\psi : \mathbb{Z}[x] \rightarrow \mathbb{Z}_p[x]$ (see **64 coefficient ring homomorphism induces polynomial ring homomorphism**). Since ψ is a ring homomorphism, we have:

$$\psi(p \cdot h(x)) = \psi(p)\psi(h(x)) \underset{p=0}{=} \underset{\text{mod } p}{=} 0$$

Also, since $f(x)g(x) = p \cdot h(x)$:

$$0 = \psi(p \cdot h(x)) = \psi(f(x)g(x)) = \psi(f(x)) \cdot \psi(g(x))$$

Since $\mathbb{Z}_p$ is an integral domain, then $\mathbb{Z}_p[x]$ is an integral domain by **70.1 polynomial ring of integral domain is an integral domain**, so the above implies that either $\psi(f(x)) = 0$ or $\psi(g(x)) = 0$, but this is true if and only if p is a factor of the content of either $f(x)$, $g(x)$, but by supposition both contents are 1, so this is impossible, a contradiction, so $f(x)g(x)$ must be primitive. $\square$

Theorem 75 (Gauss' lemma (reducibility)).

If a non-constant polynomial $f(x) \in \mathbb{Z}[x]$ is reducible in $\mathbb{Q}$, then it is reducible in $\mathbb{Z}$.

Proof. Let $f(x)$ considered in $\mathbb{Q}[x]$, and suppose it is reducible, so $\exists \overline{g(x)}, \overline{h(x)} \in \mathbb{Q}[x]$ both non-units s.t. $f(x) = \overline{g(x)}\overline{h(x)}$. Let m, n be the lcm of the denominators of the coefficients of $\overline{g(x)}, \overline{h(x)}$ respectively, and denote $g(x) = m\overline{g(x)}, h(x) = n\overline{h(x)}$, then $g(x) \in \mathbb{Z}[x], h(x) \in \mathbb{Z}[x]$. Let c_g, c_h be the contents of $g(x), h(x)$, then by **68 primitive representative**, $\exists! \widetilde{g(x)}, \widetilde{h(x)} \in \mathbb{Z}[x]$ primitive polynomials s.t. $g(x) = c_g \widetilde{g(x)}, h(x) = c_h \widetilde{h(x)}$.

Since $\mathbb{Q}, \mathbb{Z}$ are integral domains, then $\mathbb{Q}[x], \mathbb{Z}[x]$ are integral domains by **70.1 polynomial ring of integral domain is an integral domain**, so multiplication commutes, so $g(x)h(x) = c_g c_h \widetilde{g(x)}\widetilde{h(x)}$, and $f(x) = \overline{g(x)}\overline{h(x)} = \frac{1}{mn} c_g c_h \widetilde{g(x)}\widetilde{h(x)}$. Now, by **Theorem 74 Gauss' lemma (primitivity)**, since $\widetilde{g(x)}, \widetilde{h(x)}$ are primitive then their product $\widetilde{f(x)}$ is primitive, so the content of $f(x)$ is given by $\frac{c_g c_h}{mn}$, i.e. $\frac{c_g c_h}{mn} \in \mathbb{Z}$, so $f(x)$ is reducible over $\mathbb{Z}$ with factors $\frac{c_g c_h}{mn} \widetilde{g(x)}, \widetilde{h(x)}$. $\square$

Theorem 76.

*If $p \in \mathbb{Z}$ is prime and $f(x) \in \mathbb{Z}[x]$ is a non-constant polynomial, and $\overline{f(x)}$ is the corresponding polynomial of $f(x)$ in $\mathbb{Z}_p[x]$ produced via **64 coefficient ring homomorphism induces polynomial ring homomorphism**, then if $\overline{f(x)}$ is irreducible over $\mathbb{Z}_p$, and $\deg(f(x)) = \deg(\overline{f(x)})$, then $f(x)$ is irreducible over $\mathbb{Q}$.*

Proof. Suppose by contradiction $\overline{f(x)}$ satisfying the conditions of the theorem is irreducible in $\mathbb{Z}_p[x]$ but reducible in $\mathbb{Q}$, then by **Theorem 75 Gauss' lemma (reducibility)** we can find $h(x), g(x) \in \mathbb{Z}[x]$ s.t. $f(x) = h(x)g(x)$ where $h(x), g(x)$ are not units in $\mathbb{Z}[x]$ (where $h(x), g(x)$ were attained by multiplying non-units in $\mathbb{Q}$, see the proof of that theorem for the construction of these polynomials).

Consider the corresponding polynomials in $\mathbb{Z}_p[x]$, $\overline{h(x)}, \overline{g(x)}$, then since the correspondence is a ring homomorphism $\overline{f(x)} = \overline{h(x)} \cdot \overline{g(x)}$. Since by supposition $\overline{f(x)}$ is irreducible in $\mathbb{Z}_p[x]$, then it must be that either $\overline{h(x)}$ or $\overline{g(x)}$ is a unit in $\mathbb{Z}_p[x]$.

Suppose w.l.o.g that $\overline{h(x)}$ is a unit. By **70.2 group of units of polynomial over integral domain**, we have that $\overline{h(x)} \in U(p)$, so $\deg(\overline{h(x)}) = 0$. Since $\mathbb{Z}_p[x], \mathbb{Z}[x]$ are both integral domains, $\deg(\overline{f(x)}) =$

$\deg(\overline{g(x)})$, and also $\deg(f(x)) = \deg(h(x)) + \deg(g(x))$. By supposition, $\deg(f(x)) = \deg(\overline{f(x)}) = 0 + \deg(\overline{g(x)}) = \deg(\overline{g(x)})$. Also, since the leading coefficient of $g(x)$ might be a multiple of p , then $\deg(\overline{f(x)}) = \deg(\overline{g(x)}) \leq \deg(g(x))$.

Suppose $\deg(h(x)) = 0$, and recall that by construction $h(x) = ah'(x)$ where $h'(x) \in \mathbb{Q}[x]$, so it follows that $\deg(h'(x)) = 0$ (since $\deg(h(x)) = \deg(a) + \deg(h'(x)) = 0 + \deg(h'(x))$), but then $h'(x)$ is a unit in $\mathbb{Q}[x]$, a contradiction, so $\deg(h(x))$ is greater than 0, so $\deg(g(x)) + \deg(h(x)) = \deg(f(x))$.

Putting everything together, we get $\deg(\overline{f(x)}) = \deg(f(x))$, a contradiction, so it must be that $f(x)$ is irreducible over $\mathbb{Q}$.

□

Example 76.1 (showing a polynomial is irreducible over $\mathbb{Q}$).

Using the theorem, we show $f(x) = 27x^3 + 5x^2 + 6x - 1$ is irreducible in $\mathbb{Q}$. Consider the polynomial in $\mathbb{F}_2[x]$. After reducing the coefficients mod 2, we find $\overline{f(x)} = x^3 + x^2 + 1$. It is a polynomial of degree 3, so by **Theorem 73** it is reducible over $\mathbb{F}_2$ if and only if it has a zero in $\mathbb{F}_2[x]$. Plugging $x = 0, x = 1$, we find that $\overline{f(0)} = 0 + 0 + 1 = 1, \overline{f(1)} = 1 + 1 + 1 = 1$, so $\overline{f(x)}$ is irreducible over $\mathbb{F}_2$, so by **Theorem 76** it is irreducible in $\mathbb{Q}$.

Using the theorem we can also demonstrate that polynomial in $\mathbb{Q}[x]$ are irreducible in $\mathbb{Q}$, by replacing the polynomial with rational coefficients with a polynomial with integer coefficients by multiplying the polynomial with the lcm of all denominators of coefficients, as demonstrated in **Theorem 75 Gauss' lemma (reducibility)**. See the example below:

Example 76.2 (showing a polynomial in $\mathbb{Q}[x]$ is irreducible over $\mathbb{Q}$).

Consider $f(x) = (3/7)x^4 - (2/7)x^2 + (9/35)x + 3/5$ in $\mathbb{Q}[x]$. The least common divisor of the denominators of the coefficients is 35, so $35f(x) \in \mathbb{Z}[x]$, denote it $h(x)$, then $h(x) = 35f(x) = 15x^4 + 10x^2 - 9x + 21$. Consider the polynomial in $\mathbb{Z}_2[x]$, then it reduces to $\overline{h(x)} = x^4 + x + 1$. We will show that all possible factorizations of $\overline{h(x)}$ in $\mathbb{Z}_2[x]$ involve units, thus $\overline{h(x)}$ is irreducible over $\mathbb{Z}_2$ by definition.

Consider $0, 1 \in \mathbb{Z}_2$, then $\overline{h(0)} = 1, \overline{h(1)} = 1 + 1 + 1 = 1$, so $\overline{h(x)}$ has no zeros in $\mathbb{Z}_2$, so it has no linear factors. If $g(x)$ is a factor of degree 2 of $\overline{h(x)}$ in $\mathbb{Z}_2[x]$, then there are 4 cases: $g(x) = x^2$, or $g(x) = x^2 + 1$, or $g(x) = x^2 + x$, or $g(x) = x^2 + x + 1$ (convince yourself that there are no other polynomials of degree 2 in $\mathbb{Z}_2[x]$). If $g(x)$ is a factor of $\overline{h(x)}$, then $\exists q(x) \in \mathbb{Z}_2[x]$ s.t. $g(x)q(x) = \overline{h(x)}$, and by the degree equation since $\deg(\overline{h(x)}) = 4, \deg(g(x)) = 2$, it must be that $\deg(q(x)) = 2$.

Note that, $x^2, x^2 + 1, x^2 + x$ both have zeros in $\mathbb{Z}_2$, so they cannot be factors of $\overline{h(x)}$. It follows that if $\overline{h(x)}$ has a factor of degree 2 in $\mathbb{Z}_2[x]$, then it must be $x^2 + x + 1$, so it must be that $\overline{h(x)} = (x^2 + x + 1)^2 = x^4 + 2x^3 + 3x^2 + 2x + 1 = x^4 + x^2 + 1$, but clearly this is not $x^4 + x + 1 = \overline{h(x)}$, so $\overline{h(x)}$ has no factor of degree 2.

Next, if the degree of the factor is 0, then by 70.2 group of units of polynomial over integral domain it is a unit of $\mathbb{Z}_2[x]$. Similarly, if the degree is 4, then that of the other factor is 0, so again we have a unit. If the degree is 3, then that of the other factor is 1, so it is linear, but $\overline{h(x)}$ has no linear factors as we have demonstrated. Finally, as argued above, $\overline{h(x)}$ has no factors of degree 2. We conclude that $\overline{h(x)}$ is irreducible over $\mathbb{Z}_2$ by definition, so by Theorem 76, $h(x)$ is irreducible over $\mathbb{Q}$.

Suppose $f(x)$ is reducible over $\mathbb{Q}$, then $f(x) = f_1(x)f_2(x)$ both not units, but then $35f_1(x)$ is also clearly not a unit in $\mathbb{Q}$ since it does not have degree 0 since $\deg(f_1(x)) > 0$, but then $35f_1(x) \cdot f_2(x)$ is a factorization of $h(x)$ in $\mathbb{Q}$ s.t. none of the factors is a unit, a contradiction to $h(x)$ being irreducible over $\mathbb{Q}$, so it follows that $f(x)$ must also be irreducible over $\mathbb{Q}$.

We state Eisenstein's criterion without proof here. We will prove it later, as part of [[#Unique Factorization Domain]]. Eisenstein's criterion provides a very powerful tool for determining when a polynomial with integer coefficients is reducible over $\mathbb{Q}$.

Theorem 77 (Eisenstein's criterion).

Let $f(x) = \sum_{i=0}^n a_i x^i \in \mathbb{Z}[x]$ where $n = \deg(f(x))$ and $2 \leq n$. If $\exists p \in \mathbb{N}$ prime such that:

1. $p \nmid a_n$
2. $p^2 \nmid a_0$
3. $p \mid a_j$ for $j \in [0, n-1]$

Then $f(x)$ is irreducible over $\mathbb{Q}$.

Unique Factorization

In this section, we show that if F is a field, then $F[x]$ is a unique factorization domain. Informally, a unique factorization domain is an integral domain where an analogous statement to the fundamental theorem of arithmetic holds. We define this term formally later when discussing divisibility and factorization in general integral domains. For now, we focus on showing that such statement does hold for $F[x]$.

First, we show that irreducible polynomials are building blocks of fields, and later we argue that an analogous statement to Euclid's lemma for integers holds for polynomials, where irreducible polynomials assume the role of primes.

Theorem 78.

If F is a field and $p(x) \in F[x]$, then $\langle p(x) \rangle$ is a maximal ideal in $F[x]$ if and only if $p(x)$ is irreducible over F .

Proof. Suppose F is a field, then it is also an integral domain, and by **70.1 polynomial ring of integral domain is an integral domain**, $F[x]$ is an integral group. Let $p(x) \in F[x]$.

$\implies$ Suppose $\langle p(x) \rangle$ is maximal, then $p(x)$ is nonzero and nonconstant (since the ideal is proper, see **72.2 polynomial field generator**), and suppose via contradiction that $p(x)$ is reducible over F , i.e. $\exists g(x), h(x) \in F[x]$ s.t. $g(x), h(x)$ are not units of $F[x]$, and also $p(x) = g(x)h(x)$. Since we suppose $g(x), h(x)$ are not units, then by **70.2 group of units of polynomial over integral domain** it must be that they are non-constant, non-zero polynomials, i.e. they have degree greater than 0.

Consider $\langle g(x) \rangle$. By **0.0.17 principal ideal**, $g(x)h(x) = p(x) \in \langle g(x) \rangle$, so from closure of the ideal it follows that $\langle p(x) \rangle \subseteq \langle g(x) \rangle$. By supposition, $\langle p(x) \rangle$ is maximal, so it must be that either:

1. $\langle p(x) \rangle = \langle g(x) \rangle$, in which case by **72.1** it follows that $\deg(p(x)) = \deg(g(x))$, so by **63 degree equation for polynomial ring of integer domain** it follows that $\deg(h(x)) = 0$, a contradiction to $h(x)$ not being a unit, so this case is impossible.
2. $\langle g(x) \rangle = F[x]$, in which case $\deg(g(x)) = 0$ by **72.2 polynomial field generator**, but then $g(x)$ is a unit, so this case is impossible.

Both cases led to a contradiction, so it must be that $p(x)$ is irreducible over F .

$\Leftarrow$ Suppose $p(x)$ is irreducible over F , and suppose $I \in F[x]$ is an ideal of $F[x]$ such that $\langle p(x) \rangle \subseteq I$. By 72.1, it must be that $I = \langle g(x) \rangle$ for some $g(x) \in F[x]$. Since by supposition $\langle p(x) \rangle \subseteq I = \langle g(x) \rangle$, then $p(x) = g(x)h(x)$ for some $h(x) \in F[x]$. Since by supposition $p(x)$ is irreducible over F , then either of $g(x), h(x)$ is a unit. If $g(x)$ is a unit, then $\langle g(x) \rangle = F[x]$ by 72.2 polynomial field generator. If $h(x)$ is a unit, then by the degree equation $\deg(p(x)) = \deg(g(x))$, so by 72.1 $p(x)$ is also a generator of $\langle g(x) \rangle$, so $\langle p(x) \rangle = \langle g(x) \rangle$. We conclude $\langle p(x) \rangle$ is maximal.

□

Corollary 78.1.

If F is a field and $p(x) \in F[x]$ is irreducible over F , then $F[x]/\langle p(x) \rangle$ is a field.

Proof. Suppose F is a field and $p(x) \in F[x]$ is irreducible over F . By Theorem 78, it follows that $\langle p(x) \rangle$ is a maximal ideal in $F[x]$. By Theorem 63, $F[x]/\langle p(x) \rangle$ is a field. □

Corollary 78.2 (Euclid's lemma for polynomials).

If F is a field and $p(x) \in F[x]$ is irreducible over F , and $\exists a(x), b(x) \in F[x]$ such that $p(x) | a(x)b(x)$ (i.e. $p(x)$ divides the product $a(x)b(x)$). In other words if, $p(x) | a(x)b(x)$, then either $p(x) | a(x)$ or $p(x) | b(x)$.

Proof. It should be noted that it is also relatively straightforward to prove this in a manner analogous to the classical proof of Euclid's lemma for integers, but for the sake of discussion we prove the theorem using the machinery we developed here.

By 78.1, $F[x]/\langle p(x) \rangle$ is a field, so in particular it is an integral domain, so by Theorem 62, $\langle p(x) \rangle$ is a prime ideal. Since by supposition $p(x) | a(x)b(x)$, then $a(x)b(x) = p(x)h(x)$ for some $h(x) \in F[x]$, in other words $a(x)b(x) \in \langle p(x) \rangle$. By 0.0.21 prime ideal, it follows that either $a(x) \in \langle p(x) \rangle$, or $b(x) \in \langle p(x) \rangle$, which implies either $p(x) | a(x)$ or $p(x) | b(x)$, which completes the proof.

□

It should be noted that, as in $\mathbb{Z}$, this is an if and only if relation:

Proposition 69 (converse Euclid's lemma for polynomials).

If $p(x) \in F[x]$ and $\forall a(x), b(x) \in F[x]$, we find that $p(x) | a(x)b(x)$ implies either $p(x) | a(x)$ or $p(x) | b(x)$, then $p(x)$ is irreducible over F .

Proof. As with $\mathbb{Z}$ (see [Elementary Number Theory](#)), we prove via contraposition: suppose $p(x)$ is reducible over F , then we show there exists $a(x), b(x) \in F[x]$ s.t. $p(x) \mid a(x)b(x)$ and yet $p(x) \nmid a(x)$ and also $p(x) \nmid b(x)$.

By supposition that $p(x)$ is reducible, $\exists a(x), b(x) \in F[x]$ non-unit s.t. $p(x) = a(x)b(x)$, so $p(x) \mid a(x)b(x)$ with $q(x) = 1$ and $r(x) = 0$. By the degree equation, $\deg(p(x)) = \deg(a(x)) + \deg(b(x))$, and by supposition both are non-unit so both have degree greater than 0, so $\deg(a(x)), \deg(b(x)) < \deg(p(x))$. Suppose $p(x) \mid a(x)$, then $a(x) \in \langle p(x) \rangle$ but by [72.1](#) $p(x)$ must be a polynomial of least degree in its principal ideal, so $a(x) \notin \langle p(x) \rangle$, so $p(x) \nmid a(x)$, and likewise for $b(x)$.

□

Proposition 70 (Extended Euclid's lemma for polynomials).

If $p(x) \in F[x]$ and $p(x)$ is irreducible over F and $p(x) \mid \prod_{i=1}^n f_i(x)$ where $F_i = \{f_1(x), \dots, f_n(x)\} \subseteq F[x]$, then $\exists f_i(x) \in F_i$ s.t. $p(x) \mid f_i(x)$.

Proof. We prove by induction on n . If $n = 1$ then it must be that $f_1(x) = p(x)$, so the proposition holds. If $n = 2$, the n by [78.2 Euclid's lemma for polynomials](#) the proposition holds. Suppose by induction it holds for some n , and consider $n + 1$. Since $F[x]$ is a ring, multiplication in $F[x]$ associates, and $\prod_{i=1}^{n+1} f_i(x) = (\prod_{i=1}^n f_i(x))f_{n+1}(x)$, the RHS is the product of two polynomials, so again by Euclid's lemma either $p(x)$ divides $f_{n+1}(x)$, in which case we're done, or it divides the product of the other n polynomials, in which case by the induction hypothesis it divides at least one of those polynomials, which completes the proof by induction.

□

These last results lead directly into the analogous statement of the fundamental theorem of arithmetic for polynomial rings over a field (recall that classic proofs of the fundamental theorem of arithmetic make use of Euclid's lemma for all the heavy liftings). We will later define an integral domain that satisfies a property analogous to that theorem as a unique factorization domain (UFD).

At this point we will state, without proof, that $\mathbb{Q}[x]$ satisfies this property (i.e. every polynomial in $\mathbb{Q}[x]$ admits a unique factorization into irreducible polynomials, where uniqueness is up to an ordering and multiplication by units). Later we will formally derive this result as a corollary of a more general theorem regarding PIDs, but the reader may use Euclid's lemma as stated above for polynomials over a field and follow a proof of the fundamental theorem of arithmetic and see that the same proof applies to $\mathbb{Q}[x]$. We make use of this fact and of [Theorem 75 Gauss' lemma \(reducibility\)](#) to show that all polynomials in $\mathbb{Z}[x]$ admit a unique factorization into irreducibles over $\mathbb{Z}[x]$.

Theorem 79 (fundamental theorem of arithmetic for polynomials).

If $f(x) \in \mathbb{Z}[x]$ is a non-zero polynomial, then $f(x)$ can be decomposed into the product of irreducible polynomials over $\mathbb{Z}[x]$, and this decomposition is unique up to an ordering and multiplication of the irreducibles by units.

Proof. As stated above, we make use of the fact that if we consider $f(x) \in \mathbb{Q}[x]$, then $f(x)$ can be factored into a unique (up to ordering and units) product of irreducibles over $\mathbb{Q}[x]$. If $f(x)$ is irreducible over $\mathbb{Z}[x]$, then it must also be irreducible over $\mathbb{Q}[x]$ (otherwise it would be reducible with the same factors), so irreducibles over $\mathbb{Q}[x]$ with integer coefficients are also irreducible over $\mathbb{Z}[x]$. By **Theorem 75 Gauss' lemma (reducibility)**, if $f(x)$ is reducible over $\mathbb{Q}[x]$ then the reduction can be carried out in $\mathbb{Z}[x]$, so factorization in $\mathbb{Q}[x]$ ensure factorization in $\mathbb{Z}[x]$. For uniqueness, suppose the factorization is not unique, then it also corresponds to a non-unique factorization in $\mathbb{Q}[x]$, but we stated that in $\mathbb{Q}[x]$ the factorization is unique, a contradiction, so the factorization must also be unique. $\square$

Divisibility in Integral Domains

In this section, we generalize the discussion started in [[#Polynomial Factorization]] to general integral domains. First, a few definitions, which should seem familiar:

Definition 0.0.44 (divisor).

If R is a unital ring, and $a, b \in R$ and $\exists c \in R$ s.t. $b = ca$, then we say that a divides b , or that a is a factor of b , and denote this relation $a|b$.

Definition 0.0.45 (associates, irreducible, primes).

If R is a unital ring, then:

1. $a, b \in R$ are associates if $\exists u \in U(R)$ (the group of units of R) s.t. $b = au$.
2. a is irreducible in R if $\forall c, d \in R$ s.t. $a = cd$, either $a \in U(R)$ or $b \in U(R)$.
3. a is prime if $a \neq 0$ and $\forall bc \in R$, if $a|bc$ then $a|b$ or $a|c$

In $\mathbb{Z}$, we usually define primes as numbers $p \in \mathbb{Z}$ s.t. whenever $p = cd$, then one must be 1, i.e. a unit of $\mathbb{Z}$, and the other p , which coincides with our definition of *irreducible*. Our general definition for *prime*, stated for an element of $\mathbb{Z}$, is simply Euclid's lemma. In elementary number theory, we see that the converse of Euclid's lemma is also true, i.e. in $\mathbb{Z}$ we find that *prime* and *irreducible* are equivalent.

Similarly, if F is a field and then as we have demonstrated in **78.2 Euclid's lemma for polynomials** and in **69 converse Euclid's lemma for polynomials**, in $F[x]$ we find that prime and irreducible are equivalent as well.

However, this is not always the case, as we will soon demonstrate. For now, we study an important example, which will help us work our way towards the definition of a unique factorization domain

(UFD).

Definition 0.0.46 (square-free integer).

If $n \in \mathbb{Z}$ is neither 1 nor 0, and is not divisible by any perfect square in $\mathbb{Z}$, we say that n is square-free.

Equivalently, we say that n is square-free if $n \neq 1, 0$ and also n has no prime factor with even multiplicity in its unique factorization as per the fundamental theorem of arithmetic. I leave it to the reader to become convinced that these definitions are equivalent.

Example 79.1.

Note that $25 = 5^2$, so it is not square-free. Also, note that $4^2 = 16|32$, so 32 is not square-free. However, 7 is prime, so trivially it is square-free. Also, $15 = 3 \cdot 5$, so it has no prime of even multiplicity in its factorization, so 15 is also square-free.

Definition 0.0.47 (quadratic integer ring).

If $D \in \mathbb{Z}$ is square-free, then the quadratic integer ring of D , denoted $\mathbb{Z}[\sqrt{D}]$, is given by the set of all quadratic integers of the form $a+b\sqrt{D}$ where $a, b \in \mathbb{Z}$, i.e. $\mathbb{Z}[\sqrt{D}] = \{a+b\sqrt{D} | a, b \in \mathbb{Z}\}$. We equip this set with natural extensions of addition and multiplication in $\mathbb{Z}$ to form a ring:

1. addition: $(a + b\sqrt{D}) + (c + d\sqrt{D}) = (a + c) + (b + d)\sqrt{D}$
2. multiplication: $(a + b\sqrt{D}) \times (c + d\sqrt{D}) = (ac + bdD) + (ad + bc)\sqrt{D}$

The reader can verify that this is indeed a ring under the given operations, and that it is in fact an integral domain: trivially it has unity 1 and multiplication is commutative since multiplication in $\mathbb{Z}$ is commutative. In the next proposition, we show it also has no zero divisors, so it is an integral domain:

Proposition 71 (quadratic integer ring has no zero divisors).

If $D \in \mathbb{Z}$ is square-free, then $\mathbb{Z}[\sqrt{D}]$ has no zero-divisors.

Proof. We prove by contradiction. Let $a + b\sqrt{D} \in \mathbb{Z}[\sqrt{D}]$ be a zero divisor, i.e. $a + b\sqrt{D}$ is non-zero. Consider $(a + b\sqrt{D})(a - b\sqrt{D})$, which by definition simplifies to $a^2 - b^2D$. If $a^2 - b^2D = 0$, then $a^2 = b^2D$. If $b = 0$, then $a^2 = 0$ so $a = 0$, so $a + b\sqrt{D} = 0_{\mathbb{Z}[\sqrt{D}]}$, which contradicts our supposition, so we can multiply both sides by $b^{-1} \in \mathbb{Q}$, take the square root and get $\sqrt{D} = \frac{a}{b} \in \mathbb{Q}$, but from elementary number theory we know that if D is not a perfect square then $\sqrt{D} \notin \mathbb{Q}$, and by supposition D is square-free so clearly it is not a perfect square, so $a^2 - b^2D \neq 0$, so $(a + b\sqrt{D})(a - b\sqrt{D}) \neq 0$.

By 45, since $a + b\sqrt{D}$ is a zero divisor by supposition, it follows that $a^2 - b^2D$ is also a zero-divisor, so for some $c + d\sqrt{D} \neq 0$ we find $(a^2 - b^2D)(c + d\sqrt{D}) = 0$, which by definition simplifies to $(a^2 - b^2D)c + (a^2 - b^2D)d\sqrt{D} = 0$. By supposition, $a^2 - b^2D \neq 0$, so we can multiply both sides by $(a^2 - b^2D)^{-1} \in \mathbb{Q}$ and get $c + d\sqrt{D} = 0$, but this is a contradiction to our supposition that $c + d\sqrt{D} \neq 0$.

We conclude that $\mathbb{Z}[\sqrt{D}]$ has no zero-divisors. □

Next, we introduce the concept of a *norm* (not to be confused with norms from metric spaces, although they do share some similarities). Using the concept of a norm, we will show that in the quadratic integer ring $\mathbb{Z}[\sqrt{-3}]$, irreducible and prime are not equivalent.

Definition 0.0.48 (norm).

If R is an integral domain, and $N : R \rightarrow \mathbb{N}$ is a function that preserves zero (i.e. $N(0_R) = 0$), then we say that N is a norm on R . If $a \neq 0 \implies 0N(a)$, we say that N is a positive norm.

In our study of $\mathbb{Z}[\sqrt{D}]$, we equip it with a norm inspired by the modulus of a complex number. Recall that the modulus of a complex number $a + bi$ is denoted $|a + bi|$ and defined as $\sqrt{a^2 + b^2}$, and equivalently via definition of multiplication of complex numbers, we find $(a + bi)(a - bi) = a^2 - b^2(-1) = a^2 + b^2 = |a + bi|^2$. Note that in $\mathbb{Z}[\sqrt{D}]$, we also find that $(a + b\sqrt{D})(a - b\sqrt{D}) = a^2 - b^2D$ (we have shown this in the previous proposition). (The sharp-eyed would recognize that $\mathbb{Z}[\sqrt{1}]$ is simply a subring of $\mathbb{C}$)

Definition 0.0.49 (norm of $\mathbb{Z}[\sqrt{D}]$).

Let D be square-free, then we define the norm of $\mathbb{Z}[\sqrt{D}]$ to be given by $N : a + b\sqrt{D} \mapsto |a^2 - b^2D|$.

Form the definition it is apparent that N is indeed a mapping from $\mathbb{Z}[\sqrt{D}] \rightarrow \mathbb{N}$ since a, b, D are all integers so $a^2 - b^2D \in \mathbb{Z}$ from closure under ring operations, and finally the absolute value maps from $\mathbb{Z} \rightarrow \mathbb{N}$.

It is also apparent that $N(0 + 0\sqrt{D}) = |0 - 0D| = 0$, and also that $a \neq 0 \implies 0N(a)$ since the absolute value is always non-negative (which takes care of $0 \leq N(a)$) and $a^2 - b^2D = 0$ if and only if $a^2 = b^2D$ which as we argued above leads to a contradiction to D being square free if a, b are not both 0, which shows that $N(a) = 0 \iff a = 0$, thus N as defined above is a *positive norm*. We explore more useful properties of $N(a)$ in the next theorem:

Theorem 80 (properties of the norm of $\mathbb{Z}[\sqrt{D}]$).

$\forall x, y \in \mathbb{Z}[\sqrt{D}]$,

1. $N(x) = 0 \iff x = 0$
2. $|xx^*| = N(x)$, where if $x = a + b\sqrt{D}$, then we define its conjugate x^* to be $a - b\sqrt{D}$, in a similar nature to the complex conjugate.
3. $N(xy) = N(x)N(y)$, i.e. the norm is multiplicative
4. x is a unit if and only if $N(x) = 1$
5. If $N(x)$ is prime, then x is irreducible in $\mathbb{Z}[\sqrt{D}]$

Proof. We have already proven 1 in the paragraph above, so we proceed from 2.

2. By definition $xx^* = a^2 - b^2D$ (see calculation in [71 quadratic integer ring has no zero divisors](#)). Also by definition, $N(x) = |a^2 - b^2D|$, so $|xx^*| = N(x)$.

3. Denote $x = a + b\sqrt{D}$, $y = c + \sqrt{D}$. By definition, $xy = (ac + bdD) + (ad + bc)\sqrt{D}$. So, $N(xy) = |(ac + bdD)^2 - (ad + bc)^2D|$. Consider the following:

$$\begin{aligned}
 N(x)N(y) &= \\
 |a^2 - b^2D| \cdot |c^2 - d^2D| &= \\
 |(a^2 - b^2D)(c^2 - d^2D)| &= \\
 a^2c^2 - a^2d^2D - b^2Dc^2 + b^2Dd^2D &= \\
 a^2c^2 + b^2d^2D^2 - (a^2d^2 + b^2c^2)D &= \\
 a^2c^2 + 2acbdD + b^2d^2D^2 - (a^2d^2 + b^2c^2 + 2abcd)D &= \\
 (ac + bdD)^2 - (ad + bc)^2D &= \\
 N(xy)
 \end{aligned}$$

4. $\implies$ Suppose x is a unit, then $x = a + b\sqrt{D}$ with $a, b \in \mathbb{Z}$ s.t. $\exists x^{-1} \in \mathbb{Z}[\sqrt{D}]$ s.t. $xx^{-1} = 1$ the unity of $\mathbb{Z}[\sqrt{D}]$. By (3), $N(xx^{-1}) = N(x)N(x^{-1})$, and also $xx^{-1} = 1$ so $N(xx^{-1}) = N(1) = |1| = 1$, so $N(x)N(x^{-1}) = 1$. Since $N(x), N(x^{-1}) \in \mathbb{N}$, then the only solution to the above is $N(x) = N(x^{-1}) = 1$, so $N(x) = 1$.

$\Leftarrow$ Suppose $N(x) = 1$, then since $N(x) = |xx^*|$ it follows that $1 = |xx^*|$, i.e. $xx^* = \pm 1$, then $x^{-1} = \mp x^*$.

5. Suppose $N(x)$ is prime, and suppose via contradiction x is reducible in $\mathbb{Z}[\sqrt{D}]$, i.e. $x = ab$ for some non-units $a, b \in \mathbb{Z}[\sqrt{D}]$. By (3), $N(x) = N(ab) = N(a)N(b)$, so $N(a)|N(x)$, $N(b)|N(x)$. Since by supposition $N(x)$ is prime, it follows that one of $N(a), N(b)$ must be p and the other 1, w.l.o.g suppose $N(a) = 1$, then by (4), a is a unit in $\mathbb{Z}[\sqrt{D}]$, a contradiction, so x is irreducible.

□

Of the above, it is worth noting that if N is a multiplicative positive norm of a ring then (1), (3) of the above are satisfied, which implies (4), (5) (examine the respective parts of the proof to verify this), so we can also state the following:

Theorem 81 (properties of multiplicative positive-definite norm).

If R is a ring and $N : R \rightarrow \mathbb{N}$ is a multiplicative positive-definite norm (i.e. it is a positive norm and $\forall x, y \in R (N(xy) = N(x)N(y))$ and $N(x) = 0 \iff x = 0$), then:

1. $x \in R$ is a unit if and only if $N(x) = 1$
2. If $N(x)$ is prime, then x is irreducible in R .

For now, we return to our discussion of $\mathbb{Z}[\sqrt{D}]$.

Using these results we can demonstrate that while $\mathbb{Z}[\sqrt{-3}]$ is an integral domain as argued above, we do not find that prime and irreducible are the same in this domain.

Example 81.1 ($1 + \sqrt{-3}$ is irreducible in $\mathbb{Z}[\sqrt{-3}]$, but not prime).

First we will demonstrate $1 + \sqrt{-3}$ is not prime. Observe that $(1 + \sqrt{-3})(1 - \sqrt{-3}) = 1^2 - (-3) = 4 = 2^2$, so $(1 + \sqrt{-3}) \mid 2^2$. If $1 + \sqrt{-3}$ is prime, then it must also divide 2 by definition, so we should find some $a + b\sqrt{-3}$ with a, b integers s.t. $(1 + \sqrt{-3})(a + b\sqrt{-3}) = 2$, expanding the LHS we get $(a - 3b) + (b + a)\sqrt{-3} = 2$, so it must be that $a - 3b = 2$ and also $b + a = 0$, so $2 = a - 3b = b + a - 4b = -4b$, so $b = -\frac{1}{2}$, but this b is not an integer, so the above has no solution for $a, b \in \mathbb{Z}$, so $1 + \sqrt{-3} \nmid 2$, so it is not prime.

Next, we demonstrate it is irreducible. We do this by *Theorem 80 properties of the norm of $\mathbb{Z}[\sqrt{D}]$* . We calculate $N(1 + \sqrt{-3}) = |1^2 - (-3)| = |1 + 3| = 4$. Suppose $1 + \sqrt{-3} = xy$ for some $x, y \in \mathbb{Z}[\sqrt{-3}]$, then if x, y are both not units (i.e. $1 + \sqrt{-3}$ is reducible) then since $N(xy) = N(x)N(y) = 4$, it must be that $N(x) = N(y) = 2$. Denote $x = a + b\sqrt{-3}$ and solve $|a^2 - b^2(-3)| = a^2 + 3b^2 = 2$. We find that if $b \neq 0$, then $2a^2 + 3b^2$ since $b \in \mathbb{Z}$ so $3 \leq 3b^2$ and also $0 \leq a^2$, so it must be that $b = 0$. So, $a^2 = 2$, but then $a = \sqrt{2}$, which we know to be irrational, so obviously $a \notin \mathbb{Z}$, a contradiction, so $1 + \sqrt{-3}$ is irreducible.

We see that in $\mathbb{Z}[\sqrt{-3}]$, irreducible does not imply prime.

Example 81.2 ($7 \in \mathbb{Z}[\sqrt{5}]$ is irreducible).

Next, we show it is irreducible. Suppose not, then $7 = xy$ for some non-units $x, y \in \mathbb{Z}[\sqrt{5}]$, then by **Theorem 80 properties of the norm of $\mathbb{Z}[\sqrt{D}]$** we find $49 = 7^2 = N(7) = N(x)N(y)$. Since $N(x), N(y) \in \mathbb{N}$, and both are not unit by supposition so their norms are not 1 (see property (4) in the mentioned theorem), then $N(x) = N(y) = 7$. Let $x = a + b\sqrt{5}$, then we find $N(a + b\sqrt{5}) = 7$, so $|a^2 - 5b^2| = 7$, so $a^2 - 5b^2 = \pm 7$. We can exchange this for a modular equation $a^2 - 5b^2 \equiv 0 \pmod{7}$.

Since each of a, b has 7 possible values $\pmod{7}$, there are 49 total solution candidates to consider. Via explicitly considering each solution, we find that the only solution is the trivial solution $a \equiv_7 b \equiv_7 0$. It follows that $7|a$ and also $7|b$, so $49|a^2, 49|b^2$ and we have $49|a^2 - 5b^2$, but $a^2 - 5b^2 = \pm 7$ which is clearly not divisible by 49, so there is no such $x \in \mathbb{Z}[\sqrt{5}]$, hence 7 is irreducible.

With more study of $\mathbb{Z}[\sqrt{D}]$, we can show that 7 is also prime in $\mathbb{Z}[\sqrt{5}]$. This can be demonstrated via showing that $\mathbb{Z}[\sqrt{5}]/\langle 7 \rangle$ is an integral domain, which we can show via an isomorphism with a factor group of the polynomial ring $\mathbb{Z}[x]$ which we can trade $\mathbb{Z}[\sqrt{5}]/\langle 7 \rangle$ for. This raises the question - we have seen the irreducible does not imply prime in $\mathbb{Z}[\sqrt{D}]$, but we have seen an example where some prime is also irreducible, so perhaps prime $\implies$ irreducible does hold. The next theorem affirms this suspicion:

Theorem 82 (every prime is irreducible in an integral domain).

If D is an integral domain and $p \in D$ is prime, then p is also irreducible.

Proof. Suppose D is an integral domain and $p \in D$ is prime, and suppose $p = ab$ for some $a, b \in D$, then $p|ab$, and since p is prime then either $p|a$ or $p|b$. Suppose w.l.o.g $p|a$, then $a = pc$, so $p = ab = (pc)b = p(cb)$, so $cb = 1$, so b is a unit. This is true for every factoring of p into a product of two elements, so p is irreducible. $\square$

Theorem 83.

If R is a unital commutative ring, then $\forall a, b \in R$:

1. If a, b are associates, then $\langle a \rangle = \langle b \rangle$
2. If R is an integral domain, and $\langle a \rangle = \langle b \rangle$, then a, b are associates

Proof. 1. Suppose a, b are associates, then $b = au$ for some $u \in U(R)$. We will prove $\langle a \rangle = \langle b \rangle$ by mutual inclusion. Let $ar \in \langle a \rangle$, then since $u \in U(R)$ we can find $u^{-1} \in R$ s.t. $uu^{-1} = 1$, so $r = 1r = uu^{-1}r$, so $ar = (au)(u^{-1}r) = b(u^{-1}r) \in \langle b \rangle$, so $\langle a \rangle \subseteq \langle b \rangle$. Conversely, consider that since

$u \in R$ then trivially $b = au \in \langle a \rangle$ by definition. Since $\langle a \rangle$ is an ideal, the it must contain all elements of the form br for $r \in R$ by definition, and by definition this set of elements is exactly $\langle b \rangle$, so $\langle b \rangle \subseteq \langle a \rangle$, so $\langle a \rangle = \langle b \rangle$.

2. Suppose R is an integral domain and $\langle a \rangle = \langle b \rangle$, so $a \in \langle b \rangle, b \in \langle a \rangle$, so $\exists s, t \in R$ s.t. $a = bs, b = at$, so $a = (at)s = a(ts)$. If $a = 0$ then $\langle a \rangle = \{0\}$ and so $b = 0$, so trivially a, b associate. Otherwise, since R is an integral domain then it has no zero-divisors, so by the cancellation law for ring with no zero-divisors, $1 = ts$, i.e. $t, s \in U(R)$, and recall that $a = bs, b = at$, so a, b are associates by definition.

□

Theorem 84 (in a PID, prime and irreducible are equivalent).

If D is a PID, then $a \in D$ is prime if and only if a is irreducible.

Proof. $\implies$ A PID is an integral domain, so this direction is covered by **Theorem 82 every prime is irreducible in an integral domain**.

$\impliedby$ Suppose a is irreducible, i.e. $\forall x, y$ s.t. $a = xy$, either x or y is a unit. Let $bc \in D$ s.t. $a|bc$, i.e. $bc = ak$ for some $k \in D$. Consider $I = \{\alpha a + \beta b | \alpha, \beta \in D\}$, clearly it is a subset of D . We will show it is an ideal of D via **Theorem 58 ideal test**:

1. (nonempty) For $\alpha = 1$ and $\beta = 0$ we find $a \in I$ so clearly I is nonempty.
2. (closed under subtraction) Let $\alpha_1 a + \beta_1 b, \alpha_2 a + \beta_2 b \in I$, note that $(\alpha_1 a + \beta_1 b) - (\alpha_2 a + \beta_2 b) = (\alpha_1 - \alpha_2)a + (\beta_1 - \beta_2)b \in I$ (for $\alpha = \alpha_1 - \alpha_2, \beta = \beta_1 - \beta_2$).
3. (closure under multiplication with ring elements) let $r \in D$ and $\alpha a + \beta b \in I$, then by **Theorem 54 ring arithmetic**, $r(\alpha a + \beta b) = (r\alpha)a + (r\beta)b$, which is an element of I . Since D is a PID then it is commutative, so $(\alpha a + \beta b)r = r(\alpha a + \beta b) \in I$.

We conclude that I is an ideal. Since D is a PID then by definition I is generated by some $d \in D$. As we have argued above, $b \in I$ so $a = td$ for some $t \in D$ since $I = \langle d \rangle$. By supposition, a is irreducible, so either d is a unit, or t is a unit.

If t is a unit, then by definition a, d are associates, so by **Theorem 83** $\langle a \rangle = \langle d \rangle$, so a divides all elements of I . In particular, with $\alpha = 0, \beta = 1$ we find $b \in I$, so $a|b$, in agreement with the hypothesized primality of a .

Otherwise, d is a unit, then $d^{-1} \in D$, so $dd^{-1} \in \langle d \rangle$, so $1 \in \langle d \rangle$, so $1 = \alpha a + \beta b$ for some $\alpha, \beta \in D$. Multiplying both sides by c , we find $c = \alpha ac + \beta bc$. Clearly $a|\alpha ac$, and by supposition $a|bc$ so $a|\beta bc$, so by properties of the divisor $a|(\alpha ac + \beta bc)$, so $a|c$.

We have that a irreducible in a PID and $a|bc$ implies $a|b$ or $a|c$, so a is prime.

□

This theorem lets us conflate *prime* and *irreducible* in a PID. Note that, as we have seen in our discussion of polynomial rings, if the coefficients are taken from a field F , then in $F[x]$, prime and irreducible are equivalent. Recall that by **Theorem 72 polynomial ring over a field is a PID**, if F is a field then $F[x]$ is a PID. The above theorem immediately implies this result.

Corollary 84.1 (in a PID, maximal ideal and prime ideal are interchangeable).

Let D be a PID and I an ideal of D , then I is prime if and only if I is maximal.

Proof. Note that **Theorem 62** and **Theorem 63** immediately give prime $\iff$ maximal since every field is also an integral domain, so it follows that if I is prime then $I \subseteq M$ for some maximal ideal M .

$\implies$ Suppose I is prime. Since D is a PID then $I = \langle p \rangle$, $M = \langle m \rangle$ for some $p, m \in D$. Since $p \in \langle m \rangle$ then $\exists n \in D$ s.t. $p = nm$, but p is prime so by **Theorem 84 in a PID, prime and irreducible are equivalent** it is also irreducible so either m or n is a unit.

If m is unit, then $\langle m \rangle = D$ by **51 principal ideal of a unit**, so $\langle m \rangle$ is not maximal, so this is impossible. Otherwise, n is a unit so p, m are associates, so by **Theorem 83** $\langle p \rangle = \langle m \rangle$, so $I = M$, so I is also maximal.

□

Recall Bezout's identity from number theory, which states that given $a, b \in \mathbb{Z}$, $\exists x, y \in \mathbb{Z}$ s.t. $xa + yb = \gcd(a, b)$. The identity can be generalized to PIDs. First, we pick a workable definition for the gcd in a commutative ring:

Definition 0.0.50 (greatest common divisor in a commutative ring).

Let R be a commutative ring and let $a, b \in R$, then $d \in R$ is the greatest common divisor of a, b , denoted $\gcd(a, b)$, if $\forall c \in R, c \mid a \wedge c \mid b \implies c \mid d$, i.e. it is divisible by any other common divisor of a, b (where a common divisor is some $c \in R$ s.t. $cx = a, cy = b$ for some $x, y \in R$).

Theorem 85 (Bezout's identity for PIDs).

If D is a PID and $a, b, d \in D$ s.t. $\gcd(a, b) = d$, then $\exists x, y \in D$ s.t. $xa + yb = d$.

Proof. Consider $\langle a, b \rangle$ the ideal generated by a, b . By **52** it is an ideal of D . We will show $\langle a, b \rangle = \langle d \rangle$.

Since d is a common divisor for a, b then $a, b \in \langle d \rangle$, so $\langle a, b \rangle \subseteq \langle d \rangle$. Conversely, since D is a PID, then $\exists r \in D$ s.t. $\langle a, b \rangle = \langle r \rangle$, but since $a, b \in r$ then by definition $r \mid a, r \mid b$, so r is a common divisor of a, b , so $r \mid d$, so $d \in \langle r \rangle$, so $\langle d \rangle \langle r \rangle = \subseteq \langle a, b \rangle$. By mutual inclusion, we conclude $\langle a, b \rangle = \langle d \rangle$.

Next, since $\langle a, b \rangle = \langle d \rangle$ then $d \in \langle a, b \rangle$ so $\exists x, y \in D$ s.t. $xa + yb = d$ by definition of the ideal generated by a, b .

□

Unique Factorization Domain

Recall that the fundamental theorem of arithmetic shows that in $\mathbb{Z}$, every integer admits a unique factorization into a product of primes up to a reordering and associates (i.e. if $a = b$ and $a = u_2 b u_1$ where u_2, u_1 are units and b is irreducible, we say that both factorizations are the same up to a reordering and associates, since b clearly associates with itself, in $\mathbb{Z}$ this means that we consider $2 \cdot 5$ and $2 \cdot (-1) \cdot 5 \cdot (-1) \cdot 1$ to be essentially the same factorization). In **Theorem 79 fundamental theorem of arithmetic for polynomials**, we have shown that an equivalent proposition is true for $\mathbb{Z}[x]$.

Both of these results usually follow from a similar sequence of results:

1. First, we equip our integral domain with a *division algorithm*.
2. Next, we identify that our *irreducibles*, which are those elements that, under the division algorithm, are divisible only by the unit of the ring and themselves, also satisfy the *prime* property.
3. Using these two results, we begin to factorize elements into *primes*, which are also *irreducible*. We find that we are able to write every element in terms of a product of primes.
4. Next, we find that, in the cases we have examined, this factorization is *unique* (up to a reordering and associates).

In this section, we consider this process in the abstract sense and define structures that let us more generally discuss it.

First, we need to have a general notion of an integral domain with a division algorithm. We call such domains *Euclidean domains* (ED), as defined below:

Definition 0.0.51 (Euclidean domain).

If D is an integral domain, and N is a norm s.t. $\forall a, b \in D, b \neq 0$, there exists $q, r \in D$ such that:

1. $a = qb + r$
2. Either $r = 0$, or $N(r) < N(b)$

then we say that D equipped with this norm is a Euclidean domain, and call q the quotient and r the remainder of the division of a by b .

Consider a few examples:

Example 85.1 ($\mathbb{Z}$ is a Euclidean domain).

Consider $\mathbb{Z}$ with the norm $N : z \mapsto |z|$. Note that $N(xy) = |xy| = |x| \cdot |y| = N(x)N(y)$, and also $N(x) = 0 \iff x = 0$ otherwise $0 < N(x)$, so N is a multiplicative positive norm on $\mathbb{Z}$. Consequently, by **Theorem 81 properties of multiplicative positive-definite norm**, we have that the units of $\mathbb{Z}$ are the elements that satisfy $|z| = 1$, i.e. ± 1 .

By the division algorithm in $\mathbb{Z}$, we know that $\forall a, b \in \mathbb{Z}, b \neq 0$ then $\exists! q, r \in \mathbb{Z}$ s.t. $a = qb + r$, and also $r \in [0, b)$. Since $r \in [0, b)$ then $0 \leq N(r) < N(b)$. It should be noted here that the choice of r to be non-negative is arbitrary - for every q, r pair with r non-negative we can demonstrate that $q' = q + 1, r' = r - b$ satisfies $a = q'b + r'$ ($q'b + r' = (q + 1)b + r - b = qb + b + r - b = qb + r$), and also $0 \leq N(r') < N(b)$.

Since $\mathbb{Z}$ with the natural norm admits a division algorithm as described in **0.0.51 Euclidean domain**, it is a Euclidean domain.

Example 85.2 ($F[x]$ is a Euclidean domain).

If F is a field then $F[x]$ equipped with the norm $N : 0 \mapsto 0, p(x) \mapsto \deg(p(x))$ is a Euclidean domain: first, we observe that if $f(x), g(x)$ are both non-zero then $N(f(x)g(x)) = \deg(f(x)g(x)) = \deg(f(x)) + \deg(g(x))$, so N is not multiplicative, and also $N(x) = 0$ does not necessarily give $x = 0$ (any constant polynomial also has norm 0 as defined here).

In **Theorem 71 polynomial division algorithm** we have demonstrated that $\forall a(x), b(x) \in F[x], b(x) \neq 0$, we can find $q(x), r(x) \in F[x]$ s.t. $\deg(r(x)) < \deg(b(x))$ or $r(x) = 0$, and $a(x) = q(x)b(x) + r(x)$, so using our norm we find that $N(r(x)) < N(b(x))$, or $r(x) = 0$, so $F[x]$ equipped with this norm is a Euclidean domain by definition.

Theorem 86 (every Euclidean domain is a PID).

If D is a Euclidean domain, then it is a principal ideal domain.

Proof. We follow the same structure as the proof for **Theorem 72 polynomial ring over a field is a PID**. Let D be Euclidean with norm N and let I be some ideal of D . If $I = \{0\}$, then $I = \langle 0 \rangle$ so it is principal. Otherwise, $I \neq \{0\}$. In this case, consider $N(I)$ the image of I under the norm equipped to D . Since $N(I) \in \mathbb{N}$, then by the WOP $\exists m \in I$ s.t. $\forall n \in I (N(m) \leq N(n))$, i.e. m has minimal norm.

Let $a \in I$. Since D is Euclidean with norm N , $\exists q, r \in D$ s.t. $a = qm + r$ and $r = 0$ or $N(r) < N(m)$. Since I is an ideal and $m \in I$, then $qm \in I$, and since $a \in I$ then $a - qm \in I$, so $r = a - qm \in I$. Since m has minimal norm in I then $N(r) < N(m)$ is impossible, then $r = 0$, so $a \in \langle m \rangle$, so $I = \langle m \rangle$, so D is a PID.

□

Note that this theorem is a generalization of **Theorem 72 polynomial ring over a field is a PID**, so we could have proven that theorem directly as an instantiation of **Theorem 86 every Euclidean domain is a PID** for the Euclidean domain $F[x]$.

Eventually, we are interested in domains where the factorization of elements is unique. We call these *unique factorization domains* (UFDs), as defined below:

Definition 0.0.52 (unique factorization domain (UFD)).

If D is an integral domain, and $\forall a \in D, a \neq 0$, we can factor a into a product of irreducibles, and that product is unique up to a reordering and associates, then we say that D is a unique factorization domain.

We will soon prove that every PID is a UFD, and later show that every Euclidean domain is a PID, so every Euclidean domain is a UFD, thus generalizing the fundamental theorem of arithmetic for integers. The general layout of the argument we will use to show that every PID is a UFD is very similar to a standard proof of the fundamental theorem of arithmetic (while adjusting our terminology from integers to integral domains), so it would be beneficial to recall such proof in broad strokes:

1. We proved existence in $\mathbb{N}$ via strong induction on $n \in \mathbb{N}$. The key to the proof is that if $n \in \mathbb{N}$ is prime, then it is already given in factorized form since n is irreducible, otherwise n is reducible so it reduces into the product of two lesser integers for which the induction hypothesis applies, which shows existence. Existence in $\mathbb{Z}$ follows from the fact that, if $z \in \mathbb{Z}$ then either $z \in \mathbb{N}$ or $-z \in \mathbb{N}$, for which we have shown existence via induction.
2. We proved uniqueness by supposing that there exists two distinct factorizations of the same element, and considered one prime p that appears only in one of the two factorizations. It then followed that one of the primes q in the factorization where p does not appear must be divisible by p , but p is prime so it is not a unit, and that would imply that q is reducible, but q is prime so this is impossible, so the factorization must be unique.

Another way of looking at existence is the following: suppose $n \in \mathbb{N}$, then we show that $\forall n \in \mathbb{N}$, either n is prime or has a prime factor, in which case $n = mp$ for some prime p and other integer m that is also a non-trivial factor of n so it is lesser than n . Eventually this process must terminate since in each step we divide the number by at least 2 (the smallest prime), so eventually we must hit a prime. Collecting all primes along the way, we arrive at a prime power factorization for n . This way of thinking is more aligned with the proof we will follow in this section.

Definition 0.0.53 (ascending chain condition).

If R is a commutative ring with unity, and for every sequence $I_1, I_2, \dots$ of increasing (ascending) ideals in R , i.e. $I_1 \subseteq I_2 \subseteq \dots$, the sequence must terminate (or stabilize), i.e. $\exists N \in \mathbb{N}$ s.t. $\forall n \geq N, I_n = I_N$, then we say that R satisfies the ascending chain condition (ACC).

Equivalently, some writers require the ideals be strictly increasing, i.e. $I_1 \subset I_2 \subset \dots$, in which case the requirement is that the sequence be finite in length.

It is trivial to demonstrate the definitions are equivalent: removing all duplicate ideals in the first definition gives exactly the second definition.

Lemma 86.1 (union of ascending ideals is an ideal).

If $I_1, I_2, \dots$ is an ascending chain of ideals in a ring R , i.e. $I_1 \subseteq I_2 \subseteq \dots$, then their union $I = \cup_{j \in \mathbb{N}} I_j$ is an ideal.

Proof. We prove by **Theorem 58 ideal test**. Clearly I is non-empty.

1. (closure under subtraction) let $a, b \in I$, then $a \in I_n, b \in I_m$ for some $n, m \in \mathbb{N}$. Suppose w.l.o.g. $n \leq m$, then since the chain is ascending, $a \in I_n$ implies $a \in I_m$, so we have $a, b \in I_m$, and since I_m is an ideal then $a - b \in I_m$, so I is closed under subtraction.
2. (closure under multiplication with ring elements) suppose $r \in R, s \in I$, then $s \in I_n$ for some $n \in \mathbb{N}$, and since I_n is an ideal $rs \in I_n, sr \in I_n$, but by definition $I_n \subseteq I$, so $rs, sr \in I$.

So I is an ideal.

□

Rings that satisfy the ACC have a special name:

Definition 0.0.54 (Noetherian ring).

If R is a commutative ring with unity, then R is Noetherian if it satisfies the ascending chain condition.

Theorem 87 (PIDs are Noetherian).

If D is a PID, then D is Noetherian.

Proof. We will demonstrate that D satisfies the ACC. Let $I_1, I_2, \dots$ be a strictly ascending sequence of ideals. By **86.1 union of ascending ideals is an ideal** their union I is also an ideal. Since D is a PID, then $I = \langle d \rangle$ for some $d \in D$, so $d \in I$, but also $d \in I_N$ for some $N \in \mathbb{N}$ by definition of I as a union. By **Theorem 59 ideal is a superset of the principal ideals of its elements**, it follows that $\langle d \rangle \subseteq I_N$, but then

$I \subseteq I_N$, but by definition $I_1, \dots$ is strictly ascending, so it must terminate with I_N , so D satisfies the ACC, so it is Noetherian. $\square$

Lemma 87.1.

In a PID, every element is either irreducible, or has an irreducible factor.

Proof. Let D be a PID and let $a_1 \in D$. If a is irreducible, then we are done. Otherwise, $a_1 = a_2 b_2$ for non-units $a_2, b_2 \in D$. If a_2 is irreducible, then we are done. Otherwise, $a_2 = a_3 b_3$ for non-units $a_3, b_3 \in D$. We continue this sequence and consider that for each $\langle a_n \rangle$, we have $a_n = a_{n+1} b_{n+1}$, so by definition $a_n \in \langle a_{n+1} \rangle$, so $\langle a_n \rangle \subset \langle a_{n+1} \rangle$, so we have an ascending chain of ideals $\langle a_1 \rangle \subset \langle a_2 \rangle \subset \dots$, so by **Theorem 87 PIDs are Noetherian** the chain must terminate, so eventually we must find some a_n term for $n \in \mathbb{N}$ s.t. a_n is irreducible, and back-substituting all other expressions we find $a_1 = a_n \prod_{i=2}^n b_i = a_n k$, so a_1 has an irreducible factor. $\square$

We can now prove that every PID is a UFD:

Theorem 88 (PIDs are UFDs).

If D is a PID, then D is a UFD.

Proof. Let $a \in D$ and suppose D is a PID.

Existence: by **87.1** we have that $a = p_1 k_1$ for some p_1 irreducible and $k_1 \in D$. If k_1 is not a unit, we follow the same procedure for k_1 and so by the same lemma we have $k_1 = p_2 k_2$ for some p_2 irreducible and $k_2 \in D$. If k_2 is not a unit, we again employ **87.1** to write $k_2 = p_3 k_3$ for some $p_3, k_3 \in D$ with p_3 irreducible. Notice that $k_1 \in \langle k_2 \rangle$ and $k_2 \in \langle k_3 \rangle$ and so on, so by an identical argument to the one in **87.1** we have an ascending chain of ideals $\langle k_1 \rangle \subset \langle k_2 \rangle \subset \dots$ which must terminate by **Theorem 87 PIDs are Noetherian**, so eventually we have $a = \prod_{p_i \in P} p_i$ for some $P = \{p_1, \dots, p_n\}$ irreducibles in D .

Uniqueness: Suppose the above factorization is not unique, so we can find $P = \{p_1, \dots, p_n\}$ and $Q = \{q_1, \dots, q_m\}$ both sets of irreducibles in D s.t. $\prod_{p \in P} p = \prod_{q \in Q} q = a$, and also $P \neq Q$. Suppose w.l.o.g. $\exists p_i \notin Q$, then by an analogous theorem to **70 Extended Euclid's lemma for polynomials** we find that since $p_i \mid a = \prod_{q \in Q} q$ then $\exists q_j \in Q$ s.t. $p_i \mid q_j$, but since q_j is prime and also not 1 then it must be that $p_i = q_j$, so $p_i \in Q$, so $P = Q$, so the factorization is unique. $\square$

Corollary 88.1 ($F[x]$ with F field is a UFD).

If F is a field then its polynomial ring $F[x]$ is a UFD.

Proof. By **Theorem 72 polynomial ring over a field is a PID**, $F[x]$ is a PID, so by **Theorem 88 PIDs are UFDs** it is a UFD. □

Corollary 88.2 ($\mathbb{Q}[x]$ is a UFD).

$\mathbb{Q}[x]$ is a UFD.

Proof. $\mathbb{Q}$ is a field, hence by **88.1 $F[x]$ with F field is a UFD** we find that $\mathbb{Q}[x]$ is a UFD. □

In **Theorem 79 fundamental theorem of arithmetic for polynomials**, we used the above fact to show that $\mathbb{Z}[x]$ is a UFD. Recall that, since $\mathbb{Z}$ is not a field, it is not immediately a corollary of **88.1 $F[x]$ with F field is a UFD** that $\mathbb{Z}[x]$ is a UFD, so we needed to involve **Theorem 75 Gauss' lemma (reducibility)** as well to complete that proof.

Corollary 88.3 (EDs are UFDs).

If D is a Euclidean domain, then it is a UFD.

Proof. By **Theorem 86 every Euclidean domain is a PID**, D is a PID. By **Theorem 88 PIDs are UFDs**, it is a UFD. □

Corollary 88.4 (fields are UFDs).

If F is a field then F is a UFD.

Proof. By **Theorem 60 fields are PIDs**, and by **Theorem 88 PIDs are UFDs**. □

It is important to internalize that an integral domain is not guaranteed to be a UFD - only a Euclidean domain is, and finding examples where an ED fails to be a UFD does not require anything exotic. This led to many false proofs of Fermat's last theorem throughout the decades. The next example demonstrates one such case:

Example 88.1 ($\mathbb{Z}[\sqrt{-5}]$ is an integral domain but a UFD).

Consider $\mathbb{Z}[\sqrt{-5}]$. We know that that quadratic integer rings are integral domains, and we also have a positive multiplicative norm as defined in 0.0.49 norm of $\mathbb{Z}[\sqrt{D}]$.

By Theorem 80 properties of the norm of $\mathbb{Z}[\sqrt{D}]$, it quickly follows that the units of $\mathbb{Z}[\sqrt{-5}]$ are ± 1 (since those are the only solution to $N(a + b\sqrt{-5}) = 1$). Note that $46 = 2 \cdot 23$, but also $46 = 1 + 45 = 1 + 9 \cdot 5 = 1 - 3^2\sqrt{-5}^2 = (1 + 3\sqrt{-5})(1 - 3\sqrt{-5})$.

Suppose $2 = xy$ for some $x, y \in \mathbb{Z}[\sqrt{-5}]$, then $N(xy) = N(x)N(y) = N(2) = 4$. If both of x, y are not units then the only solutions are $N(x) = N(y) = 2$, but then we have $2 = N(x) = N(a + b\sqrt{-5}) = |a^2 + 5b^2| = a^2 + 5b^2$, and since $a, b \in \mathbb{Z}$ this equation has no solutions, so this is impossible, so 2 is irreducible in $\mathbb{Z}[\sqrt{-5}]$. A similar argument shows that 23 is also irreducible in this ring, so $2 \cdot 23$ is a proper factorization of 46 to irreducible elements in $\mathbb{Z}[\sqrt{-5}]$.

Suppose $1 + 3\sqrt{-5} = xy$, and note that $N(1 + 3\sqrt{-5}) = 1 + 5 \cdot 9 = 46$, so $N(x)N(y) = 46$, and the only solution where $N(x), N(y) \neq 1$ is $\{N(x), N(y)\} = \{2, 23\}$, but we have already shown that $N(x) = 2$ has no solution for $x \in \mathbb{Z}[\sqrt{-5}]$, so $1 + 3\sqrt{-5}$ is irreducible in $\mathbb{Z}[\sqrt{-5}]$. Similarly, $N(1 - 3\sqrt{-5}) = 1 + 5 \cdot 9 = N(1 + 3\sqrt{-5})$, so it too is irreducible, so $(1 + 3\sqrt{-5})(1 - 3\sqrt{-5})$ is also a proper factorization to irreducibles in $\mathbb{Z}[\sqrt{-5}]$.

We have found that 46 admits two distinct factorizations in $\mathbb{Z}[\sqrt{-5}]$, so it is not a UFD, so it is also not Euclidean nor a PID.

Using this last result, we can prove Theorem 77 Eisenstein's criterion (credited to Richard Singer in the source I follow)

Proof. Recall the conditions of the theorem: if $f(x) \in \mathbb{Z}[x]$ is a polynomial with degree n at least 2, and $p \in \mathbb{N}$ is a prime, and:

1. $p \nmid a_n$
2. $p^2 \nmid a_0$
3. $\forall i \in [0, n-1] (p \mid a_i)$

Then $f(x)$ is irreducible over $\mathbb{Q}$. We prove the theorem by contradiction: suppose all conditions hold, yet $f(x)$ is reducible over $\mathbb{Q}$, so it is reducible over $\mathbb{Z}$ by Theorem 75 Gauss' lemma (reducibility), so we have $f(x) = g(x)h(x)$ for some $g(x), h(x) \in \mathbb{Z}[x]$ that are not units, and have degree at least 1, so both have degree less than n (see construction in Theorem 76 for more detail).

Let $\psi : \mathbb{Z}[x] \rightarrow \mathbb{Z}_p[x]$ be the natural homomorphism, and denote $f'(x) = \psi(f(x)), g'(x) = \psi(g(x)), h'(x) = \psi(h(x))$ be the reduced polynomials in $\mathbb{Z}_p[x]$. Also, note that $\deg(\psi(a(x))) \leq$

$\deg(a(x))$ for every non-zero polynomial $a(x) \in \mathbb{Z}[x]$, or $\psi(a(x)) = 0$.

By supposition. all non-leading coefficients of $f(x)$ are divisible by p , so $f'(x) = a_n x^n$. Since $\mathbb{Z}_p$ is a field by 48 $\mathbb{Z}_p$ is a field, then by 88.1 $F[x]$ with F field is a UFD $\mathbb{Z}_p[x]$ is a UFD. Note that x is irreducible on $\mathbb{Z}_p[x]$, so the factorization of $f'(x)$ necessarily has x with multiplicity n as a factor, and likewise for the factorizations of $g'(x), h'(x)$ since $f'(x) = \psi(f(x)) = \psi(g(x)h(x)) = \psi(g(x))\psi(h(x)) = g'(x)h'(x)$ (since ψ is a homomorphism), so $g'(x) = bx^r, h'(x) = cx^{n-r}$ for some $b, c \in \mathbb{Z}_p, r \in \mathbb{N}$ s.t. $r \in [0, n]$ and $bc = a_n$.

If $r = 0$ or $r = n$, then one of $g'(x), h'(x)$ is a polynomial of degree n , but both polynomials must have degree less than n since the degree of a polynomial does not increase under ψ , so both cases are impossible, i.e. $g'(x) = bx^r, h'(x) = cx^{n-r}$ both with degree at least 1, so both have x as a factor, so both have 0 as a zero, i.e. $g(0) \equiv 0 \pmod{p}, h(0) \equiv 0 \pmod{p}$, so $p^2 \mid g(0)h(0)$, but $g(0)h(0) = f(0) = a_0$, so $p^2 \mid a_0$, a contradiction, so it must be that $f(x)$ is irreducible over $\mathbb{Q}$.

□

We have reached a milestone where it is worth considering the hierarchy of the algebraic structures we have seen so far. The following graph should be read as follows: an arrow indicates that the structure at the tail of the arrow always satisfies the properties of the structure at the head of the arrow, but this is not an IFF relation (for example, $\mathbb{Z}[x]$ is a UFD although it is not a PID).

field $\longleftarrow$ ED $\longleftarrow$ PID $\longleftarrow$ UFD $\longleftarrow$ $\int$ -domain $\longleftarrow$ comm ring $\longleftarrow$ ring $\longleftarrow$ rng $\longleftarrow$ abelian group

Extension Fields

Consider $\mathbb{Z}$ (an integral domain) and $\mathbb{Q}$ (a field). From previous knowledge, we know that $\mathbb{Q}$ contains a copy of $\mathbb{Z}$ as a subring, although when formalizing this relation via construction of $\mathbb{Q}$ as per 0.0.29 field of fractions, we find that $\mathbb{Q}$ does not actually contain $\mathbb{Z}$ as a subring, but rather contains an isomorphic copy of $\mathbb{Z}$ as a subring, however it is easier to think of $\mathbb{Z}$ as a proper subring of $\mathbb{Q}$ without explicitly mentioning the homomorphism $\mathbb{Z} \rightarrow \mathbb{Q}, z \mapsto \frac{z}{1}$. It is useful to think of $\mathbb{Q}$ as an *extension* of the integral domain $\mathbb{Z}$ into a field.

The same goes for $\mathbb{Q}$ versus $\mathbb{R}$ (both fields). While in formal constructions of $\mathbb{R}$ from $\mathbb{Q}$ via e.g. Dedekind cuts (which we have talked about before in [Introduction to Set Theory](#)) we find that $\mathbb{Q}$ is not a proper subfield of $\mathbb{R}$, it is useful to think of $\mathbb{R}$ as an *extension* of the field $\mathbb{Q}$.

This brief discussion motivates the following definition.

Definition 0.0.55 (extension field).

If F is a field and E is a field that has a subfield isomorphic to F , then we call E an extension field of F , and F the base field of E .

Via abuse of language, we tend to think of the above as E is an extension field of F if E restricted to F is equal to F .

Example 88.2 (extension fields).

$\mathbb{C}$ is an extension of $\mathbb{R}$, which itself is an extension of $\mathbb{Q}$. Note that $\mathbb{C}$ is also an extension of $\mathbb{Q}$, and $\mathbb{R}$ is also an extension of itself - extension fields define a natural equivalence relation.

We also see that $\mathbb{Q}[\sqrt{D}]$, defined in a similar manner to 0.0.47 quadratic integer ring only with rational coefficients rather than simply ring coefficients, is an extension of $\mathbb{Q}$, and that $\mathbb{R}$ is itself an extension of $\mathbb{Q}[\sqrt{D}]$.

Lemma 88.1.

If F is a field and $f(x) \in F[x]$ is an irreducible polynomial over F and $f(x) \neq 1$, then $F[x]/\langle f(x) \rangle$ is an extension field of F .

Proof. Note that since F is a field, $F[x]$ is a PID by Theorem 72 polynomial ring over a field is a PID. By supposition, $f(x)$ is irreducible, and since $F[x]$ is a PID then by Theorem 84 in a PID, prime and irreducible are equivalent it is also prime, and by 84.1 in a PID, maximal ideal and prime ideal are interchangeable $\langle f(x) \rangle$ is maximal, so by Theorem 63 $E = F[x]/\langle f(x) \rangle$ is a field. It remains to show that F is embedded in E , i.e. find an injective field homomorphism $F \rightarrow E$, and indeed we have a natural injective ring homomorphism $\phi : F \rightarrow E, r \mapsto r + \langle f(x) \rangle$ (more rigorously, it should be noted that r on the RHS is actually the constant polynomial r in $F[x]$ and not $r \in F$, but at this point the reader should be able to fill such gaps themselves).

It is trivial to verify ϕ is indeed a field homomorphism (a field homomorphism is simply a ring homomorphism between fields). To verify it is injective, suppose not, then $r + \langle f(x) \rangle = s + \langle f(x) \rangle$ for some $r, s \in F, r \neq s$, so by Theorem 20 properties of a coset $r - s \in \langle f(x) \rangle$, but $0 \leq \deg(r - s) \leq \max\{\deg(r), \deg(s)\} = 0$ since r, s are constant polynomials, and $f(x)$ is irreducible and not the unit so $1 \leq \deg(f(x))$, so we have a polynomial of degree less than $f(x)$ in the principal ideal generated by $f(x)$, but then $r - s = f(x)h(x)$ for some $h(x) \in F[x]$ and by 63 degree equation for polynomial ring of integer domain $0 = \deg(r - s) = \deg(f(x)) + \deg(h(x))$ which is at least $\deg(f(x))$ which is greater than 0 by supposition, a contradiction, so it must be that $r = s$ (in which case the degree of $r - s$ is, of course, undefined), so ϕ is injective.

We conclude that ϕ is an injective field homomorphism $F \rightarrow E$, so E is an extension field of F .

□

Theorem 89 (fundamental theorem of field theory (Kronecker, 1887)).

If F is a field and $f(x) \in F[x]$ is a non-zero non-constant polynomial, then there exists an extension field E of F in which $f(x)$ has a zero.

Proof. Since F is a field then $F[x]$ is a UFD so we can write $f(x) = g(x)h(x)$ where $g(x)$ is irreducible, so by 88.1, $E = F[x]/\langle g(x) \rangle$ is an extension field of F . Recall that, in coset notation, $(r + \langle A \rangle)^n = r^n + \langle A \rangle$ and also $(r + \langle A \rangle)(s + \langle A \rangle) = (r + s) + \langle A \rangle$ by definition of coset addition.

Let $\alpha = x + \langle g(x) \rangle$, and write $g(x)$ explicitly as $g(x) = \sum_{i=0}^n a_i x^i$, then via the evaluation map of $g(x)$ in $E = F[x]/\langle g(x) \rangle$, we find:

$$\begin{aligned} g(\alpha) &= \\ g(x + \langle g(x) \rangle) &= \\ \sum_{i=0}^n a_i (x + \langle g(x) \rangle)^i &= \\ \sum_{i=0}^n a_i (x^i + \langle g(x) \rangle) &= \\ \sum_{i=0}^n a_i x^i + \langle g(x) \rangle &= \\ g(x) + \langle g(x) \rangle &=_{g(x) \in \langle g(x) \rangle} 0 \end{aligned}$$

So α is a zero of $g(x)$ in E , and since $g(x)$ is a factor of $f(x)$, then $f(0) = g(0)h(0) = 0h(0) = 0$, so it is a zero of $f(x)$ in E . □

The proof of the above provides a recipe for construction of such extension fields, as we will see in the next few examples:

Example 89.1 (Constructing $\mathbb{C}$ from $\mathbb{R}$).

Consider $f(x) = x^2 + 1 \in \mathbb{R}$. It is a polynomial of degree 2 with no zeros in $\mathbb{R}$, so it is irreducible over $\mathbb{R}$. Consider $E = \mathbb{R}[x]/\langle x^2 + 1 \rangle$. By the above theorem, $\alpha = x + \langle x^2 + 1 \rangle$ is a zero of $f(x)$ in E , and indeed $f(\alpha) = (x + \langle x^2 + 1 \rangle)^2 + (1 + \langle x^2 + 1 \rangle) = (x^2 + 1) + \langle x^2 + 1 \rangle = \langle x^2 + 1 \rangle = 0_E$.

By 67 zero implies reducible, we have $f(x) = (x - \alpha)h(x)$, and since $\deg(f(x)) = 2$ then $\deg(h(x)) = 1$ so the only choice for $h(x)$ is $x + \alpha$, so we have $f(x) = (x - \alpha)(x + \alpha)$ (it is not immediately obvious, but can be shown using the fact that $x^2 + \langle x^2 + 1 \rangle = -1 + \langle x^2 + 1 \rangle$), so we have $\pm\alpha$ are zeros of $x^2 + 1$ in E (where we agreed to identify 1 with $1 + \langle x^2 + 1 \rangle$ implicitly). Note that these zeros satisfy $(\pm\alpha)^2 = -1$, i.e. they agree with our definition for i as $\sqrt{-1}$, and indeed in 72.1 $\mathbb{R}[x]/\langle x^2 + 1 \rangle \cong \mathbb{C}$ we saw that this factor field is exactly the field of complex numbers.

The above construction provides a more natural way of forming $\mathbb{C}$ from $\mathbb{R}$.

Example 89.2.

Consider $f(x) = x^4 - 9x^2 + 18 = (x^2 - 6)(x^2 - 3)$ in $\mathbb{Q}[x]$. We know that it has no zeros in $\mathbb{Q}$ since 6, 3 are both not perfect squares. **Theorem 89 fundamental theorem of field theory (Kronecker, 1887)** provides us with instructions on how to extend $\mathbb{R}$ into fields where $f(x)$ has zeros by considering one of the irreducible factors and constructing a new field in which $f(x)$ has a zero originating from that factor.

We can consider $E_1 = \mathbb{Q}[x]/\langle x^2 - 6 \rangle$ and in that field we find $\alpha = x + \langle x^2 - 6 \rangle$ s.t. $f(\pm\alpha) = 0$, and likewise in $E_2 = \mathbb{Q}[x]/\langle x^2 - 3 \rangle$ we find that for $\beta = x + \langle x^2 - 3 \rangle$, we have $f(\pm\beta) = 0$.

In $E_1[x]$, we can factor $f(x)$ to $(x - \alpha)(x + \alpha)(x^2 - 3)$ and in $E_2[x]$, we can factor $f(x)$ to $(x^2 - 6)(x - \beta)(x + \beta)$ (recall that E_1, E_2 are extensions of $\mathbb{Q}$, but the polynomials themselves reside in the polynomial ring of those extension fields, i.e. $E_1[x], E_2[x]$). By combining both extension fields, we can find a factorization of $f(x)$ in terms of 4 linear factors:

First, instead of using x as the indeterminate, we use t to avoid confusion, and consider $E_1[t] = (\mathbb{Q}[x]/\langle x^2 - 6 \rangle)[t]$. As mentioned earlier, in this polynomial ring we find $f(t) = (t - \alpha)(t + \alpha)(t^2 - 3)$. We observe that $t^2 - 3$ is irreducible over E_1 , so again by considering $E_{12} = E_1[t]/\langle t^2 - 3 \rangle$ we find that $\beta = t + \langle t^2 - 3 \rangle$ is a zero of $f(t)$ in E_{12} , and similarly $-\beta$ is also a zero since $\deg(t^2 - 3) = 2$, so in conclusion we find that in the $E_{12} = (\mathbb{Q}[x]/\langle x^2 - 6 \rangle)[t]/\langle t^2 - 3 \rangle$, $z^4 - 9z^2 + 18 \in E_{12}[z]$ can be factored into $(z - \alpha)(z + \alpha)(z - \beta)(z + \beta)$, where $\pm\alpha, \pm\beta$ are zeros of $f(z)$.

Definition 0.0.56 (algebraic and transcendental).

If E is an extension field of a field F , then $a \in E$ is called:

1. algebraic over F if $\exists f(x) \in F[x], f(x) \neq 0$ for which $f(a) = 0$ (where f is evaluated in $\mathbb{C}$ via the evaluation map)
2. transcendental over F otherwise.

Example 89.3 (algebraic and transcendental).

Consider $i \in \mathbb{C}$. Since $\mathbb{C}$ is an extension field of $\mathbb{Q}$, and $f(x) = x^2 - 1$ is a polynomial in $\mathbb{Q}[x]$, and $f(i) = 0$ (evaluated in $\mathbb{C}$), so i is algebraic over $\mathbb{Q}$.

Let $r \in \mathbb{R}$, then trivially for $f(x) = x - r$ we find $f(r) = 0$ evaluated in $\mathbb{R}$, so every element in $\mathbb{R}$ is algebraic over $\mathbb{R}$. This is true in general, and not a special property of $\mathbb{R}$. As an concrete example, we find π and e are algebraic over $\mathbb{R}$ since $x - \pi, x - e \in \mathbb{R}[x]$ have zeros in π, e respectively.

On the other hand, by the [Lindemann–Weierstrass theorem](#) we can show that e, π are transcendental over $\mathbb{Q}$. Showing a number is transcendental is much more involved then showing it is algebraic.

$\sqrt{2}$ is algebraic over $\mathbb{Q}$, since it is a zero of $x^2 - 2 \in \mathbb{Q}[x]$.

If $a \in \mathbb{C}$ is a number constructed via a sequence of additions, subtraction, multiplication, and positive roots, it can be shown that a is always algebraic over $\mathbb{Q}$. The proof for this is rather straightforward via induction: essentially, every single operation can be “unravelling” via a matching inverse operation when constructing a polynomial: for example, if we took a positive root of a , we raise the matching polynomial expression to the exponent of that root, or if we added, we can subtract, etc. The next example demonstrates this process.

Example 89.4 (proving $2 + 5\sqrt{3 - 2\sqrt[3]{5}}$ is algebraic over $\mathbb{Q}$).

Write $a = 2 + 5\sqrt{3 - 2\sqrt[3]{5}}$. We proceed as follows:

$$\begin{aligned} a &= 2 + 5\sqrt{3 - 2\sqrt[3]{5}} \\ a - 2 &= 5\sqrt{3 - 2\sqrt[3]{5}} \\ \frac{1}{5}a - \frac{2}{5} &= \sqrt{3 - 2\sqrt[3]{5}} \\ \left(\frac{1}{5}a - \frac{2}{5}\right)^2 &= 3 - 2\sqrt[3]{5} \\ \left(\frac{1}{5}a - \frac{2}{5}\right)^2 - 3 &= -2\sqrt[3]{5} \\ \left(\left(\frac{1}{5}a - \frac{2}{5}\right)^2 - 3\right)^3 &= -8 \cdot 5 \\ \left(\left(\frac{1}{5}a - \frac{2}{5}\right)^2 - 3\right)^3 + 40 &= 0 \end{aligned}$$

Consider $f(x) = \left(\left(\frac{1}{5}x - \frac{2}{5}\right)^2 - 3\right)^3 + 40 \in \mathbb{Q}[x]$ (it is clear that this is a polynomial since it consists of multiplication, addition and subtraction of polynomials, and polynomials over a field are a ring (in fact, an PID) so they are closed under these operations). By the above calculation, it is obvious that $f(a) = 0$, so $a = 2 + 5\sqrt{3 - 2\sqrt[3]{5}}$ is indeed algebraic over $\mathbb{Q}$.

Theorem 90.

If F is a field with extension E , and $a \in E$, then a is transcendental over F if and only if the evaluation map $\phi_a : F[x] \rightarrow E$ (see 65.2 evaluation map) is injective.

Proof. By definition a is transcendental over F if and only if $\forall f(x) \in F[x], f(x) \neq 0 \implies f(a) \neq 0$, where $f(a)$ is simply $\phi_a(f(x))$. By definition, $\ker(\phi_a) = \{f(x) \in F[x] \mid \phi_a(f(x)) = 0\}$, so a is transcendental over F if and only if $\ker(\phi_a) = \{0\}$, and by Theorem 17 this is true if and only if ϕ_a is injective (recall that ϕ_a is a ring homomorphism). $\square$

It is worth stating that by Theorem 65 first isomorphism theorem for rings, this theorem also provides that $F[x] \cong \phi_a(F[x])$.

In number theory when a number is called *algebraic* or *transcendental*, it is considered in the context of the field $\mathbb{Q}$:

Definition 0.0.57 (algebraic and transcendental number).

$a \in \mathbb{C}$ is called:

1. A transcendental number if it is transcendental over $\mathbb{Q}$.
2. An algebraic number otherwise.

Hence, we say π and e are *transcendental numbers*.

Theorem 91.

If E is an extension of a field F , and $\alpha \in E$ is algebraic over F , then:

1. $\exists p(x) \in F[x]$ s.t. $p(x)$ is irreducible over F
2. If $f(x) \in F[x]$, $f(x) \neq 0$ s.t. $f(\alpha) = 0$, then $p(x) \mid f(x)$, and
3. $\deg(p(x)) \leq \deg(f(x))$, i.e. this $p(x)$ is the unique polynomial of least degree s.t. $p(\alpha) = 0$ up to a multiplication by a constant.

Proof. Suppose $\alpha \in E$ is algebraic over F , then the set $P = \{p(x) \in F[x] \mid p(\alpha) = 0, p(x) \neq 0\}$ is nonempty, so by the WOP on the degrees of the elements of P , there exists a polynomial of least degree in P , denote it $p(x)$. This proves (3).

If $p(x)$ is reducible over F , then $p(x) = h(x)g(x)$ for some non-unit $h(x), g(x) \in F[x]$, and also $p(\alpha) = h(\alpha)g(\alpha)$ hence $h(\alpha)g(\alpha) = 0$. Since F is a field, then either $h(\alpha) = 0$ or $g(\alpha) = 0$ since F has no zero-divisors, so α is a zero of one of the factors of $p(x)$, but since by assumption both are non-units then $\deg(h(x)), \deg(g(x)) < \deg(p(x))$ and yet w.l.o.g. $h(\alpha) = 0$, violating the minimality of $\deg(p(x))$, so $p(x)$ is irreducible over F , which proves (1).

To prove (2), we will show that P is an ideal: clearly it is non-empty since we assume α to be algebraic over F , so we proceed by **Theorem 58 ideal test**:

1. (closure under subtraction) let $a(x), b(x) \in P$, then $(a(x) - b(x))(\alpha) = a(\alpha) - b(\alpha) = 0$ (here we used the fact that the evaluation map is a ring homomorphism)
2. (closure under multiplication with elements of $F[x]$) let $a(x) \in P, f(x) \in F[x]$, then $a(x)f(x) = f(x)a(x)$ since $F[x]$ over a commutative ring is commutative, so it suffices to show $(a(x)f(x))(\alpha) = 0$, and indeed under the evaluation map we find $(a(x)f(x))(\alpha) = a(\alpha)f(\alpha) = 0f(\alpha) = 0$, so $a(x)f(x) \in P$.

We conclude P is an ideal. Since F is a field then $F[x]$ is a PID by **Theorem 72 polynomial ring over a field is a PID**, so $P = \langle t(x) \rangle$ for some $t(x) \in P$ s.t. $t(x)$ is a polynomial of least degree in P , so $P = \langle p(x) \rangle$, so $\forall f(x) \in P, f(x) = g(x)p(x)$ by definition of the principal ideal generated by an element, and by construction P is the set of all polynomials in $F[x]$ that have α as a zero over F , so every such polynomial is divisible by $p(x)$, which proves (2). □

Note that every polynomial $f(x) \in F[x]$ s.t. $f(\alpha) = 0$ and $f(x) \neq 0$ can also be written as a monic polynomial by dividing each coefficient by the leading coefficient, while maintaining the zero and the degree of the polynomial (since this is essentially multiplying by a constant polynomial). For example, $f(x) = 2x^2 + 2$ has i as a zero, and dividing all coefficients by 2 (the leading coefficient of $f(x)$) gives $g(x) = x^2 + 1$, which is monic, has same degree, and also has i as a zero. This fact, along with the theorem, justify the next definition:

Definition 0.0.58 (irreducible polynomial for α over F).

If E is an extension field of F and $\alpha \in E$ is algebraic over F , then the unique monic polynomial of least degree $p(x)$ which has α as a zero, is called the irreducible polynomial for α over F , or the minimal polynomial of α over F , and is denoted $\text{irr}(\alpha, F)$. The degree of $\text{irr}(\alpha, F)$ is called the degree of α over F , denoted $\deg(\alpha, F)$.

The previous theorem justifies the uniqueness of this polynomial and the use of the term *irreducible.

Example 91.1.

In $\mathbb{Q}$, we find that $x^5 - 2$ is the polynomial of least degree that has $\sqrt[5]{2}$ as a zero, so we have $\text{irr}(\sqrt[5]{2}, \mathbb{Q}) = x^5 - 2$ and also $\deg(\sqrt[5]{2}, \mathbb{Q}) = 5$.

Simple Extensions

Definition 0.0.59 (field adjunction).

If E is a field extension of F and $S \subseteq E$ where $S = \{\alpha_1, \dots, \alpha_n\}$ a finite subset of elements of E , then the smallest subfield of E that contains the set $F \cup S$ is called the field F with S adjoined (read “ F adjoin S ”) and denoted $F(\alpha_1, \dots, \alpha_n)$.

It is crystal clear that $F(\alpha_1, \dots, \alpha_n)$ is the intersection of all subfields of E that contain $F \cup S$.

Definition 0.0.60 (simple extension).

If in [0.0.59 field adjunction](#) we adjoin a set S with order 1 where $S = \{\alpha\}$ for some $\alpha \in E$, then $F(\alpha)$ as defined in that definition is called a simple extension of F , and we read it as “ F adjoin α ”.

At this point we have seen two similar notations: we used square brackets when discussing [0.0.38 polynomial ring](#) and [0.0.47 quadratic integer ring](#), and parenthesis when discussing [0.0.60 simple extension](#). In the case of a polynomial ring, we can think of $R[x]$ as R adjoined with the indeterminate x , and in the case of the quadratic integers $\mathbb{Z}[\sqrt{D}]$ we similarly think of it as $\mathbb{Z}$ adjoined with $\sqrt{D}$. Notice that this is closely ties to our definition of $F(\alpha)$. As we will see in the next theorem, the definitions agree if α is algebraic over F .

Note that the notation is somewhat different, since each notation carries different meaning: the square brackets carry the information that the base structure is a ring, while the parenthesis are usually used to denote that the base structure is a field, i.e. $T[x]$ usually hints that we think of T as a *ring*, while $T(x)$ hints that we think of T is a *field*. We will follow this rule throughout our discussion. In short:

1. When we write $T[x]$, we refer to an extension of a *ring*.
2. When we write $T(x)$, we refer to an extension of a *field*.

Theorem 92 ($F(\alpha)$ is isomorphic to the field of fractions of $F[\alpha]$).

Let F be a field and E be an extension over F and $\alpha \in E$, then $F(\alpha)$ is isomorphic to the field of fractions of $F[\alpha]$ (i.e. isomorphic to the field of rational functions in α).

Proof. In this proof we abuse the subset notation to denote “included via isomorphism”, i.e. $A \subseteq B$ means “ B contains an isomorphic copy of A ”.

By definition $F(\alpha)$ is the smallest subfield of E such that $F \cup \{\alpha\} \subseteq F(\alpha)$. Clearly $F[\alpha]$ contains an isomorphic copy of $F \cup \{\alpha\}$ in the form of its constant polynomials and the linear polynomial $f(\alpha) = \alpha$. Let $Q = \text{Frac}(F[\alpha])$ be the field of fractions of $F[\alpha]$ as defined in **0.0.29 field of fractions**, i.e. it is the field of rational functions in α . We know that Q is a field, and also $F[\alpha]$ is embedded in Q via the natural homomorphism from an integral domain to its field of fractions, so $F \cup \{\alpha\} \subseteq F[\alpha] \subseteq Q$. Via minimality of $F(\alpha)$, it follows that $F(\alpha) \subseteq Q$.

Since $F(\alpha)$ is a field and thus a ring containing F and α then it must contain all expressions of the form $\sum_{i=0}^n a_i \alpha^i$ for $a_i \in F$, so $F[\alpha] \subseteq F(\alpha)$. Since $F(\alpha)$ is a field, then it must contain all quotients of elements in its subring $F[\alpha]$, which is exactly Q , so $Q \subseteq F(\alpha)$.

We have $F(\alpha) \subseteq Q$ and $Q \subseteq F(\alpha)$, so via mutual inclusion $F(\alpha) = Q$. □

Theorem 93 ($F(\alpha) \cong F[\alpha]$ if α algebraic over F).

If E is a field extension of F and $\alpha \in E$ is algebraic over F , then $F(\alpha)$ as defined in **0.0.60 simple extension** is isomorphic to $F[\alpha]$ which is the polynomial ring over F with α as the indeterminate.

Proof. Consider $\phi : F[x] \rightarrow F[\alpha]$ given by the substitution homomorphism (i.e. we substitute x for α as the indeterminate of the univariate polynomial). Clearly, $\text{Im}(\phi) = F[\alpha]$, so by **Theorem 65 first isomorphism theorem for rings** $F[x]/\text{Ker}(\phi) \cong F[\alpha]$. Note that $\text{Ker}(\phi)$ is exactly the set of polynomials $f(x) \in F[x]$ s.t. $f(\alpha) = 0$.

There are two cases:

1. α is algebraic over F , then $\text{Ker}(\phi) = \langle \text{irr}(\alpha, F) \rangle$, so $F[\alpha] \cong F[x]/\langle \text{irr}(\alpha, F) \rangle$. In **Theorem 94** $F(\alpha) \cong F[x]/\langle \text{irr}(\alpha, F) \rangle$ (the next theorem), we show that the latter is isomorphic to $F(\alpha)$, so via the transitive property, $F[\alpha] \cong F(\alpha)$.
2. α is transcendental over F , then $\text{Ker}(\phi) = \{0\}$, so $F[\alpha] \cong F[x]$. Taking the field of fractions $\text{Frac}(F[\alpha])$, we find that it is isomorphic to the field of fractions of $F[x]$ by **Theorem 68 fraction fields of isomorphic integral domains are isomorphic**, which is just the field of rational functions in x . By **Theorem 92** $F(\alpha)$ is isomorphic to the field of fractions of $F[\alpha]$, we have that $F(\alpha)$ is isomorphic to the field of fractions of $F[x]$. □

Theorem 94 ($F(\alpha) \cong F[x]/\langle \text{irr}(\alpha, F) \rangle$).

If E is a field extension of F and $\alpha \in E$ is algebraic over F , then $F(\alpha) \cong F[x]/\langle \text{irr}(\alpha, F) \rangle$.

Proof. Consider $\phi : F[x] \rightarrow F(\alpha)$, $f(x) \mapsto f(\alpha)$, i.e. ϕ is the evaluation map over $F(\alpha)$ for $x = \alpha$. We know that ϕ is a ring homomorphism, so consider its kernel: let $f(x) \in \text{Ker}(\phi)$, then $f(\alpha) = 0$, so $f(x)$ is either $\text{irr}(\alpha, F)$, or it has $\text{irr}(\alpha, F)$ as a factor, so in all cases we find $f(x) \in \langle \text{irr}(\alpha, F) \rangle$, and obviously all polynomials which have $\text{irr}(\alpha, F)$ as a factor belong to $\text{Ker}(\phi)$, so $\text{Ker}(\phi) = \langle \text{irr}(\alpha, F) \rangle$, so by **Theorem 65 first isomorphism theorem for rings**, $F[x]/\langle \text{irr}(\alpha, F) \rangle \cong \text{Im}(\phi)$. It is clear that $\phi(x) = \alpha$ and that $\forall c \in C(\phi(c) = c)$, so since $F \cup \{\alpha\} \in \text{Im}(\phi)$ then by definition $F(\alpha) \subseteq \text{Im}(\phi)$.

Let $y \in \text{Im}(\phi)$, then $\phi(f(x)) = y$ for some $f(x)$, so $y = \sum_{i=0}^n a_i \alpha^i$ for some $\{a_i\} \in F$ by definition of the evaluation homomorphism, so clearly $y \in F(\alpha)$, so $\text{Im}(\phi) \subseteq F(\alpha)$, so $F(\alpha) = \text{Im}(\phi)$. We conclude $F(\alpha) \cong F[x]/\langle \text{irr}(\alpha, F) \rangle$.

□

Corollary 94.1.

If F is a field and $p(x) \in F[x]$ is irreducible over F and α, β are zeros of $f(x)$ in some extensions E_a, E_b respectively, then $F(\alpha) \cong F(\beta)$.

Proof. By **Theorem 94** $F(\alpha) \cong F[x]/\langle \text{irr}(\alpha, F) \rangle$ since $\text{irr}(\alpha, F) = \text{irr}(\beta, F) = p(x)$, then $F(\alpha) \cong F(\beta)$ by the transitive property of isomorphism as an equivalence relation. □

The next corollary lets us think of $F(\alpha)$ in terms of a finite dimensional vector space.

Corollary 94.2 ($\dim(F(\alpha)) = \deg(\alpha, F)$).

If E is a field extension of F and $\alpha \in E$ is algebraic over F and $1 \leq \deg(\alpha, F)$, then $F(\alpha)$ is a vector space (where F is the field of scalars) with dimension $\dim(F(\alpha)) = \deg(\alpha, F)$.

Proof. We leave it to the reader to show that $F(\alpha)$ satisfies all vector space axioms. We will show that $\deg(F(\alpha)) = \deg(\alpha, F)$. Let $\beta \in F(\alpha)$, then by **Theorem 94** $F(\alpha) \cong F[x]/\langle \text{irr}(\alpha, F) \rangle$, β corresponds to some unique coset $f(x) + \langle \text{irr}(\alpha, F) \rangle$ (in the sense that $\phi_\alpha(f(x) + \langle \text{irr}(\alpha, F) \rangle) = \beta$).

Since F is a field, then by **Theorem 71 polynomial division algorithm** we can find unique $q(x), r(x) \in F[x]$, $\deg(r(x)) < \deg(\alpha, F)$ s.t. $f(x) = q(x)\text{irr}(\alpha, F) + r(x)$, so by the properties of a coset, we find $f(x) + \langle \text{irr}(\alpha, F) \rangle = r(x) + \langle \text{irr}(\alpha, F) \rangle$.

Denote $\deg(\alpha, F) = n$ and $\text{irr}(\alpha, F) = p(x)$ and recall that α is identified with $x + \langle p(x) \rangle$, i.e. $\phi_\alpha(x + \langle p(x) \rangle) = \alpha$. First, since $\deg(r(x)) \leq n-1$, then we can write $r(x) = \sum_{i=0}^{n-1} a_i x^i$ for some $a_i \in F$. Next, since $\phi_\alpha(r(x) + \langle p(x) \rangle) = \beta$ by supposition, and ϕ_α is the evaluation homomorphism, then

$$\beta =$$

$$\phi_\alpha(r(x) + \langle p(x) \rangle) =$$

$$\sum_{i=0}^{n-1} \phi_\alpha(a_i x^i + \langle p(x) \rangle) =$$

$$\sum_{i=0}^{n-1} \phi_\alpha(a_i) \phi_\alpha(x + \langle p(x) \rangle)^i =$$

$$\sum_{i=0}^{n-1} \phi_\alpha(a_i) \alpha^i =$$

$$\sum_{i=0}^{n-1} b_i \alpha^i$$

So β is given by a linear combination of $e = \{1, \alpha, \alpha^2, \dots, \alpha^{n-1}\}$, so $\text{span}(e) = F(\alpha)$, so $\dim(F(\alpha)) \leq n$.

Next, we show e consists of a set of linearly independent vectors. Let A be a diagonal matrix where $A_{ii} = e_i$, and let $t \in F^n$, and consider At . Suppose $At = 0$ for some $t \neq 0$, then it follows that At corresponds to some polynomial $t(x)$ of degree $n-1$ s.t. $t(\alpha) = 0$ (because At is exactly $\phi_\alpha(t(x))$), but by supposition the least degree of a polynomial that has α as a zero is n , a contradiction, so the only solution to $At = 0$ is $t = 0$, so by definition the columns of A are linearly independent, i.e. e is a set of linearly independent vectors in $F(\alpha)$.

We have a set of n linearly independent vectors that span the vector field, so e is a basis of $F(\alpha)$, so $\dim(F(\alpha)) = |e| = n$.

□

Example 94.1 ($\mathbb{C}$ as a vector space).

By **Theorem 94** $F(\alpha) \cong F[x]/\langle \text{irr}(\alpha, F) \rangle$ we see that $\mathbb{R}[x]/\langle x^2 + 1 \rangle$ is isomorphic to $\mathbb{R}(i)$, and by a previous result we have seen that it is also isomorphic to $\mathbb{C}$, so $\mathbb{C}$ and $\mathbb{R}(i)$ are in a sense identical. Note that $\deg(i, \mathbb{R}) = 2$ (since $\text{irr}(i, \mathbb{R}) = x^2 + 1$), so by **94.2** $\dim(F(\alpha)) = \deg(\alpha, F)$ it follows that $\mathbb{C}$ is a 2 dimensional vector space. The corollary also provides us with the basis $\{1, i\}$ for $\mathbb{C}$.

Definition 0.0.61 (point-wise fixing mapping).

If $C \subseteq A, C \subseteq B$ and $\phi : A \rightarrow B$ is a mapping s.t. $\phi_C : C \rightarrow C$ is id_C , then we say that ϕ fixes F point-wise.

Corollary 94.3.

If all conditions of 94.1 are met, then there exists an isomorphism $\phi : F(\alpha) \rightarrow F(\beta)$ s.t. $\alpha \mapsto \beta$ and ϕ fixes F pointwise.

Proof. Suppose α, β are both zeros of an irreducible $f(x) \in F[x]$ over some extension fields. By 94.2 $\dim(F(\alpha)) = \deg(\alpha, F)$, and since $\text{irr}(\alpha, F) = \text{irr}(\beta, F) = f(x)$, then it follows $\dim(F(\beta)) = \dim(F(\alpha))$, and furthermore that corollary also gives an explicit base for $F(\alpha), F(\beta)$: let $n = \dim(F(\alpha))$, then $\{1, \dots, \alpha^{n-1}\}$ is a basis of $F(\alpha)$, and $\{1, \dots, \beta^{n-1}\}$ is a basis of $F(\beta)$.

Consider the mapping $\phi : F(\alpha) \rightarrow F(\beta)$ expressed in the following terms: for every $v \in F(\alpha)$, write $v = \sum_{i=0}^{n-1} a_i \alpha^i$, then $\phi(v) = \sum_{i=0}^{n-1} a_i \beta^i$. Note that this mapping is well-defined, since from

linear algebra we know that every vector has a unique coordinates $\begin{pmatrix} a_0 \\ \vdots \\ a_{n-1} \end{pmatrix}$ (suppose otherwise, then

$v = \sum_{i=0}^{n-1} a_i \alpha^i = \sum_{i=0}^{n-1} b_i \alpha^i$, so $\sum_{i=0}^{n-1} (a_i - b_i) \alpha^i = 0$, and since $\{1, \dots, \alpha^{n-1}\}$ is a set of n linearly independent vectors then $a_i = b_i$ for every i by definition otherwise it violates linear independence of the basis).

Clearly, $\phi(\alpha) = \beta$ and $\forall f \in F$, we have $f = 1 \cdot f$ so $\phi(f) = f$, so ϕ acts of $F(\alpha)$ as desired. It is easy to verify that ϕ also preserves addition, multiplication and unity, so it is a ring homomorphism, and clearly $\phi(a) = 0 \iff a = 0$ so it is injective, and finally it is trivially surjective since for every $y = \sum_{i=0}^{n-1} a_i \beta^i$ in $F(\beta)$, we have $v = \sum_{i=0}^{n-1} a_i \alpha^i \in F(\alpha)$ s.t. $\phi(v) = y$. So, we conclude ϕ is an isomorphism that satisfies the specified properties. □

We later provide a different proof for a more general claim to the one above, in our discussion of splitting fields.

Theorem 95.

If E is an extension field of F and $S \subseteq E, S = \{a_1, \dots, a_n\}$, then:

1. $F(a_1, a_2) = F(a_1)F(a_2) = F(a_2)(a_1)$
2. $F(a_1, \dots, a_n) = F(a_1)(\dots)(a_n)$

Proof. 1. By definition, $F(a_1, a_2)$ is the smallest subfield of E containing a_1, a_2 and F , so clearly $F(a_1, a_2) \subseteq F(a_1)(a_2)$. Also by definition of a simple extension, $F(a_1) \subseteq F(a_1, a_2)$, so we have $F(a_1) \subseteq F(a_1, a_2) \subseteq F(a_1, a_2)$, but by definition $F(a_1)(a_2)$ is the smallest subset containing both $F(a_1)$ and a_2 , and clearly $F(a_1, a_2)$ contains $F(a_1)$ and also a_2 , so $F(a_1)(a_2) \subseteq F(a_1, a_2)$, so we conclude $F(a_1)(a_2) = F(a_1, a_2)$. An identical argument for $F(a_2)(a_1)$ shows $F(a_2)(a_1) = F(a_1, a_2)$, which completes the proof of (1).

2. Trivially from (1) via induction, left as an exercise.

□

The above theorem lets us forget about the order in which we adjoined S to F . For example, to add $\sqrt{2}, \sqrt{3}$ to $\mathbb{Q}$ and form $\mathbb{Q}$ adjoined with $\sqrt{2}, \sqrt{3}$, we can:

1. Extend $\mathbb{Q}$ to $\mathbb{Q}(\sqrt{2})$, then extend $\mathbb{Q}(\sqrt{2})$ to $\mathbb{Q}(\sqrt{2})(\sqrt{3})$.
2. Extend $\mathbb{Q}$ to $\mathbb{Q}(\sqrt{3})$, then extend $\mathbb{Q}(\sqrt{3})$ to $\mathbb{Q}(\sqrt{3})(\sqrt{2})$.
3. Extend $\mathbb{Q}$ to $\mathbb{Q}(\sqrt{2}, \sqrt{3})$.

By the theorem, all three constructions are the same.

Splitting Fields

Definition 0.0.62 (splitting field).

If E is an extension field of F and $f(x) \in F[x]$ has degree at least 1, then we say that $f(x)$ splits in E if $\exists a \in F$ and $\alpha_1, \dots, \alpha_n \in E$ s.t. $f(x) = a \prod_{i=1}^n (x - \alpha_i)$. If $E = F(\alpha_1, \dots, \alpha_n)$, then we say that E is a splitting field for $f(x)$ over F .

Equivalently, we say that E is a splitting field for $f(x)$ over F if E is an extension of F and $f(x)$ can be factored into linear factors over E .

Example 95.1 $\mathbb{R}(i)$ is a splitting field for $x^2 + 1$ over $\mathbb{R}$.

Consider our main example again of $\mathbb{R}[x]/\langle x^2 + 1 \rangle$. We have seen that $\mathbb{R}[x]/\langle x^2 + 1 \rangle \cong \mathbb{R}(i)$, and that in $\mathbb{R}(i)[x]$, we can write $x^2 + 1$ as $(x + i)(x - i)$, i.e. $x^2 + 1$ can be factored into linear factors over $\mathbb{R}(i)$, so by definition $\mathbb{R}(i)$ is a splitting field of $x^2 + 1$ over $\mathbb{R}$.

Example 95.2 ($\mathbb{Q}(\sqrt{3}, \sqrt{6})$ is a splitting field of $x^4 - 9x^2 + 18$ over $\mathbb{Q}$).

Consider 89.2 in which we considered $f(x) = x^4 - 9x^2 + 18 = (x^2 - 6)(x^2 - 3)$ over $\mathbb{Q}$. We have seen that in $\mathbb{Q}(\sqrt{3})(\sqrt{6}) = \mathbb{Q}(\sqrt{3}, \sqrt{6})$ (by Theorem 95), $f(x)$ is splittable, so $\mathbb{Q}(\sqrt{3}, \sqrt{6})$ is a splitting field for $f(x)$ over $\mathbb{Q}$.

Note that the base field is obviously important here - since $\pm\sqrt{3}, \pm\sqrt{6} \in \mathbb{R}$, then $\mathbb{R}$ also splits $f(x)$ over itself (but it is not a splitting field of $f(x)$ since it contains other elements such as $\sqrt{5}$ which is not in $\mathbb{Q}(\sqrt{3}, \sqrt{6})$, however in $\mathbb{Q}$ we need to consider a proper extension to allow for $f(x)$ to be splittable.

Theorem 96 (splitting field existence theorem).

If F is a field and $f(x) \in F[x]$ has degree at least one, then there exists a splitting field E for $f(x)$ over F .

Proof. We prove by induction on $\deg(f(x))$. If $\deg(f(x)) = 1$, then we're done as $f(x)$ is already splittable over F since $f(x) = ax + b$ where $a, b \in F$ and $-b^{-1}a \in F$ is the only zero of $f(x)$ (by 66 a polynomial of degree n has at most n zeros). Suppose the proposition holds for all degrees less than some $1n$. By Theorem 89 fundamental theorem of field theory (Kronecker, 1887), there exists an extension field E of F in which $\exists \alpha_n \in E$ s.t. $f(\alpha_n) = 0$, so we can write $f(x) = (x - \alpha_n)g(x)$ in $E[x]$. By 63 degree equation for polynomial ring of integer domain, $\deg(g(x)) = \deg(f(x)) - 1 = n - 1$, so by the induction hypothesis there exists a splitting field $E_{n-1} = F(\alpha_1, \dots, \alpha_{n-1})$. Consider E_{n-1} adjoined with α_n , then we have a new field $F(\alpha_1, \dots, \alpha_{n-1})(\alpha_n) = F(\alpha_1, \dots, \alpha_n)$, where $\alpha_1, \dots, \alpha_n$ are n zeros of $f(x)$, so $F(\alpha_1, \dots, \alpha_n)$ is a splitting field for $f(x)$ over F . $\square$

Essentially, the previous theorem constructs the splitting field of a polynomial by recursively applying Theorem 89 fundamental theorem of field theory (Kronecker, 1887) until no non-linear irreducible factors are left for $f(x)$.

Splitting fields play a pivotal role in Galois theory.

Theorem 97 (zeros of an irreducible over a splitting field).

If $f(x) \in F[x]$ is irreducible over F and E is a splitting field for $f(x)$ over F , then all zeros of $f(x)$ in E have same multiplicity.

Proof. Suppose α, β are zeros of $f(x)$, then $f(x) = (x - \alpha)^n(x - \beta)^m g(x)$ by definition for some $n, m \in \mathbb{N}^+$, $g(x) \in E[x]$. By 94.3 there exists an isomorphism $\phi : F(\alpha) \rightarrow F(\beta)$ that exchanges the roles of α, β , so it must be that α has multiplicity at least m . Likewise, since $\phi^{-1} : F(\beta) \rightarrow F(\alpha)$ is an

isomorphism satisfying the same property, we have that the multiplicity of β is at least that of α , so we have $n \leq m$ and $m \leq n$, so $m = n$, so all zeros of $f(x)$ have same multiplicity. □

Suppose E, K are both splitting fields for $f(x) \in F[x]$ over F . The question arises, is $E \cong K$? The answer is yes, as we will now show.

Lemma 97.1.

Let F, F' be fields and E, K be extension fields of F, F' respectively and $\alpha \in E, \beta \in K$, then if:

1. $F \cong F'$ via $\phi : F \rightarrow F'$
2. α is algebraic over F with minimal polynomial $f(x) \in F[x]$
3. β is a zero of the image of $\bar{\phi}(f(x)) \in F'[x]$, where $\bar{\phi}$ is the natural extension of ϕ to a mapping between the polynomial rings, given by $\bar{\phi} : F[x] \rightarrow F'[x], \sum_{i=0}^n a_i x^i \mapsto \sum_{i=0}^n \bar{\phi}(a_i) x^i$.

Then $F(\alpha) \cong F'(\beta)$, and furthermore $\exists! \tilde{\phi} : F(\alpha) \rightarrow F'(\beta)$ s.t. $\tilde{\phi}|_F = \phi$ (i.e. $\tilde{\phi}$ agrees with ϕ on F) and $\tilde{\phi}(\alpha) = \beta$.

Proof. By **Theorem 94** $F(\alpha) \cong F[x]/\langle \text{irr}(\alpha, F) \rangle$, we have that $F(\alpha) \cong F[x]/\langle p(x) \rangle$ via some isomorphism $\sigma : F[x]/\langle p(x) \rangle \rightarrow F(\alpha)$, and $F'(\beta) \cong F'[x]/\langle \bar{\phi}(p(x)) \rangle$ via an isomorphism $\tau : F'[x]/\langle \bar{\phi}(p(x)) \rangle \rightarrow F'(\beta)$. By **Theorem 64 properties of ring homomorphism** that $\langle \bar{\phi}(p(x)) \rangle = \bar{\phi}(\langle p(x) \rangle)$. Denote $\bar{p}(x) = \bar{\phi}(p(x))$, and consider a mapping $\psi : F[x] \rightarrow F'[x]/\langle \bar{p}(x) \rangle$ given by $\psi : f(x) \mapsto \bar{\phi}(f(x)) + \langle \bar{p}(x) \rangle$. It is trivial to see that ψ is a ring homomorphism from the properties of $\bar{\phi}$ as a homomorphism and the definition of addition and multiplication in factor rings. Also, $\psi(f(x)) = 0 \iff f(x) \in \langle \bar{p}(x) \rangle$ by **Theorem 20 properties of a coset**, so $\text{Ker}(\psi) = \langle \bar{p}(x) \rangle$, so by **Theorem 65 first isomorphism theorem for rings** $F[x]/\langle p(x) \rangle \cong F'[x]/\langle \bar{\phi}(p(x)) \rangle$. So by the transitive property, $F(\alpha) \cong F'(\beta)$.

We can find the implicit isomorphism that arises from this construction: let $\bar{\psi} : F[x]/\langle p(x) \rangle \rightarrow F'[x]/\langle \bar{\phi}(p(x)) \rangle$ be the isomorphism that arises naturally from ψ , then $\bar{\psi} : f(x) + \langle p(x) \rangle \mapsto \bar{\phi}(f(x)) + \langle \bar{\phi}(p(x)) \rangle$ (see proof for **Theorem 65 first isomorphism theorem for rings** for the construction of this isomorphism), then since the composition of isomorphisms is an isomorphism, we have that $\omega = \tau \circ \bar{\psi} \circ \sigma^{-1} : F(\alpha) \rightarrow F'(\beta)$ is an isomorphism. Let $f \in F$, then $\omega(f) = \tau(\bar{\psi}(\sigma^{-1}(f))) = \tau(\bar{\psi}(f)) = \tau(\bar{\phi}(f)) = \phi(f)$, so ω agrees with ϕ on F , and also $\omega(\alpha) = \tau(\bar{\psi}(\sigma^{-1}(\alpha))) = \tau(\bar{\psi}(x)) = \tau(x) = \beta$, so $\omega(\alpha) = \beta$, as desired. (in the theorem, we denoted this ω by $\tilde{\phi}$).

Next, we show uniqueness. Since every element of $F(\alpha)$ is given by $x = \sum_{i=0}^n a_i \alpha^i$ for $a_i \in F$, then by properties of a homomorphism any isomorphism that agrees with ϕ on F and sends α to β must map x to $\sum_{i=0}^n \phi(a_i) \beta^i$, so such mapping is unique, so ω is unique.

The following commutative diagram illustrates the relations between these structures:

$$\begin{array}{ccccc}
 & & F(\alpha) \xrightarrow{\omega=\tau\circ\bar{\psi}\circ\sigma^{-1}} F'(\beta) & & \\
 & \nearrow \sigma & & \nwarrow \tau & \\
 F[x]/\langle p(x) \rangle & \xrightarrow{\bar{\psi}} & F'[x]/\langle \bar{\phi}(p(x)) \rangle & & \\
 & \nwarrow & & \nearrow & \\
 & & F[x] \xrightarrow{\bar{\phi}} F'[x] & & \\
 \uparrow & & \uparrow & & \\
 F & \xrightarrow{\phi} & F' & &
 \end{array}$$

□

Theorem 98 (splitting fields of polynomials under isomorphism are isomorphic).

Let F, F' be isomorphic via $\phi : F \rightarrow F'$ and $p(x) \in F[x]$ be a non-constant polynomial and E, K be splitting fields for $p(x)$ over $F, \bar{\phi}(p(x))$ (defined naturally via ϕ) over F' , then $E \cong K$.

Proof. Let $\alpha_1, \dots, \alpha_n$ be the zeros of $p(x)$ in the splitting field E and let $\beta_1, \dots, \beta_n$ be the zeros of $q(x) = \bar{\phi}(p(x))$ in the splitting field K , so by definition $E = F(\alpha_1, \dots, \alpha_n)$ and $K = F'(\beta_1, \dots, \beta_n)$.

We prove by induction on n : for $n = 1$, by 97.1 $F(\alpha_1) \cong F'(\beta_1)$. Suppose the theorem holds for some n , and consider $n + 1$, then by Theorem 95, $F(\alpha_1, \dots, \alpha_{n+1}) = F(\alpha_1, \dots, \alpha_n)(\alpha_{n+1})$ and also $F'(\beta_1, \dots, \beta_{n+1}) = F'(\beta_1, \dots, \beta_n)(\beta_{n+1})$. By the induction hypothesis, $F(\alpha_1, \dots, \alpha_n) \cong F'(\beta_1, \dots, \beta_n)$, so again we have fields of the form $P(\alpha')$ and $P'(\beta')$ where $P \cong P'$ so again by 97.1, $P(\alpha') \cong P'(\beta')$, so by induction $F(\alpha_1, \dots, \alpha_n) \cong F'(\beta_1, \dots, \beta_n)$, so $E \cong K$.

This construction is expressed below:

$$\begin{array}{ccc}
 E & \xrightarrow{\bar{\omega}} & K \\
 \uparrow & & \uparrow \\
 F(\alpha) & \xrightarrow{\omega} & F'(\beta) \\
 \uparrow & & \uparrow \\
 F & \xrightarrow{\phi} & F'
 \end{array}$$

□

Corollary 98.1 (uniqueness of splitting field).

If F is a field and $p(x) \in F[x]$ is non-constant, and E is a splitting field for $p(x)$ over F , then it is unique up to an isomorphism, which fixes F .

Proof. By **Theorem 98** **splitting fields of polynomials under isomorphism are isomorphic**, substituting F' for F , we find that every two splitting fields for $p(x)$ over F are isomorphic, so the splitting field is unique up to an isomorphism that also fixes F . $\square$

Let $p(x)$ be irreducible over F , then in this section we have established that there is a unique (up to isomorphism) splitting field E over F for $p(x)$. We have also seen that clearly, a polynomial may split in a field which is not its splitting field. Consider a simple example: let $x^2 - 2 \in \mathbb{Q}$, then $\mathbb{Q}(\sqrt{2}, -\sqrt{2}) = \mathbb{Q}(\sqrt{2})$ is a splitting field for $x^2 - 2$ over $\mathbb{Q}$, however $x^2 - 2$ also splits in $\mathbb{R}$, so we expect that $\mathbb{R}$ also contains a subfield K s.t. K is a splitting field for $x^2 - 2$ over $\mathbb{Q}$, and also that K is isomorphic to $\mathbb{Q}(\sqrt{2})$ (by **98.1 uniqueness of splitting field**). More generally, we expect that if $p(x) \in F[x]$ splits over an extension E over F , and K is a splitting field for $p(x)$, then K is isomorphic to a subfield of E . The next theorem shows that this is true:

Theorem 99.

If F is a field and $p(x) \in F[x]$ splits in an extension E over F , and K is a splitting field for $p(x)$ over F , then K is isomorphic to a subfield of E .

Proof. Suppose $p(x)$ splits in E and also that $F \subseteq E$ (through an isomorphism, since we suppose E is an extension of F), and let $\alpha_1, \dots, \alpha_n$ be the zeros of $p(x)$ in E (not necessarily distinct), then $F(\alpha_1, \dots, \alpha_n)$ is a subfield of E and also by definition it is a splitting field for $p(x)$, so by **98.1 uniqueness of splitting field**, $F(\alpha_1, \dots, \alpha_n) \cong K$. $\square$

Now, suppose E is an extension over some field where every polynomial that has a zero in E also splits in E . We call such field a *normal extension field*. We make use of this definition later, when we discuss Galois theory.

Definition 0.0.63 (normal extension).

If E is an extension of a field F , and $\forall p(x) \in F[x]$, either $p(x)$ is irreducible over E or $p(x)$ splits over E (equivalently, if $p(x)$ has a zero in E then it splits over E), we say that E is a normal extension of F .

Example 99.1 ($\mathbb{Q}(\sqrt[3]{2})$ is not a normal extension of $\mathbb{Q}$).

Note that $\text{irr}(\sqrt[3]{2}, \mathbb{Q}) = x^3 - 2$ (by Eisenstein's criteria it is irreducible over $\mathbb{Q}$). By De Moivre's formula, we find that the other zeros of $x^3 - 2$ are complex, yet $\mathbb{Q}(\sqrt[3]{2}) \subset \mathbb{R} \subset \mathbb{C}$, so $x^3 - 2$ doesn't split over $\mathbb{Q}(\sqrt[3]{2})$ even though clearly this polynomial is reducible over that field, so this field is not a normal extension.

In the next section, we study the multiplicity of the zeros of (mostly irreducible) polynomials.

Multiple Zeros of a Polynomial

In [[#Zeros and Division]] we discussed the zeros of a polynomial and derived some elementary results. In 66 a polynomial of degree n has at most n zeros we say that $\deg(f(x)) = n$ implies that $f(x)$ has at most n zeros counting multiplicity. Here we discuss necessary and sufficient conditions for a zero of a polynomial to have multiplicity greater than 1, and for a polynomial to have such zero. We call such zero a *multiple zero*.

Definition 0.0.64 (multiple and simple zero).

If α is a zero of a polynomial $f(x)$ and α has multiplicity more than 1, then we say α is a multiple zero of $f(x)$. If a zero of $f(x)$ has multiplicity 1, we say it is a simple zero of $f(x)$.

Definition 0.0.65 (separable polynomial).

A polynomial $f(x)$ is separable if all its zeros are simple, i.e. if it has number of distinct zeros equal to its degree.

Example 99.2.

In $(x - 3)^2$, 3 has multiplicity 2 since $(x - 3)^2$ splits to $(x - 3)(x - 3)$ over e.g. $\mathbb{Z}$, so the zeros of $(x - 3)^2$ are $\{3, 3\}$ i.e. 3 has multiplicity 2, so it is a multiple zero of $(x - 3)^2$. Also, $(x - 2)(x + 5)(x - 7)$ is separable, since it has degree 3 and 3 distinct roots.

Consider $f(x) \in \mathbb{R}[x]$. If $\alpha \in \mathbb{R}$ is a zero of $f(x)$ with multiplicity greater than 1, then $f(x) = (x - \alpha)^n g(x)$ where $n \in \mathbb{N}$, $1 \leq n$. From calculus, we know by the product rule that $f'(x)$ (here we use $f'(x)$ to denote the *derivative* of $f(x)$, unlike in other sections where we used it freely to denote something that related to $f(x)$ but isn't $f(x)$) is given by $(n - 1)(x - \alpha)^{n-1}g(x) + (x - \alpha)^n g'(x)$. Clearly, we find that if $1 \leq n$ then $f'(\alpha) = 0$, so we find that if α is a multiple zero of $f(x)$ then $f(\alpha) = f'(\alpha) = 0$. Building upon this result it is easy to show by Bezout's identity for PIDs that $f(x)$, $f'(x)$ have a common factor. We would like to state this argument for a general polynomial ring over a field, but as the notion of a derivative from calculus makes use of limits which we cannot define on an arbitrary field, so instead we present a definition for the derivative of a polynomial which does not make use of limits, and agrees with the definition from calculus in case a limit can be defined on the polynomials at a point.

Definition 0.0.66 (formal derivative).

If F is a field and $f(x) = \sum_{i=0}^n a_i x^i \in F[x]$, then the formal derivative of $f(x)$, denoted $f'(x)$, is given by $f'(x) = \sum_{i=1}^n i \cdot a_i x^{i-1}$.

Example 99.3 (formal derivative of $x^3 - 5x^2 + 9$ over $\mathbb{Z}[x]$).

By definition, the formal derivative of $x^3 - 5x^2 + 9$ is given by $(2) \cdot (-5x^{2-1}) + 3 \cdot x^{3-1} = -10x + 3x^2$ over $\mathbb{Z}[x]$. Observe that this agrees with the derivative of $x^3 - 5x^2 + 9$ over $\mathbb{R}$ from calculus.

Theorem 100 (properties of the formal derivative).

If $f(x), g(x) \in F[x]$, $\alpha \in F$, then:

1. $(f(x) + g(x))' = f'(x) + g'(x)$ (linearity)
2. $(\alpha f(x))' = \alpha f'(x)$ (homogeneity)
3. $(f(x)g(x))' = f'(x)g(x) + g'(x)f(x)$ (product rule)

We will not prove this theorem here, and it should not come as a surprise to anyone familiar with elementary calculus. (1) and (2) follow immediately from the definition, and (3) follows from (1) and induction on $\deg(f(x))$.

Now we can complete the argument we started before presenting this definition and state it for any polynomial ring over a field:

Theorem 101 (criterion for multiple zeros).

Let F be a field and $f(x) \in F[x]$, then $f(x)$ has a multiple zero over some extension E over F if and only if $f(x), f'(x)$ have a common non-constant factor in $F[x]$.

Proof. $\implies$ Suppose $\alpha \in E$ is a multiple zero of $f(x)$, then $f(x) = (x - \alpha)^n g(x)$ for some $g(x) \in F[x]$, $1 \leq n$. By the product rule from **Theorem 100 properties of the formal derivative**, we have $f'(x) = n(x - \alpha)^{n-1}g(x) + (x - \alpha)^n g'(x)$. Clearly, $f'(\alpha) = 0$. If $f(x), f'(x)$ have no common non-constant factor in $F[x]$, then $\gcd(f(x), f'(x)) = 1$ and so $a(x)f(x) + b(x)f'(x) = 1$ for some $a(x), b(x) \in F[x]$ by **Theorem 72 polynomial ring over a field is a PID** and **Theorem 85 Bezout's identity for PIDs**. Evaluating $a(x)f(x) + b(x)f'(x)$ at $x = \alpha$, we have $0 = 1$, which is impossible, so $f(x), f'(x)$ must have a common non-constant factor in $F[x]$.

$\impliedby$ Suppose $\exists h(x) \in F[x]$ s.t. $1 \leq \deg(h(x))$ and also $h(x) \mid f(x), h(x) \mid f'(x)$, then $\exists g(x) \in F[x]$ s.t. $f(x) = h(x)g(x)$. Let α be a zero of $h(x)$ over some extension field E , then $h(x) = (x - \alpha)t(x)$ for some $t(x) \in F[x]$, so $f(x) = (x - \alpha)t(x)g(x)$. Denote $t(x)g(x) = a(x)$, then by the product rule $f'(x) = t(x) + (x - \alpha)t'(x)$. Observe that $f'(\alpha) = t(\alpha)$, but by supposition $h(x) \mid f'(x)$ so as $h(\alpha) = 0$ then $f'(\alpha) = 0$, so $t(\alpha) = 0$, so $x - \alpha \mid t(x)$, so it follows that $h(x) = (x - \alpha)^2 b(x)$ for some $b(x) \in F[x]$, so $h(x)$ has α as a multiple zero, so $f(x)$ has α as a multiple zero. □

Lemma 101.1.

If $f(x) \in F[x]$ is a non-constant irreducible over a field F , and $f(x)$ has a multiple zero, then $f'(x) = 0$ constantly.

Proof. Suppose $f(x) \in F[x]$ is a non-constant irreducible and has a multiple zero, then by **Theorem 101 criterion for multiple zeros**, it follows that $f(x), f'(x)$ have some $g(x) \in F[x]$ as a common factor s.t. $1 \leq \deg(g(x))$, but since $f(x)$ is irreducible over F then it must be that $f(x)$ and $g(x)$ associate, so

$f(x) = \alpha g(x)$ for some $\alpha \in F$, so homogeneity of the derivative $f'(x) = \alpha g'(x)$, but also we know that $f'(x) = g(x)h(x)$ for some $h(x) \in F[x]$ by supposition. If $f'(x)$ is non-zero, then $\deg(f'(x)) = \deg(g'(x)) = \deg(g(x)) + \deg(h(x))$ by the degree equation, but by definition of the derivative it is obvious that $\deg(g'(x)) < \deg(g(x))$ for every non-constant $g(x)$, a yet here we have $\deg(g(x)) \leq \deg(g'(x))$, a contradiction, so it follows that $h(x) = 0$, so $f'(x) = 0$. $\square$

Theorem 102 (multiple zeros of an irreducible).

Let $f(x) \in F[x]$ be irreducible over F , then:

1. If $\text{char}(F) = 0$, then $f(x)$ has no multiple zeros (i.e. it is separable)
2. Otherwise, $f(x)$ has a multiple zero only if $f(x) = g(x^{\text{char}(F)})$ for some $g(x) \in F[x]$.

Proof. Suppose $f(x) \in F[x]$ and suppose $f(x)$ has a multiple zero in some extension field E over F , then by 101.1 it follows that $f'(x) = 0$. Let $n = \deg(f(x))$, then $f(x) = \sum_{i=0}^n a_i x^i$ for some $a_i \in F$, and by definition $f'(x) = \sum_{i=1}^n i \cdot a_i x^{i-1}$. It follows that $f'(x) = 0$ if and only if $\forall i \in [1, n]$, $i \cdot a_i = 0$. If $\text{char}(F) = 0$, then this is impossible by definition of the characteristic. Otherwise, denote $\text{char}(F) = p$, then $i \cdot a_i = 0$ implies that i divides p , since this must hold for every non-zero coefficient a_i then it must be that $f(x) = g(x^p)$ for some $g(x) \in F[x]$. $\square$

The above theorem motivates studying the structure of fields that share this property. Such fields are called *perfect fields*.

Definition 0.0.67 (perfect field).

A field F is perfect if $\text{char}(F) = 0$ or if $\text{char}(F) = p \neq 0$ and $F^p = \{a^p | a \in F\} = F$.

The next theorem shows that all finite fields are perfect.

Theorem 103 (finite fields are perfect).

Every finite field is perfect.

Proof. Let F be a field and suppose F is finite, then F has characteristic p for some prime $p \in \mathbb{N}$ (the characteristic cannot be 0 because then it is impossible for F to be finite). Let $\phi_p : F \rightarrow F, a \mapsto a^p$. We will show that ϕ_p is an automorphism on F , i.e. ϕ_p is an isomorphism from F to itself.

First, we show ϕ_p is a ring homomorphism: let $a, b \in F$, then $\phi_p(a + b) = (a + b)^p$. We expand $(a + b)^p$ in F by the binomial theorem:

$$(a + b)^p = \frac{p!}{0!p!} a^p b^0 + \frac{p!}{1!(p-1)!} a^{p-1} b^1 + \cdots + \frac{p!}{(p-1)!1!} a^1 b^{p-1} + \frac{p!}{p!0!} a^0 b^p$$

Since p is prime (by 56.2 characteristic of integral domain since $\text{char}(F) \neq 0$ then it is prime), then

$p \mid p!$ and also $p \nmid k!$ for any $k \in [0!, (p-1)!]$, it follows that all binomial coefficients of the form $\binom{p}{k}$ where $k \neq 0, p$ are divisible by p , so all terms with these coefficients vanish since $\text{char}(F) = p$, so the only terms that remain are the first and the last (since in both cases we have $\binom{p}{0} = \binom{p}{p} = 1$, which is not divisible by p), and so

$$\phi_p(a+b) = (a+b)^p = a^p + b^p = \phi_p(a) + \phi_p(b)$$

This result is sometimes jokingly referred to as [Freshman's dream](#).

Next, we show ϕ_p preserves multiplication:

$$\phi_p(ab) = (ab)^p = a^p b^p = \phi_p(a) \phi_p(b)$$

Finally, we find that ϕ_p preserves unity, since trivially $\phi_p(1) = 1^p = 1$. So, ϕ_p is a ring (field) homomorphism from F to itself.

Next, we show ϕ_p is injective by showing that its kernel is 0. Suppose $\phi_p(a) = 0$, then $a^p = 0$, and by definition $aa^{p-1} = 0$ (since $1 \neq 0$ since p is prime). Since F is a field, then it has no zero-divisors, so either $a = 0$ or $a^{p-1} = 0$. By induction on p it is clear that it must be that $a = 0$ (since for every $1 \leq p$ it follows that if $a \neq 0$ then $a^{p-1} = 0$, and eventually we reach $p = 2$ for each $a = a^{p-1} = a^1$), so $\ker(\phi_p) = \{0\}$, so ϕ_p is injective.

Note that the results above hold for any field with characteristic $p \neq 0$.

Finally, we will show ϕ_p is surjective. Since ϕ_p is injective and F is finite, then $|F| = |\text{Im}(\phi_p)|$, and since $\text{Im}(\phi_p) \subseteq F$, then obviously $\text{Im}(\phi_p) = F$ (note that this is *not* the case if F is not finite - consider for example the mapping $f : (0, 2) \rightarrow (0, 1)$, $x \mapsto \frac{x}{2}$, clearly it is injective yet the sets $(0, 2)$ and $(0, 1)$ are not the same, even though $(0, 1) \subseteq (0, 2)$).

Since ϕ_p is injective, surjective, and a field homomorphism from F to itself, then it is an *automorphism* from F to itself, and clearly $\text{Im}(\phi_p) = F^p$ as defined in [0.0.67 perfect field](#), so F is a perfect field. $\square$

The mapping we defined in the proof of [Theorem 103 finite fields are perfect](#) is called the *Frobenius endomorphism*.

Definition 0.0.68 (Frobenius endomorphism).

The Frobenius endomorphism is a mapping $\phi_p : R \rightarrow R$, $a \mapsto a^p$, where R is a commutative ring with prime characteristic p .

By the proof of [Theorem 103 finite fields are perfect](#), the Frobenius endomorphism is indeed a commutative ring endomorphism (an *endomorphism* is a homomorphism from a structure to itself).

Theorem 104 (equivalent definitions of a perfect field).

The following are equivalent:

1. F is a perfect field.
2. F is a field with characteristic 0, or the Frobenius endomorphism is an automorphism on F .

Proof. $2 \implies 1$ is covered by **Theorem 102 multiple zeros of an irreducible**.

$1 \implies 2$: If F has characteristic 0 then it is perfect by definition. Otherwise, suppose $\text{char}(F) = p$ for some prime p and that $\phi_p : a \mapsto a^p$ is an automorphism on F , so $F = F^p$ since an automorphism is a surjection, so F is perfect. □

Theorem 105 (irreducible polynomials over a perfect field are separable).

If $f(x)$ is an irreducible polynomial over a perfect field F , then $f(x)$ is separable.

Proof. Suppose $f(x)$ is irreducible over a perfect field F . If $\text{char}(F) = 0$, then already it follows $f(x)$ has no multiple zeros by **Theorem 102 multiple zeros of an irreducible**. Otherwise, $\text{char}(F) = p$ for some prime $p \in \mathbb{N}$. Suppose $f(x)$ has a multiple zero, then by the theorem $f(x) = g(x^p)$ for some $g(x) \in F[x]$. Also, since F is perfect, then $\forall a \in F \exists b \in F$ s.t. $a = b^p$, so all coefficients a_i of $g(x)$ can be written as $a_i = b_i^p$ for some $b_i \in F$, so

$$g(x^p) = \sum_{i=0}^n a_i (x^p)^i = \sum_{i=0}^n b_i^p (x^p)^i = \sum_{i=0}^n (b_i x^i)^p = (*)$$

Since F has prime characteristic, then the Frobenius endomorphism is indeed an endomorphism on F by our proof of **Theorem 102 multiple zeros of an irreducible**, and so immediately we find that $(*) = \sum_{i=0}^n (b_i x^i)^p = \sum_{i=0}^n \phi_p(b_i x^i) = \phi_p(\sum_{i=0}^n b_i x^i)$, where the last transition is due to addition being preserved under a homomorphism. Denote $\sum_{i=0}^n b_i x^i = h(x)$, then $f(x) = g(x^p) = \phi_p(h(x)) = (h(x))^p$, but then $h(x)$ is a factor of $f(x)$, contradicting $f(x)$ being irreducible.

We conclude that it must be that $f(x)$ has no multiple zeros, so it is separable □

The last theorem motivates the following definition:

Definition 0.0.69 (separable extension).

If E is an extension of a field F , and $\alpha \in E$, its minimal polynomial $\text{irr}(\alpha, F)$ exists (i.e. α is algebraic over F) and is separable, then E is a separable extension field of F .

Algebraic Extensions

Definition 0.0.70 (algebraic extension).

An extension E of a field F is called algebraic if $\forall \alpha \in E$, α is algebraic over F . Otherwise, the extension is called a transcendental extension.

The above simplifies **0.0.69 separable extension**: we say that E is a separable extension over F if E is an algebraic extension and every minimal polynomial for $\alpha \in E$ is separable (the difference is that the requirement that such minimal polynomial exists is substituted with requiring that E be algebraic over F).

Theorem 106 (algebraic extension of a perfect field is separable).

If E is an algebraic extension of a field F , and F is a perfect field, then E is a separable extension of F .

Proof. Let $\alpha \in E$, then since E is algebraic, $\text{irr}(\alpha, F)$ exists. By **Theorem 105 irreducible polynomials over a perfect field are separable**, it is separable, hence by definition E is separable. $\square$

Definition 0.0.71 (finite extension).

If an extension E of a field F forms a vector space of finite dimension n over F , then we say that E is a finite extension of degree n and write $[E : F] = n$.

Note that we used a similar notation to denote the index of a subgroup as a part of our discussions on Lagrange's theorem and orbits and stabilizers. The context should make our intention using the notation clear.

Proposition 72.

If E is an extension of F , then $[E : F] = 1 \iff E = F$.

Proof. $\implies$ Suppose $[E : F] = 1$, then $\dim(E) = 1$ (where F is the field of scalars). Obviously, $1 \in E$, so 1 is a basis of E , but in **Theorem 52 F is a vector space over itself with dimension 1** we see that 1 is also a basis of F over itself, so $E = F$.

$\impliedby$ Suppose $E = F$, then by **Theorem 52 F is a vector space over itself with dimension 1** clearly $\dim(E) = \dim(F) = 1$, so $[E : F] = 1$. $\square$

Theorem 107 (finite extensions are algebraic).

If E is a finite extension over F , then E is algebraic over F .

Proof. Suppose E is a finite extension, then $[E : F] = n$ for some finite n , and let $\alpha \in E$. Consider the set $\{1, \alpha, \dots, \alpha^n\}$. It is a set of $n + 1$ vectors in E , and since $\deg(E) = n$ it is linearly dependent, so we can find some $c_0, \dots, c_n \in F$ s.t. $\sum_{i=0}^n c_i \alpha^i = 0$. Consider the polynomial $f(x) = \sum_{i=0}^n c_i x^i \in F[x]$. Clearly, $f(\alpha) = \sum_{i=0}^n c_i \alpha^i = 0$, so $f(x)$ has α as a zero, so α is algebraic over F , so E is algebraic over F . $\square$

Note that an algebraic extension is not always separable, but since most fields we are interested in are either finite (for example, fields that we identify with the congruence classes and modular arithmetic of a prime modulus p) or have characteristic 0 (for example, $\mathbb{Q}, \mathbb{R}, \mathbb{C}$) then most natural examples we'll see are of separable extensions as per **Theorem 106 algebraic extension of a perfect field is separable**. The theorem fails for infinite fields with nonzero characteristic, as we will show in the next example:

Example 107.1 (algebraic extension that is inseparable).

Consider $\mathbb{F}_2 = \{0, 1\}$ under modular arithmetic mod 2, and let t be transcendental over $\mathbb{F}_2$, then we can consider $\mathbb{F}_2(t)$ the group of rational functions in the indeterminate t (see discussion in **Theorem 92 $F(\alpha)$ is isomorphic to the field of fractions of $F[\alpha]$**). This field is infinite - suppose it is finite, and let $n = |\mathbb{F}_2(t)|$, then $A = \{0, t, t^2, \dots, t^n\}$ has $n + 1$ distinct elements and yet $A \subseteq \mathbb{F}_2(t)$, so it is infinite. However, $\text{char}(\mathbb{F}_2(t))$ is still 2 since $2 \cdot 1 \equiv 0$ (alternatively, since **56.1 unital subrings preserve characteristic** and $\mathbb{F}_2$ is a unital subring of $\mathbb{F}_2(t)$ and has characteristic 2, then $\text{char}(\mathbb{F}_2(t)) = 2$). It follows that $\mathbb{F}_2(t)$ is not perfect, so **Theorem 106 algebraic extension of a perfect field is separable** does not apply.

We now consider $\mathbb{F}_2(t)[x]$, i.e. the polynomial ring of $\mathbb{F}_2(t)$ in the indeterminate x , so $x^2 - t \in \mathbb{F}_2(t)[x]$. Also, recall that in the proof for **Theorem 103 finite fields are perfect** we have shown that every field of finite characteristic (not necessarily a finite field) exhibits a property which we called the Freshman's dream when calculating $(a + b)^p$ where p is the field characteristic: $(a + b)^p = a^p + b^p$. Since $\text{char}(\mathbb{F}_2(t)) = 2$, then $(x - \alpha)^2 = x^2 - \alpha^2$, so by equation coefficients we find $t = \alpha^2$, i.e. $\alpha = \pm\sqrt{t}$ (here we use $\sqrt{t}$ as notation for an element that, multiplied with itself, yields t , and we find that in the splitting field, $\sqrt{t} = -\sqrt{t} = \alpha$).

Suppose $\sqrt{t} \in \mathbb{F}_2(t)$, i.e. $\sqrt{t}$ is a rational function in t , so $\sqrt{t} = \frac{p(t)}{h(t)}$ for some $p(t), h(t) \in \mathbb{F}_2[t]$, so $t \cdot h(t) \cdot h(t) = p(t) \cdot p(t)$. By the degree equation, $\deg(t \cdot h(t) \cdot h(t)) = 1 + 2\deg(h(t))$, so it is odd, however $\deg(p(t) \cdot p(t)) = 2\deg(p(t))$, so it is even. Since parity partitions $\mathbb{Z}$, a number cannot be both even and odd, so we have a contradiction, so $\sqrt{t}$ is not a rational function, so $\sqrt{t} \notin \mathbb{F}_2(t)$.

It follows that $x^2 - t$ is irreducible over $\mathbb{F}_2(t)$, but we have seen that $\text{irr}(\alpha, \mathbb{F}_2(t)) = x^2 - t$ and clearly $\alpha = \sqrt{t} = -\sqrt{t}$ is a multiple zero of $x^2 - t$, so $\mathbb{F}_2(t)$ is inseparable since $x^2 - t \in \mathbb{F}_2(t)[x]$ is inseparable.

Consider the cubic $x^3 - 2 \in \mathbb{Q}$. We know that it is irreducible over $\mathbb{Q}$ by **Theorem 77 Eisenstein's criterion** with $p = 2$, and also from number theory we know $\sqrt[3]{2} \notin \mathbb{Q}$ so $\mathbb{Q}(\sqrt[3]{2})$ is a simple extension over $\mathbb{Q}$ with degree 3 (since $\deg(x^3 - 2) = 3$). However, as we have discussed in **99.1 $\mathbb{Q}(\sqrt[3]{2})$ is not a normal extension of $\mathbb{Q}$** , $x^3 - 2$ does not split in this field although it has a zero in it, so there exists an extension E over $\mathbb{Q}$ in which $x^3 - 2$ splits. The question about the relation of $[E : \mathbb{Q}]$, $[E : \mathbb{Q}(\sqrt[3]{2})]$ and $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}]$ naturally arises. We discuss it over the next few theorems:

Theorem 108 ($[K : F] = [K : E][E : F]$).

If E is a finite extension over F , and K is finite extension over E , then $[K : F] = [K : E][E : F]$.

Proof. Suppose F is a field, E is a finite extension over F , and K is a finite extension over E , and denote $[K : E] = n$, $[E : F] = m$. Consider E as a vector space over F , then by definition $[E : F] = \dim E$. Let $e_E \subseteq E$ be a basis of E , then $|e_E| = \dim E = [E : F] = m$, and $\forall \alpha \in E$ we can write α as a linear combination of vectors from e_E . Likewise, K is a vector space over E , and $e_K \subseteq K$ is a basis of K s.t. $|e_K| = \dim K = [K : E]$.

Let $\beta \in K$, then $\beta = \sum_{i=1}^n b_i e_{K,i}$, and also each $e_{K,i} \in E$ so $b_i = \sum_{j=1}^m a_{i,j} e_{E,j}$ for some $a_{i,j} \in F$, i.e.

$$\beta = \sum_{i=1}^n b_i e_{K,i} = \sum_{i=1}^n \left(\sum_{j=1}^m a_{i,j} e_{E,j} \right) e_{K,i} = \sum_{i=1}^n \sum_{j=1}^m a_{i,j} (e_{E,j} e_{K,i})$$

Let $e_{K'} = \{e_{E,j} e_{K,i} | i \in [1, n], j \in [1, m]\}$. Clearly $|e_{K'}| = nm = [K : E][E : F]$, and clearly by the above $\text{span}\{e_{K'}\} = K$ over F . We will show $e_{K'}$ is a set of linearly independent vectors in K over the field F . Suppose not, then for some $c_i, \dots, c_{nm}$ not all zero, we have $0 = \sum_{i=1}^{nm} c_i e_{K',i}$, which would imply $\sum_{i=1}^n d_i e_{K,i} = 0$ for some $d_i \in E$, but by supposition e_K is a basis so it must be that all d_i terms are 0, but these d_i terms are simply a linear combination of e_E with coefficients from F and since e_E is a linearly independent set then it must be that all coefficients of these terms are 0, so in conclusion $\sum_{i=1}^{nm} c_i e_{K',i} = 0$ has no non-zero solutions, so $e_{K'}$ is a linearly independent set over F , so $e_{K'}$ is a basis of K of F , so $[K : F] = nm = [K : E][E : F]$.

□

The above theorem can naturally be extended by induction to any ascending chain of extensions, so in particular we derive the following corollary:

Corollary 108.1 (degree of splitting field).

If $p(x)$ is irreducible over F , and $n = \deg(p(x))$, and E is a splitting field of $p(x)$ over F , then $[E : F] \leq n!$.

Proof. By definition of the splitting field of a polynomial and since $\deg(p(x)) = n$, then $E = F(\alpha_1, \dots, \alpha_n)$ where $\alpha_1, \dots, \alpha_n$ are zeros of $p(x)$ over E . By **98.1 uniqueness of splitting field** it suffices to consider one such splitting field E to prove the theorem. First, note that since $p(x)$ is irreducible over F , then $\alpha_1 \notin F$, so by **94.2 $\dim(F(\alpha)) = \deg(\alpha, F)$** and by **0.0.71 finite extension** we have $[F(\alpha_1) : F] = \deg(p(x)) = n$. Now, in $F(\alpha)$, we have that $p(x) = (x - \alpha_1)q(x)$ by **65**, so $[F(\alpha_1, \alpha_2) : F(\alpha)] \leq n - 1$. The above argument generalizes trivially to show that $[F(\alpha_1, \dots, \alpha_{i+1}) : F(\alpha_1, \dots, \alpha_i)] \leq n - i$.

Now, by the general form of **Theorem 108** $[K : F] = [K : E][E : F]$, we have

$$[E : F] =$$

$$[F(\alpha_1, \dots, \alpha_n) : F] =$$

$$\prod_{i=0}^{n-1} [F(\alpha_1, \dots, \alpha_{i+1}) : F(\alpha_1, \dots, \alpha_i)] \leq$$

$$\prod_{i=0}^{n-1} (n - i) =$$

$$n!$$

Which completes the proof.

□

Another corollary of **Theorem 108** $[K : F] = [K : E][E : F]$ is given below:

Corollary 108.2.

If E is an extension of F and $\alpha \in E$ is algebraic over F , and $\beta \in F(\alpha)$, then $\deg(\beta, F) \mid \deg(\alpha, F)$.

Proof. Clearly $F(\alpha)$ is a finite extension over F so $\deg(\alpha, F) = [F(\alpha) : F]$. Also, $F(\beta)$ is a finite extension over F so $\deg(\beta, F) = [F(\beta) : F]$. Since $\beta \in F(\alpha)$ then clearly $F(\beta) \subseteq F(\alpha)$ by definition of a simple extension so $F(\alpha)$ is a finite extension over $F(\beta)$, so by the theorem $\deg(\alpha, F) = [F(\alpha) : F] = [F(\alpha) : F(\beta)][F(\beta) : F] = [F(\alpha) : F(\beta)]\deg(\beta, F)$, so by definition $\deg(\beta, F) \mid \deg(\alpha, F)$. □

The above corollary plays an important role in determining whether certain numbers are constructible or not, and in answering classical construction problems that deal with constructing geometric shapes using a straightedge and a compass. The corollary also lets us prove that certain polynomials have no solution over a field, as illustrated below.

Example 108.1 ($x^3 - 2$ has no zero in $\mathbb{Q}(\sqrt{2})$).

We prove by contraposition. Suppose β is a zero of $x^3 - 2$ in $\mathbb{Q}(\sqrt{2})$. Note that $\deg(\sqrt{2}, \mathbb{Q}) = 2$. Observe that $x^3 - 2$ is irreducible over $\mathbb{Z}_2$ so by **Theorem 76** it is irreducible over $\mathbb{Q}$, so $\deg(\beta, \mathbb{Q}) = 3$. But, $2 \nmid 3$, so $\beta \notin \mathbb{Q}(\sqrt{2})$ by **108.2**, so $x^3 - 2$ has no zero in $\mathbb{Q}(\sqrt{2})$.

Theorem 109.

Let E be an algebraic extension of a field F , then $E = F(\alpha_1, \dots, \alpha_n)$ for some $S = \{\alpha_1, \dots, \alpha_n\}$ if and only if E is a finite extension of F .

Proof. $\implies$ Suppose $E = F(\alpha_1, \dots, \alpha_n)$. Consider the following iterative process:

1. Set $E_0 = F, i = 0$
2. Let $E_{i+1} = E_i(\alpha_{i+1})$ (such simple extension must exist since by supposition E is an algebraic extension of F), if $\alpha_{i+1} \in E_i$ then obviously $E_i(\alpha_{i+1}) = E_i$, so $[E_{i+1} : E_i] = 1$ by **72**. Otherwise, by **94.2** $\dim(F(\alpha)) = \deg(\alpha, F)$ it follows that E_{i+1} is a finite extension of E_i , and by transitivity of F .
3. Set $i = i + 1$ and repeat step 2 until $i = n$

We now have $E_n = F(\alpha_1)(\dots)(\alpha_n)$ and by step (2), E_n is a finite extension of F . By **Theorem 95**, we have that $F(\alpha_1)(\dots)(\alpha_n) = F(\alpha_1, \dots, \alpha_n) = E$, so E is a finite extension of F .

$\Leftarrow$ Suppose E is a finite extension of F , then by **Theorem 107** finite extensions are algebraic, E is algebraic over F . If $E = F$, then we're done. Otherwise, $E \neq F$, so since E is a finite extension then $1[E : F]$, then $\exists \alpha_1 \in E \setminus F$, so $F(\alpha_1)$ has dimension greater than 1 since $F(\alpha_1) \neq F$ so by **72** it follows that $1[F(\alpha_1) : F]$. If $E = F(\alpha_1)$, then we're done, otherwise we pick some $\alpha_2 \notin F(\alpha_1)$. By **Theorem 108** $[K : F] = [K : E][E : F]$ we have $[F(\alpha_1, \alpha_2) : F] = [F(\alpha_1, \alpha_2) : F(\alpha_1)][F(\alpha_1) : F]$. Since both terms on the LHS are greater than 1, then $[F(\alpha_1) : F][F(\alpha_1, \alpha_2) : F]$. If $F(\alpha_1, \alpha_2) = E$ then we're done, otherwise we continue this construction and since for every n , $[F(\alpha_1, \dots, \alpha_n) : F][F(\alpha_1, \dots, \alpha_n, \alpha_{n+1}) : F]$ as argued above, and $[E : F]$ is finite, then eventually we find some $F(\alpha_1, \dots, \alpha_m)$ contained in E and also with same degree as E over F , so $E = F(\alpha_1, \dots, \alpha_m)$.

□

We conclude with a theorem which identifies necessary conditions for a field adjoined set $F(\alpha_1, \dots, \alpha_n)$ to be identical to a simple extension $F(\beta)$ for some $\beta \in F(\alpha_1, \dots, \alpha_n)$.

Theorem 110 (primitive element theorem (Steinitz, 1910)).

If F is a field with characteristic 0, and α_1, α_2 are algebraic over F , then $\exists \beta \in F(\alpha_1, \alpha_2)$ s.t. $F(\alpha_1, \alpha_2) = F(\beta)$. We say that β is a primitive element of $F(\alpha_1, \alpha_2)$.

Proof. Let $\beta = \alpha_1 + d\alpha_2$. There are two cases:

1. β is a primitive element of $F(\alpha_1, \alpha_2)$, i.e. $F(\alpha_1, \alpha_2) = F(\beta)$, then we are done.
2. β is not a primitive element of $F(\alpha_1, \alpha_2)$, so it follows that $\alpha_2 \notin F(\beta)$, otherwise we have $\beta - d\alpha_2 = \alpha_1 \in F(\beta)$, so $F(\alpha_1, \alpha_2) = F(\beta)$ by definition.

So, suppose β is not a primitive element of $F(\alpha_1, \alpha_2)$ and also $\alpha_2 \notin F(\beta)$.

Let $p(x) = \text{irr}(\alpha_1, F(\beta))$ and $q(x) = \text{irr}(\alpha_2, F(\beta))$, and let K be a splitting field for $p(x), q(x)$ (which exists by **Theorem 96 splitting field existence theorem**). If $\deg(q(x)) = 1$, then by **72** it follows that $F(\beta) = F(\beta)(\alpha_2) = F(\beta, \alpha_2)$ and therefore β is a primitive element of $F(\alpha_1, \alpha_2)$, so suppose $1 < \deg(q(x))$, then $q(x)$ has at least two zeros counting multiplicity, but since $\text{char}(F) = 0$ then by **Theorem 105 irreducible polynomials over a perfect field are separable** then all zeros of $q(x)$ over K must be simple. By choice of $q(x)$, we know that α_2 is a zero of $q(x)$, so $\exists \alpha_3 \in K$ s.t. α_3 is a zero of $q(x)$ and $\alpha_3 \neq \alpha_2$.

By **94.3**, $\exists \phi : F(\beta)(\alpha_2) \rightarrow F(\beta)(\alpha_3)$ a ring isomorphism that fixes all elements of $F(\beta)$ and takes α_2 to α_3 which can be extended to an automorphism $\phi_K : K \rightarrow K$, s.t. $\phi_K(\beta) = \beta$, i.e. $\phi_K(\alpha_1 + d\alpha_2) = \alpha_1 + d\alpha_2$, but also $\phi_K(\alpha_1 + d\alpha_2) = \phi_K(\alpha_1) + d\alpha_3$. Solving for d , we find that $d = (\alpha_3 - \alpha_2)^{-1}(\phi_K(\alpha_1) - \alpha_1)$.

Since $\phi_K(\alpha_1)$ must map α_1 to one of the other zeros of $p(x)$ over K , and α_3 must be a zero of $q(x)$ over K , and there are at most $\deg(p(x)), \deg(q(x))$ such zeros respectively, then we find that there are finitely many candidates for $d \in F$ s.t. β is not a primitive element of $F(\alpha_1, \alpha_2)$. Since F has characteristic 0 it is not finite, so there are infinitely many possible values for d , so $\exists \beta$ of the form $\alpha_1 + d\alpha_2$ where $d \in F$, s.t. d is not of the form $(a - \alpha_2)(b - \alpha_1)$ where a, b are zeros of $\text{irr}(\alpha_2, F), \text{irr}(\alpha_1, F)$ respectively and $a \neq \alpha_2, b \neq \alpha_1$. By the above argument, it follows that all argument that satisfy this condition will be primitive elements of $F(\alpha_1, \alpha_2)$. □

The primitive element theorem extends naturally by induction to any finite algebraic extension over a separable field (note that this is a stronger statement than the one shown above for field with characteristic 0, working through this augmented base case is left as an exercise, since a separable field can have prime characteristic).

Galois Theory

The motivation for Galois theory is the study of solutions to complex polynomials, then is, to $p(x) \in \mathbb{C}[x]$. To achieve this, Galois theory, named after French mathematician Evariste Galois, establishes a connection between field theory and group theory. Using Galois theory, one can prove the insolubility of the quintic by radicals (that is, that there's no closed formula in terms of the coefficients of a polynomial of degree 5 and in fact, any other higher degree, in $\mathbb{C}[x]$ and thus in $\mathbb{R}[x]$), as well as the fundamental theorem of algebra. Our goal for this section will be to derive these two results.

In this section, we also adopt a new notation - instead of writing E is an extension of the field F , we write E/F . Furthermore, we may refer to the zeros of a polynomial as the *roots* of a polynomial.

Galois Group

Definition 0.0.1 (F -fixing automorphism group).

Let E/F be an extension field, then we define $\text{Aut}(E/F)$ to be the set of automorphisms on E that fix F pointwise, i.e. $\text{Aut}(E/F) = \{\sigma \in \text{Aut}(E) \mid \forall f \in F (\sigma(f) = f)\}$.

We say F is a *fixed field* under $\text{Aut}(E/F)$.

Theorem 111 ($\text{Aut}(E/F)$ is a group).

$\text{Aut}(E/F)$ is a group under composition.

Proof. First, recall from group theory that the automorphisms of a group form a group and function composition. Our proof for this theorem (found in 22) can trivially be generalized to show that the automorphisms of a field form a group under composition, so by the one-step subgroup test, since $\text{id} : E \rightarrow E$ is trivially in $\text{Aut}(E/F)$, we find that $\forall \sigma, \tau \in \text{Aut}(E/F)$, $(\sigma \circ \tau^{-1})(x) = \sigma(\tau^{-1}(x))$, and for $x \in F$ since $\sigma, \tau \in \text{Aut}(E/F)$ and hence fix F , we find that $\sigma(\tau^{-1}(x)) = \sigma(x) = x$, hence $\sigma \circ \tau^{-1} \in \text{Aut}(E/F)$, hence it forms a group under composition. $\square$

Some writers call $\text{Aut}(E/F)$ the *Galois group* of E/F . We reserve this term for the case where $|\text{Aut}(E/F)| = [E : F]$ (in which case we will call E/F a *Galois extension*).

At this point it is worth appreciating that we have already dealt with elements of $\text{Aut}(E/F)$. For example, we found in 94.3 that if $p(x)$ is irreducible over $F[x]$ and α, β are zeros of $p(x)$ in a splitting field E/F , then there exists a field automorphism $\phi : E \rightarrow E$ s.t. $\phi|_F = \text{id}_F$, and also $\phi(\alpha) = \beta$.

Theorem 112 ($\text{Aut}(E/F)$ defines a permutation on the zeros of a polynomial from $F[x]$).

If E/F is a splitting field of a polynomial $f(x) \in F[x]$, and $f(\alpha) = 0$, then $\forall \sigma \in \text{Aut}(E/F)$, $f(\sigma(\alpha)) = 0$.

Proof. Note that $\sigma(0) = 0$, and also $\sigma(\sum_{i=0}^n a_i \alpha^i) = \sum_{i=0}^n a_i (\sigma(\alpha))^i$ (here we used the properties of σ as a ring homomorphism, followed by the fact that F is fixed under σ), and clearly the RHS is $f(\sigma(\alpha))$ for $\sigma(\alpha) \in E$, so $\sigma(\alpha)$ is a zero of $f(x)$ over E , so σ sends zeros of $f(x)$ over E to zeros of $f(x)$ over E , so by definition it defines a permutation on those zeroes. $\square$

Corollary 112.1 ($\text{Aut}(E/F)$ does not exchange zeros of irreducible factors of reducible polynomials in perfect field).

If F is a perfect field and E/F is a splitting field of $f(x) \in F[x]$, and $f(x)$ is reducible in $F[x]$, then if α is a zero of an irreducible factor $p(x)$ of $f(x)$, then $\text{orb}_{\text{Aut}(E/F)}(\alpha)$ is exactly the set of zeros of $p(x)$.

Proof. Let $h(x)$ be another irreducible factor of $f(x)$. There must be such $h(x)$ with degree at least 1 since $f(x)$ is reducible by supposition. Suppose by contradiction that $\exists \sigma \in \text{Aut}(E/F)$ s.t. $h(\sigma(\alpha)) = 0$ for $\alpha \in E$ s.t. $p(\alpha) = 0$. By **Theorem 112** ($\text{Aut}(E/F)$ defines a permutation on the zeros of a polynomial from $F[x]$), we have that $p(\sigma(\alpha)) = 0$, so it follows that $h(x), p(x)$ have a shared linear factor by 65, but then $f(x)$ has a multiple zero. By **Theorem 102** (multiple zeros of an irreducible), $f(x)$ has no multiple zeros since F is a perfect field, a contradiction, so $\forall \sigma \in \text{Aut}(E/F)$ and $\forall f(x) \in F[x]$ s.t. $p(x) \in F[x]$ is an irreducible factor of $f(x)$, σ sends zeros of $p(x)$ to zeros of $p(x)$, and to zeros of $p(x)$ only (and not to other factors of $f(x)$). $\square$

We define a *Galois group* as a particular type of group of F -fixing automorphism:

Definition 0.0.2 (Galois group).

If E/F is a field extension over F , then $\text{Aut}(E/F)$ is called a Galois group if $|\text{Aut}(E/F)| = [E : F]$. In this case, we denote the automorphism group by $\text{Gal}(E/F)$.

Definition 0.0.3 (Galois extension).

If E/F is a finite field extension, then E/F is a Galois extension if $[E : F] = |\text{Aut}(E/F)|$, i.e. if $\text{Aut}(E/F)$ is a Galois group.

The next few definitions and theorems provide us with a better understanding of Galois groups.

Given an extension field E/F , we naturally find a group $\text{Aut}(E/F)$ corresponding to this extension field. Here, we demonstrate this relation in the other direction: given a subgroup of the automorphism $\text{Aut}(E)$, we can find a field K s.t. K consists of all elements fixed by $\text{Aut}(E)$.

Definition 0.0.4 (fixed field of $\text{Aut}(E)$).

Let $G \leq \text{Aut}(E)$, then $\{a \in E \mid \forall \sigma \in G (\sigma(a) = a)\}$ is called the fixed field of G , and is denoted by E^G .

Theorem 113 (E^G is a subfield of E).

IF $G \leq \text{Aut}(E)$, then E^G is a subfield of E , justifying calling it a fixed field.

Proof. Clearly $0 \in E^G$ since every automorphism must fix 0 (and also 1), so $E^G \neq \emptyset$. We proceed by **Theorem 53 subring test**. Let $a, b \in E^G$, and let $\sigma \in G$, then $\sigma(a) = a, \sigma(b) = b$, so $\sigma(a)\sigma(b) = ab$, but since σ is a homomorphism then $\sigma(ab) = \sigma(a)\sigma(b)$, so $ab \in E^G$, so it is closed under multiplication. Similarly, $\sigma(a - b) = \sigma(a) - \sigma(b) = a - b$, so it is closed under subtraction, so it is a subring of E . Commutativity of multiplication in E^G follows from commutativity of multiplication in E , so it remains to show that if $a \in E^G, a \neq 0$, then $a^{-1} \in E^G$. Again by the properties of a homomorphism, $\sigma(a^{-1}) = \sigma(a)^{-1} = a^{-1}$, so $a^{-1} \in E^G$, so E^G has the inverse property for every nonzero element, so it is a subfield of E . $\square$

Next, we consider the relation between $[E : E^G]$ and $|G|$ (where G is a subgroup of the automorphisms of E).

Lemma 113.1.

Any finite subgroup G of $\text{Aut}(E)$ is linearly independent, i.e. the only solution to $\sum_{\sigma_i \in G} a_i \sigma_i(x) = 0$ for all $x \in E$, is the trivial solution (all a_i coefficients are zero).

Proof. Let $|G| = n$. Suppose by contradiction G is linearly dependent, and let m be minimal s.t. $\sum_{i=1}^m a_i \sigma_i(x) = 0$ for all $x \in E$ and for $a_i \in E$ not all zero (where $\sigma_1, \dots, \sigma_m$ is a subset of G). Clearly $1m$ because for every $\sigma \in G, a\sigma(x) = 0 \iff a = 0 \vee \sigma(x) = 0$, and the latter is true if and only if $x = 0$ since $\sigma(0) = 0$ and σ is injective, so it must be that $a = 0$, so $1m$. We will show that it is possible to remove σ_m and still have a linearly dependent set, this time with $m - 1$ automorphisms, violating minimality of m , proving that G is linearly independent.

Let $x_0 \in E$ be an element on which σ_m disagrees with all other $\sigma_1, \dots, \sigma_{m-1}$ (one must exist otherwise σ_m is not distinct from the other functions). Clearly, $x_0 \neq 0$ since an automorphism must send 0 to 0. Let $y \in E$, since $\sum_{i=1}^m a_i \sigma_i(x) = 0$ for every $x \in E$, then it is zero for $x = x_0 y$, so:

$$\begin{aligned} \sum_{i=1}^m a_i \sigma_i(x_0 y) &= 0 \\ \sum_{i=1}^m a_i \sigma_i(y) \sigma_i(x_0) &= 0 \end{aligned}$$

Where we used the fact that σ_i is a homomorphism. Again since the initial sum is zero for every x , then it is 0 for y in particular, so $\sum_{i=1}^m a_i \sigma_i(y) = 0$. Multiplying both sides by $\sigma_m(x_0)$, we have $\sum_{i=1}^m a_i \sigma_i(y) \sigma_m(x_0) = 0$. Subtracting the last equation from 113.1, and recalling $\sigma_i(x_0) = \sigma_m(x_0)$ if and only if $i = m$, we get

$$\begin{aligned}
& \sum_{i=1}^m a_i \sigma_i(y) (\sigma_i(x_0) - \sigma_m(x_0)) = 0 \\
& \sum_{i=1}^{m-1} a_i \sigma_i(y) (\sigma_i(x_0) - \sigma_m(x_0)) + a_m \sigma_m(y) (\sigma_m(x_0) - \sigma_m(x_0)) = 0 \\
& \sum_{i=1}^{m-1} a_i \sigma_i(y) (\sigma_i(x_0) - \sigma_m(x_0)) = 0 \\
& \sum_{i=1}^{m-1} a_i (\sigma_i(x_0) - \sigma_m(x_0)) \sigma_i(y) = 0
\end{aligned}$$

Denote $b_i = a_i(\sigma_i(x_0) - \sigma_m(x_0))$. Note that $\sigma_i(x_0) - \sigma_m(x_0) \neq 0$ for $i \in [1, m-1]$ since $x_0 \neq 0$ and σ_i and σ_m disagree on x_0 by choice of x_0 , so $b_i = 0 \iff a_i = 0$ (since E is a field so has no zero divisors). So, we have $\sum_{i=1}^{m-1} b_i \sigma_i(y) = 0$ for any $y \in E$, so we have an $m-1$ linear combination of elements of G that are linearly dependent, violating minimality of m , so G is linearly independent. $\square$

We use the lemma to prove the following:

Theorem 114 ($[E : E^G] = |G|$).

Let E be a field and $G \leq \text{Aut}(E)$, then $[E : E^G] = |G|$.

Proof. We prove by contradiction. Suppose $[E : E^G] \neq |G|$, then either $[E : E^G] |G|$, or $[E : E^G] |G|$. Denote $[E : E^G] = m$, $|G| = n$.

Suppose $[E : E^G] |G|$, i.e. mn . Let $e = \{e_1, \dots, e_m\}$ be a basis of E over E^G , and consider the system of equations given below in matrix form:

$$\begin{pmatrix} \sigma_1(e_1) & \sigma_2(e_1) & \dots & \sigma_n(e_1) \\ \sigma_1(e_2) & \sigma_2(e_2) & \dots & \sigma_n(e_2) \\ \dots & \dots & \dots & \dots \\ \sigma_1(e_m) & \sigma_2(e_m) & \dots & \sigma_n(e_m) \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{pmatrix} = 0$$

Where the column vector $\vec{x} = \begin{pmatrix} x_1 \\ \dots \\ x_n \end{pmatrix} \in E^n$ (an n -tuple of elements of E).

Note that the system has n variables and m equations, and by supposition mn , so from linear algebra

we know that it has a nontrivial solution, i.e. a solution s.t. $\vec{x} \neq 0$. Let $\vec{x}$ be this solution vector. Now, let $a \in E$, then since e is a basis for E over E^G , then $a = \sum_{i=1}^m \lambda_i e_i$ for scalars $\lambda_i \in E^G$.

By writing out each equation in the system explicitly, and multiplying the i^{th} equation by λ_i , and then summing all equations, we get

$$\sum_{i=1}^m \sum_{j=1}^n \lambda_i \sigma_j(e_i) x_j = 0$$

Note that since E^G is the field fixed by G , then $\lambda_i = \sigma_j(\lambda_i)$ for every $\sigma_j \in G$, so $\lambda_i \sigma_j(e_i) = \sigma_j(\lambda_i e_i)$ (we also used the fact that σ_j preserves multiplication). Grouping [Theorem 114](#) $[E : E^G] = |G|$ by x_j , and using the relation we mentioned, and the fact that σ_j preserves addition as well, we get

$$\sum_{j=1}^n x_j \sigma_j \left(\sum_{i=1}^m \lambda_i e_i \right) = 0$$

But, $\sum_{i=1}^m \lambda_i e_i = a$ which was an arbitrary element of E , so we have $\sum_{j=1}^n x_j \sigma_j(a) = 0$ for every $a \in E$, we coefficients x_j not all 0 since $\vec{x}$ is a nontrivial solution to the system, so we have found that G is linearly dependent, which contradicts [113.1](#), so, $[E : E^G] \nmid |G|$.

Suppose then $[E : E^G] \mid |G|$, i.e. mn . Since mn , we can find $n + 1$ linearly independent vectors $a_1, \dots, a_{n+1} \in E/E^G$. Consider the homogeneous system of linear equations expressed in matrix form below:

$$\begin{pmatrix} \sigma_1(a_1) & \sigma_1(a_2) & \dots & \sigma_1(a_{n+1}) \\ \sigma_2(a_1) & \sigma_2(a_2) & \dots & \sigma_2(a_{n+1}) \\ \dots & \dots & \dots & \dots \\ \sigma_n(a_1) & \sigma_n(a_2) & \dots & \sigma_n(a_{n+1}) \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \dots \\ x_{n+1} \end{pmatrix} = 0$$

It is a system of n equations in $n + 1$ variables, so it has a non-trivial solution $\vec{x}$. Of these non-trivial solutions, let $\vec{\beta}$ be the solution with least nonzero elements (one must exist by the existence of a non-trivial solution and the well-ordering principle). Suppose this least amount is k , so, via a reordering, we can rewrite the equation as

$$\begin{pmatrix} \sigma_1(a_1) & \sigma_1(a_2) & \dots & \sigma_1(a_k) \\ \sigma_2(a_1) & \sigma_2(a_2) & \dots & \sigma_2(a_k) \\ \dots & \dots & \dots & \dots \\ \sigma_n(a_1) & \sigma_n(a_2) & \dots & \sigma_n(a_k) \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \dots \\ \beta_k \end{pmatrix} = 0$$

Finally, since each β_i is non-zero by our choice, and $\beta_i \in E$, then since E is a field $\exists \beta_i^{-1} \in E$, so we can multiply the equation by β_k^{-1} and get a new solution with k non-zero values s.t. the k^{th} value is 1. Via abuse of notation, we refer to these solution by $\beta_1, \dots, \beta_k$ as well, where $\beta_k = 1$.

From here, we proceed in a similar manner to the proof for 113.1, following a similar chain of manipulations, to find a solution with only $k - 1$ non-zero values, contradicting minimality of k , which will show that $m \nmid$ by contradiction.

Consider $\vec{\beta} = \beta_1, \dots, \beta_k, 0, \dots$ where we pad $\vec{\beta}$ with 0 to make it an $n + 1$ -tuple. Each i^{th} equation in the (original) system is $\sum_{j=1}^{n+1} \sigma_i(a_j) \beta_j = 0$. Since $G \leq \text{Aut}(E/E^G)$, then $\text{id}_E \in G$, so $\sigma_i = \text{id}_E$ for some i , so for this i the equation becomes $\sum_{j=1}^{n+1} a_j \beta_j = 0$, i.e. it is an $n + 1$ linear combination of $a_1, \dots, a_{n+1}$ with not all zero coefficients $\beta_1, \dots$ s.t. its sum is 0. If $\beta_1, \dots, \beta_k$ are all in E^G , this will contradict linear independence of $a_1, \dots, a_{n+1}$. It follows that $\exists \beta_j \in \vec{\beta}$ s.t. $\beta_j \notin E^G$. For this β_j , we must find some $\sigma_t \in G$ s.t. $\sigma_t(\beta_j) \neq \beta_j$, otherwise β_j is fixed by G so it is a part of E^G by definition of E^G . w.l.o.g let this β_j be β_1 and denote the matching automorphism by σ .

The key fact we will use in the next part of the proof is that $\sigma(\beta_1) - \beta_1 \neq 0$ for every $\sigma_i \in G$ s.t. $\sigma_i \neq \sigma$.

First, we act on every equation in the system by σ , so every equation becomes $\sigma(\sum_{j=1}^k \sigma_i(a_j) \beta_j) = \sigma(0)$. By the properties of a homomorphism, this becomes $\sum_{j=1}^k (\sigma \star \sigma_i)(a_j) \sigma(\beta_j) = 0$. Next, we use the fact that **Theorem 40 group actions are equivalent to group homomorphism to symmetric group**, so σ acting on G is a permutation of G , so in fact $\{\sigma_1, \dots, \sigma_n\} = \{\sigma \star \sigma_1, \dots, \sigma \star \sigma_n\}$. Since all equations differ only in σ_i , and since σ acts as a permutation on G , then we can identify each i^{th} equation in the original system with an equation of the form $\sum_{j=1}^k \sigma_i(a_j) \sigma(\beta_j) = 0$ in the new system (acquired by acting on each equation by σ). Now, subtracting the corresponding equations, we get

$$\begin{aligned} \sum_{j=1}^k \sigma_i(a_j) \sigma(\beta_j) - \sum_{j=1}^k \sigma_i(a_j) \beta_j &= 0 \\ \sum_{j=1}^k \sigma_i(a_j) (\sigma(\beta_j) - \beta_j) &= 0 \\ \sum_{j=1}^{k-1} \sigma_i(a_j) (\sigma(\beta_j) - \beta_j) + \sigma_i(a_k) (\sigma(\beta_k) - \beta_k) &= 0 \end{aligned}$$

Recall that $\beta_k = 1$ in our solution, so σ as a field automorphism must send 1 to itself, so the above equation becomes

$$\sum_{j=1}^{k-1} \sigma_i(a_j)(\sigma(\beta_j) - \beta_j) = 0$$

For every i in $[1, n]$. Note that, by construction, $\sigma(\beta_1) \neq \beta_1$, so we have a new solution vector with only $k - 1$ elements that is not the trivial solution (each j^{th} element of this new solution is given by $\sigma(\beta_j) - \beta_j$), so k is not minimal, a contradiction, so, $m \neq k$.

It follows that it must be that $[E : E^G] = |G|$.

□

We now consider $\text{Aut}(E/F)$ again.

Corollary 114.1 ($|\text{Aut}(E/F)| \leq [E : F]$).

Let E be a finite extension field over F , then $|\text{Aut}(E/F)| \leq [E : F]$.

Proof. Suppose E is an extension over F , then by definition F is fixed by $\text{Aut}(E/F)$. Let $K = E^{\text{Aut}(E/F)}$ be the field fixed by $\text{Aut}(E/F)$, then $F \subseteq K \subseteq E$. Since E is a finite extension over F then so is K , and by **Theorem 108** $[K : F] = [K : E][E : F] = [E : K][K : E]$. Now, by **Theorem 114** $[E : E^G] = |G|$, $[E : K] = |\text{Aut}(E/F)|$, and $[E : K] \leq [E : F]$, so $|\text{Aut}(E/F)| \leq [E : F]$. □

Corollary 114.2 ($\text{Aut}(E/E^G) = G$).

If $G \subseteq \text{Aut}(E)$, and E^G is the field fixed by G , then $\text{Aut}(E/E^G) = G$.

Proof. We will show $\text{Aut}(E/E^G) = G$ by mutual inclusion. First, suppose $\sigma \in G$, then σ fixes E^G by definition of E^G , so $\sigma \in \text{Aut}(E/E^G)$, so $G \subseteq \text{Aut}(E/E^G)$. Next, by **114.1** $|\text{Aut}(E/F)| \leq [E : F]$ we have $|\text{Aut}(E/E^G)| \leq [E : E^G]$. By **Theorem 114** $[E : E^G] = |G|$, $[E : E^G] = |G|$, so $|\text{Aut}(E/E^G)| \leq |G|$, and also $|G| \leq |\text{Aut}(E/E^G)|$, so we conclude $|G| = |\text{Aut}(E/E^G)|$ and since G is a subset of $\text{Aut}(E/E^G)$, it follows that $\text{Aut}(E/E^G) = G$. □

The following corollary provides an equivalent

Corollary 114.3 (equivalent definition of Galois group).

$\text{Aut}(E/F)$ is a Galois group if and only if $E^{\text{Aut}(E/F)} = F$.

Proof. $\implies$ Suppose $G = \text{Aut}(E/F)$ is a Galois group, and let E^G be the field fixed by G , then by **Theorem 108** $[K : F] = [K : E][E : F]$, $[E : F] = [E : E^G][E^G : F]$. Since G is Galois, then by **0.0.2 Galois group**, $[E : F] = |\text{Aut}(E/F)| = |G|$. By **Theorem 114** $[E : E^G] = |G|$, $[E : E^G] = |G|$, so we have $|G| = |G|[E^G : F]$, so by **72**, $E^G = F$, so $E^{\text{Aut}(E/F)} = F$.

$\Leftarrow$ Suppose $E^{\text{Aut}(E/F)} = F$. By definition, by **Theorem 114** $[E : E^G] = |G|$, $[E : F] = |\text{Aut}(E/F)|$, so by **0.0.2 Galois group**, $\text{Aut}(E/F)$ is Galois.

□

Consider $\text{Aut}(E/F)$. Although it is true that, by definition, $\text{Aut}(E/F)$ fixes F , it is not immediately obvious that F is the field fixed by $\text{Aut}(E/F)$ as defined in **0.0.4 fixed field of $\text{Aut}(E)$** : clearly $F \subseteq E^{\text{Aut}(E/F)}$, but in general it can be a proper extension of F . Below we see that if E is the splitting field of a separable polynomial in $F[x]$, then $E^{\text{Aut}(E/F)} = F$.

Theorem 115 (splitting field of separable polynomial admits a Galois group).

If $f(x) \in F[x]$ is separable, and E is a splitting field of $f(x)$, then $\text{Aut}(E/F)$ is a Galois group.

Proof. We prove by induction on $[E : F]$. If $[E : F] = 1$ then $E = F$ by **72**, so $\text{Aut}(E/F) = \{\text{id}_E\}$, so $|\text{Aut}(E/F)| = [E : F]$, so $\text{Aut}(E/F)$ is a Galois group.

Let $p(x)$ be the minimal polynomial whose zeros are all the zeros of $f(x)$ s.t. they are not in F . If no such zeros exist, then $f(x)$ splits over F so we have $E = F$, which we handled above, so there must exist such $p(x)$. This $p(x)$ is a factor of $f(x)$ and so it is also separable (since $f(x)$ is separable), and so it also splits over E . Let $d = \deg(p(x))$ (so $1 \nmid d$ otherwise $p(x)$ splits in F), and let $\alpha \in E \setminus F$ be a zero of $p(x)$ (i.e. $p(\alpha) = 0$).

By **Theorem 108** $[K : F] = [K : E][E : F]$, we have $[E : F] = [E : F(\alpha)][F(\alpha) : F]$. Since by choice $\text{irr}(\alpha, F) = p(x)$, then $[F(\alpha) : F] = \deg(p(x)) = d$ by definition, so we have $[E : F] = d[E : F(\alpha)]$. Since $1 \nmid d$ then $[E : F(\alpha)] \mid [E : F]$, so by the induction hypothesis $\text{Aut}(E/F(\alpha))$ is a Galois group, i.e. there are exactly $[E : F(\alpha)]$ automorphisms on E that fix $F(\alpha)$. We will show that every $\sigma \in \text{Aut}(E/F(\alpha))$ admits exactly $\deg(p(x)) = d$ distinct F -fixing automorphisms on E , which will then yield $[E : F] = d[E : F(\alpha)] = d|\text{Aut}(E/F(\alpha))| \leq |\text{Aut}(E/F)|$.

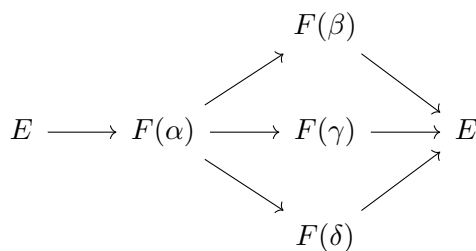
Let $\sigma \in \text{Aut}(E/F(\alpha))$, then $\sigma : E \rightarrow E$ fixes $F(\alpha)$ pointwise, so in particular σ fixes F pointwise. Suppose β is a zero of $p(x)$ in $E \setminus F$. Since $p(x)$ is irreducible over F and separable, there are exactly $\deg(p(x)) = d$ such zeros. By **97.1**, there exists a unique isomorphism $\psi : F(\alpha) \rightarrow F(\beta)$ s.t. $\psi|_F = \text{id}_F$ and $\psi(\alpha) = \beta$. This isomorphism naturally expands to an F preserving automorphism on E , $\bar{\psi} : E \rightarrow E$ by **Theorem 98 splitting fields of polynomials under isomorphism are isomorphic**.

Observe that $\bar{\psi} \circ \sigma : E \rightarrow E$ is a composition of automorphisms so it is an automorphism and it fixes F pointwise since $\bar{\psi}, \sigma$ both fix F pointwise, so $\bar{\psi} \circ \sigma \in \text{Aut}(E/F(\alpha))$, and also it has the property

that it sends α to β , i.e. it is uniquely determined by our choice of σ and our choice of β . Since $p(x)$ is separable irreducible over F with degree d , there are d choices for β , so in turn d distinct extensions $\bar{\psi} \circ \sigma$ for every $\sigma \in \text{Aut}(E/F(\alpha))$, so $d|\text{Aut}(E/F(\alpha))| \leq |\text{Aut}(E/F)|$.

We have $[E : F] \leq |\text{Aut}(E/F)|$ by the argument presented earlier. Since $|\text{Aut}(E/F)| \leq [E : F]$ by 114.1 $|\text{Aut}(E/F)| \leq [E : F]$, then $[E : F] = |\text{Aut}(E/F)|$, so $\text{Aut}(E/F)$ is a Galois group.

The key idea in this proof is illustrated below, where β, γ, δ are zeros of $p(x)$ in $E \setminus F$.



□

Corollary 115.1.

If $p(x) \in F[x]$ is separable and E/F is a splitting field for $p(x)$, then $E^{\text{Aut}(E/F)} = F$.

Proof. By Theorem 115 splitting field of separable polynomial admits a Galois group we know that $\text{Aut}(E/F)$ is a Galois group, and by 114.3 equivalent definition of Galois group this is true if and only if $E^{\text{Aut}(E/F)} = F$, hence the corollary. □

Theorem 116 (finite Galois extension is finite, normal and separable).

Let E/F be a finite Galois extension, then it is a finite, normal and separable extension.

Proof. Let E/F be a finite Galois extension, so $[E : F] = |\text{Gal}(E/F)|$. We denote $G = \text{Gal}(E/F)$.

First, we show E/F is normal.

Let $p(x) \in F[x]$ be irreducible over F . If $\exists \alpha \in E$ s.t. $p(\alpha) = 0$, then we want $p(x)$ splits in E . w.l.o.g. suppose $p(x)$ is monic (otherwise we simply divide all coefficients by the leading coefficient to retrieve a monic polynomial).

By Theorem 112 $\text{Aut}(E/F)$ defines a permutation on the zeros of a polynomial from $F[x]$, then $\text{orb}_G(\alpha)$ is a set of zeros of $p(x)$. Let $f(x) = \prod_{a_i \in \text{orb}_G(\alpha)} (x - a_i)$. Clearly, $\deg(f(x)) \leq \deg(p(x))$ since $f(x)$ is a separable polynomial whose zeros are all zeros of $p(x)$.

Let $\sigma \in G$. Consider σ naturally extended to an automorphism $\bar{\sigma} : E[x] \rightarrow E[x]$, $\sum_{i=0}^n b_i x^i \mapsto \sum_{i=0}^n \sigma(b_i) x^i$. Consider $f(x)$ under $\bar{\sigma}$, since $\bar{\sigma}$ is a homomorphism, then we get $\bar{\sigma}(f(x)) = \prod_{a_i \in \text{orb}_G(x)} (x - \sigma(a_i))$, but since σ is a permutation on $\text{orb}_G(x)$ then $f(x)$ is fixed under $\bar{\sigma}$ (since $\bar{\sigma}$ simply reorders the product $(x - a_1)(x - a_2) \dots (x - a_n)$), so all coefficients of $f(x)$ written as a formal power series are fixed by σ , and this holds for all $\sigma \in G$, so all coefficients are fixed by G , and since G is a Galois group of E over F then by **114.3 equivalent definition of Galois group** this means that all coefficients are in F , so $f(x) \in F[x]$.

Since $f(x) \in F[x]$ and also $f(x)$ has α as a zero and $p(x)$ is by construction $\text{irr}(\alpha, F)$, then $\deg(p(x)) \leq \deg(f(x))$, and we have also seen that $\deg(f(x)) \leq \deg(p(x))$ by construction of $p(x)$, hence $p(x) = f(x)$.

It follows that if $p(x) \in F[x]$ is irreducible in F , and $p(x)$ has a zero in E , then $p(x)$ splits over E . So, E/F is a normal extension by **0.0.63 normal extension**.

Next, we show E/F is separable. Let $\alpha \in E$. Since E/F is finite by supposition, then by **Theorem 107 finite extensions are algebraic**, the minimal polynomial for α in F exists. There are two cases:

1. $\alpha \in F$, in which case the minimal polynomial is simply $x - \alpha$, which is clearly separable.
2. $\alpha \notin F$, in which case by the above argument we have found that $\text{irr}(\alpha, F)$ is separable in E .

So E/F is separable, and we conclude that if E/F is a finite Galois extension, then it is a finite, normal, separable extension.

□

Theorem 117.

If E/F is a finite normal and separable extension, then E/F is the splitting field of a separable polynomial $f(x) \in F[x]$.

Proof. Suppose E/F is a finite, normal and separable extension. By the general form of **Theorem 110 primitive element theorem (Steinitz, 1910)** (here we use the fact that E/F is finite hence algebraic, and also the fact that it is a separable extension), we have that $E = F(\alpha)$ for some $\alpha \in E$. Since E/F is finite, it is also algebraic by **Theorem 107 finite extensions are algebraic**, so its minimal polynomial $f(x) \in F[x]$ exists. Since E/F is normal and $\alpha \in E$ is a zero of $f(x)$, then $f(x)$ splits in E/F , and also $F(\alpha)$ is the smallest extension of F containing α , so it is a splitting field for $f(x)$. □

Putting the last few theorems together, we get a nice set of equivalent definitions for a Galois group:

Theorem 118 (equivalent definitions for Galois group).

Let E/F be a finite field extension, TFAE:

1. $E^{\text{Aut}(E/F)} = F$
2. $|\text{Aut}(E/F)| = [E : F]$
3. E/F is a finite, normal and separable extension
4. E/F is the splitting field of a separable polynomial $f(x) \in F[x]$

Proof. 1. We have seen $1 \iff 2$ in 114.3 equivalent definition of Galois group.
 2. We have seen $2 \implies 3$ in Theorem 116 finite Galois extension is finite, normal and separable.
 3. We have seen $3 \implies 4$ in Theorem 117 .
 4. We have seen $4 \implies 1$ in Theorem 115 splitting field of separable polynomial admits a Galois group.

□

Corollary 118.1 (Galois group is isomorphic to subgroup of S_n).

If E/F is a Galois extension, and $[E : F] = n$, then its Galois group $\text{Gal}(E/F)$ is isomorphic to a subgroup of S_n the symmetric group on n letters.

Proof. Suppose $[E : F] = 1$, then $E = F$ by 72, in which case $\text{Gal}(E/F) = \{\text{id}\}$, which is the trivial group (0.0.9 trivial group) up to an isomorphism, and so is S_1 (since $|S_n| = n!$ and for $n = 1$ we have $|S_1| = 1$), so the proposition holds.

Suppose $[E : F] \neq 1$, then since E/F is finite, then it is the splitting field of a separable polynomial $f(x) \in F[x]$ by Theorem 118 equivalent definitions for Galois group (since E/F is a Galois extension by supposition). Let $p(x)$ be an irreducible factor for $f(x) \in F[x]$ and $\alpha \in E$ be a zero of $p(x)$, then $\deg(p(x)) = \deg(\alpha, F) = [E : F] = n$. We know that elements of $\text{Gal}(E/F)$ are uniquely defined by their action on the zeros of $p(x)$.

Consider an enumeration $\Omega = \alpha_1, \dots, \alpha_n$ on the zeros of $p(x)$, and consider the mapping $\phi : \text{Gal}(E/F) \rightarrow S_\Omega, \sigma \mapsto \sigma|_\Omega$. Clearly σ restricted to Ω is a mapping $\Omega \rightarrow \Omega$ since Theorem 112 $\text{Aut}(E/F)$ defines a permutation on the zeros of a polynomial from $F[x]$, so $\sigma|_\Omega \in S_\Omega$, so ϕ is well-defined. Also, since each σ is defined by its action on Ω (see proof of Theorem 98 splitting fields of polynomials under isomorphism are isomorphic), then this ϕ is injective. Next, we verify ϕ is a group homomorphism:

Let $\sigma, \tau \in \text{Gal}(E/F)$, then $\phi(\sigma) \circ \phi(\tau) = \sigma|_\Omega \circ \tau|_\Omega = (\sigma \circ \tau)|_\Omega = \phi(\sigma \circ \tau)$, so ϕ is a group homomorphism. Since ϕ is injective, its kernel is exactly id , so by Theorem 26 first isomorphism theorem, $\text{Gal}(E/F)/\{\text{id}\} \cong \phi(\text{Gal}(E/F))$, which by Theorem 18 properties of a group isomorphism is a subgroup of $\text{Domain}(\phi) = S_\Omega$, which itself is isomorphic to $S_{|\Omega|} = S_n$. By 25.2, $\text{Gal}(E/F) \cong$

$\text{Gal}(E/F)/\{\text{id}\}$. Putting it all together, we get that $\text{Gal}(E/F)$ is isomorphic to a subgroup of S_n , where $n = [E : F]$. □

Corollary 118.2 ($|\text{Gal}(E/F)| \leq n!$).

If E/F is a Galois extension of degree n , then the order of its Galois group is at most $n!$

Proof. Immediately from 118.1 Galois group is isomorphic to subgroup of S_n , since $|S_n| = n!$ □

Fundamental Theorem of Galois Theory

We have seen that Galois extensions/Galois groups form a correspondence between automorphism subgroups, and fields. Let E/F be an extension and let G be a subgroup of $\text{Aut}(E/F)$, then we know by Theorem 113 E^G is a subfield of E that $F \subseteq E^G \subseteq E$ and that E^G is a subfield of E . If E/F is Galois, the fundamental theorem of Galois theory tells us that the above extends to a one-to-one correspondence between all subfields of E/F to all subgroups of $\text{Gal}(E/F)$.

Theorem 119 (fundamental theorem of Galois theory (FTGT)).

If E/F is a Galois extension, and K is a subfield of E s.t. $F \subseteq K \subseteq E$, then:

1. (subgroups and subfields are in one-to-one correspondence) Let A be the set of all such subfields K (i.e. all finite extensions from F to E and B be the set of all subgroups of $\text{Gal}(E/F)$, then $\phi : A \rightarrow B, K \mapsto \text{Gal}(E/K)$ is a bijection.
2. $[E : K] = |\text{Gal}(E/K)|$ (i.e. E is also a Galois extension over K), and also $[K : F] = [\text{Gal}(E/F) : \text{Gal}(E/K)]$ (where the LHS is the degree of the extension and the RHS is the index of the subgroup).
3. (inclusion-inverting) $F \subset K \subset L \subset E \iff \text{id} \subset \text{Gal}(E/L) \subset \text{Gal}(E/K) \subset \text{Gal}(E/F)$.
4. K/F is a Galois extension of $F \iff \text{Gal}(E/K) \trianglelefteq \text{Gal}(E/F)$, and in this case $\text{Gal}(K/F) \cong \text{Gal}(E/F)/\text{Gal}(E/K)$.

Proof. 1. We will show $\phi : K \mapsto \text{Gal}(E/K)$ is bijective by showing it is injective and surjective. First, we show $\text{Gal}(E/K)$ is well defined for every extension $F \subseteq K \subseteq E$: clearly the identity fixes K , so $\text{Gal}(E/K)$ consists of at least the identity, so it exists for every sub-extension K between F and E .

For injectivity, suppose $\phi(K) = \phi(L)$, i.e. $\text{Gal}(E/K) = \text{Gal}(E/L)$. By definition, $E^{\text{Gal}(E/K)} = K$, and also $E^{\text{Gal}(E/L)} = L$, and since $\text{Gal}(E/K) = \text{Gal}(E/L)$, then we have $K = L$, so ϕ is injective.

For surjectivity, let $G \in B$, so $G \leq \text{Gal}(E/K)$, so $G \leq \text{Aut}(E)$, so by 114.3 equivalent definition of Galois group we have E^G is a Galois extension, so $G = \text{Gal}(E/K)$, so $E^G = K$, so $\phi(E^G) = G$.

We have found that ϕ is injective and surjective, so it is a bijection.

2. $[E : K] = |\text{Gal}(E/K)|$ since by (1) we have that K is a Galois extension for every sub-extension K between F and E . By **Theorem 108** $[K : F] = [K : E][E : F]$, we have that $[E : F] = [E : K][K : F]$, so $[K : F] = \frac{[E:F]}{[E:K]} = \frac{|\text{Gal}(E/F)|}{|\text{Gal}(E/K)|}$. Since $\text{Gal}(E/K) \leq \text{Gal}(E/F)$, then the RHS is the index of $\text{Gal}(E/K)$ in $\text{Gal}(E/F)$ (recall **0.0.2 index of a subgroup**), hence $[E : F] = [\text{Gal}(E/F) : \text{Gal}(E/K)]$.

3. Recall that $\phi : K \mapsto \text{Gal}(E/K)$ is a bijection from the subfields K of E/F s.t. K is an extension of F , to the subgroups of $\text{Gal}(E/F)$. Since ϕ is bijective, it suffices to show the argument in one direction, and the argument in the other direction follows by a symmetric argument. We choose to prove $\implies$: suppose $F \subset K \subset L \subset E$. Clearly, $\phi(E) = \text{Gal}(E/E) = \{\text{id}_E\}$. For L , $\phi(L) = \text{Gal}(E/L)$, which obviously contains the identity since it is a group, so $\{\text{id}_E\} \leq \text{Gal}(E/L)$. For K , $\phi(K) = \text{Gal}(E/K)$. Let $\sigma \in \text{Gal}(E/L)$, then since σ fixes L and $K \subset L$, then obviously σ fixes K , so $\sigma \in \text{Gal}(E/K)$, so $\text{Gal}(E/L) \leq \text{Gal}(E/K)$. By an identical argument, we find that $\text{Gal}(E/K) \leq \text{Gal}(E/F)$.

The following diagram illustrates the proof of this proposition, and should also clarify why we say that this relation is *inclusion-inverting*.

$$\begin{array}{ccc}
 E & \xrightarrow{\phi} & \{\text{id}_E\} \\
 \updownarrow & & \downarrow \\
 L & \xrightarrow{\phi} & \text{Gal}(E/L) \\
 \updownarrow & & \downarrow \\
 K & \xrightarrow{\phi} & \text{Gal}(E/K) \\
 \updownarrow & & \downarrow \\
 F & \xrightarrow{\phi} & \text{Gal}(E/F)
 \end{array}$$

If we were to properly define the lattice of a group/field via a partial order defined as inclusion of subgroups/subfields, it will immediately follow from (3) that the FTGT states that if E/F is a Galois extension, then the lattices of E/F and $\text{Gal}(E/F)$ are the same, but inverted. We will show an example of this later.

4. $\implies$ Suppose K/F is a Galois extension over F for $F \subseteq K \subseteq E$. We want to show that $\text{Gal}(E/K) \trianglelefteq \text{Gal}(E/F)$. We already know that $\text{Gal}(E/K) \leq \text{Gal}(E/F)$ since every automorphism on E that fixes K must also fix F . To show it is a normal subgroup, we proceed by **29**. Let $\sigma \in \text{Gal}(E/K)$ and let $\psi \in \text{Gal}(E/F)$. We will show that $\tau = \psi^{-1} \circ \sigma \circ \psi \in \text{Gal}(E/K)$.

Clearly $\tau \in \text{Aut}(E)$ since it is a composition of automorphisms of E , so it remains to show that τ fixes K . Let $\alpha \in K$, then $\tau(\alpha) = \psi^{-1}(\sigma(\psi(\alpha)))$. If $\alpha \in F$, then ψ and σ both fix α , so $\tau(\alpha) = \alpha$. Suppose $\alpha \in K \setminus F$. Since K/F is a Galois extension, then it is a normal, separable extension (by

Theorem 116 finite Galois extension is finite, normal and separable) and so $\text{irr}(\alpha, F)$ exists in $F[x]$ (since K/F is separable), denote it $p(x)$. Since $\psi \in \text{Gal}(E/F)$, then $\psi(\alpha)$ is also a zero of $F[x]$ by **Theorem 112** $\text{Aut}(E/F)$ defines a permutation on the zeros of a polynomial from $F[x]$. Since K/F is normal, then this zero is also in K , so $\psi(\alpha) \in K$. Since $\sigma \in \text{Gal}(E/K)$, then σ fixes $\psi(\alpha)$, so $\tau(\alpha) = \psi^{-1}(\sigma(\psi(\alpha))) = \psi^{-1}(\psi(\alpha)) = \alpha$, so τ fixes α . We conclude that τ fixes K , so $\tau \in \text{Gal}(E/K)$, hence $\text{Gal}(E/K) \trianglelefteq \text{Gal}(E/F)$.

$\Leftarrow$ Suppose $\text{Gal}(E/K) \trianglelefteq \text{Gal}(E/F)$. We want to show K/F is a Galois extension. To do that, we will show that $K^{\text{Aut}(K/F)} = F$ (by **114.3 equivalent definition of Galois group**).

Let $\sigma \in \text{Gal}(E/K)$ and $\psi \in \text{Gal}(E/F)$, then since $\text{Gal}(E/K) \trianglelefteq \text{Gal}(E/F)$, by **0.0.3 normal subgroup** it follows that $\exists \bar{\psi} \in \text{Gal}(E/F)$ s.t. $\sigma \circ \bar{\psi} = \psi \circ \sigma$. Let $\alpha \in K$, then $\bar{\psi}(\sigma(\alpha)) = \bar{\psi}(\alpha)$, so $\sigma(\bar{\psi}(\alpha)) = \psi(\alpha)$, hence $(\sigma \circ \bar{\psi})|_K = \psi|_K$ is an element on $\text{Aut}(K/F)$ (since ψ is an automorphism on E that fixes F).

Let $\beta \in K^{\text{Aut}(K/F)}$, then β is fixed by all automorphisms in $\text{Aut}(K/F)$ and in particular, it is fixed by $\psi|_K$, but this is true for every $\psi \in \text{Gal}(E/F)$, and since E/F is a Galois extension then it must be that $\psi|_K(\beta) \in F$, so $K^{\text{Aut}(K/F)} = F$, so K/F is a Galois extension over F .

Finally, it remains to show that if either of these equivalent conditions holds, then $\text{Gal}(K/F) \cong \text{Gal}(E/F)/\text{Gal}(E/K)$.

Let $\sigma \in \text{Gal}(E/F)$, then $\sigma|_K \in \text{Gal}(K/F)$ by the above (in the $\Leftarrow$ direction). Consider the mapping $\phi : \text{Gal}(E/F) \rightarrow \text{Gal}(K/F)$, $\sigma \mapsto \sigma|_K$. As we have just argued the domain and codomain of ϕ are properly defined. Let $\tau \in \text{Ker}(\phi)$, then $\phi(\tau) = \text{id}_K$, so τ must fix K pointwise, so $\text{Ker}(\phi) = \text{Gal}(E/K)$, so by **Theorem 65 first isomorphism theorem for rings** we have $\text{Im}(\phi) \cong \text{Gal}(E/F)/\text{Gal}(E/K)$. It remains to show $\text{Im}(\phi) = \text{Gal}(K/F)$.

We already know $\text{Im}(\phi) \leq \text{Gal}(K/F)$. By (2), we know that $|\text{Gal}(K/F)| = [K : F] = [\text{Gal}(E/F) : \text{Gal}(E/K)]$, and the RHS is by definition the order of the factor group $\text{Gal}(E/F)/\text{Gal}(E/K)$ which by the first isomorphism theorem is isomorphic to $\text{Im}(\phi)$, so by **Theorem 18 properties of a group isomorphism**, we have that $|\text{Im}(\phi)| = |\text{Gal}(K/F)|$, so since it is a subgroup of $\text{Gal}(K/F)$ with the same order as the group it must be the group itself, so we have $\text{Im}(\phi) = \text{Gal}(K/F)$, which completes the proof. □

By recalling that a finite Galois extension is a separable and normal extension, proposition (4) in **Theorem 119 fundamental theorem of Galois theory (FTGT)** should provide better motivation for why named the property described in **0.0.63 normal extension, normal**.

The correspondence we see between subfields of a Galois extension E/F and subgroups of its Galois field $\text{Gal}(E/F)$ is an example of a more general property, called a **Galois correspondence**.

Consider $f(x) = x^3 - 2 \in \mathbb{Q}$, first we analyze its Galois extension.

Example 119.1 (Analyzing Galois extension of $f(x) = x^3 - 2$).

Consider $f(x) = x^3 - 2 \in \mathbb{Q}[x]$. Let $E/\mathbb{Q}$ be a splitting field for $f(x)$. Since $E/\mathbb{Q}$ is a finite extension of $f(x)$, by **Theorem 107 finite extensions are algebraic** it is algebraic, and since $\text{char}(\mathbb{Q}) = 0$ then by $\mathbb{Q}$ is a perfect field and by **Theorem 106 algebraic extension of a perfect field is separable**, hence $f(x)$ is separable, so $E/\mathbb{Q}$ is a Galois extension by **Theorem 118 equivalent definitions for Galois group**.

Denote the zeros of $f(x)$ by $\sqrt[3]{2}, \sqrt[3]{2}\omega, \sqrt[3]{2}\omega^2$ where $\omega = e^{\frac{2\pi i}{3}}$ the third root of unity, then $E = F(\sqrt[3]{2}, \sqrt[3]{2}\omega, \sqrt[3]{2}\omega^2)$

Consider $\mathbb{Q}(\sqrt[3]{2})$. In this extension field, we find $f(x) = (x - \sqrt[3]{2})h(x)$ where $\deg(h(x)) = 2$ (by **63 degree equation for polynomial ring of integer domain**). We know that $h(x)$ is irreducible in $\mathbb{Q}(\sqrt[3]{2})$ since the other two zeros of $f(x)$ are complex and $\mathbb{Q}(\sqrt[3]{2})$ is isomorphic to a subfield of $\mathbb{R}$ (see **99.1 $\mathbb{Q}(\sqrt[3]{2})$ is not a normal extension of $\mathbb{Q}$**).

Consider $E/\mathbb{Q}(\sqrt[3]{2})$. We know that complex conjugation $\sigma : a \mapsto a^*$ is an automorphism on E and also clearly it preserves $\mathbb{Q}(\sqrt[3]{2})$, so $\sigma \in \text{Aut}(E/\mathbb{Q}(\sqrt[3]{2}))$, so by **Theorem 112 $\text{Aut}(E/F)$ defines a permutation on the zeros of a polynomial from $F[x]$** it defines a permutation on the zeros of $h(x)$ in E , so $\sqrt[3]{2}\omega, \sqrt[3]{2}\omega^2$ are conjugates. Since complex conjugation is achieved via the normal field operations, then this means that $\mathbb{Q}(\sqrt[3]{2}, \sqrt[3]{2}\omega) = \mathbb{Q}(\sqrt[3]{2}, \sqrt[3]{2}\omega, \sqrt[3]{2}\omega^2) = E$.

By **Theorem 108 $[K : F] = [K : E][E : F]$** , we have

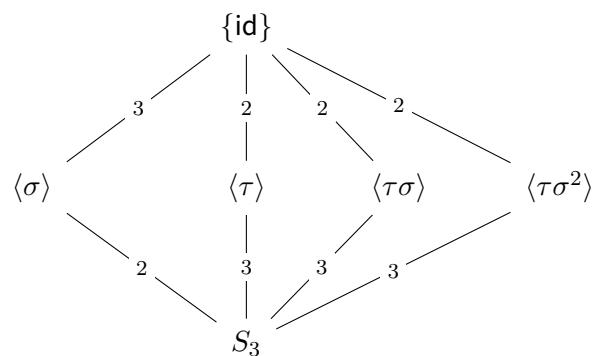
$$[E : \mathbb{Q}] = [\mathbb{Q}(\sqrt[3]{2}, \sqrt[3]{2}\omega) : \mathbb{Q}] = [\mathbb{Q}(\sqrt[3]{2}, \sqrt[3]{2}\omega) : \mathbb{Q}(\sqrt[3]{2})][\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}]$$

Since $\text{irr}(\sqrt[3]{2}, \mathbb{Q}) = f(x) = x^3 - 2$, then $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$. Also, $\text{irr}(\sqrt[3]{2}\omega, \mathbb{Q}(\sqrt[3]{2})) = h(x)$ which has degree 2, so $[\mathbb{Q}(\sqrt[3]{2}, \sqrt[3]{2}\omega) : \mathbb{Q}(\sqrt[3]{2})] = 2$.

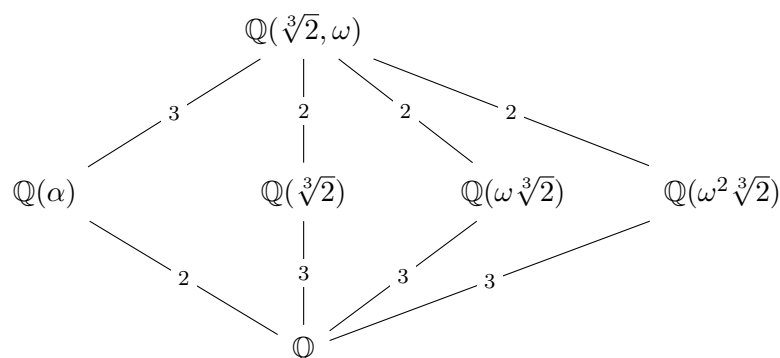
It follows that $[E : \mathbb{Q}] = 6$. Also, we have seen that $[E : \mathbb{Q}]$ is generated by two cyclic subgroups of order 2 (corresponding to complex conjugation) and 3 (corresponding to all possible mappings from $\sqrt[3]{2}$ to the other zeros of the polynomial in its splitting field), so $[E : \mathbb{Q}] \cong S_3$ the symmetric group on 3 letters.

since $\mathbb{R}$ has characteristic 0, then it is a perfect field and by **Theorem 106 algebraic extension of a perfect field is separable** the splitting

Let's see **Theorem 119 fundamental theorem of Galois theory (FTGT)** in action. First, consider the (inverted) lattice diagram for S_3 , where the numbers on the arrows indicate the index of the subgroup (upper end of the arrow) in the group (lower end of the arrow):



Now, consider the following lattice, which presents all sub-extensions from $\mathbb{Q}$ to $\mathbb{Q}(\sqrt[3]{2}, \omega)$, as broadly discussed in **119.1 Analyzing Galois extension of $f(x) = x^3 - 2$** , where the numbers are the degree of the extension:



Notice that this is *exactly* the inverted lattice diagram for S_3 , which is to be expected by the Galois correspondence between Galois extensions and Galois groups, which we have derived in **Theorem 119 fundamental theorem of Galois theory (FTGT)**.

Roots of Unity

Definition 0.0.5 (root of unity).

An n^{th} root of unity is a solution to the equation $x^n = 1$, i.e. it is a zero of the polynomial $f(x) = x^n - 1$ over some field F .

Theorem 120 (roots of unity form a cyclic abelian group).

If F is a field with characteristic 0, then the n^{th} roots of unity in $x^n - 1 \in F[x]$ form a cyclic abelian group under multiplication.

Proof. Let E be a splitting field for $x^n - 1$ over F , then since E has characteristic zero find that $x^n - 1$ has n distinct zeros in E (by **Theorem 101 criterion for multiple zeros**, since $(x^n - 1)' = nx^{n-1}$ which does not share a common factor with $x^n - 1$ (since such factor would need to have nonzero constant coefficient))

, denote them $\Omega = \{\alpha_1, \dots, \alpha_n\}$. Consider the multiplicative group of E , which is $E \setminus \{0\}$, then Ω is a subset of this multiplicative group since trivially $0^n \neq 1$ so $0 \notin \Omega$. We will show Ω is a group by **Theorem 6 one-step subgroup test for finite group**:

1. Clearly, $|\Omega| = n$ so it is non-empty.
2. Let $\alpha_i, \alpha_j \in \Omega$, then $(\alpha_i \alpha_j)^n = \alpha_i^n \alpha_j^n$ by **Theorem 2 exponent laws (power rules)** since the multiplicative group of F is abelian (since multiplication in a field commutes), and by supposition $\alpha_i^n = 1, \alpha_j^n = 1$, and since 1 is the identity then $\alpha_i^n \alpha_j^n = 1$, so $\alpha_i \alpha_j \in \Omega$, so Ω is closed under multiplication.

By the one-step subgroup test, Ω is a group, and also abelian since it is a subgroup of an abelian group.

Let $d \mid n$, then $x^d - 1$ has at most d zeros by **66 a polynomial of degree n has at most n zeros**, so for every such d there are at most d elements $\alpha \in \Omega$ s.t. $\alpha^d = 1$, so by **Theorem 10 sufficient condition for finite group to be cyclic** we have that Ω is cyclic. □

We call a generator of this group a *primitive n^{th} root of unity*:

Definition 0.0.6 (primitive n^{th} root of unity).

If F is a field with characteristic 0, then a generator of the n^{th} roots of unity of $x^n - 1 \in F[x]$ is called a primitive n^{th} root of unity, and denoted ω .

Theorem 120 roots of unity form a cyclic abelian group promises the existence of such primitive root for every $n \in \mathbb{N}$.

The next corollary deals with the zeros of a polynomial of the form $x^n - a \in F[x]$ where $\text{char}(F) = 0$.

Corollary 120.1 (zeros of $x^n - a \in F[x]$ with field characteristic 0).

If F is a field with characteristic 0 and $a \in F, a \neq 0$, and α is a zero of the polynomial $x^n - a$ in a splitting field E/F , then the zeros of $x^n - a$ are exactly $\{\alpha\omega^0, \alpha\omega^1, \dots, \alpha\omega^{n-1}\}$ where ω is a primitive n^{th} root of unity.

Proof. Since the formal derivative of $x^n - a$ is nx^{n-1} and $a \neq 0$ then it cannot share a factor with $x^n - a$ hence by **Theorem 101 criterion for multiple zeros**, $x^n - a$ has no multiple zeros, hence it has exactly n distinct zeros in E/F . Since ω generates the n^{th} roots of unity, then $|\omega| = n$, hence all of $\{\omega^0, \omega^1, \dots, \omega^{n-1}\}$ are distinct, so $\{\alpha\omega^0, \dots, \alpha\omega^{n-1}\}$ are also distinct and consists of exactly n elements, so it remains to show that every element of the form $\alpha\omega^k$ for $k \in [0, n-1]$ are zeros of $x^n - a$. Since F is a field, $(ab)^n = a^n b^n$ for every $a, b \in F$ (the multiplicative group of $F \setminus \{0\}$ is abelian), we have $(\alpha\omega^k)^n = \alpha^n (\omega^k)^n$. By **Theorem 2 exponent laws (power rules)** we have $(\omega^k)^n = (\omega^n)^k = 1^k = 1$, and also $\alpha^n = a$ since α is a zero of $x^n - a$ by supposition, hence $\alpha\omega^k$ is a zero of $x^n - a$, which completes the proof. $\square$

Radical Extension

We work our way towards one of our main results - the insolubility of the quintic by radicals. First, a definition of what it means for a polynomial to be “solvable by radicals” is in order.

As an motivating example, let us consider a general second degree polynomial over a field F with characteristic 0, this polynomial is of the form $ax^2 + bx + c = 0$, where $a \neq 0$. Recall that in our general overview of abstract algebra, we presented a derivation of the quadratic formula. Let us study this derivation, presented in its entirety below:

In this example we abuse notation and use x both as the indeterminate and as a proper element of the splitting field of the polynomial, hopefully context helps the reader avoid confusion.

$$\begin{aligned}
 ax^2 + bx + c &= 0 \\
 x^2 + \frac{b}{a}x + \frac{c}{a} &= 0 \\
 x^2 + \frac{b}{a}x &= -\frac{c}{a} \\
 x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} &= -\frac{c}{a} + \frac{b^2}{4a^2} \\
 \left(x + \frac{b}{2a}\right)^2 &= \frac{b^2 - 4ac}{4a^2} \\
 x + \frac{b}{2a} &= \pm \frac{\sqrt{b^2 - 4ac}}{2a} \\
 x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}
 \end{aligned}$$

Since we consider $ax^2 + bx + c \in F[x]$ then $a, b, c \in F[x]$, and since by supposition $a \neq 0$ (otherwise

we don't have a quadratic), then we know that $a^{-1} \in F$. The first step is simply claiming that, for a zero x of $f(x) = ax^2 + bx + c$ in a splitting field E/F , we have that f evaluated at this x is 0, which is the definition of x being a zero of the polynomial. Then, we multiply both sides by a^{-1} and proceed according to ring arithmetic, until eventually we have an expression of the form $(x + \frac{b}{2a})^2 = \frac{b^2 - 4ac}{4a^2}$. Since $a, b, c \in F$ and $a \neq 0$, then the RHS is also an element of F , so $(x + \frac{b}{2a})^2 \in F$ as well (since they are the same), but, it is not necessarily true that $x + \frac{b}{2a} \in F$ (since $x \in E/F$ but not necessarily in F). For example, if we consider the case where $F = \mathbb{Q}$ and $f(x) = x^2 - 2$, then we already know that the zeros of this polynomial are $\pm\sqrt{2}$ which are irrational so not in $\mathbb{Q}$.

The next step of the derivation requires just that - that we can take the second root (any second root, but when dealing with $\mathbb{R}$ by convention we take this to be the non-negative root) of $b^2 - 4ac$ (since $2a \in F$ (here $2a$ is a acted on by $2 \in \mathbb{Z}$) and multiplication in a field commutes then $(2a)^2 = 4a^2$ so we do have a square root for the denominator), i.e. we want that $\exists y \in E$ s.t. $y^2 = b^2 - 4ac$, which we equivalently denote by $y = \sqrt{b^2 - 4ac}$. Consider an extension $K = F(\sqrt{b^2 - 4ac})$, i.e. F adjoined with a square root of $b^2 - 4ac$, then in this field we can continue our derivation of the quadratic formula, and arrive at the end result.

We call such $F(\sqrt{b^2 - 4ac})$ a *radical extension*.

Definition 0.0.7 (radical extension).

If F is a field, then E is a radical extension over F if $E = F(\sqrt[k]{\alpha})$ for some $\alpha \in F, k \in \mathbb{N}$. Equivalently, E is a radical extension over F if E is F adjoined with an element β s.t. $\beta^k \in F$ for some $k \in \mathbb{N}$.

Definition 0.0.8 (radical series).

If F is a field, and E is an extension of F , and there exists a finite series of extensions $F = F_0 \subset F_1 \subset F_2 \subset \dots \subset F_n = E$ s.t. $F_{i+1} = F_i(\sqrt[k_i]{\alpha_i})$ for some $k_i \in \mathbb{N}, \alpha_i \in F_i$ (i.e. each F_{i+1} is a radical extension over F_i), then we say that $F_0 \subset F_1 \subset \dots \subset F_n$ is a radical series and we say that E is an extension by radicals over F .

In our example of the quadratic, we found that for $f(x) = ax^2 + bx + c \in F[x]$, the field $E = F(\sqrt{b^2 - 4ac})$ is a radical extension over F , and also since we can derive the quadratic formula in F , then clearly $f(x)$ splits in E , i.e. if K is a splitting field for $f(x)$ then by definition and by uniqueness (98.1 uniqueness of splitting field), $K \subseteq E$ via an subfield isomorphism. This motivates the following definition:

Definition 0.0.9 (solvability by radicals).

If F is a field and $f(x) \in F[x]$, then we say that $f(x)$ is solvable by radicals if there exists an extension by radicals E over F s.t. the splitting field K for $f(x)$ is contained (via subfield isomorphism) in E .

Practically, solvability by radicals means that the zeros of the polynomial can be found by manipulating

the coefficients of the polynomial, allowing for a *finite sequence* of:

1. Multiplication and subtraction.
2. Division by the leading coefficient.
3. Root taking (which led to radical extensions)

Insolvability of the Quintic

Recall that Galois theory deals with connecting fields and groups, letting us reduce questions about fields to questions about groups. So, while our end goal is to answer the question of when is a polynomial solvable by radicals, we will first define and examine a few properties of a class of groups called *solvable groups*. As the name hints, our main theorem in this section will be that if a polynomial is solvable by radicals, then its Galois group is solvable. We will then see why quadratic polynomials are not generally solvable by radicals.

Definition 0.0.10 (solvable group).

A group G is solvable if

1. there exists a normal series $\{e\} = H_0 \trianglelefteq H_1 \trianglelefteq H_2 \trianglelefteq \cdots \trianglelefteq H_n = G$, and
2. Each factor group H_{i+1}/H_i in this normal series is abelian

Theorem 121 (subgroup of solvable group are solvable).

If G is a solvable group and $H \leq G$, then H is solvable.

Proof. Suppose G is a solvable group and $H \leq G$. Since G is solvable, then we can consider a normal series $\{e\} = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_n = G$. So, by **Theorem 37 second isomorphism theorem**,

$$(H \cap G_i)G_{i-1}/G_{i-1} \cong H \cap G_i/H \cap G_{i-1}$$

(here we used the fact the the normalizer of a normal group is the group (**Theorem 35 properties of the normalizer**), and also that since $G_{i-1} \trianglelefteq G_i$ then $(H \cap G_i) \cap G_{i-1} = H \cap G_{i-1}$)

In particular, we find that $H \cap G_{i-1} \trianglelefteq H \cap G_i$. Note that:

1. $H \cap G_0 = H \cap \{e\} = \{e\}$
2. $H \cap G_n = H \cap G = H$ (since $H \leq G$)

So, letting $H_i = H \cap G_i$, we have $H_0 \trianglelefteq H_1 \trianglelefteq \cdots \trianglelefteq H_n = H$ a normal series of H . We will show that H_{i+1}/H_i is abelian. By the fourth isomorphism theorem (which we have't covered here), we have $(H \cap G_{i+1})/(H \cap G_i) \leq G_{i+1}/G_i$, hence $H_{i+1}/H_i \leq G_{i+1}/G_i$. Since G is solvable with normal sequence $G_0 \trianglelefteq \cdots \trianglelefteq G_n$, then G_{i+1}/G_i is abelian, so H_{i+1}/H_i is abelian as a subgroup of an abelian group.

We conclude that H is solvable. □

Proposition 73 (simple and solvable iff abelian).

Suppose G is a simple group, then it is solvable if and only if it is abelian.

Proof. Suppose G is simple, then by 0.0.4 simple group the only normal subgroups of G are G and $\{e\}$. Since $G/\{e\} \cong G$ by 25.2 then if G is solvable, it must be that G is abelian, and if G is abelian then G must be solvable by definition. □

Corollary 121.1 (A_n is not solvable for $5 \leq n$).

The alternating group on n letters A_n for n greater than 5 is not solvable.

Proof. Let $5 \leq n$. By Theorem 24 $A_{5 \leq n}$ is simple, A_n is simple. By 73 simple and solvable iff abelian, if A_n is solvable then it is abelian, yet by 14 $\text{Sym}(A)$ non-abelian if $2|A|$, A_n is non-abelian since $5 \leq n$ hence $2n$, so A_n is not solvable. □

Corollary 121.2 (S_n is not solvable for $5 \leq n$).

The symmetric group on n letters S_n for n greater than 5 is not solvable.

Proof. Let $n \geq 5$ and suppose by contradiction S_n is solvable, so since $A_n \leq S_n$ then by Theorem 121 subgroup of solvable group are solvable it follows that A_n is solvable, but by 121.1 A_n is not solvable for $5 \leq n$ we know such A_n is not solvable, a contradiction, hence S_n is not solvable for $n \geq 5$. □

Proposition 74 (abelian groups are solvable).

If G is abelian, then it is solvable.

Proof. Suppose G is abelian, then by 25.2 we have that $\{e\} \trianglelefteq G$ is a normal series of G and also $G/\{e\} \cong G$, and since G is abelian then so is $G/\{e\}$, hence G is solvable by definition. □

Theorem 122 (Galois group of splitting field for $x^n - a$ is solvable).

If F is a field of characteristic zero and E is a splitting field for the polynomial $x^n - a$, then $\text{Gal}(E/F)$ is solvable (and is well-defined, i.e. $\text{Aut}(E/F)$ is indeed a Galois group).

Proof. By 120.1 zeros of $x^n - a \in F[x]$ with field characteristic 0, we have that $F \subseteq F(\omega) \subseteq E$ where ω is a primitive n^{th} root of unity. If $F = F(\omega) = E$ then $\text{Gal}(E/F) = \{\text{id}\}$ which is the trivial group and clearly it is abelian and solvable, so suppose $x^n - a$ does not split in F .

There are two cases: suppose $\omega \in F$, i.e. $F = F(\omega)$, and let $\alpha \in E$ be a zero of $x^n - a$, then $\alpha \notin F(\omega)$ otherwise by **120.1 zeros of $x^n - a \in F[x]$ with field characteristic 0** we would find that $x^n - a$ splits in F which is a contradiction to our supposition. So, E is the splitting field of an irreducible polynomial in F which has characteristic 0 so this polynomial is separable, so by **Theorem 118 equivalent definitions for Galois group** $\text{Aut}(E/F) = \text{Gal}(E/F)$. By **Theorem 112** $\text{Aut}(E/F)$ defines a permutation on the zeros of a polynomial from $F[x]$, we have that the automorphisms on $\text{Gal}(E/F)$ are uniquely determined by their action on α . Let $\sigma, \tau \in \text{Gal}(E/F)$, then by **120.1 zeros of $x^n - a \in F[x]$ with field characteristic 0**, $\sigma(\alpha) = \alpha\omega^k, \tau(\alpha) = \alpha\omega^t$ for some $k, t \in \mathbb{N}$. Since $\omega \in F$ by supposition, then all n^{th} roots of unity are in F , and since σ, τ are homomorphisms that preserve F , then:

$$\begin{aligned} (\sigma \circ \tau)(\alpha) &= \\ \sigma(\tau(\alpha)) &= \sigma(\alpha\omega^t) = \sigma(\alpha)\omega^t = \\ \alpha\omega^k\omega^t &= \alpha\omega^t\omega^k = \\ \tau(\alpha)\tau(\omega^k) &= \tau(\alpha\omega^k) = \tau(\sigma(\alpha)) = \\ &= (\tau \circ \sigma)(\alpha) \end{aligned}$$

Since $\sigma \circ \tau$ is defined by its action on σ , we find that $\sigma \circ \tau = \tau \circ \sigma$, i.e. $\text{Gal}(E/F)$ is abelian, hence by **74 abelian groups are solvable**, $\text{Gal}(E/F)$ is solvable.

Othewise, $\omega \notin F$, then $F \subset F(\omega) \subset E$, so $x^n - a$ must have an irreducible factor in F since it can have at most one zero in F . Since F has characteristic 0, and E is a splitting field for this factor, then E/F is a Galois extension, hence $\text{Gal}(E/F)$ is a Galois extension, and by **Theorem 119 fundamental theorem of Galois theory (FTGT)** we have that $\text{Gal}(E/F(\omega)) \leq \text{Gal}(E/F)$. Also, since $F(\omega)$ is the splitting field for $x^n - 1$ which is separable, then $F(\omega)/F$ is a Galois extension, so again by FTGT, we have $\text{Gal}(E/F(\omega)) \trianglelefteq \text{Gal}(E/F)$, and also $\text{Gal}(F(\omega)/F) \cong \text{Gal}(E/F)/\text{Gal}(E/F(\omega))$, so we have a normal series $\{\text{id}\} \trianglelefteq \text{Gal}(E/F(\omega)) \trianglelefteq \text{Gal}(E/F)$, and it remains to show $\text{Gal}(F(\omega)/F)$ is abelian. The proof is analogous to the one presented in the previous case: every $\sigma, \tau \in \text{Gal}(F(\omega)/F)$ is uniquely defined by its action on ω and it must send ω to another n^{th} root of unity, so $\exists t, k \in \mathbb{N}$ s.t. $\sigma(\omega) = \omega^k, \tau(\omega) = \omega^t$, and $\sigma \circ \tau, \tau \circ \sigma$ are uniquely defined by their action on ω as well, which agree via computation analogous to the one from the previous case, hence they are the same, hence $\text{Gal}(F(\omega)/F)$ is abelian, hence $\text{Gal}(E/F)$ is solvable

□

Theorem 123.

If F is a field of characteristic 0, and E is an extension by radicals of F , then there exists a normal series of radical extensions $F = K_0 \subseteq K_1 \subseteq \dots \subseteq K_n = K$, s.t. $E \subseteq K$.

Proof. Since E is an extension by radical of F , then by definition there exists a series $F = F_0 \subseteq F_1 \subseteq$

$F_2 \cdots \subseteq F_n = E$, s.t. $F_{i+1} = F_i(\sqrt[k_i]{\alpha_i})$ for some $\alpha_i \in F_i$ and some $k_i \in \mathbb{N}$.

First, let $F_0 = K_0$. For each F_{i+1} , we will find a matching K_{i+1} s.t. K_{i+1} is a normal extension over K_i by taking K_{i+1} to be a splitting field for $f_i(x) = x^{k_i} - \alpha_i$. Let ω_i be a generator of the k_i^{th} roots of unity (such ω_i exists by [Theorem 120 roots of unity form a cyclic abelian group](#)), then by [120.1 zeros of \$x^n - a \in F\[x\]\$ with field characteristic 0](#) we find that for some zero β_i of $f_i(x)$, we have that $F_i(\beta_i \omega_i)$ is a splitting field for $x^{k_i} - \alpha_i$, so it is a normal extension of F_i (by [Theorem 118 equivalent definitions for Galois group](#)). For $i = 1$, we let $K_1 = F_1(\beta_1 \omega_1)$, clearly K_1 is a normal extension of K_0 as well. Continuing along this fashion, we find a series of normal extensions $K_0 \subseteq K_1 \subseteq \cdots \subseteq K_n = K$, where K itself is an extension (not necessarily proper) of E , which completes the proof. $\square$

Theorem 124 (solvability by radicals implies Galois group is solvable).

If F is a field with characteristic 0 and $f(x) \in F[x]$ is solvable by radicals, then its Galois group is solvable.

Proof. Suppose $f(x)$ is solvable by radicals, and let E be its splitting field, then E is an extension by radicals over F by definition, so by [Theorem 123](#) there exists a normal series of radical extensions $F = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_n = K$ s.t. $E \subseteq K$. Since F has characteristic 0 and E is a splitting field for $f(x) \in F[x]$, then E/F admits a Galois extension by [Theorem 118 equivalent definitions for Galois group](#) hence it is also normal over F , so we can consider a normal series $F = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_n = E$.

Since this is a normal series, each F_{i+1} is a normal extension of F_i hence F_{i+1}/F_i is a Galois extension (since $\text{char}(F) = 0$), so by [Theorem 119 fundamental theorem of Galois theory \(FTGT\)](#) we have a normal series $\{\text{id}\} \trianglelefteq \text{Gal}(E/F_{n-1}) \trianglelefteq \cdots \trianglelefteq \text{Gal}(E/F)$. Again by [Theorem 119 fundamental theorem of Galois theory \(FTGT\)](#), each $\text{Gal}(E/F_i)/\text{Gal}(E/F_{i-1}) \cong \text{Gal}(F_{i-1}/F_i)$.

Recall that in our construction of the normal radical extensions F_i/F_{i-1} in [Theorem 123](#) we constructed F_i to be the splitting field for a polynomial of the form $x^{k_i} - \alpha_i \in F_{i-1}[x]$ for some $k_i \in \mathbb{N}$, $\alpha_i \in F_{i-1}$, then by [Theorem 122 Galois group of splitting field for \$x^n - a\$ is solvable](#) $\text{Gal}(F_{i-1}/F_i)$ is solvable, hence $\text{Gal}(E/F)$ is solvable, which completes the proof. $\square$

In fact, a stronger version of the theorem exists, which we will state without proof (the proof does not require any new machinery not covered here):

Theorem 125 (solvable by radicals iff Galois group is solvable).

If F is a field and $\text{char}(F) = 0$, and $f(x) \in F[x]$, then $f(x)$ is solvable by radicals if and only if its Galois group is solvable.

[Theorem 124 solvability by radicals implies Galois group is solvable](#) proves the $\implies$ direction. We do not make use of this theorem to demonstrate insolubility of the quintic, so the proof for the

other direction is omitted. We do, however, use it below, to provide a nice proof for the solvability of polynomials of degree less than 5 that does not make use of the general formulas themselves:

Corollary 125.1 (polynomials of degree less than 5 are solvable by radicals).

If F is a field and $\text{char}(F) = 0$, and $f(x) \in F[x]$ has $\deg(f(x)) < 5$, then $f(x)$ is solvable by radicals.

Proof. By 118.1 Galois group is isomorphic to subgroup of S_n and Theorem 121 subgroup of solvable group are solvable, it suffices to show that S_n is solvable for $n = \deg(f(x))$, then by Theorem 125 solvable by radicals iff Galois group is solvable it would follow that $\text{Gal}(E/F)$ the Galois group of $f(x)$ is solvable.

Note that S_1, S_2 are simply the trivial group and the group on two elements $\mathbb{Z}_2$ under an isomorphism, both of which abelian, hence by 74 abelian groups are solvable both are solvable.

For the cases where $n = 3, n = 4$, recall Theorem 28 $S_n/A_n \cong \mathbb{Z}_2$. Since $A_n \trianglelefteq S_n$ and also $\mathbb{Z}_2$ is abelian, then if A_n is solvable then we would have that S_n is also solvable.

For $n = 3$, we have that the alternating group on 3 letters A_3 consists of exactly 3 elements (verify this by listing out S_3 explicitly, to find that $A_3 = \{(1), (123) = (13)(12), (132) = (12)(13)\}$). Since 3 is prime there exists one group of order 3 up to an isomorphism, so $A_3 \cong \mathbb{Z}_3$, whose Cayley table we have already studied and we have found that $\mathbb{Z}_3$ is abelian, so it is solvable, so as we argued above S_3 is solvable (from solvability of A_3).

For $n = 4$, we find that $|A_4| = 12$, so it suffices to find a normal subgroup of A_4 of order 4, since then the order of their quotient would be $12 \div 4 = 3$, so their quotient will be isomorphic to $\mathbb{Z}_3$ which is abelian hence solvable, so it would follow that A_4 is solvable, hence S_4 is solvable. Indeed, $H = \{(1), (12)(34), (13)(24), (14)(23)\}$ is a normal subgroup of A_4 (this can be verified by direct calculation) and has order 4, so A_4 is solvable, so S_4 is solvable.

We have demonstrated that S_1, S_2, S_3, S_4 are all solvable, so polynomial with coefficients from a field of characteristic 0 with degree up to 5 are solvable by radicals.

Note that this line of reasoning fails for $5 \leq n$, since we relied on showing that A_n is solvable to show that S_n is solvable, yet by 121.1 A_n is not solvable for $5 \leq n$, A_n is not solvable in those cases.

□

Proposition 75.

If F is a field of characteristic 0 and $f(x) \in F[x]$ is a polynomial of degree at least 5, and the Galois field for $f(x)$ is isomorphic to $S_{\deg(f(x))}$, then $f(x)$ is insolvable by radicals.

Proof. Suppose by contradiction $f(x)$ is solvable by radicals, then by Theorem 124 solvability by radicals implies Galois group is solvable, its Galois group is solvable, so $S_{\deg(f(x))}$ is solvable (since

by supposition the two are isomorphic, and it is left as an exercise to the reader to demonstrate that solvability of a group is preserved under group isomorphism), which contradicts 121.2 S_n is not solvable for $5 \leq n$, hence $f(x)$ is insolvable by radicals. $\square$

We use this proposition as a guideline for finding a quintic $f(x)$ that is insolvable by radicals: first, recall that since 5 is prime then by Theorem 16 it suffices to find that $\text{Gal}(E/F)$ contains a transposition on the zeros of $f(x)$ and a cycle of order 5 on the zeros of $f(x)$, since then we would have by the theorem that $\text{Gal}(E/F) \cong S_5$, so by 75 $f(x)$ is insolvable by radicals.

A transposition is easily achievable by taking a polynomial $f(x) \in \mathbb{Q}[x]$ s.t. $f(x)$ has a complex root, then since complex conjugation preserves $\mathbb{Q}$ and is an automorphism on its splitting field E , it would follow that $\text{Gal}(E/\mathbb{Q})$ has a transposition (which exchanges the two complex roots of the polynomial). Next, if $f(x)$ is irreducible in $\mathbb{Q}$, and $\alpha \in \mathbb{R}$ is a real root for $f(x)$, then $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 5$ (note that this is *not* a Galois extension since $f(x)$ doesn't split in $\mathbb{Q}(\alpha)$), and since $[E : \mathbb{Q}] = [E : \mathbb{Q}(\alpha)][\mathbb{Q}(\alpha) : \mathbb{Q}]$, then $[E : \mathbb{Q}]$ must be divisible by 5, hence $\text{Gal}(E/\mathbb{Q})$ is a finite group with order divisible by 5. By Theorem 47 Cauchy's theorem since 5 is prime and $5 \mid |\text{Gal}(E/\mathbb{Q})|$, then $\exists \tau \in \text{Gal}(E/\mathbb{Q})$ that has order 5, so τ is a permutation on all 5 distinct zeros of $f(x)$, so τ corresponds to a cycle of order 5 (see also 18 order of a cycle).

The following example brings it all together:

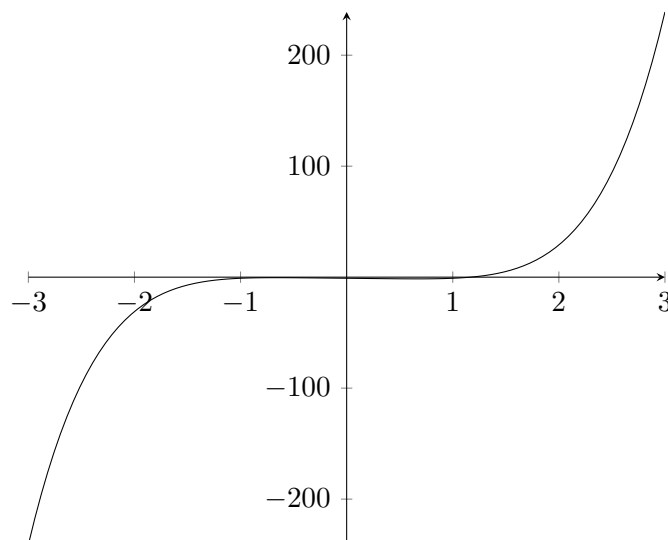
Example 125.1 ($x^5 - x - 1 \in \mathbb{Q}[x]$ is insolvable by radicals).

By **Theorem 76** with $p = 5$ we find that $x^5 - x - 1 \in \mathbb{Q}[x]$ is irreducible over $\mathbb{Q}$. From real analysis, we know $f(x) = x^5 - x - 1 \in \mathbb{R}[x]$ is continuous, and since $f(0) = -1$ and $f(2) = 2^5 - 2 - 1 = 29$, by the intermediate value theorem we know that $\exists \alpha$ in the real interval $[0, 2]$ s.t. $f(\alpha) = 0$ (since $f(0) = -1$, $f(2) = 29$ and -1029). It follows that $f(x)$ has a real zero α , hence $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 5$.

Again from real analysis, we find that $f'(x) = 5x^4 - 1$ which has two zeros (this can be proven by **125.1 polynomials of degree less than 5 are solvable by radicals** by substituting x^2 for y , then we have the quadratic $5y^2 - 1 = 0$), which by Rolle's theorem means that there are at most two extremas of the function in $\mathbb{R}$, so there are at most 3 real solutions (this is also a result from real analysis that follows from the IVT and Rolle's theorem). Since $\mathbb{Q}$ has characteristic 0 and $x^5 - x - 1$, then it has 5 distinct zeros by **Theorem 101 criterion for multiple zeros**. By the fundamental theorem of algebra, which we will soon prove independently of this example, $x^5 - x - 1$ splits in $\mathbb{C}$, hence the two remaining results are complex.

Let $E/\mathbb{Q}$ be the splitting field for $x^5 - x - 1$, then by **Theorem 118 equivalent definitions for Galois group** it is a Galois extension, and as argued above $\exists \sigma \in \text{Gal}(E/\mathbb{Q})$ which is complex conjugation since $x^5 - x - 1$ has complex zeros, so $\text{Gal}(E/\mathbb{Q})$ has a transposition on the zeros of the polynomial. Also, since $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 5$, then as argued before the example $\text{Gal}(E/\mathbb{Q})$ contains a 5-cycle, hence by **Theorem 16**, $\text{Gal}(E/\mathbb{Q}) \cong S_5$, so by **75**, $x^5 - x - 1$ is insolvable by radicals.

The plot for $x^5 - x - 1$ is presented below for clarity:



Fundamental Theorem of Algebra

We end our discussion, fittingly, with a proof of the fundamental theorem of algebra. First, we collect a result from real analysis, using the IVT:

Lemma 125.1 (every polynomial of odd degree with real coefficients has a zero in $\mathbb{R}$).

If $f(x) \in \mathbb{R}[x]$ is a polynomial then if $f(x)$ has odd degree, it has a zero in $\mathbb{R}$.

Proof. Let $f(x) \in \mathbb{R}[x]$ have odd degree. From limit arithmetic we know that the limit of $f(x)$ as $x \rightarrow \pm\infty$ is dominated by its leading term $a_n x^n$. Since n is odd, then for any $x \in \mathbb{R}$, we have $a_n x^n = -a_n(-x)^n$. Suppose w.l.o.g that a_n is positive (otherwise we prove the same result for $-f(x)$ which will have positive leading coefficient, and every zero of $-f(x)$ is also a zero of $f(x)$ since they decompose into the same factors up to the sign), then $f(x) \lim_{x \rightarrow \infty} \infty$, and $f(x) \lim_{x \rightarrow -\infty} -\infty$ (since $a_n x^n$ is odd for n odd), so for large enough x we have that $f(x) > 0$, $f(-x) < 0$. Since every polynomial is continuous then from the IVT, we know that for every $y \in [f(-x), f(x)]$, in particular for $y = 0$, $\exists x_0 \in [x, -x]$ which is an interval in $\mathbb{R}$, s.t. $f(x_0) = y = 0$, hence $f(x)$ has a zero in $\mathbb{R}$. $\square$

Essentially, the above result stems from the completeness of $\mathbb{R}$.

Lemma 125.2 ($ax^2 + bx + c \in \mathbb{R}[x]$ splits in $\mathbb{C}$).

Let $f(x) \in \mathbb{R}[x]$ s.t. $\deg(f(x)) = 2$, then $f(x)$ splits in $\mathbb{C}$.

Proof. Let $f(x) \in \mathbb{R}[x]$ be a polynomial of degree 2, then $f(x) = ax^2 + bx + c$ for some $a, b, c \in \mathbb{R}[x]$, then by the quadratic formula the zeros of this equation are $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Consider $b^2 - 4ac$. If $b^2 - 4ac$ is non-zero then its root is in $\mathbb{R}$ by completeness of $\mathbb{R}$ hence $f(x)$ splits in $\mathbb{R}$ so it obviously splits in $\mathbb{C}$. Suppose $b^2 - 4ac = 0$, then $4ac = b^2$ and $(-1)(4ac - b^2) = b^2 - 4ac$. Since $4ac - b^2$ is greater than zero then $\sqrt{4ac - b^2} \in \mathbb{R}$ again by completeness so $\sqrt{4ac - b^2} \in \mathbb{C}$, and also $i \in \mathbb{C}$ satisfies $i^2 = -1$, hence $i\sqrt{4ac - b^2} \in \mathbb{C}$, which squares to $b^2 - 4ac$, hence $\sqrt{b^2 - 4ac} \in \mathbb{C}$, hence $f(x)$ splits in $\mathbb{C}$. $\square$

Lemma 125.3 ($ax^2 + bx + c \in \mathbb{C}[x]$ splits in $\mathbb{C}$).

If $f(x) \in \mathbb{C}[x]$ is a polynomial of degree 2, then it splits in $\mathbb{C}$.

Proof. Let $f(x) = ax^2 + bx + c \in \mathbb{C}[x]$. Following a similar outline to that of **125.2** $ax^2 + bx + c \in \mathbb{R}[x]$ splits in $\mathbb{C}$, it suffices to show that every complex number has a square root in $\mathbb{C}$, hence there exists an element $\alpha \in \mathbb{C}$ s.t. $\alpha = \sqrt{b^2 - 4ac}$, so by the quadratic formula $f(x)$ splits in $\mathbb{C}$. Consider a general element in $\mathbb{C}$ expressed in polar coordinates in exponent form as $re^{i\theta}$ for some $r \in \mathbb{R}^+ \cup \{0\}$, $\theta \in [-\pi, \pi]$. Consider the element $\sqrt{r}e^{i\frac{\theta}{2}}$, since $\mathbb{C}$ is abelian then by **Theorem 2 exponent laws (power rules)**, $(\sqrt{r}e^{i\frac{\theta}{2}})^2 = (\sqrt{r})^2 e^{i\frac{\theta}{2} \cdot 2} = re^{i\theta}$. From completeness of $\mathbb{R}$, it follows that $\sqrt{r} \in \mathbb{R}$ so also in

$\sqrt{r}e^{i\frac{\phi}{2}} \in \mathbb{C}$, so we can take the square root of a number in $\mathbb{C}$, so we can carry the quadratic formula in $\mathbb{C}$ for every polynomial in $\mathbb{C}$, which completes the proof. $\square$

We now return to Galois theory to prove the following lemma:

Lemma 125.4.

Let K be a field. If $\forall f(x) \in K[x]$ of odd degree, $f(x)$ has a zero $\alpha \in K$, and L/K is a finite proper Galois extension of K , then $\text{Gal}(L/K)$ is a 2-group. (i.e. has order divisible by 2)

Proof. Let K be a field and let L/K be a finite Galois extension of K , and suppose $L \neq K$ (otherwise L/K is not a proper extension), then by **Theorem 110 primitive element theorem (Steinitz, 1910)** there exists $\beta \in L$ s.t. $L = K(\beta)$. Since any finite extension is algebraic (**Theorem 107 finite extensions are algebraic**) then there exists a minimal polynomial $f(x) \in K[x]$ s.t. $f(\beta) = 0$, and $\deg(f(x)) = [K(\beta) : K]$. Since by supposition every polynomial of odd degree with coefficients in K has a zero in K , then $\deg(f(x))$ is even, hence $2 \mid [K(\beta) : K]$, hence $[L : K] = 2^n q$ for some $n, q \in \mathbb{N}$ s.t. q is odd.

Since L/K is a Galois extension, then by **Theorem 119 fundamental theorem of Galois theory (FTGT)**, $\text{Gal}(L/K)$ has order $2^n q$ as well, denote it G . By **Theorem 48 Sylow's first theorem** it has a 2-subgroup $H \leq G$. By **Theorem 49 Sylow's third theorem**, $[G : H] = q$, i.e. the index of H in G is odd.

Let E^H be the field fixed by H , again by **Theorem 119 fundamental theorem of Galois theory (FTGT)** we have $K \subseteq E^H \subseteq L$ and $[E^H : K] = [\text{Gal}(L/K) : \text{Gal}(L/E^H)]$, and by **114.2 Aut(E/E^G) = G** , $\text{Gal}(L/E^H) = H$ (the subgroup of $\text{Gal}(L/K)$ that fixes H) hence $[E^H : K] = [G : H] = q$, so $[E^H : K]$ is odd. Suppose $E^H \neq K$, then since E^H/K is also a finite Galois extension (by FTGT), then it is finite and separable hence by the primitive element theorem, $E^H = K(\gamma)$ for some γ algebraic over K , so $[E^H : K] = \deg(g(x))$ where $\text{irr}(\gamma, K) = g(x)$, but then we have irreducible polynomial of odd degree, which is a contradiction, so $E^H = K$, so $q = 1$.

Hence, we have that $|\text{Gal}(L/K)| = 2^n$, so it is a 2-group by definition.

The following diagram illustrates the relations between the fields and groups in the proof:

$$\begin{array}{ccc}
 L & \longrightarrow & \{\text{id}\} \\
 \updownarrow & & \updownarrow \\
 E^H & \longrightarrow & H \\
 \updownarrow & & \updownarrow \text{2-subgroup} \\
 K & \longrightarrow & G
 \end{array}$$

$\square$

Definition 0.0.1 (algebraic closure).

A field F is called algebraically closed if every non-constant polynomial in $F[x]$ has a zero in F . An extension E/F is called an algebraic closure of F if every non-constant polynomial in $F[x]$ has a zero in E .

Proposition 76 (no finite field is algebraically closed).

If F is a finite field, then F is not algebraically closed

Proof. Let F be a finite field and consider an enumeration $a_1, \dots, a_n$ on its elements, then consider the polynomial $f(x) = \prod_{a_i \in F} (x - a_i) + 1 \in F[x]$. Suppose $f(x)$ has a zero $\alpha \in F$, then $(x - \alpha)$ is one of $x - a_i$ for some i , so $f(x)$ is in the ideal generated by $(x - a_i)$, so $x - a_i$ divides $f(x)$, but evidently the remainder of this division is 1 by construction of $f(x)$, a contradiction, hence $f(x)$ is irreducible in F . It follows that F is not algebraically closed. $\square$

Theorem 126 (fundamental theorem of algebra).

$\mathbb{C}[x]$ is algebraically closed.

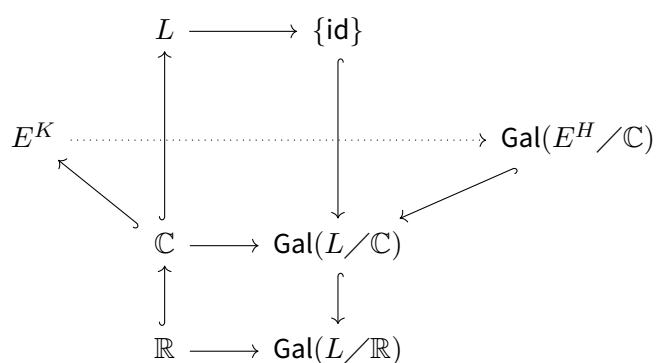
Proof. It suffices to show that there are no irreducible non-constant polynomials in $\mathbb{C}[x]$. Suppose by contradiction $\exists f(x) \in \mathbb{C}[x]$ s.t. $f(x)$ is irreducible over $\mathbb{C}$, and let L be a splitting field for $f(x)$ over $\mathbb{C}$ (L exists by **Theorem 96 splitting field existence theorem**), then $L/\mathbb{C}$ is a Galois extension (since $f(x)$ is separable since $\text{char}(\mathbb{C}) = 0$ and L is a splitting field for $f(x)$). Note that this implies $\mathbb{C} \subset L$ and in particular $\mathbb{C} \neq L$.

Since $\mathbb{R} \subset \mathbb{C} \subset L$ and $\mathbb{C}$ is a splitting field for $x^2 + 1$ over $\mathbb{R}$, then L is a splitting field for $(1 + x^2)f(x)$ hence $L/\mathbb{R}$ is also a Galois extension since it is the splitting field of a separable polynomial (separable since $\text{char}(\mathbb{R}) = 0$). By **125.1 every polynomial of odd degree with real coefficients has a zero in $\mathbb{R}$** , $\mathbb{R}$ qualifies for **125.4**, so we have that $\text{Gal}(L/\mathbb{R})$ is a Sylow 2-group.

By **Theorem 119 fundamental theorem of Galois theory (FTGT)**, $\text{Gal}(L/\mathbb{C}) \leq \text{Gal}(L/\mathbb{R})$, so its order must divide the order of $\text{Gal}(L/\mathbb{R})$ by **Theorem 21 Lagrange's theorem**, so it is also a 2-group. Since $\text{Gal}(L/\mathbb{C})$ is a 2-group then in particular $2 \mid |\text{Gal}(L/\mathbb{C})|$, so it has a subgroup of order exactly 2 by **Theorem 47 Cauchy's theorem**.

Let K be that subgroup, and let E^K be the field fixed by K , by **114.2 $\text{Aut}(E/E^G) = G$** and **114.3 equivalent definition of Galois group**, $K = \text{Gal}(E^K/\mathbb{C})$, so $E^K/\mathbb{C}$ is a finite Galois extension and by FTGT $[E^K : \mathbb{C}] = |K| = 2$, so by **Theorem 118 equivalent definitions for Galois group**, E^K is the splitting field of an irreducible polynomial of degree 2 in $\mathbb{C}$, but by **125.3 $ax^2 + bx + c \in \mathbb{C}[x]$ splits in $\mathbb{C}$** , there is no such polynomial, a contradiction, so $L = \mathbb{C}$, another contradiction, hence $\mathbb{C}[x]$ is algebraically closed.

The following diagram illustrates the Galois correspondences and structures involved in the proof:



□

Which also immediately provides the following weaker statement:

Corollary 126.1 ($\mathbb{C}$ is an algebraic closure of $\mathbb{R}$).

$\mathbb{C}$ is an algebraic closure of $\mathbb{R}$.

Proof. Immediately from [Theorem 126 fundamental theorem of algebra](#) and [0.0.1 algebraic closure](#).

□